Book I of Euclid's Elements

Reconstruction

CGJteamLab

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Contents

Introduction

From Hilbert's Foundations to Synthetic Practice

This volume documents a formal reconstruction of the first part of Euclid's *Elements*, Book I, over a Hilbert-style foundation in Lean 4.

The original objective was comparatively modest: to determine whether classical Euclidean proofs could be reconstructed faithfully in a modern proof assistant without replacing synthetic geometry by coordinates, real-number measurement, or algebraic computation.

The reconstruction of Propositions I.1–I.32 has revealed a more specific problem.

Hilbert's axioms provide a foundation for geometry. They separate incidence, order, congruence, and continuity, and they make explicit many assumptions that remain implicit in Euclid. But there is still a substantial distance between such a foundational system and the language in which ordinary synthetic proofs are actually carried out.

A mathematician moves almost invisibly between statements such as:

- three points are collinear in a different order;
- one point lies between two others;
- two descriptions determine the same ray;
- an angle may be transported along congruent rays;
- a proper part of a segment is smaller than the whole;
- equal parts may be added to or removed from segment inequalities;
- a configuration suggested by a diagram really has the required order and separation properties.

None of these operations is mysterious. But neither are they merely syntactic details. In a formal proof they must be represented by definitions, lemmas, and explicit chains of inference.

This is the layer that the present reconstruction has progressively made visible.

Book Zero

Beeson, Narboux, and Wiedijk use the informal name *Book Zero* for fundamental preliminary theorems that are used in Euclid's Book I but do not themselves depend on propositions of Book

I. Their Book Zero includes elementary results about congruence, betweenness, collinearity and noncollinearity, segment order, rays, laying off segments, angle comparison, and plane separation.[1]

The CGJteamLab development adopts this terminology. In the present Hilbert-style reconstruction, Book Zero plays the same structural role, but now specifically as an intermediate theory between Hilbert's foundational axioms and the level at which ordinary synthetic arguments are naturally conducted. The formalization of Book I suggests the following interpretation.

Book Zero functions as an intermediate language between Hilbert's axioms and classical synthetic reasoning:

$$\boxed{\text{Hilbert axioms} \longrightarrow \text{Book Zero} \longrightarrow \text{classical synthetic geometry}}$$

Its purpose is not to replace Hilbert's foundation and not to introduce a second axiomatic system. Rather, it packages the elementary consequences of that foundation at the level at which geometric arguments naturally operate.

This role becomes particularly clear in Proposition I.21. There the proof requires what looks informally like arithmetic with segments: parts are added, removed, compared, and inequalities are transported through congruence. No numerical length function is necessary. Betweenness, congruence, and strict segment comparison already carry the required arithmetic through geometric witnesses.

Thus Book Zero is not merely a library for shortening Lean proofs. It is an attempt to identify the missing operational layer between foundational geometry and working synthetic mathematics.

Euclid and Hilbert Do Not Have the Same Dependency Graph

The reconstruction also makes clear that the order of Euclid's propositions cannot simply be identified with the logical dependency order of a Hilbert-style theory.

Several different situations occur.

First, some Euclidean propositions correspond to principles that are already foundational in Hilbert geometry. Euclid's Proposition I.4, for example, proves SAS by superposition, whereas SAS belongs to the Hilbert congruence structure. Similarly, the angle-copying construction of Proposition I.23 is already contained in Hilbert's angle-construction principle.

Second, some Euclidean propositions genuinely require more than neutral Hilbert geometry. Propositions I.1 and I.22 expose intersection or continuity assumptions that cannot be obtained from incidence, order, and congruence alone. A different boundary appears at Proposition I.29. The criteria I.27 and I.28 derive parallelism from angle relations in neutral geometry, whereas I.29 reverses the direction and requires the Euclidean uniqueness of parallels.

The distinction is sharper than the numerical order of the *Elements* suggests. Proposition I.31 comes after I.29, but its existence statement is already neutral: an angle can be transported and I.27 then proves that the constructed line is parallel. Only uniqueness of that parallel requires the Euclidean parallel principle. Thus the sequence of propositions and the logical layers of the theory cross one another.

Third, and most importantly for the present project, many propositions require no stronger

[1]Michael Beeson, Julien Narboux, and Freek Wiedijk, "Proof-checking Euclid," *Annals of Mathematics and Artificial Intelligence* 85 (2019), 213–257, doi:10.1007/s10472-018-9606-x.

axiom but depend on a considerable intermediate theory of order, rays, segment comparison, angle comparison, noncollinearity, separation, and transport.

It is this third class that explains much of the apparent complexity of formal synthetic geometry.

What Formalization Reveals

A classical diagram compresses information.

It may show immediately that a point lies inside an angle, that two points occur on the same ray, that one point is between two others, or that an auxiliary construction has a particular orientation. A proof assistant cannot obtain these facts by looking at the figure.

Consequently formalization repeatedly exposes obligations that are easy to overlook in textbook prose:

$$\text{diagrammatic configuration} \quad \not\Rightarrow \quad \text{formal order information.}$$

Proposition I.24 provides a particularly clear example. Euclid's construction suggests one relative position of the auxiliary points, but that position is not forced merely by the stated construction. The formal proof must analyze the possible configurations explicitly.

The same phenomenon reappears in the theory of parallels. In I.27 the informal argument must exclude an intersection on either side of the transversal. In I.29 the decisive step is not another angle calculation but uniqueness: a neutrally constructed candidate parallel must be identified with the given parallel. Proposition I.31 then shows that existence and uniqueness, often compressed into one familiar statement, belong to different logical layers.

Such phenomena should not be regarded simply as inconveniences caused by Lean. They reveal logical structure already present in the mathematics.

The Purpose of This Documentation

The chapters that follow are therefore not intended merely as line-by-line explanations of Lean source files.

For each proposition we record several distinct objects:

- Euclid's classical construction and proof;
- the corresponding reconstruction over the Hilbert foundation;
- the actual dependency architecture of the Lean theorem;
- the Hilbert and Book Zero components used in the proof;
- formal obligations hidden by the traditional diagram;
- the relationship of the proposition to earlier and later propositions of Book I.

The appendices compare this reconstruction with the Hilbert-style developments of Greenberg and Hartshorne and with Wilkins's detailed study of the internal dependency structure of

Euclid's Book I. These comparisons are not included merely as historical commentary. They help distinguish genuine foundational dependencies from intermediate lemmas that a formal development must make explicit, and they expose alternative proof paths, converse relations, and configuration cases hidden by the linear order of the *Elements*.

Current Scope

The present volume covers Euclid Book I through Proposition I.32.

Thirty-two propositions are still too small a sample from which to claim a final architecture for synthetic geometry. They do, however, include the first transition from neutral geometry to the Euclidean theory of parallels. The present development should therefore be understood as an experiment in progress, but one that now crosses a genuine foundational boundary.

Nevertheless, the evidence is already substantial enough to identify a recurring structure.

Hilbert supplied the foundational theory. Euclid supplied the classical sequence of geometric problems and proofs. Between them lies a layer of elementary synthetic operations that is normally left implicit because mathematicians use it automatically.

Formal reconstruction forces that layer into view.

The principal question of the project is therefore no longer only

> Can Euclid's proofs be formalized over Hilbert geometry?

It has become the more structural question

> What intermediate theory is required to turn Hilbert's axioms into a practical language for classical synthetic geometry?

The reconstruction of Book I is one way of investigating that question.

Chapter 1

Proposition I.1

Euclid's first proposition in Book I is usually presented as the straightforward construction obtained from the intersection of two circles. A formal reconstruction, however, reveals two logically independent problems: the existence of a common point of the corresponding circles and the proof that the resulting point does not lie on the line determined by the given base.

The proof of non-collinearity relies on the classical idea that a proper part of a segment cannot be congruent to the entire segment. This argument is well known from modern axiomatic reconstructions of geometry, including the work of Michael Beeson and his collaborators.

The purpose of this note is not to present a new geometric proof. Rather, it is to demonstrate that, once Book Zero has been developed, this argument becomes a direct application of the previously established theory. Proposition I.1 therefore serves as the first experimental test of Book Zero as an intermediate language for synthetic proofs.

1.1 The Classical Configuration and Construction

Given a nondegenerate segment AB, Euclid constructs an equilateral triangle on it.

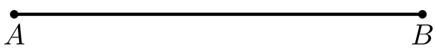


Figure 1.1: The data of Proposition I.1: a nondegenerate segment AB.

Draw the circle centered at A through B, and the circle centered at B through A. Choose an intersection point C, and join C to A and B.

The metric conclusion is immediate from the definitions of the two circles:

$$AC \cong AB, \qquad BC \cong AB.$$

Therefore the three sides are equal and ABC is equilateral.

Diagrammatic audit. The four successive classical snapshots have been reduced to the initial segment and the completed two-circle construction. The intermediate states merely add the first circle and then the second; they are not distinct geometric cases.

Wilkins draws attention to the genuinely nontrivial point hidden by the familiar picture: Euclid tacitly assumes that the two circles intersect. Thus the diagram should not be read as a proof of

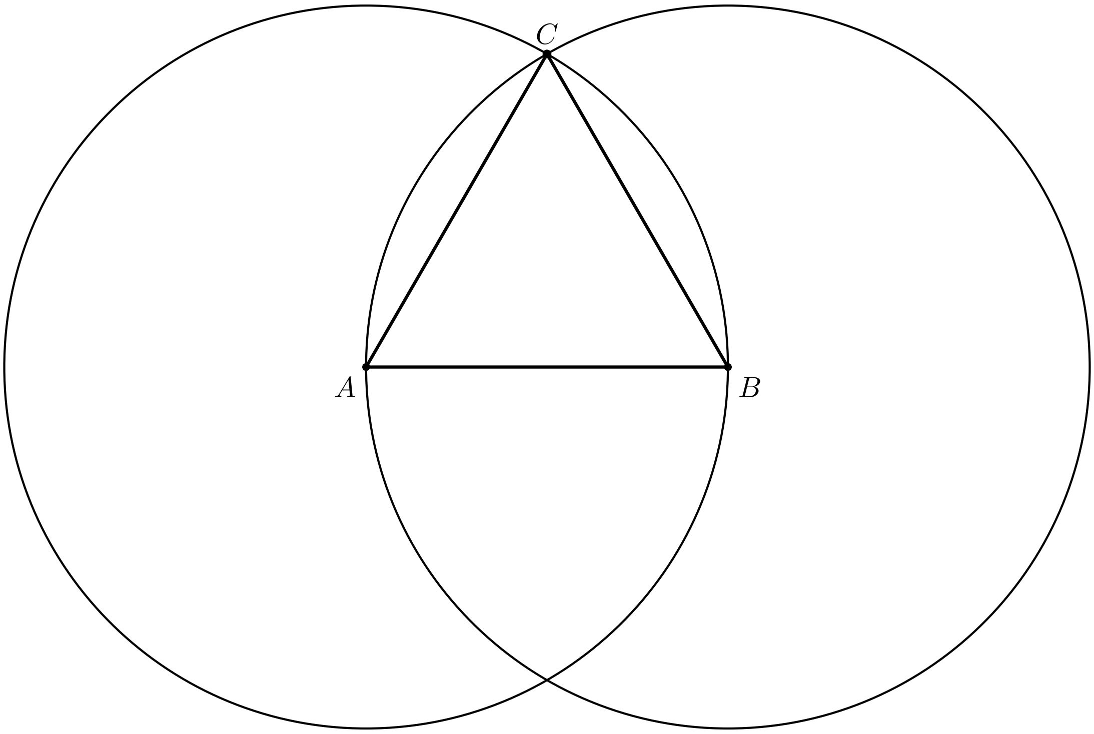


Figure 1.2: The complete classical construction. Since C lies on both circles, $AC = AB$ and $BC = AB$; hence $AC = BC$ by Common Notion 1.

intersection. The existence of C requires an additional circle-intersection or continuity principle beyond the mere permission to draw the two circles.

1.2 Separating the Two Problems

The reconstruction reveals two logically independent stages.

1. **Existence stage.** One must obtain a point C such that

$$AC \cong AB, \qquad BC \cong AB.$$

   In the current version of the project, this stage is provided by the temporary principle `hilbert_equidistant_point_exists`. Ultimately, it is intended to be derived from an appropriate theory of continuity in Hilbert geometry.

2. **Synthetic stage.** From the congruences above one must derive

$$\neg \operatorname{Collinear}(A, B, C).$$

   This stage no longer requires any theory of circles.

1.3 Theorem

Theorem. Let $A \neq B$. Assume that there exists a point C such that

$$AC \cong AB, \qquad BC \cong AB.$$

Then the points A, B, C are not collinear.

1.4 Proof

The formal proof separates existence of an equidistant point from proof that the point is genuinely off the base line.

First `hilbert_equidistant_point_exists` supplies a point C with

$$AC \cong AB, \qquad BC \cong AB.$$

The proof then excludes the degenerate possibilities $C = A$ and $C = B$ using the null-segment theorem.

It remains to prove noncollinearity. Assume that A, B, C are collinear. Since the three points are pairwise distinct, betweenness trichotomy gives exactly three possibilities:

$$A - B - C, \qquad B - A - C, \qquad A - C - B.$$

In every case one of the congruent segments would be a proper part of the other. Book Zero #45, `bookZero_45_partNotEqualWhole`, rules this out. Hence A, B, C are not collinear.

This is a significant foundational difference from Euclid. The formal proof does not construct C as an intersection of two circles; existence is provided by the Hilbert interface, and Book Zero order theory is then used to prove that the witness cannot be collinear with A, B.

□

1.5 Proof Architecture of the Reconstruction

```
Euclid
------
segment AB
   |
circle center A through B
+ circle center B through A
   |
intersection C
   |
AC = AB and BC = AB
   |
Common Notion 1
   |
equilateral ABC


Hilbert / Lean
--------------
A != B
   |
hilbert_equidistant_point_exists
   |
C with AC ~= AB and BC ~= AB
   |
exclude C=A and C=B
```

```
   |
assume A,B,C collinear
   |
betweenness trichotomy
   |
bookZero_45_partNotEqualWhole
in each of the three cases
   |
not collinear A B C
   |
euclid_proposition_1
```

1.6 Significance of the Reconstruction

The geometric result itself is not new. Constructing an equilateral triangle on a given segment is one of the classical theorems of elementary geometry.

Likewise, the idea underlying the proof of non-collinearity is not new. Modern axiomatic reconstructions, in particular the work of Michael Beeson and his collaborators, employ an analogous argument based on the theorem that a proper part of a segment cannot be congruent to the entire segment.

The novelty of the present reconstruction lies elsewhere.

The objective of the CGJteamLab project was not to discover a new proof of Proposition I.1, but to investigate whether the theory developed in Book Zero is sufficiently rich to serve as an intermediate language for synthetic proofs.

The experiment shows that, once the existence problem for an equidistant point has been isolated, the entire argument can be expressed almost exclusively in terms of the theorems previously established in Book Zero.

From the perspective of the architecture of the theory, this is more significant than Proposition I.1 itself. It demonstrates that Book Zero is not merely a collection of auxiliary lemmas, but rather forms a new layer of abstraction between Hilbert's axioms and classical synthetic reasoning.

Schematically, the situation may be represented as

$$\text{Hilbert's axioms} \longrightarrow \text{Book Zero} \longrightarrow \text{Proposition I.1.}$$

In this sense, Proposition I.1 constitutes the first experimental test confirming that Book Zero can serve as a language of synthetic proofs for the further reconstruction of Euclidean geometry.

1.7 Relationship with the Foundations

Proposition I.1 has no earlier proposition of Book I on which it can depend. It therefore exposes the foundational boundary of the reconstruction more directly than any later proposition.

Euclid begins Book I with a circle-based construction: two circles with equal radii are drawn from the endpoints of the given segment, and an intersection point supplies the third vertex of the equilateral triangle.

In the Hilbert reconstruction, however, the essential issue is not segment congruence. Constructing points at prescribed distances is already supported by the congruence layer. The difficult step is to obtain a point realizing both distance conditions simultaneously and in the required nondegenerate configuration.

Thus I.1 immediately tests the interaction of

$$\text{incidence} \quad + \quad \text{order} \quad + \quad \text{congruence}$$

at the very beginning of Book I.

1.8 Hilbert and Book Zero Components Used

1.8.1 Segment Congruence

The equilateral conclusion is expressed synthetically:

$$AB \cong AC \qquad \text{and} \qquad AB \cong BC.$$

No numerical length function is used.

1.8.2 Betweenness Case Analysis

The reconstruction separates the possible collinear orders of three points. The three cases

$$A - B - C, \qquad B - A - C, \qquad A - C - B$$

are handled explicitly.

This is one of the earliest examples of a recurring formalization phenomenon: a configuration that is visually excluded in Euclid's diagram must be ruled out logically from order and congruence.

1.8.3 Book Zero Order Lemmas

The contradiction in each collinear branch depends on elementary results expressing that a proper part of a segment cannot be congruent to the whole segment.

These low-level theorems are precisely the kind of facts collected in Book Zero so that Book I proofs do not repeatedly reconstruct elementary betweenness arithmetic.

1.8.4 Noncollinearity

An equilateral triangle must be a genuine triangle. The formal construction therefore requires not only three pairwise congruent side relations but also a proof that the third point does not lie on the given line.

This nondegeneracy obligation is explicit in Lean and becomes an important pattern throughout the rest of Book I.

1.9 Formalization Notes

The complete implementation is available in the repository. Two excerpts show the mathematical structure.

Existence and the two equal-distance facts arrive together:

```
obtain <C, hAC, hBC> :=
  hilbert_equidistant_point_exists Geo A B hAB
```

After degeneracies are excluded, collinearity is split into the three betweenness cases:

```
rcases
    hilbert_between_trichotomy
      Geo A B C hAB hBCne hACne hCol with
  hABC | hBAC | hACB
```

Each branch is closed by `bookZero_45_partNotEqualWhole`, with the necessary congruence orientation supplied by symmetry or endpoint reversal. Those branch details are verification bookkeeping; the common geometric idea is simply that a proper part cannot equal the whole.

1.10 Dependency Summary

```
Classical:
Postulate III + circle intersection + Common Notion 1 -> I.1

Formal:
hilbert_equidistant_point_exists
        +
null-segment exclusion
        +
betweenness trichotomy
        +
bookZero_45_partNotEqualWhole
        |
        v
euclid_proposition_1
```

The formal theorem therefore replaces Euclid's circle-intersection mechanism by an abstract equidistant-point existence theorem and an explicit proof of noncollinearity.

1.11 Conclusion

Proposition I.1 is the natural starting point of Euclid's Book I, but its formal reconstruction immediately reveals that "starting point" does not mean "foundationally primitive."

Euclid's two-circle diagram suppresses elementary facts about point order, segment comparison, and nondegeneracy. Once these are made explicit, the proof reaches below Book I into the Hilbert and Book Zero layers.

The reconstruction therefore establishes an important theme for the entire project at the very first proposition: Book I is the visible classical development, while Book Zero supplies the elementary synthetic language needed to make its diagrammatic reasoning precise.

From this point onward, later propositions can reuse that language instead of rebuilding it. Proposition I.1 is consequently both the first Euclidean construction and the first demonstration of why the formal reconstruction requires a substantial interface beneath Euclid's text.

Chapter 2

Proposition I.2

This note compares Euclid's original construction for Proposition I.2 with its reconstruction over the Hilbert foundation. In the CGJteamLab development the proposition follows immediately from Book Zero #49 (Layoff), illustrating the difference between the architectural roles of segment construction in Euclid's geometry and the Hilbert axiomatic framework.

2.1 Statement

Given a point A and a segment BC, construct a point D such that

$$AD \cong BC.$$

2.2 Euclid's Classical Construction

Let A be the prescribed endpoint and BC the given segment. Euclid constructs a segment beginning at A and equal to BC.

Figure 2.1: The data of Proposition I.2: a prescribed point A and a given segment BC.

Join A to B, and construct the equilateral triangle ABD on AB by Proposition I.1. Draw the circle centered at B through C, and produce DB until it meets that circle at G. Next draw the circle centered at D through G, and produce DA until it meets this second circle at L.

The proof can be summarized by

$$BC = BG, \qquad DG = DB + BG, \qquad DL = DA + AL,$$

together with

$$DB = DA, \qquad DG = DL.$$

Subtracting the equal sides DB and DA from the equal radii DG and DL gives $BG = AL$. Since $BG = BC$, it follows that

$$AL = BC.$$

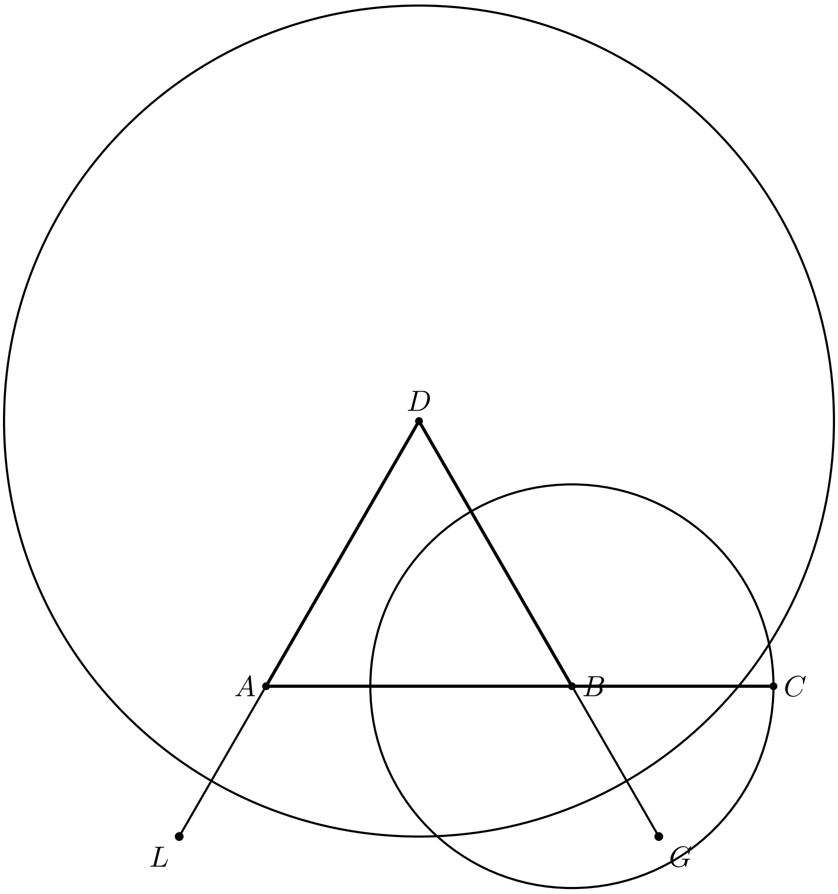


Figure 2.2: The complete classical construction. The first circle gives $BG = BC$; the second gives $DG = DL$. Since ABD is equilateral, $DB = DA$, and subtraction of equals yields $AL = BG = BC$.

Diagrammatic audit. The six successive construction snapshots in the previous version have been reduced to two figures: the initial data and the completed construction. The intermediate pictures added points and circles one at a time but did not represent distinct geometric cases.

Wilkins's account also exposes a foundational subtlety that deserves mention. The sides DB and DA must be produced far enough to meet the relevant circles. This uses a strong reading of Euclid's Second Postulate together with a tacit continuity or intersection principle. That issue belongs to the feasibility of Euclid's construction, not to the Hilbert/Book Zero proof used below.

2.3 Hilbert Reconstruction

The formal theorem is not a literal transcription of Euclid's construction. Its statement includes a chosen nondegenerate ray AE. The task is to place a copy of BC on that ray.

This is exactly Book Zero #49, `bookZero_49_layoff`. Given

$$A \neq E, \qquad B \neq C,$$

the theorem produces a point D such that

```
HilbertSameRay Geo A E D
```

and

$$AD \cong BC.$$

Thus the equilateral triangle, both circles, the two produced sides, and the subtraction argument of Euclid disappear from Proposition I.2 itself. Their role is replaced by the already established abstract operation of laying off a segment on a prescribed ray.

2.4 Formalization

The complete implementation is available in the repository. In this case the mathematical content of the wrapper is literally one theorem call:

```
exact
  bookZero_49_layoff
    Geo A E B C hAE hBC
```

There is no additional incidence or congruence bookkeeping in `Proposition02.lean`. The two nondegeneracy assumptions are precisely the hypotheses required by the Book Zero layoff theorem.

2.5 Proof Architecture of the Reconstruction

```
Euclid
------
point A + segment BC
        |
join AB
        |
I.1: equilateral ABD
        |
circle centered B through C
+ produce DB to G
        |
circle centered D through G
+ produce DA to L
        |
subtract equal segments
        |
AL = BC


Hilbert / Lean
--------------
chosen ray AE + segment BC
        |
bookZero_49_layoff
        |
D on ray AE and AD ~= BC
        |
euclid_proposition_2
```

2.6 Discussion

Proposition I.2 is an especially clear example of compression by the Book Zero interface. Euclid devotes a nontrivial construction to transporting a length to a prescribed endpoint. The formal development treats that operation as an already established synthetic primitive.

This does not mean that the classical construction is dispensable mathematically. Rather, the proof burden has moved to the theorem `bookZero_49_layoff`. The Book I wrapper can therefore express the operation at the level at which later geometric proofs actually use it.

2.7 Relationship with Earlier Propositions

2.7.1 Proposition I.1

Proposition I.2 is the first substantial reuse of the equilateral triangle construction of I.1.

Euclid constructs equilateral triangles as auxiliary devices for transporting a given length to a prescribed point. The equilateral property creates rigid pairs of congruent segments from which the target segment congruence can be obtained.

Thus I.1 is not an isolated construction: its symmetry becomes working infrastructure immediately in I.2.

2.7.2 Transition to Proposition I.3

Proposition I.2 establishes free segment transport: given a segment and a prescribed point, a congruent segment can be constructed from that point.

Proposition I.3 adds an order constraint. When a longer segment is given, the transported length must be cut from inside it.

Hence the progression is

$$\text{I.2}: \text{ copy a length} \qquad \longrightarrow \qquad \text{I.3}: \text{ cut that length from a longer segment.}$$

2.8 Hilbert and Book Zero Components Used

2.8.1 Book Zero #49: Layoff

The sole substantive dependency of the formal proposition is `bookZero_49_layoff`. It packages existence of a point on a chosen ray together with congruence to a prescribed nondegenerate segment.

2.8.2 Same-Ray Structure

The conclusion records position by

```
HilbertSameRay Geo A E D.
```

This is stronger and more useful than merely asserting collinearity: it fixes the direction from A in which the copied segment is laid off.

2.8.3 Segment Congruence

The metric part of the conclusion remains purely synthetic:

$$AD \cong BC.$$

No numerical length function is introduced.

2.9 Formalization Notes

Proposition I.2 reveals an early difference between Euclid's constructional development and the Hilbert foundation.

Euclid has not yet postulated a general segment-layoff operation. Consequently he constructs the transported length indirectly using equilateral triangles and joins.

In Hilbert geometry, segment construction is already part of the congruence structure. The central mathematical content of I.2 is therefore available at a lower foundational level.

The dependency comparison is

$$\text{Euclid: I.1 + auxiliary construction} \qquad \text{versus} \qquad \text{Hilbert: segment layoff.}$$

This is analogous to what occurs much later in Proposition I.23, where Euclid derives angle transport through earlier triangle constructions while Hilbert III.4 provides angle construction directly.

Thus I.2 is the segment-level version of a recurring phenomenon in the reconstruction: a Euclidean construction theorem may become a thin interface theorem when the corresponding construction has been made foundational.

2.10 Dependency Summary

```
Classical:
I.1 + Postulates II, III
+ circle/line intersection
+ Common Notions
        |
        v
I.2

Formal:
bookZero_49_layoff
        |
        v
euclid_proposition_2
```

The formal theorem is therefore a thin Book I interface over a construction already established in Book Zero.

2.11 Conclusion

Proposition I.2 establishes the transport of segment length to a prescribed point.

For Euclid this is a genuine derived construction built from the equilateral-triangle machinery of I.1. In the Hilbert reconstruction, the corresponding segment-layoff principle belongs to the foundational congruence interface.

The contrast is mathematically informative. It shows, already at the second proposition of Book I, that the historical order of Euclid's constructions need not coincide with the dependency order of a modern axiomatic reconstruction.

I.2 immediately becomes infrastructure for I.3, where free transport is combined with order to cut a prescribed length from a longer segment. It is therefore the first general segment-construction interface of the Book I development.

Chapter 3

Proposition I.3

3.1 Introduction

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Classically, this construction is obtained by combining Proposition I.2 with an additional circle construction. A segment equal to the shorter one is first laid off from an endpoint of the greater segment, and a circle is then used to determine the required point on the greater segment.

In the present reconstruction over Hilbert geometry, the logical structure is substantially different. The theory developed in Book Zero already contains an abstract notion of one segment being less than another. This notion is defined by the existence of a point between the endpoints of the greater segment such that the corresponding subsegment is congruent to the smaller one.

Consequently, the existential content of Euclid's Proposition I.3 is already embodied in the definition of segment inequality. The reconstruction therefore consists not in repeating Euclid's geometric construction, but in exposing the witness contained in the definition of `HilbertSegmentLess`.

This proposition provides the first clear example where a classical Euclidean construction becomes an immediate consequence of the abstract language developed in Book Zero.

3.2 The Classical Configuration

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Unlike Proposition I.2, the present construction does not introduce a new geometric technique. Instead, it combines Proposition I.2 with one additional circle construction.

The construction proceeds as follows.

3.2.1 Final Configuration

The objective is to determine a point E on the greater segment AB such that

$$AE \cong CD.$$

The final geometric configuration is shown in Figure 3.1.

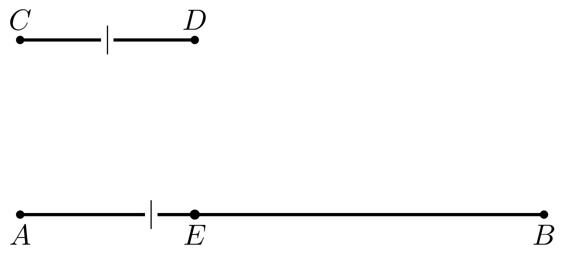


Figure 3.1: The final configuration of Proposition I.3: $A - E - B$ and $AE \cong CD$.

From the construction we obtain

$$AE \cong CD.$$

The remaining task is to justify that the point E lies between the endpoints of the greater segment.

3.2.2 Construction Algorithm

The construction uses Proposition I.2 as an already established construction primitive.

Step 1. Given Segments

Let AB be the greater segment and CD the given shorter segment.

Step 2. Apply Proposition I.2

By Proposition I.2, construct a segment AF congruent to the given segment CD.

Thus

$$AF \cong CD.$$

Step 3. Draw the Circle Centered at A

Draw the circle centered at A passing through F.

Step 4. Determine the Required Point

Let E be the intersection of the circle with the greater segment.

Since

$$AE \cong AF$$

and

$$AF \cong CD,$$

it follows that

$$AE \cong CD.$$

The required segment has therefore been obtained.

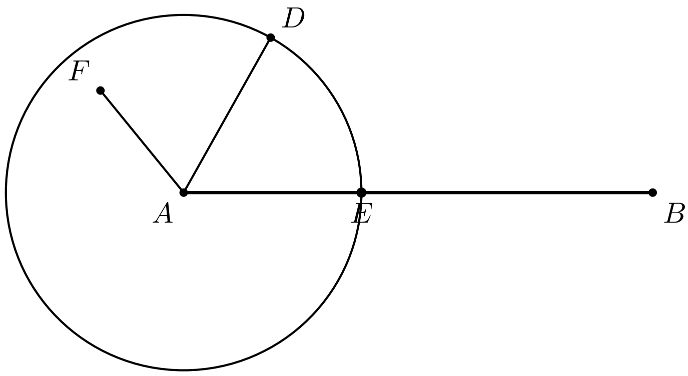


Figure 3.2: The classical cutting step. Proposition I.2 first provides a point D with AD equal to the given shorter segment. The circle centered at A through D meets the greater segment AB at E, so $AE = AD$. The auxiliary point F belongs to the I.2 construction and need not carry further metric markings here.

3.3 Classical Proof

Let AB be the greater segment and CD the shorter one. By Proposition I.2 construct a point D' such that

$$AD' \cong CD.$$

Draw the circle with center A through D'. Because $CD < AB$, this circle meets the segment AB at a point E between A and B. Every radius of the circle is congruent, so

$$AE \cong AD'.$$

Together with $AD' \cong CD$, the First Common Notion gives

$$AE \cong CD.$$

Hence the required part AE has been cut from the greater segment.

□

Diagrammatic audit. Two figures are useful. The first shows only the target relation $A - E - B$ and $AE \cong CD$. The second shows the actual classical mechanism: the transferred length from I.2 and the circle centered at A whose intersection with AB determines E. Separate snapshots for each construction instruction would add no geometric information.

Wilkins emphasizes that I.3 completes a strategy distributed over the first three propositions: I.1 supplies the equilateral-triangle construction used inside I.2, I.2 transports a length, and I.3 uses the resulting radius to cut the required part from the longer segment.

3.4 Proof Architecture of the Reconstruction

```
Euclid
------
CD < AB
   |
I.2: construct D with AD = CD
   |
circle centered at A through D
   |
intersection E with AB
   |
AE = AD = CD
   |
I.3


Hilbert / Lean
--------------
HilbertSegmentLess Geo C D A B
   |
unpack witness E
   |
A-E-B and CD ~= AE
   |
congruence symmetry
   |
A-E-B and AE ~= CD
   |
euclid_proposition_3
```

3.5 Hilbert Reconstruction

The Book Zero formulation changes the logical status of the proposition. The hypothesis

`HilbertSegmentLess Geo C D A B`

already contains a witness E satisfying

$$A - E - B \qquad \text{and} \qquad CD \cong AE.$$

Consequently Proposition I.3 does not invoke a segment-layoff theorem at all. It merely extracts that witness and reverses the orientation of the congruence to obtain

$$AE \cong CD.$$

This is stronger than saying that Book Zero supplies convenient infrastructure for Euclid's construction. With the present definition of strict segment comparison, the existential conclusion of I.3 is literally part of the hypothesis.

3.6 Lean Formalization

The complete implementation is available in the repository. The entire formal argument is worth showing because it exposes the definitional collapse of the classical construction:

```
rcases hCDAB with <E, hAEB, hCDAE>

exact <E, hAEB,
  hilbert_congruent_symmetry
    Geo C D A E hCDAE>
```

There is no hidden construction step in `Proposition03.lean`. Unpacking `hCDAB` produces exactly the required point and betweenness statement; only congruence symmetry is needed to match the orientation of the Euclidean conclusion.

3.7 Relationship with Earlier Propositions

3.7.1 Proposition I.2

Proposition I.3 is the first direct application of the segment-transfer construction established in I.2.

I.2 says that a segment congruent to a given segment can be placed at a prescribed point. I.3 adds an order constraint: when the target segment is longer than the given one, the transferred length can be realized as an initial part of that longer segment.

Thus I.3 converts free segment transport into controlled segment cutting.

3.7.2 Proposition I.1

Proposition I.1 enters indirectly through I.2. Euclid's segment-transfer construction uses equilateral triangles as part of the mechanism for moving a length to a new position.

By the time I.3 is reached, this construction has already been packaged as Proposition I.2 and can be used as a reusable operation.

3.7.3 Transition to Proposition I.4

Propositions I.1–I.3 form the initial construction block of Book I: equilateral triangle construction, segment transport, and segment cutting.

Proposition I.4 changes the character of the development. It is no longer primarily a construction theorem but a rigidity theorem for triangles: SAS.

Thus I.3 closes the first elementary segment-construction sequence and prepares the congruence theory beginning with I.4.

3.8 Hilbert and Book Zero Components Used

3.8.1 Strict Segment Comparison

The decisive object is `HilbertSegmentLess`. In the present Book Zero language, $CD < AB$ is witnessed by an interior point E of AB whose initial subsegment is congruent to CD. Thus order and congruence are already coupled in the definition.

3.8.2 Betweenness

The witness contains the positional fact

$$A - E - B.$$

This is the formal counterpart of the classical requirement that the circle meet the greater segment itself, not merely its supporting line.

3.8.3 Congruence Symmetry

The witness supplies $CD \cong AE$, while the proposition is stated with $AE \cong CD$. The only theorem used after unpacking the witness is segment-congruence symmetry.

3.9 Formalization Notes

Proposition I.3 is mathematically simple but formally instructive.

In modern numerical language one might describe the theorem as follows: if

$$|AB| > |CD|,$$

then there is a point E on AB such that

$$|AE| = |CD|.$$

The reconstruction deliberately avoids this numerical formulation. "Greater than" is represented by a synthetic segment-order relation, and the conclusion consists of two separate geometric facts:

$$A - E - B \qquad \text{and} \qquad AE \cong CD.$$

This distinction is important. Congruence alone would allow a copy of CD to be placed anywhere; Proposition I.3 requires the copy to occur inside the prescribed longer segment.

The formal theorem therefore combines the two foundational layers that will recur throughout Book I:

$$\text{congruence} \quad + \quad \text{order}.$$

In the Hilbert reconstruction the actual segment-layoff operation is already available at the interface level. Consequently the public Euclidean theorem is relatively short: its main role is to package layoff together with the betweenness condition dictated by the strict segment comparison.

3.10 Dependency Summary

```
Classical:
I.1 -> I.2 -> circle construction -> I.3

Formal:
HilbertSegmentLess
        |
        +-- witness E
        +-- A-E-B
        +-- CD ~= AE
        |
congruence symmetry
        |
euclid_proposition_3
```

The contrast is especially sharp: the classical proposition performs a construction, whereas the formal theorem exposes existential data already encoded by the Book Zero segment-order relation.

3.11 Conclusion

Proposition I.3 completes the first elementary segment-construction block of Book I.

Proposition I.2 established free transport of a segment length to a prescribed point. Proposition I.3 adds order: the copied segment is cut from a longer prescribed segment and its endpoint must lie between the original endpoints.

The formalization makes this distinction particularly clear. The theorem is not simply about equality of lengths; it is about the interaction of segment congruence with betweenness.

This combination of congruence and order becomes fundamental throughout the later development. I.3 therefore serves as a compact early example of the two Hilbert layers working together before Book I turns in Proposition I.4 to the rigidity theory of triangles.

Chapter 4

Proposition I.4

4.1 Introduction

Euclid's fourth proposition is the first theorem on triangle congruence.

Unlike the preceding propositions, it does not describe a geometric construction. Instead, it establishes that two triangles are congruent whenever two corresponding sides and the included angle are equal.

Euclid's original proof relies on the method of superposition, one of the most discussed arguments in the *Elements*. Modern axiomatic systems generally avoid superposition as a primitive method of proof.

In the present reconstruction over Hilbert geometry, the proposition is obtained from the Side–Angle–Side (SAS) congruence theorem, which has already been established within the Hilbert theory.

Thus, while the mathematical content coincides with Euclid's Proposition I.4, the logical structure of the proof is fundamentally different.

4.2 The Classical Configuration

Unlike the first three propositions of Book I, Proposition I.4 is not a geometric construction. Instead, it establishes a criterion for the congruence of two triangles.

The proposition considers two triangles satisfying three geometric hypotheses:

- two pairs of corresponding sides are congruent;
- the included angles are congruent.

The objective is to prove that the triangles are congruent. As a consequence, the remaining corresponding side and the remaining corresponding angles are also congruent.

4.2.1 Final Configuration

The geometric configuration is shown in Figure 4.1.

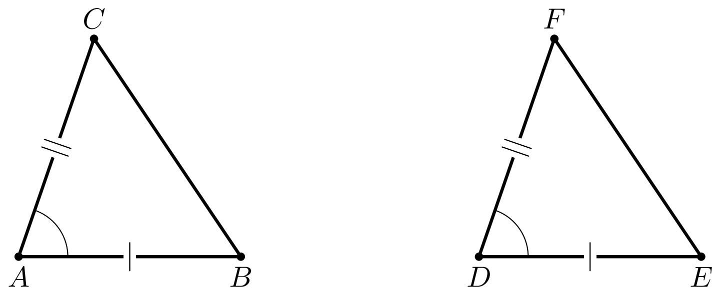


Figure 4.1: The SAS data of Proposition I.4: $AB \cong DE$, $AC \cong DF$, and $\angle BAC \cong \angle EDF$.

The hypotheses are

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF.$$

The desired conclusions are

$$BC \cong EF,$$

together with

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

4.3 Statement

Proposition I.4.

If two triangles have two corresponding sides congruent and the included angles congruent, then the triangles are congruent.

More precisely, let ABC and DEF be two non degenerate triangles:

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF.$$

Then

$$BC \cong EF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

4.4 Classical Proof

Euclid's proof is a superposition argument. Conceive the triangle ABC as being applied to DEF so that A coincides with D and the ray AB coincides in direction with the ray DE. Since

$$AB \cong DE,$$

the point B must then coincide with E.

There is a further positional condition that is easy to miss in the bare classical diagram. The transported point C must be chosen on the same side of the line DE as F. With that choice, the equality

$$\angle BAC \cong \angle EDF$$

makes the ray AC coincide with the ray DF. Since also

$$AC \cong DF,$$

the point C coincides with F.

The endpoints of the remaining sides therefore coincide pairwise, so BC coincides with EF. The two triangles coincide under the superposition, and hence

$$BC \cong EF, \qquad \angle ABC \cong \angle DEF, \qquad \angle ACB \cong \angle DFE.$$

Euclid also concludes that the two triangles are equal in area. That additional magnitude statement belongs to the classical proposition but is not part of the present `TriangleCongruenceResult` interface.

$\square$

Diagrammatic audit. Only one geometric figure is needed. Proposition I.4 has no auxiliary construction: the proof acts on the two triangles already present in the hypotheses. The figure therefore records exactly the SAS data and nothing more. Wilkins's discussion is useful here not because it supplies additional construction diagrams, but because it makes explicit the same-side condition on the transported third vertex that Euclid leaves implicit in words.

4.5 Proof Architecture of the Reconstruction

```
Euclid
------
AB ~= DE
AC ~= DF
angle BAC ~= angle EDF
        |
apply triangle ABC to triangle DEF
A -> D, ray AB -> ray DE
        |
B -> E
        |
choose the same side of DE for C and F
        |
equal included angle + AC ~= DF
        |
C -> F
        |
triangles coincide
        |
BC ~= EF and corresponding angles equal
```

```
Hilbert / Lean
--------------
two noncollinear triangles
+ two side congruences
+ included-angle congruence
        |
TriangleCongruentFromSAS
        |
TriangleCongruenceResult
        |
euclid_proposition_4
```

4.6 Hilbert Reconstruction

The foundational organization is different. Superposition is not replayed inside Proposition I.4. The required rigidity is already available through the Hilbert SAS interface.

For two noncollinear triangles ABC and DEF, the hypotheses

$$AB \cong DE, \qquad AC \cong DF, \qquad \angle BAC \cong \angle EDF$$

are precisely the inputs of `TriangleCongruentFromSAS`. Its result is a `TriangleCongruenceResult`, which packages the corresponding side and angle congruences.

Thus the difference from Euclid is foundational rather than mathematical: Euclid proves SAS by superposition; the Hilbert development has already established SAS before the Book I wrapper is reached.

4.7 Lean Formalization

The complete implementation is available in the repository. Here the entire mathematical step is a single interface call, so only that call is worth displaying:

```
exact
  TriangleCongruentFromSAS
    Geo
    A B C
    D E F
    hABC
    hDEF
    hAB
    hAngle
    hAC
```

The explicit hypotheses $hABC$ and $hDEF$ record nondegeneracy, which the classical picture leaves implicit. No auxiliary points or intermediate geometric constructions occur in `Proposition04.lean`.

4.8 Relationship with Earlier Propositions

4.8.1 Proposition I.2

The classical proof of SAS is expressed by Euclid through superposition rather than by a sequence of earlier construction propositions. Nevertheless, the preceding segment-transport theorem already establishes the general idea that congruent lengths can be reproduced independently of their original position.

In the Hilbert reconstruction, segment congruence is part of the foundational congruence structure, so no separate segment-copying construction is needed inside I.4.

4.8.2 Proposition I.3

Proposition I.3 develops the controlled cutting of one segment by a length congruent to another. It belongs to the same early congruence infrastructure, although it is not the decisive logical ingredient in SAS.

The contrast is useful: I.2 and I.3 are construction theorems about segments, whereas I.4 is the first major rigidity theorem for complete triangles.

4.8.3 Transition to Proposition I.5

Proposition I.5 uses SAS repeatedly. The auxiliary construction in the isosceles-triangle proof creates pairs of triangles with two corresponding sides and the included angle congruent, and I.4 propagates that local symmetry into further side and angle congruences.

Thus I.4 becomes one of the principal reusable congruence principles of Book I immediately after it is established.

4.9 Hilbert and Book Zero Components Used

4.9.1 Hilbert SAS

The decisive formal ingredient is the Hilbert SAS congruence axiom or its established interface theorem. Given two corresponding side congruences and congruence of the included angles, it supplies the remaining corresponding side and angle congruences.

The public Euclidean theorem therefore packages a foundational Hilbert congruence principle into the form of Proposition I.4.

4.9.2 Nondegenerate Triangles

The formal hypotheses explicitly require genuine triangles. The noncollinearity assumptions prevent the angle data from degenerating and make precise a condition that the classical diagram leaves implicit.

4.9.3 Angle Orientation

Angles are represented by ordered triples of points. Consequently the formal proof must ensure that the included angles supplied to SAS and the corresponding angles returned from it have exactly the orientations required by the Euclidean statement.

Symmetry and transport lemmas handle these changes explicitly.

4.9.4 Segment Congruence

The side hypotheses are expressed directly by the Hilbert segment congruence relation. No numerical length function or coordinate representation is introduced.

4.10 Formalization Notes

Proposition I.4 occupies a special position in the reconstruction.

For Euclid, SAS is proved by superposition: one triangle is imagined as being placed on another so that a side and the included angle coincide. The remaining equal side then forces the third vertex and hence the whole triangle to coincide.

In a Hilbert-style axiomatization, the same rigidity is foundational. SAS belongs to the congruence axioms. The Lean proof of the Euclidean proposition is therefore much shorter than Euclid's argument because the geometric content has already been placed in the Hilbert interface.

This is not a loss of information. It records a genuine difference in foundational organization:

$$\text{Euclid: superposition proof} \qquad \text{versus} \qquad \text{Hilbert: SAS congruence principle.}$$

The formal statement also makes nondegeneracy explicit. A diagram of two triangles silently suggests that each triple of vertices is noncollinear; Lean requires this information as part of the hypotheses needed to apply the angle-congruence machinery.

Proposition I.4 is therefore an early example where the Book I order and the logical order of the Hilbert foundation diverge sharply. What is a theorem for Euclid is already foundational infrastructure for the reconstruction.

4.11 Dependency Summary

```
Classical:
SAS data -> superposition -> coincidence -> I.4

Formal:
TriangleCongruentFromSAS -> euclid_proposition_4
```

No new local axiom is introduced by the Book I theorem. The proof burden has already been discharged in the Hilbert congruence layer.

4.12 Conclusion

Proposition I.4 is the first fundamental triangle-congruence theorem of Book I.

Its historical proof uses superposition, one of the most discussed features of Euclid's early geometry. In the Hilbert reconstruction the same mathematical rigidity is represented directly by the SAS congruence principle.

The resulting Lean theorem is therefore short for a substantive reason: the proof burden has moved from Book I into the axiomatic foundation.

This distinction becomes increasingly important in the propositions that follow. I.5 uses SAS as working infrastructure, and later constructions repeatedly rely on the ability to convert local side and angle congruences into congruence information about an entire triangle.

Proposition I.4 thus marks the point where the early segment constructions of I.1–I.3 give way to the systematic theory of triangle rigidity.

Chapter 5

Proposition I.5

5.1 Introduction

Proposition I.5 is the first theorem of Book I devoted specifically to the geometry of isosceles triangles. It establishes that the angles at the base of an isosceles triangle are congruent. Moreover, if the equal sides are extended beyond the base, the exterior angles formed by these extensions are also congruent.

In the Elements this proposition is one of the best known results of classical geometry and later became known under the traditional name *pons asinorum*. Although the theorem itself is elementary, it marks the beginning of a systematic study of congruence properties of triangles.

Euclid proves the proposition by combining the congruence theorem of Proposition I.4 with auxiliary constructions on the extensions of the equal sides.

In the present reconstruction the logical organization is different. The theorem is obtained by composing two previously established results of the Hilbert interface. The congruence of the base angles is provided by the theorem on isosceles triangles, while the congruence of the exterior angles follows from Hilbert's theorem on adjacent angles.

Consequently, Proposition I.5 is the first example in which a Euclidean proposition is reconstructed by combining several independent interface theorems rather than by introducing new geometric techniques.

5.2 The Classical Configuration

Euclid's proof of Proposition I.5 does not use only the original isosceles triangle and the extensions of its equal sides. It introduces additional auxiliary points and joins, producing a larger configuration to which Proposition I.4 can be applied.

Let ABC be an isosceles triangle satisfying

$$AB \cong AC.$$

Produce the equal sides AB and AC beyond the base vertices B and C. Euclid then chooses an auxiliary point on one extension, transfers a corresponding segment to the other extension, and joins the resulting points to the opposite vertices of the base.

The following sequence shows the successive stages of Euclid's configuration.

5.2.1 Initial Isosceles Triangle

The construction begins with an isosceles triangle ABC satisfying

$$AB \cong AC.$$

At this stage no auxiliary points or segments have yet been introduced.

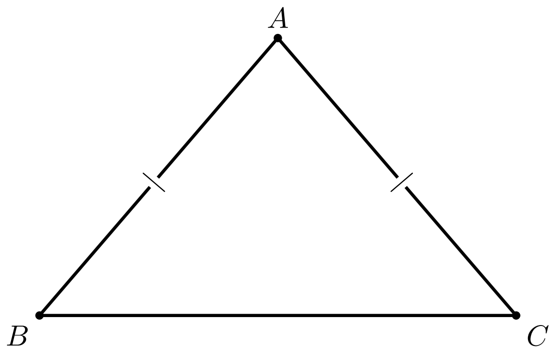


Figure 5.1: The initial isosceles triangle, with $AB \cong AC$.

5.2.2 Extension of the Equal Sides

Euclid next produces the equal sides AB and AC beyond the base vertices. Let D lie beyond B on the line AB, and let E lie beyond C on the line AC.

Thus

$$A - B - D, \qquad A - C - E.$$

5.3 Statement

Let ABC be a nondegenerate isosceles triangle satisfying

$$AB \cong AC.$$

Let the equal sides be extended beyond the base vertices so that

$$A - B - D, \qquad A - C - E.$$

Then

$$\angle ABC \cong \angle ACB,$$

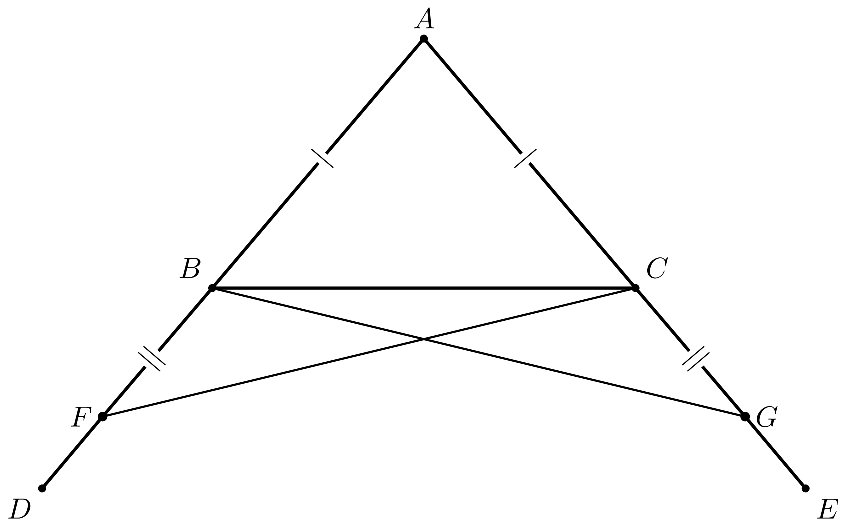


Figure 5.2: The complete classical configuration for Proposition I.5. The sides are produced beyond B, C; F is chosen on the left extension, G on the right so that $BF \cong CG$; finally FC and GB are joined. The outer points D, E record the rays used in the statement.

and

$$\angle CBD \cong \angle BCE.$$

Thus both the base angles of the isosceles triangle and the exterior angles formed after extending the equal sides are congruent.

Diagrammatic audit. The four successive construction snapshots have been replaced by one complete classical figure. For I.5 the successive states do not represent genuinely different geometric cases: they merely add F, then G, then the two joins. A single well-spaced configuration makes the proof easier to follow.

Wilkins's note confirms that the two genuinely important stages are the two SAS comparisons. First FAC is compared with GAB; this gives, among other consequences, $FC \cong GB$. Then FBC is compared with GCB, which yields the equal angles below the base. Subtracting equal angles then gives the equality of the base angles. These logical stages are described in the proof rather than represented by repeated copies of the same diagram.

5.4 Classical Proof

Euclid's proof is considerably more elaborate than the modern reconstruction presented later in this chapter. Rather than reasoning directly from the properties of an isosceles triangle, he introduces an auxiliary construction that allows repeated applications of the Side– Angle–Side congruence theorem established in Proposition I.4.

After extending the equal sides of the triangle, an arbitrary point is chosen on one extension. Proposition I.3 is then used to construct an equal segment on the opposite extension. The newly obtained points are joined to the opposite vertices of the base, completing the auxiliary configuration.

The proof proceeds by applying Proposition I.4 twice. The first application establishes congruence relations between the auxiliary triangles and yields equal corresponding sides and angles. These newly obtained equalities provide the hypotheses required for a second application of Proposition I.4.

Finally, Euclid combines the resulting congruence relations with the elementary properties of equal segments and equal angles to conclude that the interior base angles of the original isosceles triangle are congruent. The same argument immediately establishes the congruence of the corresponding exterior angles formed after extending the equal sides.

The auxiliary construction is therefore not an end in itself. Its sole purpose is to transform Proposition I.5 into two successive applications of Proposition I.4, allowing the theorem to be derived using only previously established geometric results.

5.5 Proof Architecture of the Reconstruction

The formal proof of Proposition I.5 is obtained by composing two previously established theorems of the Hilbert interface.

The first theorem establishes the congruence of the base angles of an isosceles triangle. The second theorem transfers this congruence to the adjacent exterior angles after the equal sides have been extended.

The resulting logical dependency is shown below.

```
Euclid
------
AB ~= AC
   |
produce AB, AC beyond B, C
   |
choose F; I.3 gives G with BF ~= CG
   |
join FC and GB
   |
I.4: SAS on FAC and GAB
   |
I.4: SAS on FBC and GCB
   |
subtract equal angles
   |
base angles equal
and angles below base equal


Hilbert / Lean
--------------
AB ~= AC
   |
hilbert_isosceles_base_angles
   |
angle ABC ~= angle ACB
```

```
   |
A-B-D and A-C-E
   |
hilbert_adjacent_angles_congruent
   |
angle CBD ~= angle BCE
```

Unlike the classical proof of Euclid, the formal development does not repeat the geometric argument establishing the base-angle theorem. Instead, the proposition is reconstructed by composing two independent results already available in the Hilbert interface. Consequently, Proposition I.5 illustrates the modular nature of the library: new Euclidean propositions are obtained by combining previously verified synthetic theorems.

5.6 Hilbert Reconstruction

At the level of Proposition I.5, the Hilbert proof is much shorter than Euclid's classical construction.

The first conclusion,

$$\angle ABC \cong \angle ACB,$$

is already the theorem

```
hilbert_isosceles_base_angles
```

applied to the noncollinear isosceles triangle ABC.

For the second conclusion, the hypotheses

$$A - B - D, \qquad A - C - E$$

say that the equal sides have been produced beyond the base vertices. The established theorem

```
hilbert_adjacent_angles_congruent
```

transports the congruence of the base angles to their adjacent exterior angles, giving

$$\angle CBD \cong \angle BCE.$$

Thus the formal proposition does not construct Euclid's auxiliary points F, G, and it does not replay the two SAS arguments. Those facts have already been packaged into the Hilbert interface theorems used here.

5.7 Lean Formalization

The complete implementation is available in the repository. The two interface calls contain the mathematical content of the formal proof.

The base-angle conclusion is immediate:

```
have hBase :=
  hilbert_isosceles_base_angles
    Geo A B C hABC hABAC
```

The exterior-angle conclusion is then obtained from the two extensions and the already established base-angle congruence:

```
have hExterior :=
  hilbert_adjacent_angles_congruent
    Geo
    A B C D
    A C B E
    hABD
    hACE
    hABC
    hACB
    hBase
```

The theorem returns the pair $\langle hBase, hExterior \rangle$. The only additional bookkeeping is the rotated noncollinearity hypothesis required for the second call.

5.8 Relationship with Earlier Propositions

5.8.1 Proposition I.2

Euclid's proof requires the transfer of a prescribed segment length to a new position in the auxiliary construction. Proposition I.2 supplies the underlying segment-transport mechanism.

In the Hilbert reconstruction this role is absorbed by the established segment-layoff infrastructure.

5.8.2 Proposition I.3

After extending the equal sides of the original isosceles triangle, Euclid cuts off an auxiliary segment equal to a prescribed segment. Proposition I.3 provides exactly this operation.

This construction creates the equal-side data needed for the subsequent triangle-congruence arguments.

5.8.3 Proposition I.4

SAS is the decisive congruence theorem in the classical proof. It is applied to auxiliary triangles in order to propagate the original equality of the two sides into the angle congruences required at the base.

The formal reconstruction uses the corresponding Hilbert SAS theorem and angle-congruence transport.

5.8.4 Transition to Proposition I.6

Proposition I.5 proves
$$AB \cong AC \Longrightarrow \angle ABC \cong \angle BCA.$$

Proposition I.6 proves the converse implication. Together they establish the first full equivalence between side symmetry and angle symmetry in a triangle.

5.9 Hilbert and Book Zero Components Used

5.9.1 Segment Extension and Layoff

The proof extends the equal sides beyond the base vertices and chooses auxiliary points at prescribed distances. These operations are supplied by the established Hilbert order and congruence interfaces.

5.9.2 SAS

The Hilbert SAS theorem is the central rigidity principle in the reconstruction. It converts the constructed side congruences and the appropriate included-angle congruence into further corresponding side and angle congruences.

5.9.3 Angle Congruence Transport

The auxiliary triangles naturally produce angles in orientations that do not always coincide syntactically with the final base angles. Symmetry, transitivity, and transport of angle congruence are therefore part of the formal proof rather than silent diagrammatic steps.

5.9.4 Betweenness and Collinearity

The extensions of the two equal sides are represented by explicit betweenness hypotheses. These yield the collinearity and ray information needed to identify the auxiliary angles with the angles occurring in the Euclidean statement.

5.10 Formalization Notes

Proposition I.5 marks an important transition in the reconstruction of Book I.

The earlier propositions are centred on elementary constructions and a single application of the SAS theorem. Beginning with Proposition I.5, most subsequent propositions of Book I are expected to be reconstructed by composing previously established interface theorems.

This modular organization differs substantially from the presentation of the *Elements*. Euclid develops each proposition as an independent geometric argument, whereas the present reconstruction emphasizes reusable formal components. The resulting proofs are shorter, their logical dependencies are explicit, and each proposition becomes a thin verification layer built upon the synthetic theory already developed in the Hilbert interface.

5.11 Dependency Summary

The classical and formal architectures are sharply different:

```
Classical:
I.3 + two uses of I.4 + angle subtraction  ->  I.5

Formal:
hilbert_isosceles_base_angles
          +
hilbert_adjacent_angles_congruent         ->  euclid_proposition_5
```

The Book I theorem is therefore a thin interface layer over results already proved in the Hilbert development.

5.12 Conclusion

Proposition I.5 is the first substantial theorem about the internal symmetry of a triangle.

Starting from two congruent sides, Euclid constructs an extended auxiliary configuration and uses SAS twice to prove congruence of the base angles. The theorem also establishes the corresponding congruence of the exterior angles below the base.

The formal reconstruction shows that the geometric idea remains elementary, but its successful reuse depends on a well-developed interface for segment extension, layoff, betweenness, collinearity, and angle-congruence transport.

I.5 is structurally important far beyond the immediate statement. Its converse is Proposition I.6; the pair then supplies the isosceles machinery used in the uniqueness argument of I.7 and ultimately in the SSS theorem of I.8.

Thus Proposition I.5 begins a coherent four-proposition development

$$\mathrm{I.5} \longrightarrow \mathrm{I.6} \longrightarrow \mathrm{I.7} \longrightarrow \mathrm{I.8},$$

moving from isosceles symmetry to general triangle rigidity.

Chapter 6

Proposition I.6

6.1 Introduction

Proposition I.6 establishes the converse of the preceding proposition. Where Proposition I.5 proves that equal sides of a triangle imply equal base angles, Proposition I.6 shows that equal base angles imply the equality of the opposite sides.

This result completes the fundamental correspondence between the sides and angles of an isosceles triangle. Together, Propositions I.5 and I.6 establish the equivalence between the two defining properties of an isosceles triangle.

Euclid's original proof is considerably more involved than the modern reconstruction. It proceeds by contradiction, introducing an auxiliary construction obtained from Proposition I.3 and ultimately deriving an impossible configuration by repeated use of Proposition I.4.

In the Hilbert reconstruction developed in this project, the theorem admits a substantially shorter proof. The previously established Angle–Side–Angle theorem is sufficient to derive the equality of the opposite sides directly, eliminating the auxiliary construction and the reductio ad absurdum argument required by Euclid.

6.2 The Classical Configuration

Euclid's proof begins with a triangle whose base angles are assumed to be equal. The desired conclusion is that the opposite sides are also equal.

Instead of arguing directly, Euclid assumes that the two sides are unequal and derives a contradiction. The proof therefore combines an auxiliary construction with a reductio ad absurdum argument.

The following sequence illustrates the successive stages of Euclid's construction.

6.2.1 Initial Triangle

Let ABC be a triangle satisfying

$$\angle ABC \cong \angle ACB.$$

At this stage no auxiliary construction has yet been introduced.

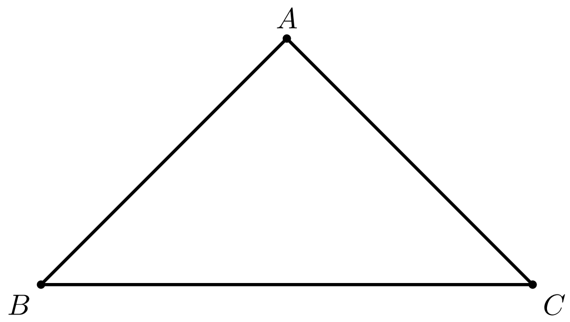


Figure 6.1: The data of Proposition I.6: the base angles satisfy $\angle ABC \cong \angle ACB$.

6.2.2 Assuming Unequal Sides

To obtain a contradiction, Euclid assumes that the two sides opposite the equal angles are not equal. Without loss of generality, suppose

$$AB > AC.$$

Using Proposition I.3, a point D is chosen on the segment AB so that

$$DB \cong AC.$$

Consequently,

$$A - D - B.$$

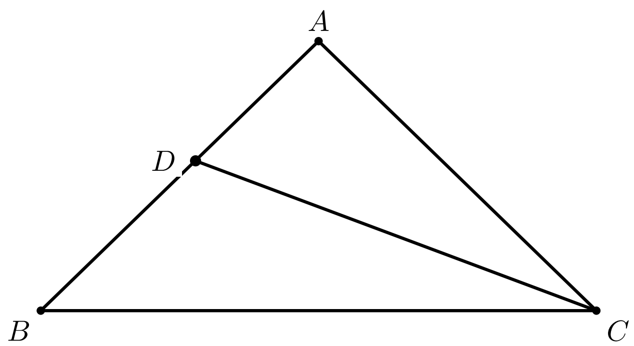


Figure 6.2: Euclid's contradiction configuration for the case $AB > AC$. Proposition I.3 chooses D on AB with $DB \cong AC$, and D is joined to C. The congruence is stated in the text rather than marked again on the diagram.

6.3 Statement

Theorem 1 (Euclid I.6)**.** *Let ABC be a nondegenerate triangle.*

If

$$\angle ABC \cong \angle ACB,$$

then

$$AB \cong AC.$$

That is, equal angles in a triangle subtend equal opposite sides.

Diagrammatic audit. The three former auxiliary pictures were successive snapshots of the same construction and have been merged into one. Two figures are sufficient: the initial triangle and the contradiction configuration with the point D and the segment DC.

Wilkins's discussion confirms that the essential classical move is exactly this one: assuming, say, $AB > AC$, choose $D \in AB$ with $DB = AC$, then compare DBC with ABC by SAS. His note also records an alternative angle-based contradiction, but that is another proof rather than another necessary diagram for the reconstruction.

6.4 Classical Proof

Euclid proves the theorem by contradiction.

Assume that the sides opposite the equal angles are unequal. Without loss of generality, suppose that AB is greater than AC. By Proposition I.3, a point is constructed on the longer side so that the remaining segment is equal to the shorter side.

Joining the auxiliary point to the remaining vertex produces a second triangle. Proposition I.4 is then applied to show that the auxiliary triangle is congruent to the original one.

The resulting congruence forces equalities that are incompatible with the assumption that one side is strictly longer than the other. Since the assumption leads to a contradiction, the two sides must be equal.

Thus equal angles in a triangle necessarily subtend equal opposite sides.

6.5 Proof Architecture of the Reconstruction

The formal proof follows a substantially different route from Euclid's classical argument. Rather than introducing an auxiliary point, constructing additional triangles, and reasoning by contradiction, the implementation exploits a higher-level theorem already available in the Hilbert interface.

The proof compares the given triangle with itself after exchanging the roles of the two equal angles. Since the common side is trivially congruent to itself, the derived Angle–Side–Angle theorem immediately implies the equality of the remaining sides.

Consequently, the entire classical construction disappears from the formal proof and is replaced by a single application of a previously established congruence theorem.

```
Euclid
------
ABC with angle ABC ~= angle ACB
        |
assume AB != AC
        |
choose one strict inequality, e.g. AB > AC
        |
I.3: choose D on AB with DB ~= AC
        |
join DC
        |
I.4: SAS
        |
auxiliary triangle equals whole triangle
        |
contradiction
        |
AB ~= AC


Hilbert / Lean
--------------
angle ABC ~= angle ACB
        |
reorient the two angle congruences
        |
BC ~= CB
        |
hilbert_asa_sides
on triangles BCA and CBA
        |
AB ~= AC
```

6.6 Hilbert Reconstruction

The formal proof uses a substantially shorter route than Euclid's contradiction argument. It applies an established ASA theorem directly.

From the hypothesis

$$\angle ABC \cong \angle ACB$$

the proof first obtains the corresponding reversed angle congruences needed for the triangles

$$\triangle BCA \qquad \text{and} \qquad \triangle CBA.$$

The side BC is common to both triangles, so

$$BC \cong CB.$$

The theorem `hilbert_asa_sides` then yields the equality of the remaining corresponding sides. After reorienting the segment congruence, this is exactly

$$AB \cong AC.$$

Thus no segment trichotomy, point D, Proposition I.3 construction, or contradiction appears in `Proposition06.lean`. Those belong to the classical Euclidean proof, not to the formal wrapper used here.

6.7 Lean Formalization

The complete implementation is available in the repository. Only the two steps that explain the formal proof are reproduced here.

First the common side and the two angle orientations are prepared. The decisive theorem call is then:

```
have hASA :=
  hilbert_asa_sides
    Geo
    B C A
    C B A
    hBCA
    hCBA
    hBC
    hAngleB
    hAngleC
```

The returned side congruence has the orientation dictated by those two triangles, so the final line merely reverses its endpoints:

```
exact
  (Geo.congruent_reverse_second
    A B C A).mp
    ((Geo.congruent_reverse_first
      B A C A).mp hASA.1)
```

The remaining code establishes the two noncollinearity variants and reorients the given angle congruence. These are formal interface obligations, not additional geometric constructions.

6.8 Relationship with Earlier Propositions

6.8.1 Proposition I.3

The contradiction argument assumes that one of the two sides is longer than the other and then cuts from the longer side a segment congruent to the shorter one.

This is precisely the segment-cutting operation established in Proposition I.3. It creates the auxiliary point from which the isosceles configuration used in the proof is obtained.

6.8.2 Proposition I.4

Once the auxiliary point has been constructed, SAS is used to compare the relevant triangles. Proposition I.4 transports the available side and angle congruences into the additional congruence information needed for the contradiction.

6.8.3 Proposition I.5

Proposition I.6 is the converse of Proposition I.5.

I.5 states that equal sides imply equal opposite angles:

$$AB \cong AC \Longrightarrow \angle ABC \cong \angle BCA.$$

I.6 reverses the implication:

$$\angle ABC \cong \angle BCA \Longrightarrow AB \cong AC.$$

The two propositions therefore establish the first clean equivalence between side symmetry and angle symmetry in a triangle.

6.8.4 Transition to Proposition I.7

Proposition I.7 uses this isosceles infrastructure inside a uniqueness argument. Equal distances to the endpoints of a base generate isosceles triangles, and the resulting angle congruences help exclude a second possible vertex on the same side of the base.

Thus I.6 is part of the rigidity machinery leading directly to I.7 and then to SSS in I.8.

6.9 Hilbert and Book Zero Components Used

6.9.1 Segment Trichotomy

The formal proof must turn the assumption

$$AB \ncong AC$$

into a usable strict comparison. Segment trichotomy separates the possibilities

$$AB < AC, \qquad AB \cong AC, \qquad AC < AB.$$

The congruent branch contradicts the hypothesis, while the two strict branches are treated symmetrically.

6.9.2 Segment Layoff

In each strict branch a point is constructed on the longer side so that the cut-off segment is congruent to the shorter side. This is the formal counterpart of Euclid's application of Proposition I.3.

6.9.3 Isosceles Base Angles

The auxiliary equal-side configuration is converted into angle congruence by the established Hilbert isosceles theorem.

6.9.4 SAS and Angle Transport

SAS and angle-congruence transitivity transport the local auxiliary equalities into the orientation required to contradict the original angle hypothesis.

6.10 Formalization Notes

Proposition I.6 provides one of the clearest examples of the difference between a classical geometric proof and its modern formal reconstruction.

Euclid establishes the theorem by introducing an auxiliary point, constructing an additional triangle, and arguing by contradiction. The proof occupies a central place in Book I because it demonstrates how previous propositions can be combined to derive a converse statement.

Within the Hilbert framework, the same mathematical result becomes an almost immediate consequence of the previously established Angle–Side–Angle theorem. Once ASA has been derived from Hilbert's axioms, the converse property of isosceles triangles follows directly, without auxiliary constructions or reductio ad absurdum.

This proposition therefore illustrates an important principle of the CGJteamLab reconstruction: as the interface theory grows, increasingly complex classical arguments collapse into short applications of high-level synthetic theorems.

6.11 Dependency Summary

The two proof paths are genuinely different:

```
Classical:
I.3  +  I.4  + contradiction  ->  I.6

Formal:
hilbert_asa_sides             ->  euclid_proposition_6
```

The formal proof is therefore not a line-by-line rendering of Euclid's argument. It reuses a stronger theorem already available in the Hilbert interface.

6.12 Conclusion

Proposition I.6 completes the converse half of Euclid's elementary isosceles-triangle theory.

Together, I.5 and I.6 establish the equivalence between equal sides and equal opposite angles. The classical proof of I.6 is by contradiction: if the sides were unequal, the longer side could be cut down to create an auxiliary isosceles configuration incompatible with the assumed angle congruence.

The formal reconstruction exposes one additional logical ingredient that is almost invisible on paper: from noncongruence of the two sides one must obtain a definite strict order. Segment trichotomy supplies that step and forces the proof to account for both possible orientations of the inequality.

This side-angle equivalence becomes immediate infrastructure for Proposition I.7 and, through I.7, for the SSS theorem of Proposition I.8.

Chapter 7

Proposition I.7

7.1 Introduction

Proposition I.7 occupies a special position in Book I of Euclid's *Elements*. Unlike the preceding propositions, its principal purpose is not to establish a new geometric construction or a direct property of triangles, but to prove a uniqueness principle.

Euclid shows that, on the same side of a given base, there cannot exist two distinct vertices determining triangles whose corresponding sides are equal. This uniqueness result immediately precedes Proposition I.8, where it is used as a crucial ingredient in the classical proof of the Side–Side–Side congruence theorem.

The Hilbert reconstruction follows a different logical order. The general Side–Side–Side theorem has already been established within the Hilbert interface. Consequently, Proposition I.7 becomes a concise uniqueness argument derived from the developed congruence theory, without reproducing Euclid's original chain of auxiliary arguments.

This proposition therefore provides one of the clearest examples of the difference between the historical organization of Euclid's *Elements* and the logical organization of a modern axiomatic development.

7.2 The Classical Configuration

Let AB be a given finite straight line and let C and D be two points lying on the same side of the line AB.

Assume that

$$AC \cong AD$$

and

$$BC \cong BD.$$

The proposition asks whether the two vertices C and D can be distinct under these assumptions.

Euclid's objective is to prove that such a configuration is impossible.

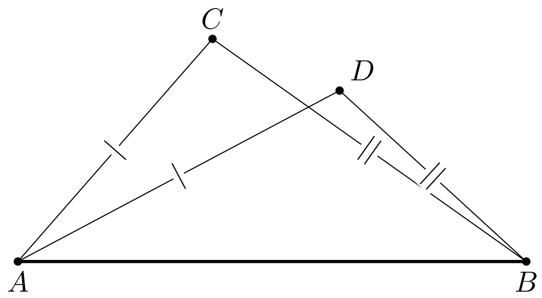


Figure 7.1: The forbidden configuration of Proposition I.7: distinct points C, D lie on the same side of the base AB, while $AC \cong AD$ and $BC \cong BD$.

7.2.1 The Contradictory Configuration

Euclid joins the two assumed vertices by the segment CD.

Since

$$AC \cong AD,$$

the triangle ACD is isosceles. By Proposition I.5,

$$\angle ACD \cong \angle CDA.$$

From the geometric position of the points, one of these angles is strictly greater than the corresponding angle formed with the base vertex B.

On the other hand, since

$$BC \cong BD,$$

the triangle BCD is also isosceles. Proposition I.5 therefore gives

$$\angle BCD \cong \angle CDB.$$

The same pair of angles is thus forced to be both equal and unequal, which is impossible. Hence the two assumed vertices cannot be distinct.

7.3 Statement

Theorem 2 (Euclid I.7)**.** *Let AB be a given base, and let C and D lie on the same side of the line AB.*

Assume

$$AC \cong AD$$

and

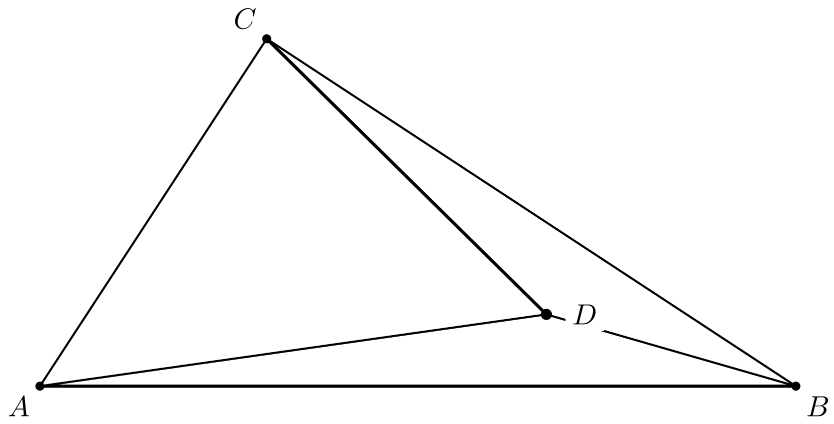


Figure 7.2: Euclid's representative configuration after joining C to D. The point D is placed deep inside the configuration so that the four side segments and the auxiliary segment CD are visually separated. The equalities $AC \cong AD$ and $BC \cong BD$ are hypotheses of the proposition and are stated in the text rather than marked again on this diagram. This is one positional case, not a diagram of every same-side configuration.

$$BC \cong BD.$$

Then

$$C = D.$$

Equivalently, on a fixed side of a given base there is at most one point having prescribed distances from the two endpoints of that base.

Diagrammatic audit. The two original introductory figures were duplicates and have been merged. One figure now records the actual hypotheses, with the two pairs of equal segments marked; a second records the configuration obtained after joining C and D.

There is an important limitation to the second picture. Wilkins emphasizes that a complete proof of I.7 requires several positional configurations, whereas Euclid explicitly treats only one. In particular, after excluding the cases in which D lies on one of the rays through C, the relevant half-plane is divided into four regions K, L, M, N. The classical diagram below should therefore be read as Euclid's representative case, not as a picture of every possible same-side configuration.

7.4 Classical Proof

Assume, for contradiction, that C and D are distinct.

Join C to D. Since

$$AC \cong AD,$$

the triangle ACD is isosceles. By Proposition I.5, its base angles are congruent:

$$\angle ACD \cong \angle CDA.$$

Similarly, since

$$BC \cong BD,$$

the triangle BCD is isosceles, and Proposition I.5 gives

$$\angle BCD \cong \angle CDB.$$

The position of the points on the same side of the base forces one of these angles to contain the other as a proper part. Hence an angle would have to be congruent to one of its proper parts, which is impossible.

Therefore the assumption that C and D are distinct leads to a contradiction. Hence

$$C = D.$$

Thus, on a fixed side of a given base, a point is uniquely determined by its distances from the two endpoints of the base.

7.5 Proof Architecture of the Reconstruction

The formal proof bears little resemblance to Euclid's original argument.

Instead of constructing auxiliary triangles and reasoning by contradiction, the implementation first applies the already established Side–Side–Side theorem of the Hilbert interface. This immediately shows that the two candidate triangles are congruent.

From the resulting congruence, the corresponding angle at the common base vertex is obtained. The uniqueness of angle construction then forces both candidate vertices to lie on the same ray issued from the base vertex.

Finally, since the two points lie on the same ray and are at the same distance from the common endpoint, the uniqueness of segment construction implies that the two points coincide.

```
Euclid
------
same-side distinct C,D
AC ~= AD, BC ~= BD
        |
join CD
        |
I.5 + angle-order argument
        |
contradiction


Hilbert / Lean
```

```
--------------
same-side C,D off base AB
        |
noncollinearity of ABC and ABD
        |
HilbertSSS on ABC and ABD
        |
equal angles at B
        |
hilbert_angle_unique_common_ray
        |
C and D lie on the same ray from B
        |
hilbert_segment_construction_unique
        |
C = D
```

7.6 Hilbert Reconstruction

The formal proof avoids the positional case analysis highlighted by the classical discussion.

Assume first that $C \neq D$. Since C and D lie on the same side of the line through A, B, both are off that line. The interface theorem `hilbert_not_collinear_of_off_line` therefore gives the noncollinearity of ABC and ABD.

The common side AB, together with

$$AC \cong AD, \qquad BC \cong BD,$$

allows Hilbert SSS to be applied to the triangles ABC and ABD. In particular,

$$\angle ABC \cong \angle ABD.$$

The same-side hypothesis now matters a second time. Angle uniqueness on the common ray at B shows that C and D determine the same ray from B. Finally $BC \cong BD$, together with uniqueness of segment construction on that ray, forces

$$C = D,$$

contradicting the assumption $C \neq D$.

Thus the Hilbert proof replaces Euclid's region-sensitive angle-order argument by two uniqueness principles: uniqueness of the ray determined by an angle on a fixed side, followed by uniqueness of a point at a prescribed distance on that ray.

7.7 Lean Formalization

The complete implementation is available in the repository. Three short excerpts expose the structure of the formal argument.

First SSS is applied to the two triangles sharing the base AB:

```
have hSSS :=
  HilbertSSS
    Geo
    A B C
    A B D
    hABC
    hABAB
    hBCBD
    hACAD
```

The angle at B is then fed into the same-ray uniqueness theorem:

```
rcases
    hilbert_angle_unique_common_ray
      Geo
      A B C D
      base
      hAB
      hAbase
      hBbase
      hCbase
      hSame
      hAngleB with
  <X, hRayXC, hRayXD>
```

Finally, the equal distances from B and the common ray identify the two points:

```
exact
  hilbert_segment_construction_unique
    Geo
    B C
    B X
    C D
    hRayXC
    hRayXD
    (hilbert_congruent_reflexive Geo B C)
    hBD_BC
```

The remaining code establishes the off-line and noncollinearity conditions required by these interface theorems. Those bookkeeping steps are important for verification but do not add a new geometric idea to the proof.

7.8 Relationship with Earlier Propositions

7.8.1 Proposition I.5

The classical proof of I.7 creates isosceles triangles from the assumed equal distances to the endpoints of the common base. Proposition I.5 then converts those segment congruences into base-angle congruences.

This is the first essential step in the contradiction: metric symmetry at the level of sides becomes angular symmetry.

7.8.2 Proposition I.6

Proposition I.6 is the converse of the isosceles-triangle theorem. It allows angle congruence to be converted back into segment congruence when the contradiction argument reaches the appropriate auxiliary triangle.

Thus I.5 and I.6 together provide the two-way bridge between side congruence and angle congruence used in the uniqueness argument.

7.8.3 Transition to Proposition I.8

The importance of I.7 becomes fully visible in Proposition I.8.

Euclid's SSS proof superposes one triangle on another. After two vertices and the common base have been matched, I.7 excludes the possibility that the third vertex could occupy a different position on the same side of the base while preserving both distances to the base endpoints.

Thus I.7 supplies the uniqueness principle hidden inside Euclid's superposition proof of SSS.

7.9 Hilbert and Book Zero Components Used

7.9.1 Same-Side Geometry

The statement of I.7 is fundamentally a same-side uniqueness theorem. The two candidate vertices are assumed to lie on the same side of the common base. This positional information is explicit in the Hilbert reconstruction rather than inferred from the drawing.

7.9.2 Segment Congruence

The hypotheses prescribe equal distances from each candidate vertex to the two endpoints of the base. The formal proof repeatedly transports and composes these congruences in order to create the isosceles configurations used in the contradiction.

7.9.3 Isosceles Angle Theorems

The established Hilbert isosceles infrastructure converts the equal-side hypotheses into angle congruences. These are the formal counterparts of Euclid's applications of I.5.

7.9.4 Order and Collinearity

The contradictory configuration depends on the relative positions of the candidate vertices and the base. Betweenness, collinearity, and same-side information replace the diagrammatic assumptions in the classical proof.

7.10 Formalization Notes

Proposition I.7 illustrates a fundamental difference between the historical development of Euclid's geometry and its modern axiomatic reconstruction.

In the *Elements*, Proposition I.7 serves as an essential lemma for the proof of the Side–Side–Side congruence theorem in Proposition I.8. Euclid therefore establishes the uniqueness result before proving the general congruence theorem.

The Hilbert reconstruction follows the opposite logical direction. Once the general Side–Side–Side theorem has been derived from the Hilbert axioms, Proposition I.7 becomes an immediate uniqueness corollary obtained by combining triangle congruence with the uniqueness of angle and segment constructions.

This proposition therefore demonstrates that the historical order of the *Elements* need not coincide with the logical organization of a modern formal theory. As the interface develops, results that were once foundational become concise consequences of more general synthetic theorems.

7.11 Dependency Summary

The geometric content of Proposition I.7 can be summarized as

$$\begin{array}{c} C, D \text{ on the same side of } AB \\ AC \cong AD, \qquad BC \cong BD \\ \downarrow \\ \text{isosceles configurations} \\ \downarrow \\ \text{I.5 and angle-congruence transport} \\ \downarrow \\ \text{contradictory angle configuration} \\ \downarrow \\ C = D. \end{array}$$

Equivalently, on a prescribed side of a base, the two distances to the base endpoints determine the vertex uniquely.

The formal reconstruction uses only established incidence, order, and congruence infrastructure. No coordinate representation or numerical distance is introduced.

7.12 Conclusion

Proposition I.7 is best understood as a rigidity or uniqueness theorem.

Its classical formulation is somewhat indirect: two different points on the same side of a base cannot have the same prescribed distances to both endpoints of that base. The formal reconstruction exposes the structural content more clearly: the intersection of two distance conditions is unique once the side of the base has been fixed.

The proof depends on the isosceles theory developed in I.5 and I.6 and turns segment congruence into a contradiction among angles and point positions.

Its main importance is prospective. Proposition I.8 uses exactly this uniqueness principle to justify the final step of the SSS superposition argument. I.7 therefore forms the bridge between the elementary isosceles results and the general triangle-rigidity theorem of I.8.

Chapter 8

Proposition I.8

8.1 Introduction

Proposition I.8 is one of the central results of Book I of Euclid's *Elements*. It establishes the Side–Side–Side criterion for triangle congruence, proving that triangles having three corresponding sides equal must also have their corresponding angles equal.

Within Euclid's development, this proposition concludes a carefully prepared sequence of results. Proposition I.7 provides the essential uniqueness argument that allows the final congruence theorem to be established.

The Hilbert reconstruction follows a different logical organization. The general Side–Side–Side theorem has already been derived from the Hilbert axioms before the Euclidean propositions are reconstructed. Consequently, Proposition I.8 becomes a direct application of this general theorem rather than the culmination of a long geometric argument.

This proposition therefore marks the point where the historical proof of Euclid and the modern Hilbert reconstruction differ most significantly while establishing exactly the same mathematical result.

8.2 The Classical Configuration

Consider two triangles,

$$\triangle ABC \quad \text{and} \quad \triangle DEF,$$

whose corresponding sides satisfy

$$AB \cong DE,$$

$$BC \cong EF,$$

and

$$AC \cong DF.$$

The objective is to prove that the corresponding angles of the two triangles are also congruent.

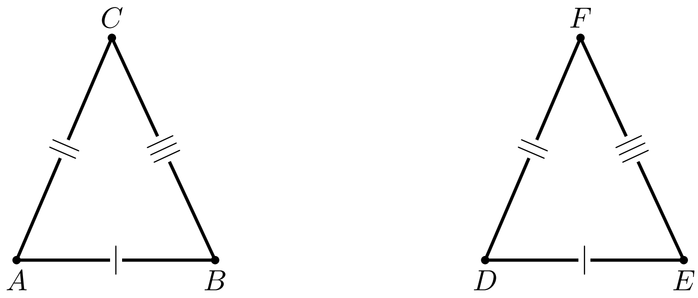


Figure 8.1: The data of Proposition I.8: $AB \cong DE$, $AC \cong DF$, and $BC \cong EF$.

8.2.1 Superposition

Euclid conceptually places one triangle upon the other so that the vertex D coincides with A and the side DE lies along the side AB.

Since

$$AB \cong DE,$$

the endpoint E coincides with B.

The remaining vertex F must then lie at the intersection of two circles centered at A and B with radii equal to the corresponding sides of the first triangle.

Proposition I.7 shows that, on a fixed side of the base, such a point is unique. Consequently, the vertex F must coincide with C.

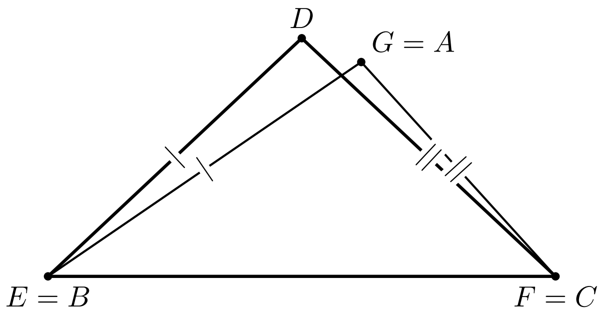


Figure 8.2: Euclid's superposition argument. After BC is placed on EF, the transported vertex G and the vertex D lie on the same side of EF. If $G \neq D$, then $EG = ED$ and $FG = FD$, contradicting Proposition I.7. Hence the vertices coincide.

8.2.2 Application of Proposition I.7

Once the triangles have been superposed, the only remaining possibility is that the third vertices do not coincide.

In that case there would exist two distinct points on the same side of the common base satisfying the same two distance conditions.

Proposition I.7 excludes precisely this situation.

Therefore the remaining vertices must also coincide, and the two triangles agree completely. Consequently, all corresponding angles are congruent.

8.3 Statement

Theorem 3 (Euclid I.8)**.** *Let*

$$\triangle ABC \quad \text{and} \quad \triangle DEF$$

be nondegenerate triangles.

Assume

$$AB \cong DE,$$

$$BC \cong EF,$$

and

$$AC \cong DF.$$

Then

$$\angle BAC \cong \angle EDF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

That is, triangles having three corresponding sides congruent also have their corresponding angles congruent.

Diagrammatic audit. Two geometric figures are useful here. The first records the three side correspondences. The second isolates the genuinely nontrivial classical move: superposition followed by Proposition I.7. Wilkins's study note makes this same logical point explicit: after the first triangle is moved so that B, C occupy E, F on the chosen side, Proposition I.7 forces the remaining vertex to coincide with D. The much longer alternative proof attributed to Philo of Byzantium introduces five positional cases; it is historically interesting, but it is not the proof formalized here and is not duplicated diagrammatically.

8.4 Classical Proof

Superpose the triangle ABC onto the triangle DEF so that the vertex B coincides with E and the side BC lies along the side EF.

Since

$$BC \cong EF,$$

the point C coincides with F.

Suppose that the remaining vertex A does not coincide with D. Then there would exist two distinct points on the same side of the base EF satisfying

$$EA \cong ED$$

and

$$FA \cong FD.$$

This contradicts Proposition I.7, which asserts the uniqueness of such a point.

Therefore the vertex A must coincide with D, and the two triangles coincide completely.

Hence all corresponding angles are congruent:

$$\angle BAC \cong \angle EDF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

This establishes the Side–Side–Side congruence theorem.

8.5 Proof Architecture of the Reconstruction

The formal reconstruction is dramatically simpler than Euclid's classical proof.

No superposition argument is required. Once the three corresponding side congruences are available, the previously established Side–Side–Side theorem of the Hilbert interface immediately provides the congruence of the two triangles together with the congruence of all corresponding angles.

The entire classical argument is therefore replaced by a single application of a high-level interface theorem.

```
Euclid
------
three corresponding side equalities
          |
superpose BC on EF
          |
I.7 excludes a distinct remaining vertex
          |
triangles coincide
          |
corresponding angles are equal


Hilbert / Lean
--------------
three corresponding segment congruences
          |
          v
HilbertSSS
          |
          v
angleA, angleB, angleC
          |
          v
euclid_proposition_8
```

8.6 Hilbert Reconstruction

The Hilbert reconstruction is structurally much shorter than Euclid's proof at the level of Proposition I.8. The hypotheses are exactly the three side congruences required by the established theorem `HilbertSSS`. Applying that theorem to ABC and DEF yields the complete triangle congruence data, including all three corresponding angle congruences.

Consequently no superposition, rigid motion, or separate appeal to Proposition I.7 appears in `Proposition08.lean`. Those issues belong to the foundational proof of SSS rather than to this Book I wrapper.

8.7 Lean Formalization

The complete implementation is available in the repository. Here the code is short enough that only its decisive interface call needs to be displayed.

```
have hSSS :=
  HilbertSSS
    Geo
    A B C
    D E F
    hABC
    hAB
    hBC
```

```
    hAC
```

The result returned by `HilbertSSS` already contains the three corresponding angle congruences. Proposition I.8 merely exposes them in the order required by the Euclidean statement:

```
exact
  <hSSS.2.angleA,
   hSSS.2.angleB,
   hSSS.2.angleC>
```

Thus the formal proposition does not reproduce Euclid's superposition argument. That foundational work has already been absorbed into the Hilbert SSS theorem.

8.8 Relationship with Earlier Propositions

8.8.1 Proposition I.4

Proposition I.4 establishes SAS: two sides and the included angle determine the remaining side and angles of a triangle.

Proposition I.8 provides the complementary rigidity theorem. If all three corresponding sides are congruent, then the corresponding angles are congruent as well.

Together I.4 and I.8 provide the two fundamental triangle-congruence principles used repeatedly in the rest of Book I.

8.8.2 Proposition I.5

The isosceles-triangle theorem supplies an early example of recovering angle information from segment congruence. Proposition I.8 generalizes that phenomenon from the special isosceles configuration to arbitrary triangles with three corresponding sides congruent.

8.8.3 Proposition I.7

Euclid uses Proposition I.7 to exclude the possibility that two different vertices on the same side of a base could realize the same two prescribed distances from the endpoints of that base.

This uniqueness result is the critical classical ingredient in I.8: after superposition, I.7 forces the remaining vertex to coincide with its counterpart.

8.8.4 Transition to Proposition I.9

Proposition I.9 uses SSS immediately. Its auxiliary construction creates two triangles with three corresponding sides congruent, and I.8 then yields equality of the two angles forming the desired angle bisector.

Thus I.8 becomes a central reusable congruence theorem as soon as it is established.

8.9 Hilbert and Book Zero Components Used

8.9.1 Hilbert SSS

The formal reconstruction packages the SSS rigidity statement as a Hilbert theorem rather than reproducing Euclid's superposition argument inside every later application.

Its conclusion supplies the corresponding angle congruences required by the public Euclidean proposition.

8.9.2 Same-Side and Opposite-Side Geometry

The classical proof relies on the phrase "apply one triangle to the other" and on the location of the transported vertex. In the Hilbert setting this positional information is expressed explicitly through side-of-line relations.

8.9.3 Angle Congruence

The target of I.8 is not merely coincidence of triangles but congruence of corresponding angles. The formal theorem therefore returns the angle-congruence information in the orientation required by subsequent Book I propositions.

8.9.4 Nondegeneracy

The triangles must be genuine triangles. The formal hypotheses record noncollinearity explicitly, preventing degenerate configurations that are silently excluded by the classical diagram.

8.10 Formalization Notes

Proposition I.8 concludes the first major block of triangle congruence theory developed in Book I.

Historically, Euclid establishes the Side–Side–Side criterion only after proving the uniqueness result of Proposition I.7 and employing the method of superposition. The classical proof therefore depends on a carefully ordered sequence of earlier propositions.

The Hilbert reconstruction follows a different logical organization. The general Side–Side–Side theorem has already been established as a theorem of the Hilbert interface. Consequently, Proposition I.8 becomes a direct application of this previously developed result, requiring no additional geometric construction.

Together, Propositions I.4 through I.8 illustrate the progressive development of triangle congruence theory. The classical arguments of Euclid are systematically replaced by applications of increasingly powerful synthetic theorems derived from the Hilbert axioms.

This transition marks the completion of the first coherent layer of the CGJteamLab Hilbert interface and demonstrates how a modern axiomatic framework can substantially simplify the formal verification of classical geometry.

8.11 Dependency Summary

The classical dependency structure is

$$\begin{gathered} AB \cong DE, \quad AC \cong DF, \quad BC \cong EF \\ \downarrow \\ \text{superposition} \\ \downarrow \\ \text{I.7}: \ \text{uniqueness of the third vertex} \\ \downarrow \\ \text{coincidence of the triangles} \\ \downarrow \\ \text{corresponding angles congruent} \\ \downarrow \\ \text{I.8.} \end{gathered}$$

In the Hilbert reconstruction, the same rigidity is isolated as the SSS theorem and used through the congruence interface.

No numerical lengths, coordinates, or new geometric axioms are required by the public Proposition I.8.

8.12 Conclusion

Proposition I.8 establishes Side-Side-Side congruence for triangles.

Historically, Euclid proves the result by superposition and uses Proposition I.7 to rule out a second possible position of the remaining vertex. The reconstruction makes the underlying rigidity theorem explicit in the Hilbert layer.

This is a major infrastructural point in Book I. From I.8 onward, three segment congruences can be converted into angle congruences without reconstructing the superposition argument each time.

The theorem is used immediately in Proposition I.9 and continues to support midpoint, perpendicular, and later triangle constructions. Thus I.8 is not only a classical proposition but one of the principal congruence interfaces of the formal development.

Chapter 9

Proposition I.9

9.1 Introduction

Proposition I.9 presents Euclid's construction of the bisector of a given rectilinear angle.

The classical proof combines several earlier propositions. A segment of one side of the angle is first copied onto the other side, after which an auxiliary equilateral triangle is constructed. The Side–Side–Side criterion established in Proposition I.8 is then used to prove that the joining segment bisects the angle.

The Hilbert reconstruction follows a different route. Rather than constructing an auxiliary equilateral triangle, it first lays off a segment congruent to one side of the angle, constructs the midpoint of the resulting segment, and finally applies the general Side–Side–Side theorem.

Although the geometric constructions differ substantially, both approaches establish the existence of an angle bisector within neutral geometry.

9.2 The Classical Configuration

Let $\angle BAC$ be the given rectilinear angle; see Figure 9.1.

The objective is to construct a ray issuing from the vertex A that divides the angle into two congruent angles.

No assumption is made on the lengths of the sides forming the angle; only the angle itself is given.

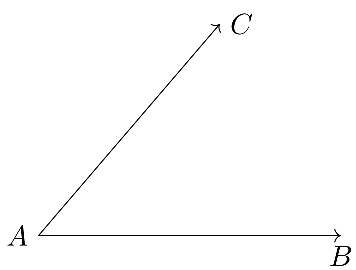


Figure 9.1: The given angle $\angle BAC$.

9.2.1 Construction

Choose an arbitrary point D on the side AB.

On the side AC, construct a point E such that

$$AE \cong AD.$$

Join the points D and E, and construct the equilateral triangle DEF on the segment DE.

Finally, join the vertex A to the vertex F; the completed construction is shown in Figure 9.2.

The claim is that the segment AF bisects the given angle.

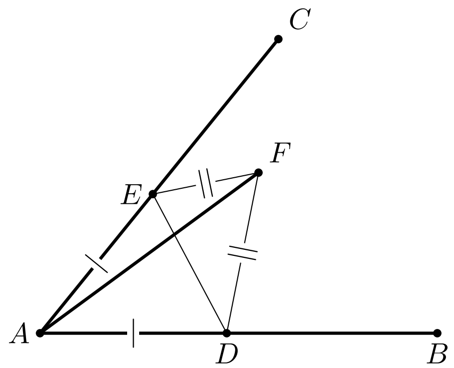


Figure 9.2: Euclid's construction for Proposition I.9. The points D, E satisfy $AD \cong AE$, and DEF is equilateral. With AF common, SSS for ADF and AEF proves that AF bisects the given angle.

9.2.2 Application of Proposition I.8

Since the triangle DEF is equilateral,

$$DF \cong EF.$$

Moreover,

$$AD \cong AE$$

by construction, while

$$AF \cong AF$$

is common to the two triangles.

Therefore the triangles

$$\triangle ADF \quad \text{and} \quad \triangle AEF$$

have three corresponding sides congruent.

By Proposition I.8,

$$\angle DAF \cong \angle FAE.$$

Since D lies on the ray AB and E lies on the ray AC, these are precisely the two parts of the original angle $\angle BAC$.

Hence AF bisects the given angle.

9.3 Statement

Theorem 4 (Euclid I.9)**.** *Let $\angle BAC$ be a nondegenerate angle.*

Then there exists a point M such that

$$\angle BAM \cong \angle MAC.$$

Equivalently, the ray AM bisects the angle $\angle BAC$.

Diagrammatic audit. Two classical figures are sufficient: the given angle and the completed auxiliary construction. Wilkins's essential diagram is the latter: points D, E are equidistant from A, DEF is equilateral, and AF is the candidate bisector. These are exactly the data used by SSS. No additional classical snapshots are needed. The Hilbert proof, however, uses a genuinely different configuration based on a midpoint of DC; that difference is recorded in the architecture diagram rather than hidden behind a copy of Euclid's picture.

9.4 Classical Proof

Choose a point D on the side AB and construct a point E on the side AC such that

$$AD \cong AE.$$

Join the points D and E, and construct the equilateral triangle DEF.

Finally, join the vertex A to the vertex F.

Since

$$AD \cong AE,$$

$$DF \cong EF,$$

and

$$AF \cong AF,$$

the triangles

$$\triangle ADF \quad \text{and} \quad \triangle AEF$$

have three corresponding sides congruent.

By Proposition I.8,

$$\angle DAF \cong \angle FAE.$$

Because D lies on the ray AB and E lies on the ray AC, these angles are exactly

$$\angle BAF \quad \text{and} \quad \angle FAC.$$

Hence the ray AF bisects the given angle.

This completes the construction.

9.5 Proof Architecture of the Reconstruction

The Hilbert reconstruction replaces Euclid's auxiliary equilateral triangle by two fundamental constructions already available in the Hilbert interface.

First, a segment congruent to one side of the angle is laid off on the other side of the angle. The midpoint of the resulting segment is then constructed, and the Side–Side–Side theorem is applied to two auxiliary triangles.

Thus the existence of the angle bisector follows entirely from previously established neutral-geometric results.

```
Euclid
------
angle BAC
   |
I.3: choose D,E with AD ~= AE
   |
I.1: construct equilateral DEF
   |
I.8: SSS on ADF and AEF
   |
AF bisects angle BAC


Hilbert / Lean
--------------
noncollinear A,B,C
   |
segment_construction: D on ray AB, AD ~= AC
   |
HilbertMidpointExists: M midpoint of DC
```

```
   |
HilbertSSS: ADM ~= ACM
   |
same-ray angle transport: DAM = BAM
   |
AM bisects angle BAC
```

9.6 Hilbert Reconstruction

The formal reconstruction uses a different symmetric configuration from Euclid's. From the noncollinear points A, B, C, segment construction produces a point D on the ray AB such that

$$AD \cong AC.$$

The midpoint theorem is then applied to DC, producing M with

$$D - M - C, \qquad DM \cong MC.$$

The triangles ADM and ACM have the three corresponding sides

$$AD \cong AC, \qquad DM \cong CM, \qquad AM \cong AM.$$

Hilbert SSS therefore yields

$$\angle DAM \cong \angle CAM.$$

Finally, because D lies on the same ray from A as B, angle transport identifies $\angle DAM$ with $\angle BAM$. Hence

$$\angle BAM \cong \angle CAM,$$

which is the required angle bisection.

This route uses neither Euclid's equilateral triangle DEF nor a direct invocation of Proposition I.8. Its symmetry is created instead by segment layoff and midpoint construction.

9.7 Lean Formalization

The complete implementation is available in the repository. The purpose of the present section is therefore not to reproduce the whole source file, but to isolate the steps that explain the formal geometry.

The first characteristic move is the replacement of Euclid's pair of equal segments $AD = AE$ by a single laid-off segment on the ray AB:

```
rcases
    HilbertCongruence.segment_construction
```

```
      (Geo := Geo)
      A C
      A B
      hAB with
  <D, hRayBD, hAD_AC>
```

Thus D lies on the ray AB and satisfies

$$AD \cong AC.$$

This is the first point at which the Hilbert reconstruction visibly departs from Euclid's construction.

The second decisive step is the use of a midpoint of DC:

```
rcases
    HilbertMidpointExists
      Geo D C hDC with
  <M, hMid>
```

The midpoint provides

$$D - M - C, \qquad DM \cong MC.$$

Together with $AD \cong AC$ and the common side AM, this creates the two triangles used for SSS.

The proof then applies

```
have hSSS :=
  HilbertSSS
    Geo
    A D M
    A C M
    hADM
    hACM
    hAD_AC
    hDM_CM
    hAM_AM
```

and extracts the equality of the angles at A. A final same-ray transport replaces the ray AD by the original ray AB, yielding

$$\angle BAM \cong \angle CAM.$$

The full theorem contains the incidence and noncollinearity bookkeeping needed to justify these steps. Those details remain in the source code; the three excerpts above are the parts that explain the mathematical structure of the Lean proof.

9.8 Relationship with Earlier Propositions

9.8.1 Proposition I.1

Euclid begins the construction by choosing points on the two arms of the given angle and constructing an equilateral triangle on the segment joining them. Proposition I.1 supplies the symmetric auxiliary configuration from which the angle bisector will be obtained.

9.8.2 Proposition I.3

The classical proof uses segment cutting to arrange equal auxiliary lengths on the two sides of the given angle. Proposition I.3 supplies this segment-layoff step.

9.8.3 Proposition I.8

Proposition I.8 is the decisive congruence theorem in Euclid's proof. The two triangles adjacent to the candidate bisector have three corresponding sides congruent, so SSS yields congruence of the two angles at the original vertex.

Thus the classical dependency pattern is

$$\mathrm{I.3} \longrightarrow \mathrm{I.1} \longrightarrow \mathrm{I.8} \longrightarrow \mathrm{I.9}.$$

9.8.4 Transition to Proposition I.10

Proposition I.10 uses the angle-bisection construction immediately: an equilateral triangle is erected on the given segment and its apex angle is bisected. SAS then shows that the point where the bisector meets the base is the midpoint.

Thus I.9 becomes constructional infrastructure as soon as it is proved.

9.9 Hilbert and Book Zero Components Used

9.9.1 Angle-Bisector Representation

The conclusion is expressed synthetically by a ray from the vertex whose two adjacent angles are congruent. No numerical angle measure is used.

9.9.2 Segment Layoff

The Hilbert segment-construction machinery supplies the equal auxiliary segments needed to create the symmetric configuration around the given angle.

9.9.3 SSS

The congruence of the two auxiliary triangles is obtained from the established Hilbert SSS theorem. The desired angle congruence is then a corresponding-angle consequence.

9.9.4 Interior and Ray Information

The constructed bisector must lie in the intended interior region of the given angle. The formal proof therefore carries ray and side-of-line information that is implicit in Euclid's diagram.

9.10 Formalization Notes

Proposition I.9 is the first construction in Book I whose Hilbert reconstruction differs substantially from Euclid's original proof.

Euclid constructs an auxiliary equilateral triangle and applies the Side–Side–Side criterion established in Proposition I.8. The proof is therefore closely tied to the historical sequence of propositions in the *Elements*.

The Hilbert reconstruction adopts a different strategy. A congruent segment is first laid off on one side of the angle, the midpoint of the resulting segment is constructed, and the general Side–Side–Side theorem is applied directly to two auxiliary triangles.

This approach illustrates a general principle of the CGJteamLab library: classical geometric constructions are not reproduced verbatim when a simpler synthetic argument is available within the existing Hilbert theory.

Proposition I.9 therefore demonstrates how a modern axiomatic development can replace a classical construction by a shorter proof while preserving the same mathematical content.

9.11 Dependency Summary

The formal reconstruction follows the same basic synthetic geometry as the classical proof:

$$\begin{array}{c} \text{nondegenerate angle} \\ \downarrow \\ \text{lay off equal auxiliary segments} \\ \downarrow \\ \text{construct symmetric auxiliary triangle} \\ \downarrow \\ \text{SSS} \\ \downarrow \\ \text{congruent adjacent angles} \\ \downarrow \\ \text{angle bisector} \\ \downarrow \\ \texttt{euclid\_proposition\_9}. \end{array}$$

The reconstruction uses only the established Hilbert and Book Zero infrastructure. It introduces neither numerical angle measure nor a new geometric axiom.

9.12 Conclusion

Proposition I.9 constructs the bisector of a given rectilinear angle.

The classical proof is an elegant combination of the early construction theorems and SSS: equal auxiliary segments create a symmetric configuration, and triangle congruence converts that symmetry into equality of the two angles at the vertex.

The formal reconstruction preserves this synthetic character while making the positional content explicit. A bisector is not merely an abstract equality of two angle magnitudes; it is represented by a specific ray lying in the appropriate geometric configuration.

I.9 is also an important infrastructural milestone. The construction is used immediately in Proposition I.10 and becomes part of the reusable angle-construction language needed throughout the later development.

Chapter 10

Proposition I.10

10.1 Introduction

Proposition I.10 constructs the midpoint of a given finite straight line.

In Euclid's development this construction depends on several earlier results. An equilateral triangle is first erected on the given segment using Proposition I.1. The vertex angle of this triangle is then bisected by Proposition I.9, and the Side–Angle–Side congruence criterion of Proposition I.4 is used to prove that the intersection point is the midpoint of the original segment.

The Hilbert reconstruction follows a fundamentally different approach. The existence of a midpoint has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this previously derived result.

This proposition therefore provides one of the clearest examples of the difference between Euclid's historical development and a modern axiomatic reconstruction.

10.2 The Classical Configuration

Let AB be the given finite straight line; see Figure 10.1.

The objective is to construct a point lying on the segment AB that divides it into two congruent parts.

Such a point is called the midpoint of the segment.

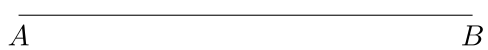


Figure 10.1: The given finite straight line AB.

10.2.1 Construction

Construct the equilateral triangle ABC on the segment AB by Proposition I.1.

Next, bisect the angle $\angle ACB$ by Proposition I.9.

Let the angle bisector meet the segment AB at the point D; the completed construction is

Figure 10.2.

The claim is that D is the midpoint of the segment AB.

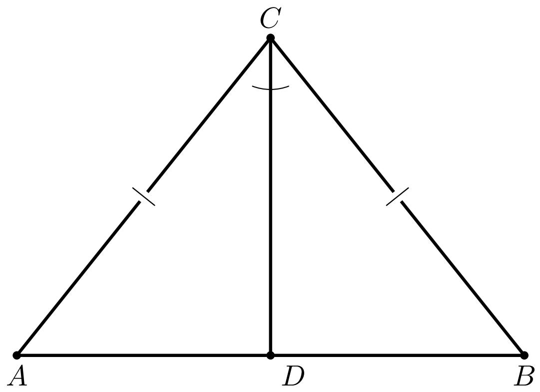


Figure 10.2: Euclid's construction for Proposition I.10. The equilateral triangle gives $AC \cong BC$; the ray CD bisects the angle at C. SAS for ACD and BCD yields $AD \cong DB$.

10.2.2 Application of Proposition I.4

Consider the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD.$$

Since the triangle ABC is equilateral,

$$AC \cong BC.$$

The segment

$$CD$$

is common to both triangles, and by construction,

$$\angle ACD \cong \angle DCB.$$

Therefore the hypotheses of Proposition I.4 are satisfied.

It follows that

$$AD \cong DB.$$

Since the point D lies on the segment AB, it is the midpoint of the given finite straight line.

10.3 Statement

Theorem 5 (Euclid I.10)**.** *Let AB be a finite straight line with*

$$A \neq B.$$

Then there exists a midpoint of the segment AB. That is, there exists a point M such that

$$A, M, B$$

are collinear,

$$M$$

lies between A and B, and

$$AM \cong MB.$$

Equivalently, every nondegenerate segment admits a midpoint.

Diagrammatic audit. Two geometric figures are sufficient. The first records the input segment; the second contains the entire Euclidean construction and the data used by SAS. Wilkins likewise presents the equilateral-triangle/angle-bisector configuration as the essential picture. His note also records a distinct historical construction attributed by Proclus to Apollonius, using two circles and SSS; that alternative is mathematically interesting but is not part of Euclid's proof adopted here and is therefore not duplicated as a second construction.

10.4 Classical Proof

Construct the equilateral triangle ABC on the segment AB by Proposition I.1.

Bisect the angle $\angle ACB$ by Proposition I.9, and let the bisecting ray meet the segment AB at the point D.

Since

$$AC \cong BC,$$

$$CD \cong CD,$$

and

$$\angle ACD \cong \angle DCB,$$

the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD$$

satisfy the hypotheses of Proposition I.4.

Hence

$$AD \cong DB.$$

Because the point D lies on the segment AB, it is the midpoint of the given finite straight line.

This completes the construction.

10.5 Proof Architecture of the Reconstruction

The Hilbert reconstruction replaces the entire classical construction by a previously established theorem of the Hilbert interface.

Rather than constructing an auxiliary equilateral triangle, bisecting an angle, and applying the Side–Angle–Side criterion, the existence of a midpoint is obtained directly from the general midpoint existence theorem.

Consequently, Proposition I.10 becomes a simple wrapper around an existing interface theorem.

```
Euclid
------
given segment AB
      |
I.1: equilateral triangle ABC
      |
I.9: bisect angle ACB
      |
I.4: SAS
      |
AD ~= DB
      |
D is midpoint of AB


Hilbert / Lean
--------------
A != B
  |
  v
HilbertMidpointExists
  |
  v
exists M, HilbertIsMidpoint Geo M A B
  |
  v
euclid_proposition_10
```

10.6 Hilbert Reconstruction

The reconstruction consists of a direct application of the general midpoint existence theorem.

Given a nondegenerate segment, the theorem `HilbertMidpointExists` immediately produces a point lying between the endpoints and equidistant from them.

No auxiliary geometric construction is required, since the existence of midpoints has already been established independently within the Hilbert development.

This represents one of the largest simplifications obtained by the layered architecture of the Hilbert interface.

10.7 Lean Formalization

The Lean theorem is correspondingly short:

```
theorem euclid_proposition_10
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B : Geo.Point)
    (hAB : A != B) :
    exists M : Geo.Point,
      HilbertIsMidpoint Geo M A B := by
  exact
    HilbertMidpointExists
      Geo A B hAB
```

Thus the proposition itself introduces no auxiliary construction. Its mathematical content has already been packaged in `HilbertMidpointExists`; I.10 is the Euclidean-facing wrapper exposing that result in the Book I development.

10.8 Relationship with Earlier Propositions

10.8.1 Proposition I.1

Euclid's construction of the midpoint begins by constructing an equilateral triangle on the given segment. Proposition I.1 therefore supplies the first auxiliary point and the equal sides needed later in the congruence argument.

10.8.2 Proposition I.4

The decisive classical step is SAS. After the angle at the apex has been bisected, the two resulting triangles share one side, have a pair of equal sides from the equilateral construction, and have congruent included angles. Proposition I.4 then yields congruence of the two triangles and hence equality of the two halves of the original segment.

10.8.3 Proposition I.9

Proposition I.9 is the immediate constructional input: it bisects the angle at the apex of the equilateral triangle.

Thus Euclid's dependency chain is particularly transparent:

$$\mathrm{I.1} \longrightarrow \mathrm{I.9} \longrightarrow \mathrm{I.4} \longrightarrow \mathrm{I.10}.$$

10.8.4 Transition to Proposition I.11

The midpoint construction becomes useful immediately in the classical proof of Proposition I.11, where a symmetric configuration about a point on a line is used to construct a perpendicular.

More generally, midpoint and segment-bisection infrastructure becomes a recurring component of later Book I constructions.

10.9 Hilbert and Book Zero Components Used

10.9.1 Midpoint Representation

The formal conclusion is expressed by the project midpoint predicate: the constructed point lies between the endpoints and the two resulting segments are congruent. Thus midpoint remains entirely synthetic:

$$A - M - B \qquad \text{and} \qquad AM \cong MB.$$

10.9.2 Interface Theorem

At the level of Proposition I.10 itself, the only substantive geometric dependency is

```
HilbertMidpointExists
```

together with the incidence and congruence structures required by that theorem. Angle bisection and SAS belong to Euclid's historical proof; they are not invoked by `euclid_proposition_10`.

10.10 Formalization Notes

Proposition I.10 provides one of the clearest illustrations of the difference between Euclid's historical development and a layered axiomatic reconstruction.

In the *Elements*, the midpoint construction depends on three earlier propositions. Euclid first constructs an equilateral triangle, then bisects one of its angles, and finally applies the Side–Angle–Side criterion to prove that the resulting point divides the original segment into two equal parts.

The Hilbert reconstruction is fundamentally different. The existence of midpoints has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this theorem, requiring no additional geometric construction.

This proposition demonstrates the principal advantage of the layered architecture developed in CGJteamLab. Once sufficiently rich interface theorems have been established, many classical propositions cease to be independent proofs and instead become concise applications of the existing theory.

Proposition I.10 is perhaps the simplest example of this philosophy, reducing an entire classical construction to a single theorem call.

10.11 Dependency Summary

The two proof architectures should be kept distinct:

```
Classical:
I.1  ->  I.9  ->  I.4  ->  I.10

Formal:
HilbertMidpointExists  ->  euclid_proposition_10
```

The formal theorem does not replay Euclid's equilateral-triangle, angle-bisection, and SAS construction. It reuses a midpoint theorem already proved in the Hilbert interface. No coordinate system, numerical length, or new geometric axiom is introduced.

10.12 Conclusion

Proposition I.10 completes the first ten propositions of Book I with the construction of a midpoint.

The classical proof is a compact synthesis of three earlier results: I.1 supplies an equilateral triangle, I.9 bisects its apex angle, and I.4 converts the resulting SAS configuration into equality of the two halves of the given segment.

The formal reconstruction preserves this synthetic character while making one feature especially explicit: a midpoint is not merely a point with two equal distances. Its position between the endpoints is part of the theorem and must be carried by the Hilbert order structure.

I.10 therefore consolidates the early construction and congruence machinery into a reusable segment-bisection result. This midpoint infrastructure becomes immediately relevant to the perpendicular constructions beginning with Proposition I.11.

Chapter 11

Proposition I.11

11.1 Introduction

Proposition I.11 states:

> To draw a straight line at right angles to a given straight line from a given point on it.

Let AB be the given straight line and let C be the given point on AB.

Euclid's construction proceeds as follows. Choose a point D on AC. By Proposition I.3, construct a point E on the opposite side of C such that

$$CD \cong CE.$$

On the segment DE construct the equilateral triangle DFE by Proposition I.1, and join C to F.

Since

$$CD \cong CE, \qquad DF \cong EF,$$

and CF is common to the two triangles DCF and ECF, Proposition I.8 gives

$$\angle DCF \cong \angle ECF.$$

The points D, C, and E lie on one straight line. Hence the two congruent angles DCF and FCE are adjacent angles formed when the straight line CF stands on the straight line DE.

By Definition 10, each of these angles is therefore a right angle. Consequently, CF is drawn at right angles to the given straight line AB at the given point C.

Thus the required perpendicular has been constructed.

11.2 The Classical Configuration

Let AB be the given straight line and let C be the prescribed point on it. The task is to construct a straight line through C perpendicular to AB.

Euclid's construction chooses a point D on one side of C on the given line, then chooses E on the opposite side so that
$$CD \cong CE.$$
An equilateral triangle DFE is constructed on the base DE.

The relevant configuration is therefore

$$D - C - E, \qquad CD \cong CE, \qquad DF \cong FE.$$

The segment CF is the candidate perpendicular. The proof compares the two triangles

$$\triangle DCF \qquad \text{and} \qquad \triangle ECF.$$

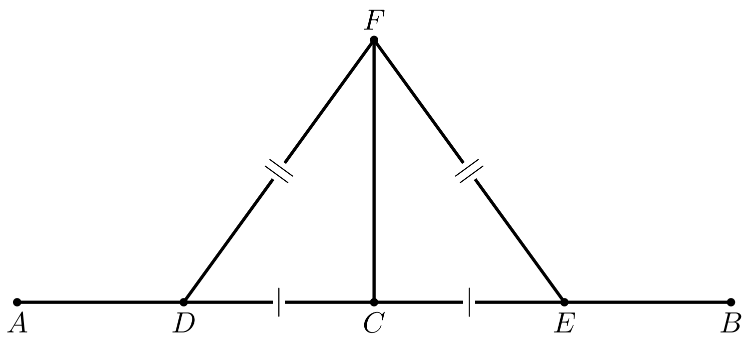


Figure 11.1: Classical construction for Proposition I.11. The points D, C, E are collinear with $CD \cong CE$, and DFE is equilateral. SSS applied to DCF and ECF makes the adjacent angles at C equal.

11.3 Statement

The synthetic content of Proposition I.11 is the existence of a perpendicular through a prescribed point of a given straight line.

Theorem 6 (Euclid I.11)**.** *Let A, C, B be three points such that*

$$A * C * B.$$

Then there exists a point X such that the angle ACX is a right angle.

In the Hilbert development, a right angle is represented directly by the Euclidean definition: an angle is right when it is congruent to its adjacent angle obtained by extending one of its sides through the vertex.

More precisely, we define
$$\text{RightAngle}(A, O, B)$$
to mean that there exists a point C such that

$$A * O * C$$

and
$$\angle AOB \cong \angle BOC.$$

Thus Proposition I.11 is represented formally by

$$A * C * B \quad \Longrightarrow \quad \exists X \ \mathrm{RightAngle}(A, C, X).$$

The corresponding Lean definition is

```
def HilbertRightAngle
(Geo : Geometry.Geo)
(A O B : Geo.Point) : Prop :=
exists C : Geo.Point,
Geo.Between A O C /\
Geo.AngleCongruent A O B B O C
```

and the proposition itself has the form

```
theorem euclid_proposition_11
[HilbertCongruence Geo]
(A C B : Geo.Point)
(hACB : Geo.Between A C B) :
exists X : Geo.Point,
HilbertRightAngle Geo A C X
```

Notice that the formal statement does not attempt to encode Euclid's particular construction of the perpendicular. It expresses the geometric result of that construction: through the prescribed point C on the given line there exists a ray CX forming a right angle with the line.

Diagrammatic audit. Only one classical geometric figure is needed here. Wilkins's diagram contains exactly the construction that carries the proof: the symmetric points D, E, the equilateral triangle DFE, and the segment CF. There is no hidden sequence of geometrically different states worth duplicating. In contrast, the Hilbert proof below uses a substantially different auxiliary construction; its complexity is primarily order-theoretic and is better exposed by the dependency architecture than by copying the classical picture repeatedly.

11.4 Classical Proof

Let AB be the given straight line and let C be the prescribed point on it.

Choose a point D on AC. By Proposition I.3, cut off from the ray starting at C in the direction of B a segment CE congruent to CD. Thus

$$CD \cong CE,$$

and the points D, C, E lie on one straight line, with C between D and E.

Construct the equilateral triangle DFE on the segment DE by Proposition I.1, and join C to F; see Figure 11.1. Since the triangle is equilateral,

$$DF \cong EF.$$

Moreover,

$$CF \cong CF$$

by reflexivity.

Consider the triangles

$$\triangle DCF \qquad \text{and} \qquad \triangle ECF.$$

Their three corresponding sides satisfy

$$CD \cong CE, \qquad DF \cong EF, \qquad CF \cong CF.$$

Therefore Proposition I.8 yields

$$\angle DCF \cong \angle ECF.$$

Since D, C, E are collinear and C lies between D and E, the rays CD and CE are opposite rays. Hence

$$\angle DCF \qquad \text{and} \qquad \angle FCE$$

are adjacent angles formed by the straight line CF standing on the straight line DE.

The two adjacent angles are congruent. By Euclid's Definition 10, each of them is therefore a right angle.

Consequently,

$$\angle DCF$$

is a right angle, and the straight line through C and F is perpendicular to the given straight line AB at the prescribed point C.

This proves Proposition I.11.

11.5 Proof Architecture of the Reconstruction

The formal reconstruction separates the notion of a right angle from the particular statement of Proposition I.11.

At the foundational level, the predicate `HilbertRightAngle` expresses Euclid's Definition 10. An angle AOB is right when there exists a point C on the extension of AO through O such that

$$\angle AOB \cong \angle BOC.$$

The substantial geometric result is then established independently as

```
hilbert_right_angle_exists
```

which proves that, whenever

$$A * C * B,$$

there exists a point X such that

$$\text{HilbertRightAngle}(A, C, X).$$

The proof of this existence theorem is considerably more elaborate than the final formalization of Proposition I.11. It uses the developed Hilbert theory of angle construction, segment construction, midpoints, triangle congruence, ray order, plane separation, and angle addition.

Once this general result has been established, Proposition I.11 itself becomes a direct interface theorem:

```
hilbert_right_angle_exists
|
v
euclid_proposition_11
```

More explicitly, the architecture is

$$\boxed{\texttt{HilbertRightAngle}}$$
$$\downarrow$$
$$\boxed{\text{Hilbert synthetic geometry}}$$

angle construction
segment construction
midpoint existence
Hilbert SSS
SameSide/OppositeSide theory
ray-order theorem
angle addition

$$\downarrow$$
$$\boxed{\texttt{hilbert\_right\_angle\_exists}}$$
$$\downarrow$$
$$\boxed{\texttt{euclid\_proposition\_11}}$$

Thus the formal dependency structure differs sharply from Euclid's historical construction. Euclid obtains the perpendicular by constructing an equilateral triangle and applying Proposition I.8. The Hilbert development instead proves a general right-angle existence theorem from the already developed synthetic theory and then obtains Proposition I.11 as an immediate consequence.

This distinction is intentional. The proposition file records the mathematical content of Euclid I.11, while the underlying Hilbert layer is free to establish that content by a more general synthetic argument.

11.6 Hilbert Reconstruction

The Hilbert proof of the existence theorem does not reproduce Euclid's equilateral-triangle construction. Instead, it derives a right angle from the general machinery already available in the Hilbert layer.

Assume

$$A * C * B.$$

Thus A, C, B lie on a common line, which we denote by b.

Choose a point D outside b. The angle-construction theorem is then applied at C to construct a point E, on the same side of b as D, such that

$$\angle ACD \cong \angle BCE.$$

The uniqueness part of angle construction determines the ray CE.

Next choose a point F on the ray CD such that

$$CF \cong CE.$$

Let X be the midpoint of FE. Hence

$$F * X * E$$

and

$$FX \cong XE.$$

Consider the triangles

$$\triangle CFX \qquad \text{and} \qquad \triangle CEX.$$

They satisfy

$$CF \cong CE, \qquad FX \cong XE, \qquad CX \cong CX.$$

Hilbert SSS therefore gives

$$\angle FCX \cong \angle ECX.$$

Since F and D lie on the same ray from C, replacing the ray CF by the ray CD does not change the corresponding angle. Consequently,

$$\angle DCX \cong \angle ECX.$$

We have therefore obtained the two component congruences

$$\angle ACD \cong \angle BCE$$

and

$$\angle DCX \cong \angle ECX.$$

The remaining problem is to justify that these component angles may be added in the required configuration.

This is the technically most substantial part of the reconstruction. The ray-order theorem applied to the noncollinear configuration determines two possible cases:

$$\text{RayMeetsSegment}(CD, EA)$$

or

$$\text{RayMeetsSegment}(CE, DA).$$

In the first case, the ray CD meets the open segment EA at a point Y. The SameSide/OppositeSide calculus shows that

$$A \text{ and } X$$

lie on opposite sides of the line CD, while

$$B \text{ and } X$$

lie on opposite sides of the line CE.

In the second case, the ray CE meets the open segment DA at a point Y. Plane-separation arguments show instead that

$$A \text{ and } X$$

lie on the same side of CD, while

$$B \text{ and } X$$

lie on the same side of CE.

Thus in both cases the two configurations have the same side pattern:

$$\text{SameSide}(A, X; CD) \iff \text{SameSide}(B, X; CE).$$

The proof also establishes that X does not lie on the original line AB. Hence both

$$\angle ACX \quad \text{and} \quad \angle BCX$$

are nondegenerate angles.

The angle-addition theorem can now be applied to

$$\angle ACD \cong \angle BCE$$

and

$$\angle DCX \cong \angle ECX.$$

It yields

$$\angle ACX \cong \angle BCX.$$

Since $A * C * B$, the rays CA and CB are opposite. Reversing the order of the sides in the second angle gives

$$\angle ACX \cong \angle XCB.$$

Therefore the two adjacent angles formed by CX with the straight line AB are congruent.

By the definition of `HilbertRightAngle`,

$$\text{HilbertRightAngle}(A, C, X).$$

Hence

$$\exists X \ \text{HilbertRightAngle}(A, C, X),$$

which is the required Hilbert form of Proposition I.11.

11.7 Lean Formalization

The formalization is divided into three levels: the definition of a right angle, the general Hilbert existence theorem, and the final Euclid I.11 interface theorem.

The notion of a right angle is represented by the following definition:

```
def HilbertRightAngle
(Geo : Geometry.Geo)
(A O B : Geo.Point) : Prop :=
exists C : Geo.Point,
Geo.Between A O C /\
Geo.AngleCongruent A O B B O C
```

This is a direct formalization of the adjacent-angle characterization of a right angle. The witness C extends the ray OA through O, and the congruence

$$\angle AOB \cong \angle BOC$$

states that the two adjacent angles are equal.

The main work is contained in the general existence theorem:

```
theorem hilbert_right_angle_exists
[HilbertCongruence Geo]
(A C B : Geo.Point)
(hACB : Geo.Between A C B) :
exists X : Geo.Point,
HilbertRightAngle Geo A C X := by
...
```

The proof first extracts the incidence and nondegeneracy information contained in `hACB`. It then chooses an auxiliary point off the line AB and constructs a congruent angle on the opposite ray at C.

A point F is subsequently constructed on the auxiliary ray with

$$CF \cong CE,$$

and the midpoint X of FE is introduced. The central triangle congruence step is encoded by an application of Hilbert SSS:

```
have hSSS :=
HilbertSSS ...
```

This produces the congruence of the two angles at C required for the second component of the final angle-addition argument.

The principal technical issue is not the congruence calculation but the relative position of the rays. The proof therefore invokes the ray-order theorem and splits into its two possible cases:

```
rcases hRayOrder with hCaseD | hCaseE
```

Each branch establishes the same logical condition required by the angle-addition theorem:

```
have hSideConfiguration :
HilbertSameSide Geo A X lineCD <->
HilbertSameSide Geo B X lineCE := by
...
```

In one branch both propositions are false; in the other both are true. This case distinction isolates the plane-separation reasoning from the final angle calculation.

After proving the necessary noncollinearity conditions, the two component angle congruences are combined by

```
have hACX_BCX :
Geo.AngleCongruent A C X B C X :=
hilbert_angle_addition
...
hSideConfiguration
```

```
...
hAngleE
hDCX_ECX
```

The second angle is then reversed:

```
have hACX_XCB :
Geo.AngleCongruent A C X X C B :=
(Geo.angle_congruent_reverse_second
A C X B C X).mp hACX_BCX
```

Since the original hypothesis already supplies

$$A * C * B,$$

the point B itself is the witness required by the definition of `HilbertRightAngle`. The existence theorem is therefore completed by

```
refine <X, ?_>
refine <B, hACB, ?_>
exact hACX_XCB
```

Once this theorem is available, the formal statement of Proposition I.11 contains no additional geometric argument:

```
theorem euclid_proposition_11
[HilbertCongruence Geo]
(A C B : Geo.Point)
(hACB : Geo.Between A C B) :
exists X : Geo.Point,
HilbertRightAngle Geo A C X := by
exact
hilbert_right_angle_exists
Geo A C B hACB
```

Thus the Lean theorem corresponding to Euclid I.11 is deliberately small. The substantial synthetic argument has been factored into `hilbert_right_angle_exists`, making right-angle existence available independently of Proposition I.11 and reusable in the subsequent development.

11.8 Relationship with Earlier Propositions

11.8.1 Proposition I.1

Euclid's classical proof invokes Proposition I.1 to construct the equilateral triangle DFE on the auxiliary segment DE.

In the Hilbert reconstruction the perpendicular construction is obtained more directly from angle construction and uniqueness, so the public proof need not reproduce the equilateral-triangle construction.

11.8.2 Proposition I.3

The classical construction requires a point E on the opposite side of C satisfying

$$CE \cong CD.$$

This is a segment-layoff step of the kind developed in the early propositions.

The Hilbert layer provides the corresponding layoff operation directly.

11.8.3 Proposition I.8

In Euclid's proof, the two triangles DCF and ECF have three corresponding sides congruent. Proposition I.8 therefore gives the congruence of the adjacent angles at C, from which their right-angle character follows.

The Hilbert reconstruction packages the same rigidity more directly through angle construction and the definition of a right angle.

11.8.4 Transition to Proposition I.12

Proposition I.11 solves the perpendicular problem when the prescribed point lies on the line. Proposition I.12 treats the complementary case of a point outside the line.

Together they form the basic perpendicular-construction block used in the subsequent straight-angle theory.

11.9 Hilbert and Book Zero Components Used

11.9.1 Segment Layoff

The construction needs a point on the opposite ray at a prescribed distance from the given point. This is supplied by the established Hilbert segment-construction machinery.

11.9.2 Hilbert III.4

Angle construction provides a ray forming an angle congruent to a prescribed angle on a selected side of the supporting line. Its uniqueness clause is particularly important in identifying the constructed perpendicular direction.

11.9.3 `HilbertRightAngle`

The final statement does not use a numerical value of 90°. Perpendicularity is expressed through the synthetic right-angle predicate already defined in the Hilbert interface.

11.9.4 Betweenness and Collinearity

The fact that the auxiliary points lie on opposite sides of C along the given line must be carried by explicit betweenness and primitive collinearity data. These replace the informal reading of point order from the classical diagram.

11.10 Formalization Notes

Proposition I.11 provides an instructive example of the distinction between the organization of Euclid's *Elements* and the organization of the Hilbert formalization.

In Euclid's development, Proposition I.11 is itself a construction. Its proof depends explicitly on earlier propositions:

$$\text{I.1}, \qquad \text{I.3}, \qquad \text{I.8},$$

together with the definition of a right angle. The perpendicular is obtained by constructing an equilateral triangle and comparing two triangles by SSS.

The formal Hilbert development reaches the same result through a different dependency structure. Proposition I.11 does not carry the construction itself. Instead, the substantial argument is isolated as the general theorem

```
hilbert_right_angle_exists
```

and Proposition I.11 merely exposes this theorem at the Euclidean interface.

This is similar in form to Proposition I.10. In both cases the final Euclid theorem is short because the required existence result has already been established in the underlying Hilbert theory. The amount of work hidden behind the interface, however, is very different.

For Proposition I.10, midpoint existence was already a natural primitive result of the developed theory. Proposition I.11 required the introduction of a new general notion,

```
HilbertRightAngle
```

and a substantial synthetic existence proof.

The proof also reveals where the formal difficulty of the result lies. The elementary congruence argument is comparatively short: after constructing F and the midpoint X of FE, SSS gives

$$\angle FCX \cong \angle ECX.$$

The difficult part is proving that the component angles occur in the correct relative configuration so that angle addition is legitimate.

This requires the developed theory of

$$\text{SameSide}, \qquad \text{OppositeSide}, \qquad \text{rays}, \qquad \text{betweenness},$$

together with the ray-order theorem. The resulting two-case analysis is substantially longer than the final congruence calculation.

Thus Proposition I.11 illustrates an important feature of the formalization. Statements that are elementary in an informal diagrammatic proof may require a significant amount of explicit order and separation theory once all geometric configurations must be justified formally.

At the same time, this complexity is not local overhead that must be repeated in later propositions. The result has been packaged as `hilbert_right_angle_exists`. Subsequent arguments may use right-angle existence directly without reconstructing the SameSide/OppositeSide analysis.

The resulting architecture is therefore

$$\text{foundational Hilbert theory} \longrightarrow \text{general synthetic lemmas} \longrightarrow \text{Euclid propositions.}$$

Proposition I.11 is a particularly clear instance of this pattern: its Euclidean interface is almost trivial precisely because the geometric content has been absorbed into the reusable Hilbert layer.

Finally, the comparison with Euclid should not be interpreted as a requirement that the formal proof reproduce Euclid's historical proof step by step. The purpose of the proposition layer is to recover the mathematical statements of Book I over the developed Hilbert theory. When that theory already supplies a stronger or more reusable result, the Euclid proposition can appropriately appear as a thin interface theorem.

11.11 Dependency Summary

The classical dependency pattern is

$$\text{segment layoff} \longrightarrow \text{I.1} \longrightarrow \text{I.8} \longrightarrow \text{equal adjacent angles} \longrightarrow \text{I.11.}$$

The Hilbert reconstruction packages the construction differently:

$$\begin{array}{c}
\text{point on a line} \\
\downarrow \\
\text{segment layoff + order data} \\
\downarrow \\
\text{angle construction / uniqueness} \\
\downarrow \\
\texttt{HilbertRightAngle} \\
\downarrow \\
\text{perpendicular through the prescribed point} \\
\downarrow \\
\texttt{euclid_proposition_11}.
\end{array}$$

No numerical angle measure and no new geometric axiom are introduced.

11.12 Conclusion

Proposition I.11 begins the explicit perpendicular-construction block of Book I.

Euclid reaches the perpendicular through an auxiliary equilateral triangle and SSS. The Hilbert reconstruction can express the same geometric requirement more directly through segment construction, angle construction, uniqueness, and the synthetic right-angle predicate.

The formalization therefore illustrates the same phenomenon already visible elsewhere in the reconstruction: the Euclidean proposition is preserved as the public milestone, while the dependency graph underneath it reflects the stronger organization of Hilbert geometry.

Together with Proposition I.12, I.11 supplies the perpendicular infrastructure needed for the straight-angle results beginning with Proposition I.13.

Chapter 12

Proposition I.12

12.1 Introduction

Euclid's Proposition I.12 asks for the construction of a perpendicular to a given straight line from a given point lying outside that line.

Classically, Euclid obtains the perpendicular by choosing a point on the opposite side of the given line, drawing a circle centered at the external point, and using the two intersections of the circle with the given line. The segment joining these intersections is bisected by Proposition I.10, and Proposition I.8 is then applied to prove that the line joining the external point to the midpoint is perpendicular to the given line.

The reconstruction over the Hilbert foundation follows a different synthetic route. No circle-line intersection theorem is used. Instead, a second point is constructed on the opposite side of the given line so that two distinct points of the line are equidistant from the original external point and the constructed point.

This leads naturally to the perpendicular-bisector configuration. The required midpoint property is isolated in the general theorem

```
hilbert_equidistant_line_bisects_segment
```

which states that if two distinct points of a line are equidistant from two points lying on opposite sides of that line, then the intersection of the joining segment with the line bisects that segment.

A second auxiliary result,

```
hilbert_collinear_angle_transport
```

encapsulates the angle-order argument needed in the proof of this perpendicular-bisector theorem.

Consequently, Proposition I.12 provides another example of the role of the Book Zero layer: a comparatively elaborate geometric configuration is separated into reusable synthetic lemmas before the final Euclidean proposition is formulated.

12.2 The Classical Configuration

Euclid's twelfth proposition asks for a perpendicular to a given straight line from a point outside it.

Let AB be the given straight line and let C be the given point outside it.

12.2.1 Final Configuration

The objective is to determine a point H on the straight line AB such that

$$CH \perp AB.$$

Euclid constructs two points E and G on AB satisfying

$$CE \cong CG,$$

and then bisects EG at H.

The final proof configuration, shown later in Figure 12.4, satisfies

$$E - H - G, \qquad EH \cong HG, \qquad CE \cong CG.$$

12.2.2 Construction Algorithm

The construction uses Proposition I.10 as an already established construction primitive.

Step 1. Given Line and External Point

Let AB be the given straight line and let C be a point not lying on AB; see Figure 12.1.

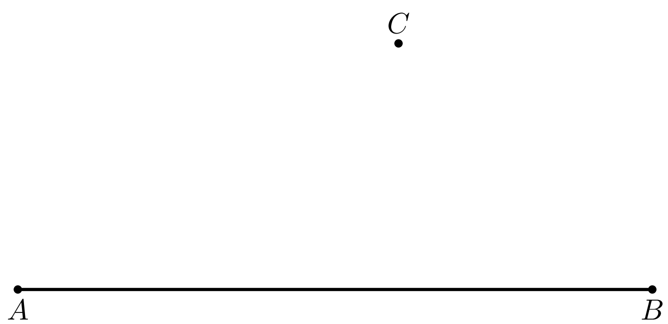


Figure 12.1: Input data for Proposition I.12: a straight line AB and a point C outside it.

Step 2. Choose a Point on the Opposite Side

Choose a point D on the opposite side of the straight line AB from C, as in Figure 12.2.

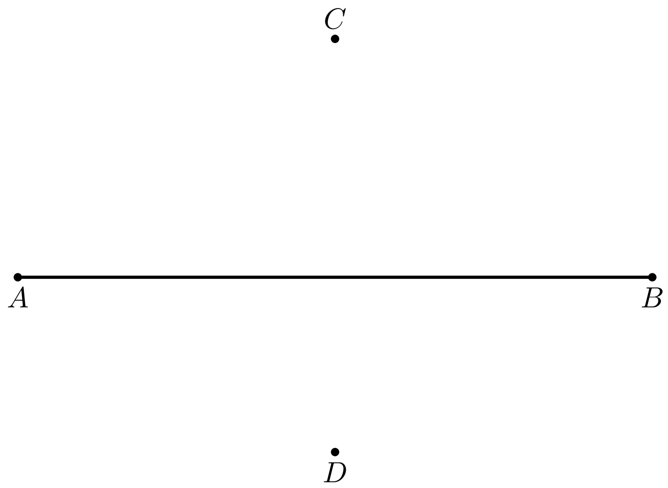


Figure 12.2: Choose D on the side of the line AB opposite C. This side-of-line information is essential for the later intersection argument.

Step 3. Draw the Circle Centered at C

Draw the circle centered at C and passing through D; the resulting intersection configuration is shown in Figure 12.3.

Let the circle meet the given straight line at E and G.

Since E and G lie on the same circle centered at C,

$$CE \cong CG.$$

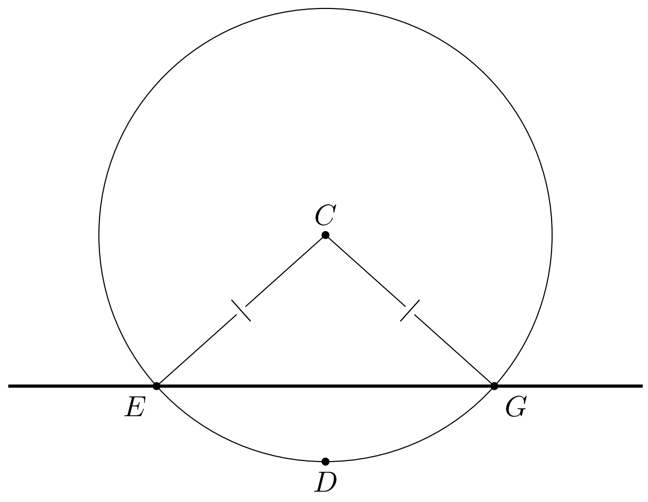


Figure 12.3: The circle centered at C through D meets the given line at two distinct points E and G, so $CE \cong CG$. The existence of two such intersections is a genuine geometric existence issue, not something proved by the picture itself.

Step 4. Bisect EG

Apply Proposition I.10 to bisect the segment EG at H.

Thus

$$E - H - G$$

and

$$EH \cong HG.$$

Step 5. Join CH

Join the external point C to the midpoint H; the completed proof configuration is Figure 12.4.

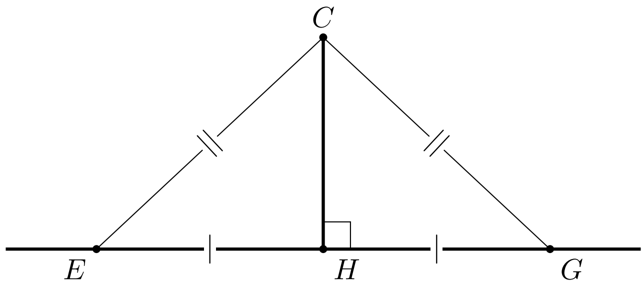


Figure 12.4: Completed Euclidean proof configuration. The midpoint relation $EH \cong HG$, the radii $CE \cong CG$, and the common side CH give SSS for triangles CEH and CGH.

The two triangles EHC and GHC now satisfy

$$EH \cong HG, \qquad HC \cong HC, \qquad CE \cong CG.$$

By Proposition I.8,

$$\angle EHC \cong \angle GHC.$$

Since E, H, G are collinear, these are adjacent angles. By Definition 10 they are right angles.

Therefore

$$CH \perp AB.$$

The required perpendicular has been constructed.

Diagrammatic audit. The construction sequence is retained only where the geometry genuinely changes: the input data, the opposite-side choice, the circle intersection, and the completed SSS configuration. The former midpoint-only snapshot is omitted because it adds only the label H to the preceding circle diagram. Conversely, the circle-intersection figure is essential: Wilkins explicitly notes that the existence of two distinct points E, G requires a tacit topological intersection principle. The picture records those intersections but does not prove their existence.

12.3 Classical Proof

Euclid's construction is obtained from the configuration described above.

Choose a point D on the opposite side of the given line AB from the external point C. Draw the circle centered at C through D, and let it meet AB at E and G. Then

$$CE \cong CG.$$

Apply Proposition I.10 to bisect EG at H. Hence

$$E - H - G \qquad \text{and} \qquad EH \cong HG.$$

Now compare the triangles

$$\triangle EHC \qquad \text{and} \qquad \triangle GHC.$$

They satisfy

$$EH \cong HG, \qquad HC \cong HC, \qquad CE \cong CG.$$

By Proposition I.8, the triangles are congruent by SSS, and therefore

$$\angle EHC \cong \angle GHC.$$

Since E, H, G are collinear, these congruent adjacent angles are right angles. Consequently

$$CH \perp AB.$$

The classical dependency pattern is therefore

$$\text{circle construction} \longrightarrow CE \cong CG \longrightarrow \text{I.10} \longrightarrow EH \cong HG \longrightarrow \text{I.8} \longrightarrow \text{equal adjacent angles} \longrightarrow CH \perp AB.$$

The formal Hilbert reconstruction below proves the same proposition by a different synthetic route and deliberately avoids the circle-line intersection used by Euclid.

12.4 Proof Architecture of the Reconstruction

12.4.1 Dependency Diagram

```
Hilbert / Book Zero dependency path
-----------------------------------

hilbert_collinear_angle_transport
              |
              v
hilbert_equidistant_line_bisects_segment
              |
              v
hilbert_perpendicular_from_point_exists
              |
              v
euclid_proposition_12
```

The formal proof separates Proposition I.12 into two reusable Book Zero lemmas and one final construction theorem.

The first auxiliary result is

```
hilbert_collinear_angle_transport
```

It transports an angle congruence when one side of the angle is replaced by another point on the same supporting line. The proof handles the two possible ray configurations: direct transport along the same ray, and transport through supplementary angles when the new ray is opposite.

The second auxiliary result is

```
hilbert_equidistant_line_bisects_segment
```

Suppose two distinct points A and B lie on a line b, while P and Q lie on opposite sides of b, and

$$AP \cong AQ, \qquad BP \cong BQ.$$

If the segment PQ meets b at H, then

$$PH \cong HQ.$$

The logical structure is therefore

```
hilbert_collinear_angle_transport
        |
hilbert_equidistant_line_bisects_segment
        |
hilbert_perpendicular_from_point_exists
        |
euclid_proposition_12
```

The main construction theorem

```
hilbert_perpendicular_from_point_exists
```

uses the perpendicular-bisector result to obtain the midpoint H of PQ. It then chooses a point R on the given line distinct from H, applies Hilbert SSS to the triangles RHP and RHQ, and uses vertical angles together with the definition of a right angle.

Finally,

```
euclid_proposition_12
```

is a thin interface theorem exposing this general Hilbert result as Euclid's Proposition I.12.

12.5 Hilbert Reconstruction

The Hilbert reconstruction preserves the geometric objective of Proposition I.12 but does not reproduce Euclid's circle construction.

Let A and B be distinct points of the given line b, and let P be the prescribed point outside b.

The first task is to construct a point Q on the opposite side of b from P such that

$$AP \cong AQ, \qquad BP \cong BQ.$$

To obtain Q, extend PA through A to a point S. Thus P and S lie on opposite sides of b.

Using Hilbert angle construction at A, construct a ray on the side of b containing S whose angle with AB is congruent to the angle formed by AP and AB. On this ray lay off a segment congruent to AP. Denote its endpoint by Q; see Figure 12.5.

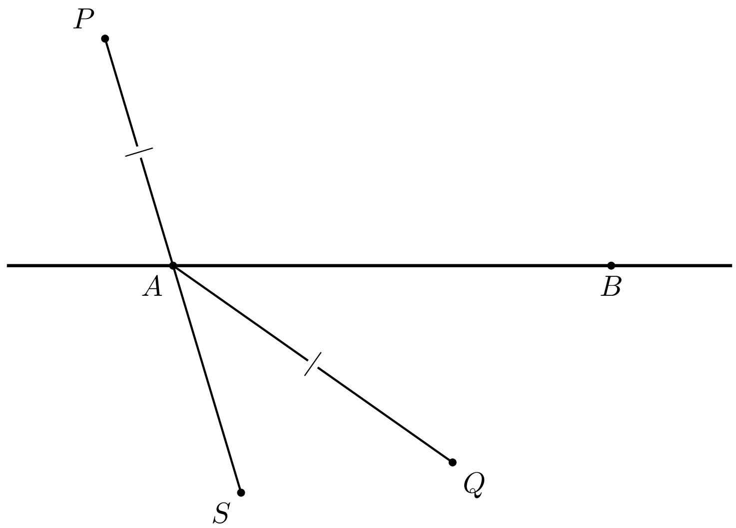


Figure 12.5: Hilbert construction of the reflected point Q. The line PA is extended through A to S; angle construction chooses the direction of AQ, and segment layoff gives $AQ \cong AP$.

The construction gives

$$AP \cong AQ.$$

The two triangles

$$\triangle APB \qquad \text{and} \qquad \triangle AQB$$

now have

$$AP \cong AQ, \qquad AB \cong AB,$$

and the included angles at A are congruent. By SAS,

$$BP \cong BQ.$$

Moreover, the construction places Q on the opposite side of b from P. Hence the segment PQ intersects b at a point H. The resulting configuration is shown in Figure 12.6.

At this point the general Book Zero theorem

```
hilbert_equidistant_line_bisects_segment
```

applies. Since

$$AP \cong AQ, \qquad BP \cong BQ,$$

and A and B are distinct points of b, the intersection point H bisects PQ:

$$PH \cong HQ.$$

Thus the difficult incidence and angle-order argument has been encapsulated before Proposition I.12 itself is completed.

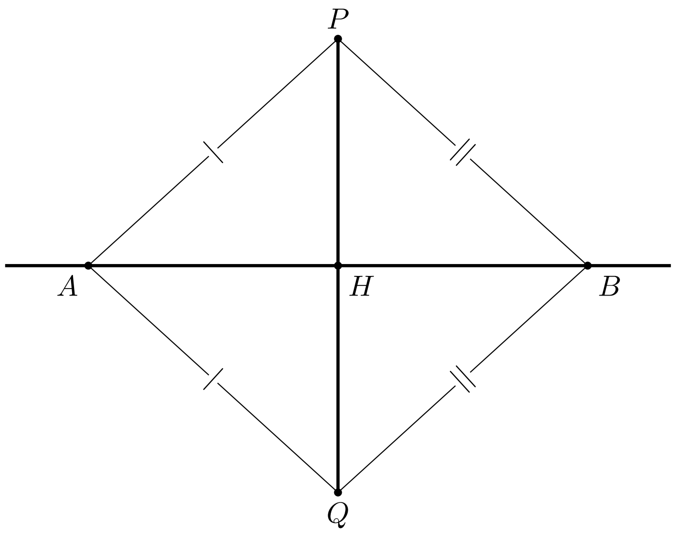


Figure 12.6: Hilbert perpendicular-bisector configuration. The constructed equalities $AP \cong AQ$ and $BP \cong BQ$, together with the intersection $P - H - Q$, allow the Book Zero bisector theorem to conclude $PH \cong HQ$.

To obtain the right angle, choose one of A and B distinct from H and call it R. Since R is one of the two equidistant points,

$$RP \cong RQ.$$

Together with

$$PH \cong HQ \qquad \text{and} \qquad RH \cong RH,$$

Hilbert SSS applied to

$$\triangle RHP \qquad \text{and} \qquad \triangle RHQ$$

gives congruent angles at H.

Since $P - H - Q$, these congruent angles determine a perpendicular to b at H. In the formal development this last step is expressed using vertical angles and the predicate

```
HilbertRightAngle
```

The reconstruction therefore replaces Euclid's use of a circle and Propositions I.10 and I.8 by a Hilbert construction based on angle construction, segment layoff, SAS, the perpendicular-bisector theorem, and SSS.

The resulting geometric object is nevertheless the same: a point H on the given line such that the line through P and H is perpendicular to the given line.

12.6 Lean Formalization

The Lean formalization separates the reusable geometric content from the final statement of Proposition I.12.

The main construction theorem is

```
theorem hilbert_perpendicular_from_point_exists
```

```
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B P : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase : HilbertIncidence.OnLine A base)
(hBbase : HilbertIncidence.OnLine B base)
(hPbase : not HilbertIncidence.OnLine P base) :
exists H R : Geo.Point,
HilbertIncidence.OnLine H base /\
HilbertIncidence.OnLine R base /\
HilbertRightAngle Geo R H P
```

Here H is the foot of the perpendicular and R is a second point of the given line used to determine its direction at H.

The presence of R in the conclusion is important. The intersection point H may coincide with either A or B, so neither of the original points can be fixed in advance as the first point in the right-angle predicate. Since

$$A \neq B,$$

at least one of them is distinct from H and can therefore be chosen as R.

The formal proof first constructs a point Q on the opposite side of the given line from P. The construction establishes

$$AP \cong AQ$$

and, by SAS,

$$BP \cong BQ.$$

The opposite-side relation supplies an intersection point H with

$$P - H - Q.$$

The reusable Book Zero theorem

```
hilbert_equidistant_line_bisects_segment
```

then gives

$$PH \cong HQ.$$

The remaining argument selects R, applies Hilbert SSS to the triangles RHP and RHQ, and converts the resulting angle congruence into a right angle by means of vertical angles.

The Euclidean proposition itself is exposed by the wrapper

```
theorem euclid_proposition_12
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B P : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase : HilbertIncidence.OnLine A base)
(hBbase : HilbertIncidence.OnLine B base)
```

```
        (hPbase : not HilbertIncidence.OnLine P base) :
        exists H R : Geo.Point,
        HilbertIncidence.OnLine H base /\
        HilbertIncidence.OnLine R base /\
        HilbertRightAngle Geo R H P := by

        exact
        hilbert_perpendicular_from_point_exists
        Geo
        A B P
        base
        hAB
        hAbase
        hBbase
        hPbase
```

Thus the theorem carrying Euclid's name is deliberately only a thin wrapper. The geometric work is performed by the Hilbert construction theorem and by the reusable results already established in the Book Zero layer.

12.7 Relationship with Earlier Propositions

12.7.1 Proposition I.8

Euclid uses Proposition I.8 at the final stage of the classical proof. Once the two radii from C and the two half-segments at H are known to be congruent, SSS gives congruent adjacent angles and hence the perpendicular.

The Hilbert reconstruction also uses SSS, but in a different configuration: the triangles RHP and RHQ arise from the perpendicular-bisector construction rather than from Euclid's circle.

12.7.2 Proposition I.10

In Euclid's proof, Proposition I.10 is the construction that bisects the chord EG.

The Hilbert reconstruction does not call I.10. Instead, the midpoint property

$$PH \cong HQ$$

is derived from the more general theorem `hilbert_equidistant_line_bisects_segment`.

Thus the Euclidean midpoint construction is replaced by a reusable perpendicular-bisector principle.

12.7.3 Proposition I.11

Proposition I.11 constructs a perpendicular at a point already lying on the given line. Proposition I.12 solves the complementary problem: the prescribed point lies outside the line and the foot of the perpendicular must itself be constructed.

Together I.11 and I.12 complete the two elementary perpendicular construction problems needed later in Book I.

12.8 Hilbert and Book Zero Components Used

12.8.1 `hilbert_collinear_angle_transport`

This helper transports angle congruence when one endpoint on a supporting line is replaced by another collinear point. It handles both the same-ray and opposite-ray configurations.

12.8.2 `hilbert_equidistant_line_bisects_segment`

This is the central reusable theorem extracted during the reconstruction of I.12. Two distinct points of a line, each equidistant from P and Q, force the intersection of PQ with that line to bisect PQ.

12.8.3 `hilbert_perpendicular_from_point_exists`

This theorem assembles the construction: it produces the opposite-side point, obtains the midpoint from the equidistant-line theorem, chooses a second point of the base line, and proves the resulting angle is right.

12.8.4 Hilbert SSS and Vertical Angles

SSS converts the three segment congruences in the triangles RHP and RHQ into angle congruence at H. Vertical-angle congruence then puts that equality into the adjacent-angle orientation required by `HilbertRightAngle`.

12.9 Formalization Notes

Proposition I.12 reveals a substantial difference between Euclid's constructional language and the Hilbert/Book Zero architecture.

Euclid obtains the perpendicular through a concrete circle construction. The circle centered at the external point produces two points of the given line at equal distance from that point. Proposition I.10 supplies their midpoint, and Proposition I.8 converts the resulting three side equalities into equality of the adjacent angles at the midpoint.

The Hilbert reconstruction does not attempt to reproduce this sequence literally. In particular, no circle is introduced. Instead, the essential geometric configuration behind Euclid's construction is isolated:

$$AP \cong AQ, \qquad BP \cong BQ.$$

Thus two distinct points A and B of the given line are equidistant from P and Q. The line through A and B therefore plays the role of the perpendicular-bisector axis of PQ.

This observation leads to the theorem

```
hilbert_equidistant_line_bisects_segment
```

which extracts the reusable geometric content of the configuration. Once the segment PQ intersects the line at H, the theorem gives

$$PH \cong HQ.$$

The proposition is therefore no longer organized around a particular circle construction. It is organized around an abstract property of two equidistant points and the line determined by them.

The proof of the perpendicular-bisector theorem itself exposed another reusable operation. When an angle is known relative to one point of a line, replacing that point by another collinear point requires attention to the order of the points: the relevant ray may remain unchanged or become the opposite ray. This argument is encapsulated in

```
hilbert_collinear_angle_transport
```

rather than being repeated inside Proposition I.12.

This is characteristic of the role of Book Zero. Elementary incidence, order, congruence, ray, and angle manipulations are packaged as synthetic lemmas. Later Euclidean propositions can then operate with larger geometric units instead of repeatedly descending to the Hilbert axioms.

There is also an important difference from the preceding elementary propositions. Proposition I.12 does not collapse to a single previously available Book Zero operation. Its reconstruction required the identification of genuinely reusable intermediate geometry. In this sense, the formalization of I.12 contributed new infrastructure to Book Zero rather than merely consuming existing infrastructure.

The final theorem

```
euclid_proposition_12
```

is consequently a wrapper around

```
hilbert_perpendicular_from_point_exists
```

The wrapper records the Euclidean proposition in the formal library, while the underlying theorem records the more general Hilbert construction on which it depends.

Proposition I.12 therefore illustrates the intended division of labor in the development:

$$\begin{array}{ccl}
\text{Hilbert axioms} & \longrightarrow & \text{elementary incidence, order, and congruence theory} \\
\text{Book Zero} & \longrightarrow & \text{reusable synthetic geometric mechanisms} \\
\text{Hilbert construction theorem} & \longrightarrow & \text{existence of the required perpendicular} \\
\text{Euclid wrapper} & \longrightarrow & \text{Proposition I.12}
\end{array}$$

The result is not a literal transcription of Euclid's proof. It is a reconstruction of the same geometric theorem over the Hilbert foundation, with the intermediate reasoning reorganized into reusable components.

12.10 Dependency Summary

The Euclidean and Hilbert dependency structures are visibly different.

For Euclid,

$$\text{circle} \longrightarrow \text{I.10} \longrightarrow \text{I.8} \longrightarrow \text{I.12}.$$

For the reconstruction,

$$\begin{array}{c}
\text{Hilbert angle construction} + \text{segment layoff} + \text{SAS} \\
\downarrow \\
AP \cong AQ, \quad BP \cong BQ, \quad P, Q \text{ on opposite sides} \\
\downarrow \\
\texttt{hilbert_equidistant_line_bisects_segment} \\
\downarrow \\
PH \cong HQ \\
\downarrow \\
\text{Hilbert SSS} + \text{vertical angles} \\
\downarrow \\
\texttt{HilbertRightAngle} \\
\downarrow \\
\texttt{hilbert_perpendicular_from_point_exists} \\
\downarrow \\
\texttt{euclid_proposition_12}.
\end{array}$$

The reconstruction introduces no new axiom. Its main contribution is the extraction of reusable synthetic lemmas from the geometry hidden inside the classical construction.

12.11 Conclusion

Proposition I.12 is an especially clear example of reconstruction rather than literal transcription.

Euclid uses a circle, Proposition I.10, and Proposition I.8. The Hilbert development identifies the underlying perpendicular-bisector geometry and proves the result without reproducing the circle-line construction.

This change of route is productive. The proof yields `hilbert_collinear_angle_transport` and, more importantly, `hilbert_equidistant_line_bisects_segment` as reusable synthetic infrastructure.

The public theorem remains Euclid I.12: from a point outside a given line, a perpendicular to that line exists. Underneath that statement, however, the formal dependency graph records a more general Hilbert construction whose intermediate results can be reused independently.

Together with I.11, Proposition I.12 completes the basic perpendicular construction block and prepares the straight-angle theory beginning with Proposition I.13.

Chapter 13

Proposition I.13

13.1 Introduction

Euclid's Proposition I.13 concerns the two adjacent angles formed when one straight line stands on another straight line.

Let C, B, D lie on one straight line, with B between C and D, and let A be a point outside that line. The two adjacent angles are

$$\angle CBA \qquad \text{and} \qquad \angle ABD.$$

Euclid states that these angles are either two right angles or are together equal to two right angles.

The classical proof distinguishes two cases. If the adjacent angles are equal, then by Definition 10 they are both right angles. If they are not equal, Euclid erects a perpendicular at B using Proposition I.11 and compares the resulting angle decompositions.

The reconstruction over the Hilbert/Book Zero layer is much shorter. Book Zero already contains an explicit relation expressing that one angle is supplementary to another:

```
BookZeroSupplement
```

For the configuration of Proposition I.13, this relation contains exactly the required geometric data. Consequently, the formal theorem does not reconstruct Euclid's case distinction or introduce an arithmetic operation on angles. It identifies the two adjacent angles directly as a supplementary pair.

Thus Proposition I.13 provides a particularly clear example of the compression obtained from the Book Zero language: a classical geometric argument becomes essentially an instance of an already available synthetic relation.

13.2 The Classical Configuration

Proposition I.13 is not a construction theorem. The geometric configuration is already given.

Let C, B, D lie on one straight line with

$$C - B - D,$$

and let A lie outside the straight line CD.

The straight line BA stands on the straight line CD as shown in Figure 13.1 and forms the adjacent angles

$$\angle CBA \qquad \text{and} \qquad \angle ABD.$$

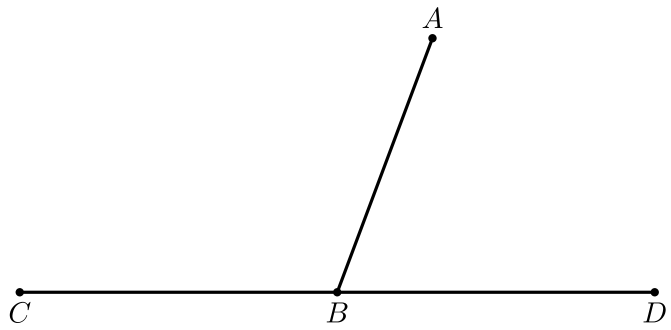


Figure 13.1: Classical configuration for Proposition I.13. The relation $C - B - D$ makes BC and BD opposite rays, while BA is the standing ray.

13.3 Statement

Assume

$$C - B - D$$

and let A be a point distinct from B.

Then the adjacent angles

$$\angle CBA \qquad \text{and} \qquad \angle ABD$$

form a supplementary pair.

In Euclid's formulation, they are therefore either two right angles or together equal to two right angles.

In the Book Zero language this conclusion is represented by

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

By definition,

$$\text{BookZeroSupplement}(C, B, A, A, D)$$

means

$$\text{SameRay}(B, A, A)$$

together with

$$C - B - D.$$

The first condition expresses the trivial fact that the ray BA coincides with itself, while the second is precisely the given straight-line configuration.

13.4 Classical Proof

Euclid proves Proposition I.13 by distinguishing two cases.

13.4.1 Case 1. The Adjacent Angles Are Equal

Suppose first that

$$\angle CBA \cong \angle ABD.$$

Since C, B, D lie on one straight line, the two angles are adjacent angles formed by the straight line BA standing on CD.

By Euclid's Definition 10, when a straight line standing on another straight line makes the adjacent angles equal, each of those angles is a right angle.

Therefore

$$\angle CBA \qquad \text{and} \qquad \angle ABD$$

are two right angles.

This is the first alternative in Proposition I.13.

13.4.2 Case 2. The Adjacent Angles Are Not Equal

Suppose now that

$$\angle CBA \ncong \angle ABD.$$

By Proposition I.11, construct through B a perpendicular BE to the straight line CD; see Figure 13.2.

Thus

$$\angle CBE \qquad \text{and} \qquad \angle EBD$$

are right angles.

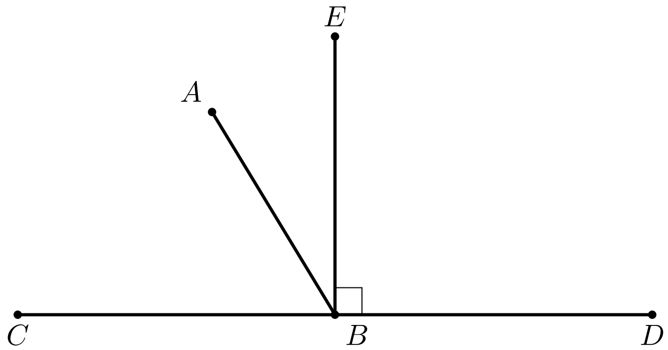


Figure 13.2: Auxiliary perpendicular configuration in the non-equal-angle case. The ray BE is perpendicular to CD, and BA lies between the rays BC and BE in the displayed arrangement.

Since BE is perpendicular to CD,

$$\angle CBE + \angle EBD$$

is equal to two right angles.

The original ray BA divides one of these right-angle configurations. In the arrangement shown in Figure 13.2,

$$\angle CBE = \angle CBA + \angle ABE.$$

Adding $\angle EBD$ gives

$$\angle CBE + \angle EBD = \angle CBA + \angle ABE + \angle EBD.$$

But

$$\angle ABE + \angle EBD = \angle ABD.$$

Hence

$$\angle CBE + \angle EBD = \angle CBA + \angle ABD.$$

The left-hand side consists of two right angles. Therefore

$$\angle CBA + \angle ABD$$

is equal to two right angles.

This is the second alternative in Proposition I.13.

Only the two numbered figures above are needed. Figure 13.1 states the given geometry; Figure 13.2 introduces the one genuinely new construction. Additional copies of the same rays would not correspond to new geometric information.

Thus, whether the two original adjacent angles are equal or unequal, they are either themselves two right angles or are together equal to two right angles.

13.5 Proof Architecture of the Reconstruction

Euclid's proof proceeds through the geometry of right angles and angle addition:

```
Euclid
------
C-B-D and standing ray BA
        |
are the adjacent angles equal?
        |
   +----+----+
   |         |
  yes        no
   |         |
Definition 10  erect perpendicular BE
   |         |
right angles  angle decomposition
   |         |
   +----+----+
        |
two right angles


Hilbert / Book Zero
-------------------
C-B-D
```

```
  +
SameRay(B,A,A)
        |
        v
BookZeroSupplement(C,B,A,A,D)
        |
        v
euclid_proposition_13
```

The two architectures should not be represented by additional geometric figures. The difference is logical rather than spatial: Euclid proves a statement about angle sums, whereas the Book Zero theorem recognizes the same straight-angle configuration directly.

The contrast is substantial. Euclid proves the proposition by introducing a perpendicular and reasoning about decompositions and combinations of angles. At the Book Zero level, the same geometric configuration has already been abstracted as the supplementary-angle relation.

No new geometric construction is therefore required, and Proposition I.11 disappears from the formal dependency path.

13.6 Hilbert Reconstruction

The Hilbert reconstruction of Proposition I.13 does not reproduce Euclid's proof.

The essential observation is that the conclusion of the proposition has already been isolated in Book Zero as the relation

```
BookZeroSupplement
```

For five points A, B, C, D, F, this relation is defined by

$$\text{BookZeroSupplement}(A, B, C, D, F)$$

if and only if

$$\text{SameRay}(B, C, D)$$

and

$$A - B - F.$$

Thus the relation records synthetically the configuration in which one side of an angle is continued through its vertex while the other side is preserved along the same ray.

For Proposition I.13 we instantiate this relation as

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

Expanding the definition gives exactly two conditions:

$$\text{SameRay}(B, A, A)$$

and

$$C - B - D.$$

The second condition is one of the hypotheses of Proposition I.13.

The first condition says only that the ray BA is the same ray as itself. Since $A \neq B$, it follows immediately from reflexivity of the same-ray relation.

Consequently,

$$\text{BookZeroSupplement}(C, B, A, A, D)$$

follows directly.

No perpendicular is constructed. No case distinction between equal and unequal adjacent angles is required. No decomposition or addition of angles occurs in the proof.

This does not mean that Euclid's geometric argument has been discarded. Rather, the geometric notion expressed by that argument has already been incorporated into the language of Book Zero. Proposition I.13 therefore becomes a recognition theorem: the given configuration is an instance of an existing synthetic relation.

13.7 Lean Formalization

The Lean theorem is correspondingly short:

```
theorem euclid_proposition_13
[HilbertIncidence Geo]
[HilbertOrder Geo]
(C B D A : Geo.Point)
(hCBD : Geo.Between C B D)
(hBA : B != A) :
BookZeroSupplement Geo C B A A D := by

constructor

· exact
hilbert_sameRay_refl
Geo B A hBA.symm

· exact hCBD
```

The proof follows the definition of

```
BookZeroSupplement
```

literally.

The first component of the conclusion is

$$\mathrm{SameRay}(B, A, A).$$

It is obtained from

```
hilbert_sameRay_refl
```

using $A \neq B$, which is supplied by the symmetric form of the hypothesis $B \neq A$.

The second component is

$$C - B - D,$$

which is exactly the hypothesis

```
hCBD
```

Thus the formal proof contains no intermediate geometric construction and requires no theorem corresponding to Euclid's use of Proposition I.11.

13.8 Discussion

Proposition I.13 is a particularly clear test of the role assigned to Book Zero in the reconstruction.

In Euclid's original development, the statement requires an actual proof. The argument distinguishes whether the two adjacent angles are equal. In the nontrivial case, Proposition I.11 is invoked to erect a perpendicular, and the conclusion is obtained by decomposing and recombining angles.

At the Hilbert level, one could reconstruct this argument explicitly. The existing theory already contains the necessary machinery for right angles, supplementary angles, and angle addition.

That reconstruction is unnecessary, however, once the Book Zero language is used.

The relation

```
BookZeroSupplement
```

was introduced independently as a reusable description of a standard synthetic configuration. Proposition I.13 is precisely an instance of that configuration.

The resulting compression is therefore structural rather than tactical. Lean is not discovering a shorter proof of Euclid's argument. The formal language has been organized at a level at which the conclusion of I.13 can already be stated directly.

This produces an unusually short dependency path:

$$\text{betweenness} \quad + \quad \text{same-ray reflexivity} \quad \longrightarrow \quad \text{BookZeroSupplement} \quad \longrightarrow \quad \text{Euclid I.13.}$$

In particular, Proposition I.11, although essential to Euclid's classical proof, is not a dependency of the reconstructed theorem.

This distinction is important. The objective of the reconstruction is not to force every Euclidean proposition to retain the dependency graph of the original text. The objective is to express Euclidean geometry over a stable synthetic interface derived from the Hilbert foundation.

Earlier propositions exhibited several possible relationships with this interface. Some consumed already available Book Zero results, while others exposed missing reusable geometry that had to be added to the interface.

Proposition I.13 exhibits a third and especially simple situation: the proposition is already expressed by the vocabulary of Book Zero.

Schematically,

$$\begin{array}{ccl} \text{Euclid} & : & \text{case distinction} + \text{I.11} + \text{angle composition} \\ \text{Hilbert/Book Zero} & : & \text{recognition of a supplementary configuration} \\ \text{Lean} & : & \text{same-ray reflexivity} + \text{given betweenness} \end{array}$$

The four-line proof is therefore not merely an accidental shortening of Proposition I.13. It is evidence that the Book Zero layer is functioning as intended: elementary synthetic arguments can be encapsulated once and subsequently used as part of the language in which later geometry is formulated.

13.9 Relationship with Earlier Propositions

13.9.1 Proposition I.11

Proposition I.11 constructs a perpendicular to a given straight line at a given point on it. In the classical development this supplies the canonical right-angle configuration against which the adjacent angles in I.13 can be compared.

The formal reconstruction can package the same geometry through the Hilbert and Book Zero angle machinery, but the conceptual role remains the same: a perpendicular provides the reference configuration for two right angles.

13.9.2 Proposition I.12

Proposition I.12 extends the perpendicular construction to a point outside a line. It is not the principal dependency of I.13, but together with I.11 it completes the elementary perpendicular infrastructure that precedes the straight-angle block.

13.9.3 Transition to Proposition I.14

Proposition I.13 is the direct theorem: when a straight line stands on another straight line, the adjacent angles are two right angles or are together equal to two right angles.

Proposition I.14 reverses this logic. From the corresponding supplementary-angle condition it recovers the straight-line configuration.

Thus I.13 and I.14 form a natural theorem/converse pair and should be read as one local unit in the development of straight-angle geometry.

13.10 Hilbert and Book Zero Components Used

The reconstruction of Proposition I.13 relies on the Book Zero relation

```
BookZeroSupplement
```

defined by

$$\mathrm{BookZeroSupplement}(A, B, C, D, F)$$

if and only if

$$\mathrm{SameRay}(B, C, D)$$

and

$$A - B - F.$$

Geometrically, this represents a pair of supplementary angles. The angle determined by the rays BA and BC has as its supplement the angle determined by the rays BD and BF, where BD lies on the same ray as BC and BF is the opposite extension of BA.

For Proposition I.13 the relation is instantiated as

$$\mathrm{BookZeroSupplement}(C, B, A, A, D).$$

Thus the required conditions become

$$\mathrm{SameRay}(B, A, A)$$

and

$$C - B - D.$$

The first is immediate from same-ray reflexivity, while the second is the given betweenness hypothesis.

The proof also uses

```
hilbert_sameRay_refl
```

which expresses reflexivity of a nondegenerate ray:

$$A \neq B \quad \Longrightarrow \quad \mathrm{SameRay}(A, B, B).$$

13.11 Formalization Notes

The formal reconstruction is considerably shorter than Euclid's classical proof because the supplementary-angle configuration has already been packaged in Book Zero.

The theorem does not reproduce Euclid's case distinction and does not construct the perpendicular from Proposition I.11. Instead, it proves

$$\mathrm{BookZeroSupplement}(C, B, A, A, D)$$

directly from the two components required by the definition:

$$\mathrm{SameRay}(B, A, A) \qquad \text{and} \qquad C - B - D.$$

The first component follows from

```
\thilbert_sameRay_refl
```

using the nondegeneracy hypothesis $A \neq B$. The second component is exactly the given betweenness hypothesis.

Thus no numerical angle measure, angle addition, right-angle case analysis, or auxiliary perpendicular is needed in the Lean proof. The formalization records the same synthetic content at a more abstract level: Proposition I.13 is recognized as an instance of the already established Book Zero notion of supplementary angles.

This is also the form needed by Proposition I.14, which uses the same supplementary-angle structure in the converse direction.

13.12 Dependency Summary

The formal dependency structure is

$$\begin{array}{c}\text{a straight line standing on a straight line}\\ \downarrow\\ \text{betweenness / collinearity data}\\ \downarrow\\ \text{adjacent-angle configuration}\\ \downarrow\\ \text{comparison with a right-angle configuration}\\ \downarrow\\ \text{equal-angle or unequal-angle branch}\\ \downarrow\\ \text{two right angles or supplementary angles}\\ \downarrow\\ \texttt{euclid\_proposition\_13}.\end{array}$$

The proof uses only the established Hilbert and Book Zero infrastructure. No new geometric axiom is introduced.

13.13 Conclusion

Proposition I.13 begins the straight-angle block of Book I.

Its familiar classical statement looks almost arithmetical: the two adjacent angles made by a line standing on another line amount to two right angles. The formal reconstruction shows that no numerical angle measure is required to express or prove this fact.

Instead, the theorem is recovered from explicit incidence, order, right-angle, and angle-congruence structure. The classical distinction between equal and unequal adjacent angles is preserved, while the diagrammatic assumptions concerning the supporting straight line are made explicit.

The proposition is also structurally important. I.14 supplies its converse, and I.15 completes the local theory by proving the congruence of vertical angles. I.13 therefore marks the beginning of a coherent three-proposition development of the geometry of intersecting straight lines.

Chapter 14

Proposition I.14

14.1 Introduction

Euclid's Proposition I.14 is the converse in spirit of the preceding Proposition I.13.

Proposition I.13 starts from a straight-line configuration and proves that the adjacent angles are together equal to two right angles. Proposition I.14 reverses this direction: if two adjacent angles are together equal to two right angles, and their outer rays lie on opposite sides of the common straight line, then those outer rays form one straight line.

In Euclid's original proof the argument is indirect. The ray opposite to one of the given rays is constructed, Proposition I.13 is applied, and equality of the resulting angles forces the constructed ray to coincide with the remaining given ray.

The Hilbert reconstruction follows the same geometric idea at a more structural level. The condition "equal to two right angles" is encoded without numerical angle measure, using an opposite ray and angle congruence. The decisive step is then supplied by the uniqueness clause of Hilbert's angle-construction axiom.

Proposition I.14 is therefore an instructive example of the interaction between Euclid's angle language, Hilbert plane separation, angle construction, and the ray calculus developed in Book Zero.

14.2 The Classical Configuration

Let AB be a straight line, and let two further straight lines BC and BD meet it at the common point B, with C and D lying on opposite sides of AB.

The two adjacent angles are therefore

$$\angle ABC \qquad \text{and} \qquad \angle ABD.$$

The hypothesis of Proposition I.14 is that these two angles are together equal to two right angles. The conclusion is that the two outer rays BC and BD are opposite rays and hence form one straight line.

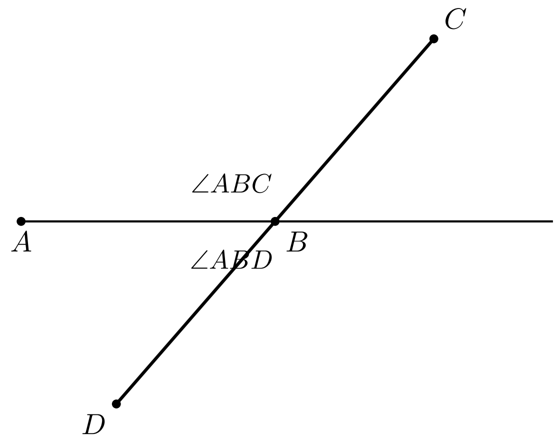


Figure 14.1: The classical configuration of Proposition I.14. The rays BC and BD lie on opposite sides of the straight line through A and B.

14.3 Statement

Theorem 7 (Euclid I.14)**.** *Let AB be a straight line, and let the straight lines BC and BD meet it at the point B on opposite sides of AB.*

If the adjacent angles

$$\angle ABC \qquad \text{and} \qquad \angle ABD$$

are together equal to two right angles, then BC and BD form one straight line.

Equivalently,

$$C - B - D.$$

The proposition may therefore be viewed as a converse to the straight-line angle relation established in Proposition I.13.

The geometric content of the conclusion is stronger than mere collinearity. The point B lies between C and D, so the rays BC and BD are opposite rays with common endpoint B.

14.4 Classical Proof

Euclid argues by extending one of the two outer rays through the common vertex; the auxiliary configuration is shown in Figure 14.2.

Extend CB through B to a point E. Then

$$C - B - E,$$

so BC and BE form one straight line.

By Proposition I.13, the adjacent angles

$$\angle ABC \qquad \text{and} \qquad \angle ABE$$

are together equal to two right angles.

But by hypothesis,

$$\angle ABC \qquad \text{and} \qquad \angle ABD$$

are also together equal to two right angles.

Therefore the remaining angles must be equal:

$$\angle ABE = \angle ABD.$$

But if BE and BD were distinct, one of these two angles would be a proper part of the other. A proper part cannot be equal to the whole. Hence the rays BE and BD cannot be distinct.

Therefore BE coincides with BD. Since

$$C - B - E,$$

it follows that

$$C - B - D.$$

Thus BC and BD form one straight line.

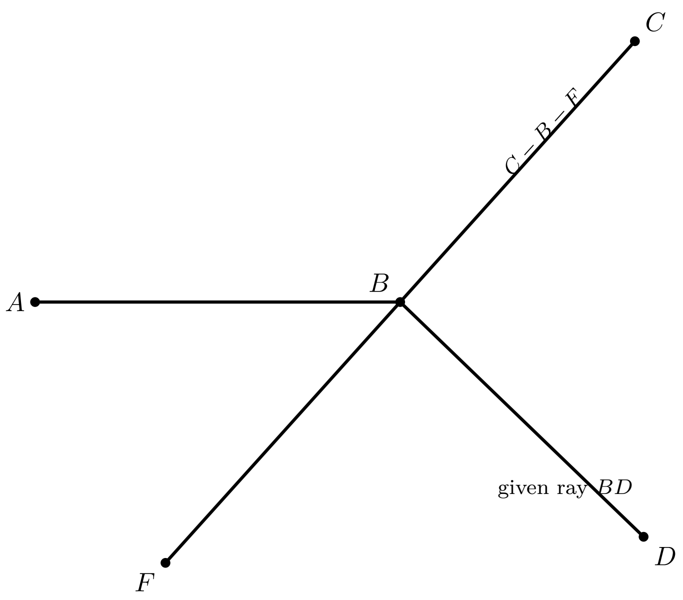


Figure 14.2: Wilkins auxiliary construction before the uniqueness step. The straight line CBF is the constructed extension through B; BD is the given ray. The proof shows that BD must coincide with the ray BF.

The essential structure of Euclid's argument is therefore

$$\text{extension} \longrightarrow \text{I.13} \longrightarrow \text{equality of the remaining angles} \longrightarrow \text{coincidence of the rays.}$$

The second diagram is genuinely useful, but it must be read as a *pre-coincidence* configuration. It would be misleading to draw BD as a visibly different ray and simultaneously treat $\angle ABF = \angle ABD$ as already geometrically realized. The equality is precisely what forces the two rays to coincide. Wilkins makes the missing uniqueness step explicit by using the fact that D and F lie on the same side of AB.

14.5 Proof Architecture of the Reconstruction

The Hilbert reconstruction preserves the central geometric idea of Euclid's proof but reorganizes its logical dependencies.

Euclid extends CB through B to a point E and uses Proposition I.13 to obtain the supplementary angle relation. He then compares this relation with the hypothesis of Proposition I.14 and concludes that the rays BD and BE must coincide.

In the formal reconstruction, the point E is incorporated directly into the synthetic representation of the hypothesis. We require

$$C - B - E$$

together with

$$\angle ABD \cong \angle ABE.$$

Thus the intermediate appeal to Proposition I.13 is no longer needed: the supplementary configuration required by Euclid's argument is already present in the formal hypothesis.

The remaining problem is to prove that BD and BE determine the same ray from B.

At this point the reconstruction divides into two cases according to whether C, D, E are collinear.

```
Hilbert hypothesis
------------------
C-B-E
angle ABD ~= angle ABE
C and D opposite sides of base AB
        |
        v
derive: C and E opposite sides of base AB
        |
        v
case split: are C,D,E collinear?
        |
   +----+----+
   |         |
  yes        no
   |         |
line         D and E are on the same side of AB
uniqueness   |
   |         v
```

```
   |     Hilbert III.4 angle uniqueness
   |         |
   |     BE and BD are the same ray
   |         |
   +----+----+
        |
transport C-B-E along the identified ray
        |
        v
C-B-D
```

The two classical figures have different jobs. The first states the problem; the second introduces the extension used in the proof. Beyond these, another geometric copy of the same crossing is unnecessary. The formal case split is better represented as a dependency diagram, because its branches are logical alternatives rather than two prescribed Euclidean constructions.

In the collinear case, incidence and line uniqueness are sufficient.

In the noncollinear case, plane separation first places D and E on the same side of the base line. The uniqueness clause of Hilbert's angle-construction axiom then identifies the rays BD and BE. Ray calculus and transport of betweenness finally yield

$$C - B - D.$$

Schematically, the two proof architectures are

$$\begin{aligned} \text{Euclid} &: \text{extension} \to \text{I.13} \to \text{angle comparison} \to \text{ray coincidence}, \\ \text{Hilbert reconstruction} &: \text{encoded supplement} \to \text{case split} \to \text{line/angle uniqueness} \to \text{betweenness}. \end{aligned}$$

14.6 Hilbert Reconstruction

The Hilbert reconstruction begins by replacing Euclid's phrase "together equal to two right angles" by a purely synthetic configuration.

No numerical measure of angles and no operation of angle addition are introduced. Instead, the supplementary relation is represented by an opposite ray together with angle congruence.

14.6.1 Encoding the Angle Hypothesis

Choose a point E on the continuation of CB beyond B, so that

$$C - B - E.$$

The rays BC and BE are therefore opposite. Consequently, $\angle ABE$ is the angle supplementary to $\angle ABC$.

The hypothesis that

$$\angle ABC \qquad \text{and} \qquad \angle ABD$$

are together equal to two right angles is represented by requiring

$$\angle ABD \cong \angle ABE.$$

The geometry is already visible in Figure 14.2: the witness E lies on the continuation of CB beyond B, and the formal hypothesis identifies the angle made by BD with the supplementary angle made by BE. A second drawing of the same rays would add no new information.

In Lean this condition is packaged as

```
def HilbertAnglesEqualTwoRightAngles
[HilbertIncidence Geo]
(A B C D : Geo.Point) : Prop :=
exists E : Geo.Point,
Geo.Between C B E /\
Geo.AngleCongruent
A B D
A B E
```

Thus the witness E contains exactly the two pieces of geometric information needed by the reconstruction:

$$C - B - E$$

and

$$\angle ABD \cong \angle ABE.$$

This definition should not be interpreted as a numerical definition of the sum of two angles. It is a synthetic encoding of the particular supplementary configuration occurring in Proposition I.14.

14.6.2 The Collinear Case

Assume first that C, D, and E are collinear.

From the definition of `HilbertAnglesEqualTwoRightAngles` we already have

$$C - B - E.$$

Hence B lies on the line determined by C and E.

Since C and D lie on opposite sides of the base line AB, the segment CD meets that line. Let X be a point of intersection, so that

$$C - X - D$$

and X lies on the base line.

No separate figure is needed for this branch. The point X is only an incidence witness: if the line through C, D, E meets the base line at X, then line uniqueness forces $X = B$. Drawing X as a second visible point near B is more confusing than informative, because the proof immediately shows that the two points coincide.

Both B and X now lie on two lines: the base line AB and the line containing C, D, E.

Suppose that

$$X \neq B.$$

By uniqueness of the line through two distinct points, the two lines would then have to coincide. In particular, C would lie on the base line.

This contradicts the hypothesis that C and D lie on opposite sides of the base line, since a point belonging to the line itself cannot be one of the two opposite-side points.

Therefore

$$X = B.$$

Substituting this equality into

$$C - X - D$$

gives

$$C - B - D.$$

Thus the collinear branch requires no angle argument at all. Once C, D, E are known to be collinear, the conclusion follows entirely from betweenness, plane separation, incidence, and uniqueness of the line through two distinct points.

14.6.3 The Noncollinear Case

Assume now that C, D, and E are not collinear.

The segment CD meets the base line because C and D lie on opposite sides of it. The segment CE also meets the base line, namely at B, since

$$C - B - E.$$

Because C, D, E form a genuine triangle, the Hilbert plane-separation theory implies that the remaining endpoints D and E lie on the same side of the base line; this is the configuration shown in Figure 14.3. Thus

$$\mathrm{SameSide}(D, E; AB).$$

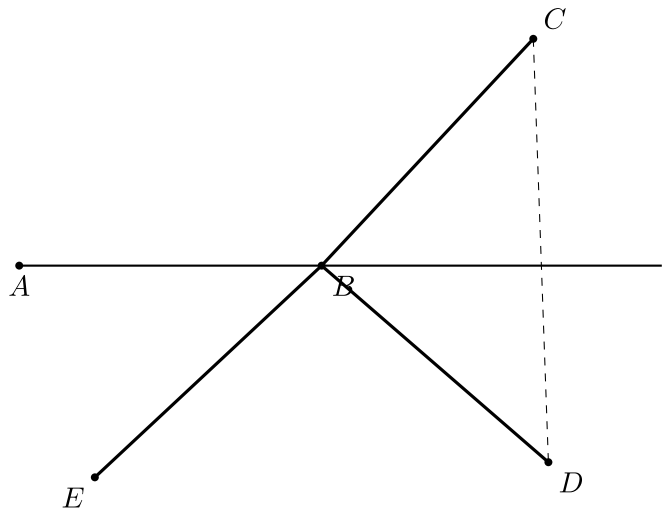


Figure 14.3: Noncollinear branch of the Hilbert reconstruction. The relation $C - B - E$ puts E opposite C across B; plane separation places D and E on the same side of the base line AB. The equality of the two angles then invokes the uniqueness clause of Hilbert III.4.

We now use Hilbert's angle-construction axiom.

The angle $\angle ABE$ is nondegenerate. Construct, on the side of the base line containing D, a ray BZ satisfying

$$\angle ABZ \cong \angle ABE.$$

Hilbert III.4 includes a uniqueness clause: on the prescribed side of the base ray there is only one ray realizing the prescribed angle.

Since E lies on the same side of the base line as D, and

$$\angle ABE \cong \angle ABE,$$

the uniqueness clause gives

$$\mathrm{SameRay}(B, Z, E).$$

On the other hand, the encoded hypothesis of Proposition I.14 gives

$$\angle ABD \cong \angle ABE,$$

and hence, by symmetry,

$$\angle ABE \cong \angle ABD.$$

The point D lies on the side on which BZ was constructed. Applying the same uniqueness clause again gives

$$\mathrm{SameRay}(B, Z, D).$$

We therefore have two same-ray relations with the common reference ray BZ:

$$\mathrm{SameRay}(B, Z, E), \qquad \mathrm{SameRay}(B, Z, D).$$

The Book Zero ray calculus now yields

$$\mathrm{SameRay}(B, E, D).$$

Finally, from the encoded supplementary configuration we already know

$$C - B - E.$$

Replacing E by another point D on the same ray from B preserves the betweenness relation. Transport of betweenness along equal rays therefore gives

$$C - B - D.$$

This completes the noncollinear branch and hence the Hilbert reconstruction of Proposition I.14.

14.7 Lean Formalization

The reconstructed proposition is formalized by the theorem

```
theorem euclid_proposition_14
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C D : Geo.Point)
(base : Geo.Line)
(hAB : A != B)
(hAbase :
HilbertIncidence.OnLine A base)
(hBbase :
HilbertIncidence.OnLine B base)
(hOppCD :
HilbertOppositeSide Geo C D base)
(hTwoRight :
HilbertAnglesEqualTwoRightAngles
Geo A B C D) :
Geo.Between C B D := by
```

The formal conclusion

$$\mathrm{Between}(C, B, D)$$

expresses directly that B lies between C and D, and hence that the rays BC and BD form one straight line.

The principal dependency structure of the proof is

```
HilbertAnglesEqualTwoRightAngles
        |
        +-- C-B-E
        +-- angle ABD ~= angle ABE
        |
HilbertOppositeSide(C,D; base AB)
        |
case split on Collinear(C,D,E)
        |
   +----+--------------------+
   |                         |
collinear                 noncollinear
   |                         |
line uniqueness           SameSide(D,E; base AB)
   |                         |
X = B                    Hilbert III.4 uniqueness
   |                         |
   |                     same ray BE, BD
   |                         |
   +-----------+-------------+
               |
hilbert_between_transport_sameRays
               |
               v
             C-B-D
```

The proof introduces no new geometric axiom. The new predicate

```
HilbertAnglesEqualTwoRightAngles
```

serves only as an interface representation of the angle hypothesis. The actual derivation uses the existing incidence, order, plane separation, angle-construction, and ray machinery.

A notable difference from Euclid's dependency graph is that the Lean theorem does not invoke Proposition I.13. The supplementary configuration needed in the classical proof has already been encoded in the hypothesis, and the identification of the relevant rays is obtained directly from Hilbert III.4.

14.8 Relationship with Earlier Propositions

14.8.1 Proposition I.13

Proposition I.13 gives the direct statement: when one straight line stands on another, the two adjacent angles are either two right angles or together equal to two right angles.

Proposition I.14 is the converse. Starting from the supplementary-angle condition, it proves that the two outer rays form one straight line.

Thus I.13 and I.14 form a natural direct/converse pair.

14.8.2 Proposition I.4

The reconstruction ultimately relies on the Hilbert congruence infrastructure whose triangle-level consequences include SAS. More importantly for I.14, Hilbert III.4 supplies uniqueness of angle construction on a prescribed side of a ray.

This uniqueness principle plays the role of a rigidity theorem: once two rays on the same side make congruent angles with the same initial ray, they must coincide.

14.8.3 Transition to Proposition I.15

Propositions I.13 and I.14 establish the supplementary and straight-line geometry surrounding an intersection. Proposition I.15 then turns to the opposite angles produced by two intersecting lines and proves their congruence.

The three propositions therefore form a coherent local block: straight-angle structure, its converse, and vertical angles.

14.9 Hilbert and Book Zero Components Used

The proof of Proposition I.14 uses several previously established components of the Hilbert and Book Zero layers. Their roles in the reconstruction are recorded here explicitly.

14.9.1 `hilbert_third_side_endpoints_sameSide`

In the noncollinear branch, the points C, D, E form a genuine triangle. The base line meets the interior of CD, because C and D lie on opposite sides of it, and it meets the interior of CE at B, since

$$C - B - E.$$

The theorem

```
hilbert_third_side_endpoints_sameSide
```

then shows that the remaining endpoints D and E lie on the same side of the base line.

Schematically,

$$\begin{matrix} C - X - D, \\ C - B - E, \\ X, B \in AB \end{matrix} \qquad \Longrightarrow \qquad \mathrm{SameSide}(D, E; AB).$$

This is the plane-separation step that makes the uniqueness clause of Hilbert's angle-construction axiom applicable to both D and E.

14.9.2 Hilbert III.4: Uniqueness of Angle Construction

Hilbert's angle-construction axiom provides not only the existence of a copy of a given angle but also uniqueness on a prescribed side of the base ray.

In Proposition I.14 a ray BZ is constructed so that

$$\angle ABZ \cong \angle ABE.$$

Both E and D lie on the prescribed side of the base line. The uniqueness clause can therefore be applied twice, yielding

$$\mathrm{SameRay}(B, Z, E)$$

and

$$\mathrm{SameRay}(B, Z, D).$$

This is the formal replacement for Euclid's final inference that equal angles on the same side determine the same straight line.

14.9.3 `bookZero_36_ray3`

The Book Zero theorem

```
bookZero_36_ray3
```

is a ray-composition principle.

In the present proof,

$$\mathrm{SameRay}(B, Z, E)$$

and

$$\mathrm{SameRay}(B, Z, D)$$

imply

$$\mathrm{SameRay}(B, E, D).$$

Thus the auxiliary ray BZ introduced by Hilbert III.4 can be eliminated, leaving the geometrically relevant statement that BE and BD are the same ray.

14.9.4 `hilbert_between_transport_sameRays`

The final step uses

```
hilbert_between_transport_sameRays
```

to transport strict betweenness along corresponding rays.

We know

$$C - B - E$$

and have established

$$\mathrm{SameRay}(B, E, D).$$

The ray from B toward C is unchanged, while E is replaced by the point D on the same ray from B. Therefore the theorem gives

$$C - B - D.$$

This is precisely the formal conclusion of Proposition I.14.

The dependency path of the noncollinear branch can therefore be summarized as

$$\begin{array}{c} \text{plane separation} \\ \downarrow \\ \mathrm{SameSide}(D, E; AB) \\ \downarrow \\ \text{Hilbert III.4} \\ \downarrow \\ \mathrm{SameRay}(B, Z, E), \mathrm{SameRay}(B, Z, D) \\ \downarrow \\ \texttt{bookZero\_36\_ray3} \\ \downarrow \\ \mathrm{SameRay}(B, E, D) \\ \downarrow \\ \texttt{hilbert\_between\_transport\_sameRays} \\ \downarrow \\ C - B - D. \end{array}$$

14.10 Formalization Notes

Proposition I.14 is a good example of a classical converse whose formal proof is governed by uniqueness rather than by numerical angle arithmetic.

The hypothesis says that two adjacent angles have the same total effect as a straight angle. In a traditional proof this is often read directly from the diagram. In the Hilbert reconstruction, the decisive step is to compare rays using the uniqueness clause of angle construction.

The proof naturally separates into two cases.

If the relevant points are already collinear, the desired straight-line conclusion follows from order and collinearity reasoning.

If they are not collinear, the angle hypothesis determines a ray that would have to occupy the same side and form the same angle as another already specified ray. Hilbert III.4 then forces the two rays to coincide. The resulting same-ray information is transported back to betweenness, contradicting the assumed noncollinear configuration.

Schematically,

$$\text{supplementary-angle hypothesis} \longrightarrow \text{same-side information} \longrightarrow \text{angle-construction uniqueness} \longrightarrow \text{same ray} \longrightarrow$$

This is structurally different from simply writing an equation such as $\alpha + \beta = 180^\circ$. No numerical angle measure is introduced. The straight-line conclusion is recovered entirely from incidence, order, angle congruence, side-of-line information, and uniqueness.

14.11 Dependency Summary

The formal dependency structure is

$$\begin{array}{c}
\text{adjacent-angle hypothesis} \\
\downarrow \\
\text{collinear / noncollinear case split} \\
\downarrow \\
\texttt{hilbert_third_side_endpoints_sameSide} \\
\downarrow \\
\text{Hilbert III.4 uniqueness} \\
\downarrow \\
\texttt{bookZero_36_ray3} \\
\downarrow \\
\texttt{hilbert_between_transport_sameRays} \\
\downarrow \\
\text{required straight-line configuration} \\
\downarrow \\
\texttt{euclid_proposition_14}.
\end{array}$$

14.12 Conclusion

Proposition I.14 completes the converse half of the supplementary-angle theory begun in I.13.

Its formal reconstruction is especially instructive because it avoids introducing angle addition or numerical angle measures. The proof instead uses Hilbert's side-of-line and angle-construction uniqueness machinery to show that the geometry of the rays is rigid enough to force a straight line.

The proposition therefore illustrates a recurring feature of the project: statements that look algebraic in modern notation can often be recovered synthetically from incidence, order, congruence, and uniqueness.

Together with I.13, it prepares the local theory of intersecting lines used immediately in Proposition I.15.

Chapter 15

Proposition I.15

15.1 Introduction

Euclid's Proposition I.15 establishes the equality of vertical angles formed by two intersecting straight lines.

If two straight lines intersect at a point, each pair of opposite angles is equal. In modern terminology, these are the vertical-angle relations.

Euclid proves the proposition by applying Proposition I.13 to two adjacent linear pairs and subtracting a common angle from two sums that are each equal to two right angles.

The Hilbert reconstruction is considerably shorter. The corresponding vertical-angle theorem has already been derived in the Hilbert layer as

```
hilbert_vertical_angles
```

from Hilbert's theory of adjacent angles. Proposition I.15 therefore becomes an application of an existing synthetic theorem rather than a new geometric derivation.

This proposition provides another example in which a classical Euclidean proof is compressed by the intermediate Hilbert interface: the mathematical content required by Euclid has already been isolated as a reusable theorem.

15.2 The Classical Configuration

Let the straight lines AB and CD intersect at the point E, with

$$A - E - B$$

and

$$C - E - D.$$

The intersection determines four angles around E.

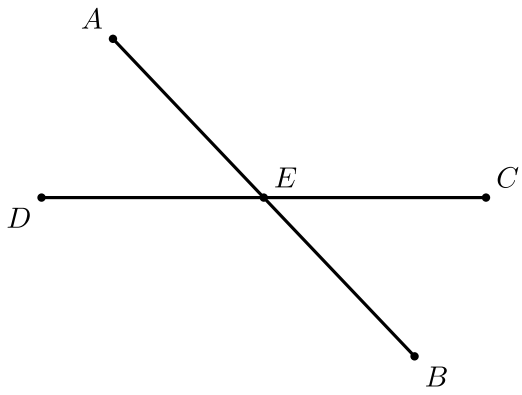


Figure 15.1: Classical configuration for Proposition I.15. The betweenness relations $A - E - B$ and $D - E - C$ encode the two pairs of opposite rays.

The opposite pairs are

$$\angle AEC \qquad \text{and} \qquad \angle BED,$$

and

$$\angle CEB \qquad \text{and} \qquad \angle DEA.$$

Proposition I.15 states that each pair is equal.

Only one geometric diagram is needed. The two conclusions are not two different configurations: they are the two opposite pairs determined by the same crossing. Repeating the intersection would add no mathematical information.

The betweenness relations

$$A - E - B, \qquad C - E - D$$

express synthetically that EA and EB are opposite rays and that EC and ED are opposite rays. Thus the classical configuration is already naturally adapted to Hilbert's order geometry.

15.3 Statement

Euclid's Proposition I.15 states:

> If two straight lines cut one another, they make the vertical angles equal to one another.

For the configuration

$$A - E - B, \qquad C - E - D,$$

the proposition gives the two vertical-angle congruences

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

In the Hilbert reconstruction we assume, in addition, that the two lines are genuinely distinct. This is expressed by

$$\neg\,\mathrm{Collinear}(A, E, C).$$

The formal statement is therefore

$$\begin{array}{l} A - E - B, \\ C - E - D, \\ \neg\,\mathrm{Collinear}(A, E, C) \end{array} \quad \Longrightarrow \quad \begin{cases} \angle AEC \cong \angle BED, \\ \angle CEB \cong \angle DEA. \end{cases}$$

The noncollinearity hypothesis is the synthetic expression of a proper intersection of two distinct straight lines. It also guarantees that the angles occurring in the proposition are nondegenerate.

Thus the Lean formulation makes explicit a condition that is implicit in Euclid's phrase "two straight lines cut one another." Strict betweenness specifies the two pairs of opposite rays, while noncollinearity excludes the case in which all five points lie on a single straight line.

15.4 Classical Proof

Euclid first proves the equality of one pair of vertical angles.

Since the straight line AE stands upon the straight line CD, Proposition I.13 gives

$$\angle CEA + \angle AED$$

equal to two right angles.

Again, since the straight line DE stands upon the straight line AB, Proposition I.13 gives

$$\angle AED + \angle DEB$$

equal to two right angles.

Therefore

$$\angle CEA + \angle AED = \angle AED + \angle DEB.$$

Subtracting the common angle $\angle AED$ from both sides, by Common Notion 3, gives

$$\angle CEA = \angle DEB.$$

Thus one pair of vertical angles is equal.

In the same way,

$$\angle CEB = \angle AED.$$

Therefore, when two straight lines cut one another, they make the vertical angles equal to one another.

Wilkins's diagram is especially effective here because the geometry is entirely local at E. The proof needs only the four rays and two linear pairs. The colored sectors in Wilkins's note help distinguish the angles, but no additional construction is involved, so the reconstruction keeps the diagram unshaded and records the equalities in the text.

15.5 Proof Architecture of the Reconstruction

The formal reconstruction and Euclid's original proof establish the same geometric fact at different levels of the theory.

```
Euclid
------
A-E-B and C-E-D
      |
      +-- CEA + AED = 2R   (I.13)
      |
      +-- DEB + AED = 2R   (I.13)
      |
      v
CEA + AED = DEB + AED
      |
Common Notion 3
      |
      v
CEA = DEB

same argument
      |
      v
CEB = AED


Hilbert / Lean
--------------
A-E-B, C-E-D, noncollinearity
      |
hilbert_vertical_angles
      |
      v
AEC ~= BED
      |
```

```
reorder the same crossing
      |
transport noncollinearity
      |
hilbert_vertical_angles
      |
      v
CEB ~= DEA
```

The formal proof therefore introduces no new geometric point and no new diagrammatic case. Its extra work is purely structural: the second pair is obtained by reorienting the same intersection and proving that the reordered triple is still noncollinear.

Euclid derives the vertical-angle theorem from Proposition I.13. Each of two adjacent angle pairs is equal to two right angles. Since the two sums contain a common angle, Common Notion 3 allows that angle to be subtracted, leaving the two vertical angles equal.

Thus the classical dependency is

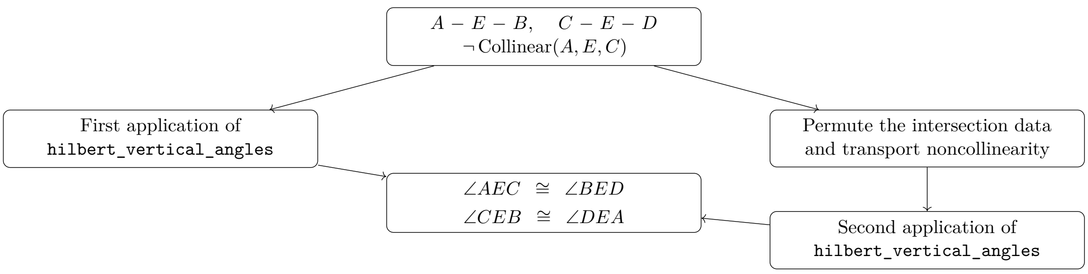
$$\text{I.13} \longrightarrow \text{two linear-pair equalities} \longrightarrow \text{Common Notion 3} \longrightarrow \text{I.15.}$$

The Hilbert development has already isolated the corresponding geometric argument in the theorem

`hilbert_vertical_angles`

Consequently, Proposition I.15 does not require a new reconstruction of angle addition or subtraction. The first pair of vertical angles follows by a direct application of this theorem.

The second pair is obtained by viewing the same intersection with the roles of the two straight lines exchanged. The only additional work is to transport the required noncollinearity hypothesis to the reordered triple of points.



Schematically, the two proof architectures are therefore

$$\begin{array}{rcl} \text{Euclid} & : & \text{I.13} \to \text{angle sums} \to \text{C.N.3} \to \text{vertical angles,} \\ \text{Hilbert reconstruction} & : & \text{intersection data} \to \texttt{hilbert\_vertical\_angles} \to \text{vertical angles.} \end{array}$$

The difference is architectural rather than mathematical. The Hilbert layer has already packaged the vertical-angle argument as a reusable synthetic theorem, so the Euclidean proposition becomes a thin interface theorem over previously established geometry.

15.6 Hilbert Reconstruction

The Hilbert reconstruction starts directly from the synthetic data of the intersection:

$$A - E - B, \qquad C - E - D,$$

together with

$$\neg\,\mathrm{Collinear}(A, E, C).$$

Unlike Euclid's proof, no explicit notion of a sum of angles equal to two right angles is required. The relevant geometric content has already been established in the Hilbert layer as the vertical-angle theorem.

15.6.1 The First Pair of Vertical Angles

The theorem

```
hilbert_vertical_angles
```

has precisely the form needed for the first pair.

Applied to

$$A - E - B, \qquad C - E - D, \qquad \neg\,\mathrm{Collinear}(A, E, C),$$

it gives directly

$$\angle AEC \cong \angle BED.$$

Thus the first conclusion of Proposition I.15 requires no auxiliary construction.

15.6.2 The Second Pair of Vertical Angles

For the second pair we view the same intersection with the two straight lines exchanged.

From

$$A - E - B$$

the symmetry of strict betweenness gives

$$B - E - A.$$

We also require the noncollinearity condition

$$\neg\,\mathrm{Collinear}(C, E, B).$$

Suppose, for contradiction, that

$$\mathrm{Collinear}(C, E, B).$$

Since $A - E - B$, the points A, E, B are collinear. Permuting the first collinearity relation and using transitivity of collinearity through the distinct points E and B would then imply

$$\mathrm{Collinear}(A, E, C),$$

contrary to the original hypothesis.

Hence

$$\neg\,\mathrm{Collinear}(C, E, B).$$

We may therefore apply `hilbert_vertical_angles` a second time, now to

$$C - E - D, \qquad B - E - A.$$

This yields

$$\angle CEB \cong \angle DEA.$$

Combining the two applications gives the complete conclusion

$$\angle AEC \cong \angle BED, \qquad \angle CEB \cong \angle DEA.$$

The Hilbert reconstruction is therefore substantially shorter than Euclid's original argument. The classical use of Proposition I.13 and Common Notion 3 has already been absorbed into the previously proved vertical-angle theorem; the only additional formal work is the permutation of the intersection data required for its second application.

15.7 Lean Formalization

The reconstructed proposition is formalized by the theorem

```
theorem euclid_proposition_15
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C D E : Geo.Point)
(hAEB : Geo.Between A E B)
(hCED : Geo.Between C E D)
(hAEC : not PrimCollinear Geo A E C) :
Geo.AngleCongruent A E C B E D /\
Geo.AngleCongruent C E B D E A := by
```

The conclusion records both pairs of vertical angles:

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

The first pair is obtained directly:

```
have hFirst :
Geo.AngleCongruent A E C B E D :=
hilbert_vertical_angles
Geo A E C B D
hAEB
hCED
hAEC
```

For the second pair, the first line is read in the opposite direction:

```
have hBEC : Geo.Between B E A :=
(HilbertOrder.between_incidence
A E B hAEB).2.2.2.2
```

The required noncollinearity condition is then derived by contradiction. If C, E, B were collinear, cyclic permutation gives E, B, C collinear. Since A, E, B are already collinear and $E \neq B$, transitivity would imply that A, E, C are collinear, contradicting `hAEC`.

In Lean this is expressed by

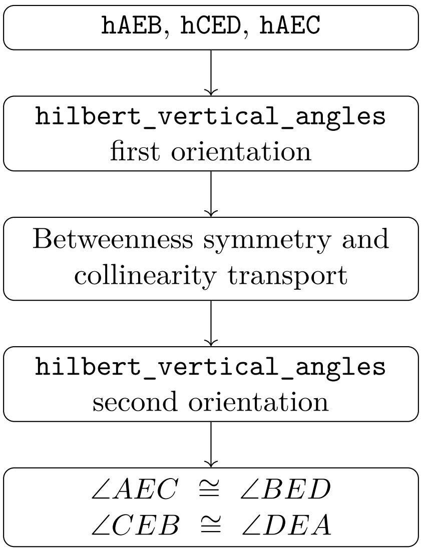
```
have hCEB :
not PrimCollinear Geo C E B := by
intro h

have hEBC :
PrimCollinear Geo E B C :=
PrimCollinearCycle Geo C E B h

have hAEBcol :
PrimCollinear Geo A E B :=
(HilbertOrder.between_incidence
A E B hAEB).2.2.2.1

have hEB : E != B :=
(HilbertOrder.between_incidence
A E B hAEB).2.1

have hAECcol :
PrimCollinear Geo A E C :=
```

```
hilbert_primCollinear_trans
Geo A E B C
hEB
hAEBcol
hEBC

exact hAEC hAECcol
```

The second pair of vertical angles is now another direct application:

```
have hSecond :
Geo.AngleCongruent C E B D E A :=
hilbert_vertical_angles
Geo C E B D A
hCED
hBEC
hCEB

exact <hFirst, hSecond>
```

Thus the formal dependency structure is



No new geometric axiom or auxiliary geometric construction is introduced in the formalization. Proposition I.15 is obtained entirely from the existing Hilbert incidence, order, collinearity, and vertical-angle theory.

15.8 Relationship with Earlier Propositions

15.8.1 Proposition I.13

Proposition I.13 establishes the supplementary-angle configuration created when a straight line stands on another straight line. That result belongs to the same local geometry of intersecting lines that underlies I.15.

The logical roles are different, however. I.13 concerns adjacent angles whose sum forms a straight angle; I.15 concerns opposite angles and proves their congruence.

15.8.2 Proposition I.14

Proposition I.14 gives a converse criterion for recognizing a straight line from a supplementary configuration. Together, I.13 and I.14 develop the straight-angle infrastructure surrounding an intersection.

Proposition I.15 then extracts a different invariant of the same configuration: opposite, or vertical, angles are congruent.

15.8.3 Transition to Proposition I.16

Vertical-angle congruence becomes an explicit transport tool in Proposition I.16. The exterior-angle proof naturally produces angles in orientations that do not always match the public Euclidean statement, and I.15 supplies the congruence needed to move between those orientations.

Thus I.15 is not merely the end of the elementary intersecting-lines block. It becomes part of the infrastructure for the strict angle-comparison theory beginning with I.16.

15.9 Hilbert and Book Zero Components Used

Proposition I.15 is formally short because its main geometric content has already been isolated in the Hilbert layer.

The proof uses one principal theorem together with elementary order and collinearity machinery required to apply it in the two orientations of the intersection.

15.9.1 `hilbert_vertical_angles`

The central result is

```
hilbert_vertical_angles
```

which states that if

$$C - E - B, \qquad D - E - F,$$

and

$$\neg\,\mathrm{Collinear}(C, E, D),$$

then

$$\angle CED \cong \angle BEF.$$

This theorem is the immediate vertical-angle corollary developed in the Hilbert layer after the theorem on adjacent congruent angles.

In Proposition I.15 it is used twice.

The first application gives

$$\angle AEC \cong \angle BED.$$

The second application, after reversing the first straight line and transporting noncollinearity, gives

$$\angle CEB \cong \angle DEA.$$

Thus essentially all of the geometric content of Euclid's Proposition I.15 is already contained in `hilbert_vertical_angles`.

15.9.2 `HilbertOrder.between_incidence`

The theorem

```
HilbertOrder.between_incidence
```

extracts the elementary order and incidence consequences of strict betweenness.

From

$$A - E - B$$

the proof obtains both

$$B - E - A$$

and the collinearity relation

$$\mathrm{Collinear}(A, E, B).$$

It also supplies the inequality

$$E \neq B,$$

which is needed for the transitivity of collinearity.

Thus a single betweenness hypothesis provides all of the auxiliary order data required for the second application of the vertical-angle theorem.

15.9.3 `PrimCollinearCycle`

The permutation theorem

```
PrimCollinearCycle
```

cyclically reorders a collinearity relation.

In the contradiction argument,

$$\text{Collinear}(C, E, B)$$

is transformed into

$$\text{Collinear}(E, B, C).$$

This places the points in exactly the orientation required by the transitivity theorem used in the next step.

15.9.4 `hilbert_primCollinear_trans`

The theorem

```
hilbert_primCollinear_trans
```

expresses the transitivity of collinearity when two distinct common points determine the same line.

In Proposition I.15 we combine

$$\text{Collinear}(A, E, B)$$

with

$$\text{Collinear}(E, B, C)$$

and the inequality

$$E \neq B.$$

The result is

$$\text{Collinear}(A, E, C),$$

contradicting the original noncollinearity hypothesis.

Hence

$$\neg\,\mathrm{Collinear}(C, E, B),$$

which is precisely the hypothesis needed for the second invocation of `hilbert_vertical_angles`.

The complete formal dependency may therefore be summarized as

$$\begin{array}{c}
A - E - B, \quad C - E - D, \quad \neg\,\mathrm{Collinear}(A, E, C) \\
\downarrow \\
\texttt{hilbert_vertical_angles} \\
\downarrow \\
\angle AEC \cong \angle BED \\
\text{betweenness symmetry} + \text{collinearity transport} \\
\downarrow \\
\neg\,\mathrm{Collinear}(C, E, B) \\
\downarrow \\
\texttt{hilbert_vertical_angles} \\
\downarrow \\
\angle CEB \cong \angle DEA.
\end{array}$$

No genuinely new Book Zero theorem is required for Proposition I.15. The proposition is therefore primarily an interface theorem over the already established Hilbert vertical-angle theory.

15.10 Formalization Notes

Proposition I.15 is classically almost immediate from the supplementary angle theory, but its formal reconstruction highlights a different issue: orientation.

At an intersection there are two pairs of vertical angles. The geometric theorem is symmetric, while the formal predicates are written with ordered point triples. Consequently the proof must explicitly recover the appropriate collinearities from betweenness, rotate or cycle them into the required orientation, and then invoke the reusable vertical-angle theorem.

The essential pattern is

$$\text{betweenness} \longrightarrow \text{primitive collinearity} \longrightarrow \text{orientation normalization} \longrightarrow \texttt{hilbert\_vertical\_angles} \longrightarrow \text{angl}$$

This is a good example of why the Book Zero and Hilbert interface layers contain many apparently small symmetry and rotation lemmas. On paper, one simply says that two lines intersect. In Lean, the exact ordering of the points on those lines determines which theorem application type-checks.

No new geometric axiom is introduced by I.15. The proposition packages the already established intersection and angle-congruence machinery into the classical Euclidean statement.

15.11 Dependency Summary

The formal dependency structure is

two intersecting straight lines
$\downarrow$
betweenness hypotheses
$\downarrow$
`HilbertOrder.between_incidence`
$\downarrow$
collinearity rotation / transitivity
$\downarrow$
`hilbert_vertical_angles`
$\downarrow$
first pair of vertical angles congruent
second pair of vertical angles congruent
$\downarrow$
`euclid_proposition_15`.

15.12 Conclusion

Proposition I.15 completes the elementary theory of angles formed by intersecting straight lines.

Its classical content is the familiar theorem that vertical angles are congruent. The formal reconstruction shows that the principal burden is not angle arithmetic but the management of incidence and orientation: betweenness must be converted into primitive collinearity, and the resulting triples must be placed in exactly the orientation expected by the vertical-angle theorem.

The proposition is also structurally important for what follows. Vertical-angle congruence becomes a reusable transport mechanism in Proposition I.16, where Book I moves from congruence statements to strict angle comparison.

Thus I.15 closes one local geometric block and simultaneously provides infrastructure for the next.

Chapter 16

Proposition I.16

16.1 Introduction

Euclid's Proposition I.16 is the exterior-angle inequality for a triangle. If a side of a triangle is extended beyond one of its vertices, the exterior angle formed at that vertex is greater than either of the two remote interior angles.

For a nondegenerate triangle ABC, extend BC through C to a point D, so that

$$B - C - D.$$

The proposition asserts

$$\angle BAC < \angle ACD \qquad \text{and} \qquad \angle ABC < \angle ACD.$$

Unlike Proposition I.15, this proposition is not merely a thin application of one previously packaged theorem. Its reconstruction requires a substantial piece of Hilbert order-and-congruence geometry: an interior ray must be transported between congruent angles, and the resulting angle comparison must be expressed without numerical angle measure.

The formal development therefore isolates the geometric machinery in the Hilbert interface and leaves the Euclidean proposition itself comparatively short. In particular, the final Lean file contains three theorems:

- `euclid_proposition_16_first`,
- `euclid_proposition_16_second`,
- `euclid_proposition_16`.

16.2 The Classical Configuration

Let ABC be a nondegenerate triangle and let the side BC be produced through C to D:

$$B - C - D.$$

The angle $\angle ACD$ is exterior to the triangle. The two interior opposite angles are $\angle BAC$ and $\angle ABC$.

The formal hypotheses are

$$\neg\,\mathrm{Collinear}(A, B, C) \qquad \text{and} \qquad B - C - D.$$

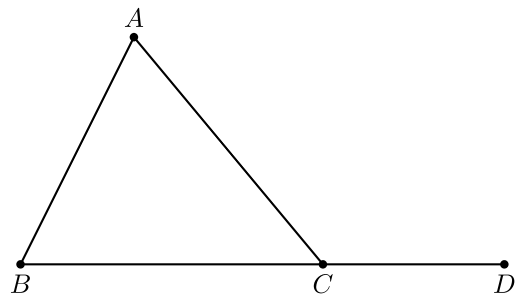


Figure 16.1: Statement configuration for Proposition I.16: BC is produced beyond C to D.

The noncollinearity hypothesis makes the triangle nondegenerate, while strict betweenness records that D lies on the continuation of BC beyond C.

16.3 Statement

Euclid states Proposition I.16 as follows:

> If one side of a triangle is produced, the exterior angle is greater than either of the two interior opposite angles.

In the Hilbert reconstruction, strict comparison of angles is represented by `HilbertAngleLess`. Thus the formal conclusion is

$$\mathrm{HilbertAngleLess}(BAC, ACD) \quad \wedge \quad \mathrm{HilbertAngleLess}(ABC, ACD).$$

More explicitly,

$$\neg\,\mathrm{Collinear}(A, B, C), \quad B - C - D \Longrightarrow \begin{cases} \angle BAC < \angle ACD, \\ \angle ABC < \angle ACD. \end{cases}$$

The relation `HilbertAngleLess` is synthetic. It does not compare numerical measures. An angle is smaller than another when a congruent copy of the first can be realized as a proper interior subangle of the second.

16.4 Classical Proof

Euclid proves the first inequality by a midpoint construction.

Let E be the midpoint of AC. Join BE and extend it beyond E to a point F such that

$$BE = EF.$$

Join CF.

Since $AE = EC$, $BE = EF$, and the vertical angles $\angle AEB$ and $\angle CEF$ are equal by Proposition I.15, the triangles AEB and CEF are congruent by Proposition I.4. Hence

$$\angle BAE = \angle ECF.$$

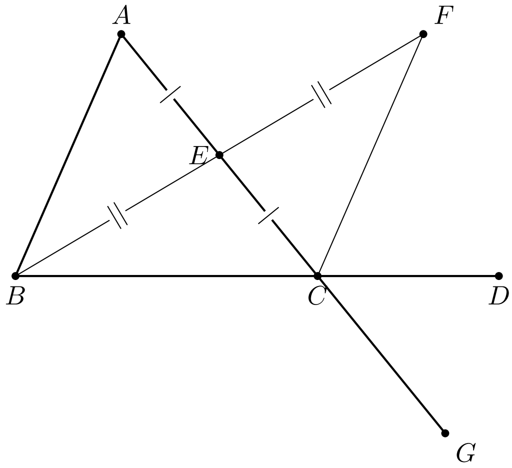


Figure 16.2: Wilkins's auxiliary construction for the first inequality: E bisects AC, BE is produced to F with $BE \cong EF$, and AC is produced beyond C to G.

Since E lies on AC, this gives

$$\angle BAC = \angle ACF.$$

But the ray CF lies inside the exterior angle ACD. Therefore

$$\angle BAC < \angle ACD.$$

This last sentence is exactly where the diagram carries hidden order information. Wilkins spells it out: F must lie on the same side of CD as A and on the same side of AC as D; these two half-plane conditions characterize the interior of $\angle ACD$. Thus the inclusion of the ray CF in the exterior angle is a theorem, not a fact that may be read off the picture.

For the second opposite angle Euclid repeats the same argument after extending another side of the triangle. The equality of the corresponding vertical exterior angles then identifies the resulting exterior angle with $\angle ACD$.

Thus the classical architecture is

$$\text{midpoint construction} \to \text{I.15} \to \text{SAS (I.4)} \to \text{interior subangle} \to \text{I.16}.$$

16.5 Proof Architecture of the Reconstruction

The Hilbert reconstruction retains the synthetic structure of Euclid's argument but packages the difficult order-theoretic steps into reusable interface theorems.

```
statement: B-C-D
        |
        +---------------- first opposite angle ----------------+
        |                                                      |
        v                                                      |
Hilbert auxiliary construction                                 |
A-M-C, B-M-E + congruence data                                 |
        |                                                      |
        v                                                      |
```

```
AB || CE                                                      |
        |                                                     |
hilbert_exterior_angle_less                                   |
        |                                                     |
        v                                                     |
BAC < ACD                                                     |
                                                              |
        +--------------- second opposite angle ---------------+
        |
extend AC beyond C: A-C-F
        |
apply first half to triangle BAC
        |
ABC < BCF
        |
vertical angles at C:
BCF ~= ACD
        |
transport strict angle order
        |
ABC < ACD
        |
        v
euclid_proposition_16
```

Here the distinction between the classical and formal diagrams matters. Wilkins's E, F construction explains Euclid's SAS argument and the nontrivial claim that F lies inside the exterior angle. The Lean helper uses its own auxiliary points M, E and packages that order geometry into the Hilbert interface. The second Lean inequality is then obtained by a genuinely new extension $A - C - F$ and vertical-angle transport.

For the first opposite angle the auxiliary construction produces points M and E with

$$A - M - C, \qquad B - M - E,$$

and the required segment and angle congruences. From these data the interface proves that AB is parallel to CE. The theorem `hilbert_exterior_angle_less` then concludes directly that

$$\angle BAC < \angle ACD.$$

Schematically:

$$\begin{gathered}
\neg\,\mathrm{Collinear}(A, B, C) \\
\downarrow \\
\texttt{hilbert_exterior_angle_aux} \\
\downarrow \\
A - M - C,\ B - M - E,\ \text{congruence data} \\
\downarrow \\
\texttt{hilbert_exterior_angle_aux_parallel} \\
\downarrow \\
AB \parallel CE \\
\downarrow \\
\texttt{hilbert_exterior_angle_less} \\
\downarrow \\
\angle BAC < \angle ACD.
\end{gathered}$$

The second inequality is deliberately reduced to the first. Extend AC through C to F. Apply the first part to triangle BAC to obtain

$$\angle ABC < \angle BCF.$$

The lines ACF and BCD intersect at C, so Proposition I.15 gives the vertical-angle congruence

$$\angle BCF \cong \angle ACD.$$

Transporting the right-hand side of the strict angle comparison across this congruence yields

$$\angle ABC < \angle ACD.$$

Thus the second half has the compact architecture

$$\text{first half of I.16} \to \text{I.15 / vertical angles} \to \text{transport of } < \to \text{second half of I.16}.$$

16.6 Hilbert Reconstruction

16.6.1 The First Opposite Interior Angle

The theorem `euclid_proposition_16_first` begins from

$$\neg\,\mathrm{Collinear}(A, B, C), \qquad B - C - D.$$

It invokes

`hilbert_exterior_angle_aux`

to obtain points M, E and the auxiliary configuration required for the exterior-angle argument. The same configuration is passed to

`hilbert_exterior_angle_aux_parallel`,

which proves

$$AB \parallel CE.$$

Finally,

`hilbert_exterior_angle_less`

combines the auxiliary construction, the extension $B - C - D$, and the parallelism to establish

$$\angle BAC < \angle ACD.$$

The Euclidean theorem itself therefore contains no unresolved construction. The substantial work has been moved into the Hilbert interface, where it can be reused independently of Proposition I.16.

16.6.2 The Second Opposite Interior Angle

For the second inequality, the proof first derives $A \neq C$ from the noncollinearity of A, B, C. Hilbert's order axiom then extends AC through C to a point F:

$$A - C - F.$$

The triangle is viewed in the order BAC. Applying `euclid_proposition_16_first` to this reordered triangle gives

$$\angle ABC < \angle BCF.$$

It remains to identify $\angle BCF$ with the desired exterior angle $\angle ACD$. From $B - C - D$ we obtain the reversed betweenness relation

$$D - C - B.$$

The two straight lines ACF and BCD therefore form a pair of vertical angles at C. The theorem `VerticalAngles` yields, after the required symmetry and reversal operations,

$$\angle BCF \cong \angle ACD.$$

The theorem

`hilbert_angleLess_transport_right`

then transports the comparison across the congruent right-hand angle:

$$\angle ABC < \angle BCF, \qquad \angle BCF \cong \angle ACD \quad \Longrightarrow \quad \angle ABC < \angle ACD.$$

16.7 Lean Formalization

The first half is formalized as follows:

```
theorem euclid_proposition_16_first
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D : Geo.Point)
    (hABC : not PrimCollinear Geo A B C)
    (hBCD : Geo.Between B C D) :
    HilbertAngleLess Geo B A C A C D := by

  rcases
      hilbert_exterior_angle_aux
        Geo A B C hABC with
    <M, E,
      hAMC,
      hAMMC,
      hBME,
      hBMME,
      hAngle>

  have hParallel :
      Geo.Parallel A B C E :=
    hilbert_exterior_angle_aux_parallel
```

```
      Geo A B C M E
      hABC
      hAMC
      hBME
      hAngle

  exact
    hilbert_exterior_angle_less
      Geo A B C D M E
      hABC
      hAMC
      hBME
      hBCD
      hAngle
      hParallel
```

The second half deliberately reuses the first:

```
have hLessBCF :
    HilbertAngleLess Geo A B C B C F :=
  euclid_proposition_16_first
    Geo
    B A C F
    hBACnc
    hACF
```

The vertical-angle relation is then constructed at C and normalized to the required orientation:

```
have hVerticalRaw :
    Geo.AngleCongruent A C D F C B :=
  VerticalAngles
    Geo
    A C D F B
    hACF
    hDCA
    hACDnc

have hVerticalSymm :
    Geo.AngleCongruent F C B A C D :=
  Geo.angle_congruent_symmetry
    A C D
    F C B
    hVerticalRaw

have hVertical :
    Geo.AngleCongruent B C F A C D :=
  (Geo.angle_congruent_reverse_first
    F C B A C D).mp
    hVerticalSymm
```

Finally the strict inequality is transported:

```
exact
  hilbert_angleLess_transport_right
```

```
    Geo
    A B C
    B C F
    A C D
    hLessBCF
    hACDnc
    hVertical
```

The complete proposition simply combines the two parts:

```
theorem euclid_proposition_16
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D : Geo.Point)
    (hABC : not PrimCollinear Geo A B C)
    (hBCD : Geo.Between B C D) :
    HilbertAngleLess Geo B A C A C D /\
    HilbertAngleLess Geo A B C A C D := by

  have hFirst :=
    euclid_proposition_16_first
      Geo A B C D hABC hBCD

  have hSecond :=
    euclid_proposition_16_second
      Geo A B C D hABC hBCD

  exact <hFirst, hSecond>
```

16.8 Relationship with Earlier Propositions

16.8.1 Proposition I.10

Euclid's classical proof uses a midpoint construction as part of the auxiliary configuration. Proposition I.10 is therefore one of the constructional ingredients behind the traditional argument.

In the Hilbert reconstruction, however, the decisive exterior-angle comparison has already been isolated in reusable infrastructure, so the public I.16 proof does not need to reproduce the entire midpoint construction locally.

16.8.2 Proposition I.15

Vertical-angle congruence is used when the auxiliary configuration produces an angle in the opposite orientation from the one required in the final statement.

The theorem `VerticalAngles` therefore acts as an orientation bridge inside the reconstruction.

16.8.3 Transition to Proposition I.17

Proposition I.16 supplies the strict exterior-angle comparison that drives Proposition I.17. In I.17, each assertion that two interior angles are together less than two right angles is obtained by extending a side and applying the exterior-angle theorem in an appropriate orientation.

Thus I.16 begins the strict angle-order block of Book I that continues through I.17–I.19.

16.9 Hilbert and Book Zero Components Used

16.9.1 `HilbertAngleLess`

The proposition requires a genuinely synthetic notion of strict angle comparison. The relation `HilbertAngleLess` expresses that a congruent copy of one angle occurs as a proper interior subangle of another. This avoids introducing numerical angle measure and keeps Proposition I.16 inside pure incidence, order, and congruence geometry.

16.9.2 `hilbert_exterior_angle_aux`

This theorem supplies the auxiliary points used in the first exterior-angle construction. It packages the midpoint-style configuration that underlies Euclid's original proof and returns the betweenness and congruence data required by the subsequent argument.

16.9.3 `hilbert_exterior_angle_aux_parallel`

From the auxiliary configuration, this theorem proves

$$AB \parallel CE.$$

The parallel relation is not an additional Euclidean postulate here; it is a derived relation inside the neutral Hilbert development used to organize the exterior-angle argument.

16.9.4 `hilbert_exterior_angle_less`

This is the principal interface theorem used by the first half of I.16. It converts the auxiliary construction and its parallelism into the strict angle comparison

$$\angle BAC < \angle ACD.$$

Its proof depends on the interior-subangle machinery developed in the Hilbert layer.

16.9.5 Interior-subangle transport

A substantial part of the work required for Proposition I.16 lies below the final Euclidean file. The Hilbert interface establishes that an interior ray can be transported between congruent angles while preserving the corresponding subangle.

The completed infrastructure includes the segment and crossbar mechanisms needed for that transport, in particular the ability to transport strict segment comparison through congruence, transport an interior point between congruent segments, and preserve ray–segment intersection when the endpoints are moved along the same two rays.

This is the part of the reconstruction where the formal proof exposes geometric structure that is largely implicit in the paper proof.

16.9.6 `VerticalAngles`

The second half uses the vertical-angle theorem already exposed by Proposition I.15. After AC is extended through C to F, the angles

$$\angle BCF \quad \text{and} \quad \angle ACD$$

are vertical. Their congruence allows the second inequality to be reduced to the first.

16.9.7 `hilbert_angleLess_transport_right`

This theorem expresses invariance of strict angle comparison under congruence of the right-hand angle. In the final step,

$$\angle ABC < \angle BCF$$

and

$$\angle BCF \cong \angle ACD$$

imply

$$\angle ABC < \angle ACD.$$

16.10 Formalization Notes

Proposition I.16 is a major transition from angle congruence to strict angle comparison.

Earlier propositions establish equalities of angles through triangle congruence, isosceles geometry, supplementary configurations, and vertical angles. I.16 requires a genuinely ordered conclusion: an exterior angle is strictly greater than either opposite interior angle.

The reconstruction therefore uses `HilbertAngleLess` rather than a numerical angle measure. The strict comparison is obtained from an explicit geometric subangle configuration and then transported through angle congruences until its endpoints match the Euclidean statement.

A second important feature is that the two opposite interior angles are not handled by pretending that orientation is irrelevant. One branch can be obtained directly from the exterior-angle infrastructure; the other requires additional transport, including vertical-angle congruence. Lean forces these orientations to be recorded explicitly.

Schematically, the formal mechanism is

$$\text{auxiliary exterior-angle configuration} \longrightarrow \text{strict subangle comparison} \longrightarrow \text{angle-congruence transport} \longrightarrow \text{I.16.}$$

This infrastructure becomes immediately reusable. Proposition I.17 uses I.16 to control sums of two interior angles without introducing angle addition, while Propositions I.18 and I.19 use the resulting strict angle-order theory to relate side order and angle order in a triangle.

16.11 Dependency Summary

The formal dependency structure can be summarized as

$$\begin{gathered}\text{Hilbert incidence} + \text{order} + \text{congruence} \\ \downarrow \\ \text{segment construction and ray/order theory} \\ \downarrow \\ \text{interior-subangle transport / crossbar machinery} \\ \downarrow \\ \texttt{hilbert\_exterior\_angle\_less} \\ \downarrow \\ \texttt{euclid\_proposition\_16\_first} \\ \downarrow \\ \text{vertical angles} + \text{angle-less transport} \\ \downarrow \\ \texttt{euclid\_proposition\_16\_second} \\ \downarrow \\ \texttt{euclid\_proposition\_16}.\end{gathered}$$

The final Euclidean theorem is therefore compact, but its compactness is meaningful: the difficult geometry has been factored into reusable Hilbert-interface results rather than hidden in an axiom or a local `sorry`.

16.12 Conclusion

Proposition I.16 marks a significant step beyond the immediately preceding propositions. It introduces a strict comparison of angles and requires a robust synthetic notion of interiority. In Euclid's presentation this is handled by a midpoint construction, SAS, and the observation that one angle is a proper part of another. In the formal Hilbert reconstruction the same content is decomposed into explicit reusable mechanisms for rays, betweenness, congruence, crossbar behavior, and transport of strict angle comparison.

The final Lean theorem is short precisely because those mechanisms have been established independently. No additional axiom is introduced in `Proposition16.lean`, and the completed development compiles without `sorry`.

Chapter 17

Proposition I.17

17.1 Introduction

Euclid's Proposition I.17 states that in every triangle any two interior angles are together less than two right angles.

For a nondegenerate triangle ABC, the three conclusions are

$$\angle BAC + \angle ABC < 2R,$$

$$\angle BAC + \angle ACB < 2R,$$

and

$$\angle ABC + \angle ACB < 2R.$$

In Euclid's classical proof, Proposition I.16 supplies an exterior-angle inequality and Proposition I.13 identifies an adjacent pair of angles with two right angles. The informal argument then adds the same angle to both sides of an inequality.

The formal reconstruction takes a different route. No numerical angle measure and no binary operation of angle addition are introduced. Instead, the phrase "two angles are together less than two right angles" is represented synthetically: the first angle is required to be strictly smaller than the supplementary angle of the second. This follows the same design principle already used in the reconstruction of Proposition I.14, where "equal to two right angles" was encoded by comparison with a supplementary angle.

The final Lean development consists of a new interface predicate, three pairwise forms of Proposition I.17, and one wrapper theorem combining them.

17.2 The Classical Configuration

Let ABC be a nondegenerate triangle. Euclid proves a representative case by extending one side of the triangle. In Wilkins's presentation, BC is produced beyond C to a point D:

$$B - C - D.$$

Proposition I.16 gives

$$\angle ABC < \angle ACD.$$

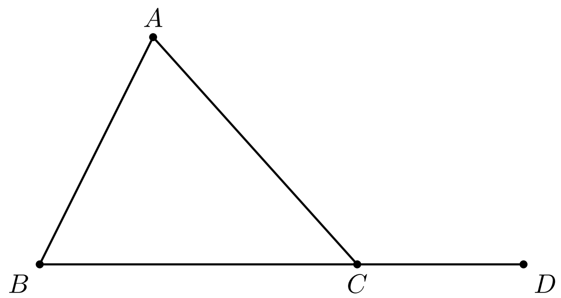


Figure 17.1: The classical extension used for the pair $\angle ABC, \angle ACB$: BC is produced beyond C.

Since CB and CD are opposite rays, the adjacent angles $\angle ACB$ and $\angle ACD$ form a supplementary pair. Proposition I.13 therefore gives the classical conclusion

$$\angle ABC + \angle ACB < 2R.$$

The formal reconstruction uses the same extension for two of the three pairwise conclusions. I.16 supplies both

$$\angle BAC < \angle ACD \qquad \text{and} \qquad \angle ABC < \angle ACD,$$

so the single witness D proves both

$$\angle BAC + \angle ACB < 2R$$

and

$$\angle ABC + \angle ACB < 2R.$$

For the remaining pair, the formal proof uses the symmetric extension at the other endpoint of the base. Produce CB beyond B to a point E:

$$C - B - E.$$

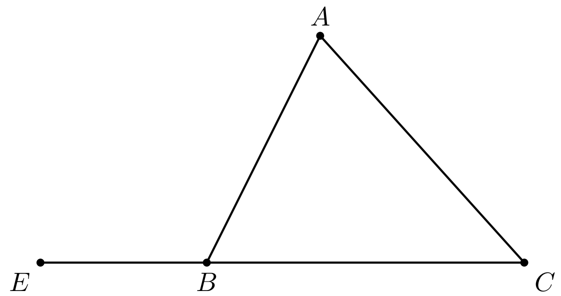


Figure 17.2: The second extension used by the Lean wrapper: CB is produced beyond B.

Applying I.16 to the reordered triangle ACB yields

$$\angle CAB < \angle ABE.$$

After reversing the first angle,

$$\angle BAC < \angle ABE,$$

and $\angle ABE$ is supplementary to $\angle ABC$. Hence

$$\angle BAC + \angle ABC < 2R.$$

The two figures therefore have genuinely different roles. They are not duplicate views of the same proof: the first extension supplies the two pairs containing $\angle ACB$, while the second supplies the remaining pair.

17.3 Classical Proof

Euclid proves one representative case and then observes that the same argument applies to the other pairs.

Extend BC beyond C to D. By Proposition I.16,

$$\angle ABC < \angle ACD.$$

Add $\angle ACB$ to both sides:

$$\angle ABC + \angle ACB < \angle ACD + \angle ACB.$$

By Proposition I.13, the adjacent angles $\angle ACD$ and $\angle ACB$ are together equal to two right angles. Hence

$$\angle ABC + \angle ACB < 2R.$$

The other two pairs follow in the same way after choosing the appropriate side extension.

The classical dependency pattern is therefore

I.16 exterior-angle inequality $\longrightarrow$ add the remaining interior angle $\longrightarrow$ I.13 supplementary pair $\longrightarrow < 2R$.

The formal reconstruction deliberately removes the middle arithmetic step.

17.4 Synthetic Representation of "Less Than Two Right Angles"

The central design choice is the predicate

`HilbertAnglesLessThanTwoRightAngles`.

For two angles α and β, instead of introducing a sum $\alpha + \beta$, choose a supplementary angle β' to β. Then

$$\alpha + \beta < 2R$$

is represented by

$$\alpha < \beta'.$$

In the formal language, if the second angle is $\angle CPD$, a witness E is chosen with

$$D - P - E.$$

Then PE is the opposite ray to PD, so $\angle CPE$ is supplementary to $\angle CPD$. The definition requires

$$\angle AOB < \angle CPE.$$

The Lean definition is:

```
def HilbertAnglesLessThanTwoRightAngles
    [HilbertIncidence Geo]
    (A O B C P D : Geo.Point) : Prop :=
  exists E : Geo.Point,
    Geo.Between D P E /\
    HilbertAngleLess Geo A O B C P E
```

This representation is entirely synthetic. It records only strict betweenness and strict comparison of angles. No real-valued measure, no degree notation, and no algebraic operation on angles is required.

17.5 Proof Architecture of the Reconstruction

The diagrammatic structure of the Lean proof is:

```
triangle ABC
    |
    +-- extend BC beyond C: B-C-D
    |       |
    |       +-- I.16 first
    |       |      BAC < ACD
    |       |      -> BAC + ACB < 2R
    |       |
    |       +-- I.16 second
    |              ABC < ACD
    |              -> ABC + ACB < 2R
    |
    +-- extend CB beyond B: C-B-E
            |
            +-- I.16 on triangle ACB
                   CAB < ABE
                   -> reverse CAB to BAC
                   -> BAC + ABC < 2R

three pairwise results
        |
        v
euclid_proposition_17
```

This is a case where two geometric diagrams are justified: the public theorem packages three conclusions, and the formal proof actually uses two different side extensions to obtain them.

The reconstruction is especially compact because Proposition I.16 already gives exactly the inequalities needed against exterior angles.

For the two pairs containing $\angle ACB$, extend BC beyond C:

$$B - C - D.$$

Then Proposition I.16 gives

$$\angle BAC < \angle ACD$$

and

$$\angle ABC < \angle ACD.$$

Since $\angle ACD$ is the supplement of $\angle ACB$, these are precisely the synthetic forms of

$$\angle BAC + \angle ACB < 2R$$

and

$$\angle ABC + \angle ACB < 2R.$$

For the remaining pair, extend CB beyond B:

$$C - B - E.$$

Apply Proposition I.16 to the reordered triangle ACB. This gives

$$\angle CAB < \angle ABE.$$

Reversal of the first angle identifies $\angle CAB$ with $\angle BAC$, while $\angle ABE$ is supplementary to $\angle ABC$. Hence

$$\angle BAC + \angle ABC < 2R.$$

Schematically:

$$\begin{array}{ccc} & \neg\,\text{Collinear}(A, B, C) & \\ & \downarrow & \\ & \text{extend } BC \text{ beyond } C & \\ & \downarrow & \\ & B - C - D & \\ & \downarrow & \\ & \text{I.16} & \\ \swarrow & & \searrow \\ \angle BAC < \angle ACD & & \angle ABC < \angle ACD \\ \downarrow & & \downarrow \\ \angle BAC + \angle ACB < 2R & & \angle ABC + \angle ACB < 2R \end{array}$$

and separately

$$\begin{array}{c} \text{extend } CB \text{ beyond } B \\ \downarrow \\ C - B - E \\ \downarrow \\ \text{I.16 on triangle } ACB \\ \downarrow \\ \angle CAB < \angle ABE \\ \downarrow \\ \angle BAC + \angle ABC < 2R. \end{array}$$

17.6 Lean Formalization

17.6.1 The Pair $\angle BAC$ and $\angle ACB$

The first theorem extends BC through C and uses the first half of Proposition I.16.

```
theorem euclid_proposition_17_BAC_ACB
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    HilbertAnglesLessThanTwoRightAngles
      Geo B A C A C B := by
```

```
  have hBCA :
      not PrimCollinear Geo B C A := by
    intro h

    have hACB :
        PrimCollinear Geo A C B :=
      PrimCollinearSymm Geo B C A h

    have hABC' :
        PrimCollinear Geo A B C :=
      PrimCollinearRotate Geo A C B hACB

    exact hABC hABC'

  have hBC : B != C :=
    hilbert_noncollinear_ne_first
      Geo B C A hBCA

  rcases
      HilbertOrder.between_extension
        B C hBC with
    <D, hBCD>

  refine <D, hBCD, ?_>

  exact
    euclid_proposition_16_first
      Geo A B C D hABC hBCD
```

The witness D simultaneously records the supplementary configuration and serves as the exterior-angle target in I.16.

17.6.2 The Pair $\angle ABC$ and $\angle ACB$

The second theorem uses the same extension but invokes the second half of I.16.

```
theorem euclid_proposition_17_ABC_ACB
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    HilbertAnglesLessThanTwoRightAngles
      Geo A B C A C B := by

  have hBCA :
      not PrimCollinear Geo B C A := by
    intro h

    have hACB :
        PrimCollinear Geo A C B :=
      PrimCollinearSymm Geo B C A h

    have hABC' :
        PrimCollinear Geo A B C :=
```

```
      PrimCollinearRotate Geo A C B hACB

    exact hABC hABC'

  have hBC : B != C :=
    hilbert_noncollinear_ne_first
      Geo B C A hBCA

  rcases
      HilbertOrder.between_extension
        B C hBC with
    <D, hBCD>

  refine <D, hBCD, ?_>

  exact
    euclid_proposition_16_second
      Geo A B C D hABC hBCD
```

Thus the two inequalities associated with the vertex C differ only in which conclusion of I.16 is selected.

17.6.3 The Pair $\angle BAC$ and $\angle ABC$

For the remaining pair, extend CB through B to E and apply I.16 to the triangle in the order ACB.

```
theorem euclid_proposition_17_BAC_ABC
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    HilbertAnglesLessThanTwoRightAngles
      Geo B A C A B C := by

  have hCBA :
      not PrimCollinear Geo C B A := by
    intro h

    have hABC' :
        PrimCollinear Geo A B C :=
      PrimCollinearSymm Geo C B A h

    exact hABC hABC'

  have hCB : C != B :=
    hilbert_noncollinear_ne_first
      Geo C B A hCBA

  rcases
      HilbertOrder.between_extension
        C B hCB with
    <E, hCBE>

  refine <E, hCBE, ?_>
```

```
  have hACB :
      not PrimCollinear Geo A C B := by
    intro h

    have hABC' :
        PrimCollinear Geo A B C :=
      PrimCollinearRotate Geo A C B h

    exact hABC hABC'

  have hLessCAB :
      HilbertAngleLess Geo C A B A B E :=
    euclid_proposition_16_first
      Geo A C B E hACB hCBE

  rcases hLessCAB with
    <hCABnc, hABEnc, X, hMeet, hAngle>

  have hBACnc :
      not PrimCollinear Geo B A C := by
    intro h
    exact hABC
      (PrimCollinearSwap Geo B A C h)

  have hAngle' :
      Geo.AngleCongruent B A C A B X :=
    (Geometry.Geo.angle_congruent_reverse_first
      Geo C A B A B X).mp hAngle

  exact
    <hBACnc,
      hABEnc,
      X,
      hMeet,
      hAngle'>
```

The only additional work here is normalization of the orientation of the first angle: I.16 returns $\angle CAB < \angle ABE$, whereas the final interface uses $\angle BAC$. The unordered-angle representation makes this a reversal operation rather than a new geometric argument.

17.6.4 The Final Wrapper

The final proposition simply combines the three pairwise results.

```
theorem euclid_proposition_17
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    HilbertAnglesLessThanTwoRightAngles Geo B A C A B C /\
    HilbertAnglesLessThanTwoRightAngles Geo B A C A C B /\
    HilbertAnglesLessThanTwoRightAngles Geo A B C A C B := by

  have h1 :=
```

```
    euclid_proposition_17_BAC_ABC
      Geo A B C hABC

  have h2 :=
    euclid_proposition_17_BAC_ACB
      Geo A B C hABC

  have h3 :=
    euclid_proposition_17_ABC_ACB
      Geo A B C hABC

  exact <h1, h2, h3>
```

17.7 Relationship with Earlier Propositions

Proposition I.17 sits at an interesting point in the reconstruction because its classical statement appears to require angle arithmetic, while the existing synthetic interface allows that arithmetic to be avoided.

17.7.1 Proposition I.13

I.13 identifies two adjacent angles on a straight line as a supplementary pair. In the Book Zero reconstruction this is represented structurally by `BookZeroSupplement`, rather than by an equation involving angle sums.

For I.17 we use the same geometric idea but do not need to invoke the theorem I.13 explicitly. The witness produced by `between_extension` already records the opposite ray required to construct the supplementary angle.

17.7.2 Proposition I.14

I.14 introduced the useful design pattern of representing "equal to two right angles" by congruence with a supplementary angle. The I.17 predicate is the strict-inequality analogue of that pattern:

$$\alpha + \beta = 2R \quad \rightsquigarrow \quad \alpha \cong \mathrm{supp}(\beta),$$

while

$$\alpha + \beta < 2R \quad \rightsquigarrow \quad \alpha < \mathrm{supp}(\beta).$$

Thus I.17 extends an already established interface convention rather than introducing a new angle algebra.

17.7.3 Proposition I.16

I.16 is the decisive geometric input. Once a side is extended, it provides strict comparison of each remote interior angle with the exterior angle. Since that exterior angle is automatically supplementary to the adjacent interior angle, the conclusions of I.16 are already in the exact form required by `HilbertAnglesLessThanTwoRightAngles`.

This gives the particularly short dependency path

$$\text{betweenness extension} \longrightarrow \text{I.16} \longrightarrow \text{synthetic } < 2R.$$

17.8 Hilbert and Book Zero Components Used

17.8.1 `HilbertAngleLess`

The strict comparison relation introduced for I.16 remains the fundamental notion of angle inequality. It is synthetic: one angle is less than another when a congruent copy of the first occurs as a proper interior subangle of the second.

17.8.2 `HilbertOrder.between_extension`

This construction supplies the points D and E on the required opposite rays:

$$B - C - D \qquad \text{or} \qquad C - B - E.$$

The witness is simultaneously the side extension used by I.16 and the supplementary-ray witness used by the definition of "less than two right angles."

17.8.3 Collinearity Symmetry and Rotation

The three applications of I.16 require the triangle vertices in different orders. The small uses of `PrimCollinearSymm`, `PrimCollinearRotate`, and `PrimCollinearSwap` normalize noncollinearity hypotheses without adding any geometry.

17.8.4 Angle Reversal

In the third pair, Proposition I.16 naturally produces the angle $\angle CAB$, while the public form of I.17 uses $\angle BAC$. The theorem `angle_congruent_reverse_first` transfers the congruence across reversal of the first angle.

17.9 Formalization Notes

Proposition I.17 is a useful example of how a familiar numerical-looking statement is represented without introducing angle measures.

The phrase "two angles of a triangle are together less than two right angles" is not formalized by adding numerical angle magnitudes. Instead, the reconstruction uses an extension of one side and compares the relevant interior angle with an exterior angle. Proposition I.16 then provides the strict comparison.

Consequently the three conclusions of I.17 are proved by three geometric instances of the same mechanism rather than by a single algebraic inequality involving an angle-sum operation.

Schematically,

extend a side $\longrightarrow$ straight-angle configuration $\longrightarrow$ I.16 $\longrightarrow$ strict comparison $\longrightarrow$ “two angles $<$ two right ang

The repetition in the Lean proof is therefore mathematically meaningful. Each pair of interior angles requires the appropriate extension, orientation of collinearity, and angle reversal. The formalization makes these orientation changes explicit rather than treating them as automatic consequences of a diagram.

This proposition also marks an important design choice in the project: there is still no need for a numerical angle measure or a general addition operation on angles. The classical statement can be recovered entirely from incidence, order, angle congruence, and strict angle comparison.

17.10 Dependency Summary

The formal dependency structure is

$$
\begin{array}{c}
\text{Hilbert incidence + order + congruence} \\
\downarrow \\
\texttt{HilbertAngleLess} \\
\downarrow \\
\text{Euclid I.16} \\
\downarrow \\
\texttt{HilbertAnglesLessThanTwoRightAngles} \\
\downarrow \\
\texttt{I.17\_BAC\_ABC} \quad \texttt{I.17\_BAC\_ACB} \quad \texttt{I.17\_ABC\_ACB} \\
\downarrow \\
\texttt{euclid\_proposition\_17}.
\end{array}
$$

No additional geometric axiom is introduced. More importantly, no general arithmetic of angles is introduced merely to reproduce Euclid’s phrase “together less than two right angles.” The proposition is expressed entirely in the existing synthetic language of rays, betweenness, supplementary configurations, and strict angle comparison.

17.11 Conclusion

Proposition I.17 is a useful example of the difference between reproducing a classical proof literally and reconstructing its mathematical content over a synthetic interface.

Euclid’s paper proof uses an implicit monotonicity principle for angle addition: from $\alpha < \beta$ one passes to $\alpha + \gamma < \beta + \gamma$. A literal formalization would therefore require an angle-sum operation and an order theory for such sums.

The Hilbert/Book Zero reconstruction avoids that additional layer. The statement “two angles are together less than two right angles” is encoded directly by comparison of one angle with the supplement of the other. Once this representation is chosen, Proposition I.16 supplies exactly the needed inequalities, and Proposition I.17 becomes a short structural consequence of side extension and exterior-angle comparison.

The completed Lean development compiles cleanly and contains no `sorry`.

Chapter 18

Proposition I.18

18.1 Introduction

Euclid's Proposition I.18 states that in a triangle the greater side is opposite the greater angle.

For a nondegenerate triangle ABC, if

$$AB < AC,$$

then

$$\angle ACB < \angle ABC.$$

The classical proof is short but structurally rich. A copy of the shorter side AB is laid off on the longer side AC. This produces a point D with

$$A - D - C \qquad \text{and} \qquad AD \cong AB.$$

The auxiliary triangle ABD is therefore isosceles. Proposition I.5 identifies its base angles, while Proposition I.16 compares one of these angles with an angle of the original triangle. Finally, the fact that an interior subangle is smaller than the whole angle completes the comparison.

The formal reconstruction follows this synthetic argument closely. It does not introduce numerical angle measures. Instead, it uses the strict comparison relation `HilbertAngleLess` already developed for Proposition I.16.

The reconstruction also isolates two general facts about strict angle comparison:

`hilbert_interior_angle_less`

and

`hilbert_angleLess_trans`.

These belong naturally to the Hilbert interface rather than to Proposition I.18 itself.

18.2 The Classical Configuration

Let ABC be a nondegenerate triangle and suppose

$$AB < AC.$$

Choose D on AC so that

$$A - D - C \qquad \text{and} \qquad AD \cong AB.$$

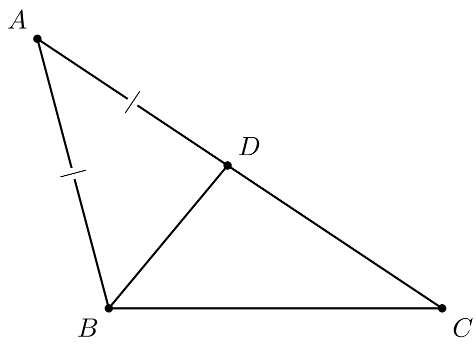


Figure 18.1: Classical configuration for Proposition I.18. The point D lies on AC, with $AD \cong AB$, and BD is joined.

The construction creates two auxiliary triangles.

First, triangle ABD is isosceles because

$$AB \cong AD.$$

Hence Proposition I.5 gives

$$\angle ABD \cong \angle ADB.$$

Second, consider triangle BCD. Since $A - D - C$, the ray DA is the continuation of DC through D. Thus $\angle BDA$ is an exterior angle of triangle BCD.

Proposition I.16 therefore gives

$$\angle BCD < \angle BDA.$$

Because A, D, C are collinear with D between A and C, the rays CD and CA coincide. Hence

$$\angle BCD = \angle BCA.$$

Combining these observations gives

$$\angle BCA < \angle BDA \cong \angle ABD.$$

Finally, the ray BD lies inside $\angle ABC$, since it meets the segment AC at the interior point D. Therefore

$$\angle ABD < \angle ABC.$$

By transitivity,

$$\angle BCA < \angle ABC,$$

which is the desired conclusion.

18.3 Classical Proof

Assume that in triangle ABC,

$$AC > AB.$$

Choose a point D on AC such that

$$AD = AB.$$

Since $AC > AB$, the point D lies strictly between A and C.

The triangle ABD is isosceles, so by Proposition I.5,

$$\angle ABD = \angle ADB.$$

Now $\angle ADB$ is an exterior angle of triangle BCD. Proposition I.16 gives

$$\angle ADB > \angle BCD.$$

Since D lies on AC,

$$\angle BCD = \angle BCA.$$

Consequently,

$$\angle ABD > \angle BCA.$$

But BD lies inside $\angle ABC$, so

$$\angle ABC > \angle ABD.$$

Therefore

$$\angle ABC > \angle BCA.$$

The classical dependency pattern is thus

$$\begin{gathered} AB < AC \\ \downarrow \\ A - D - C, \quad AB \cong AD \\ \downarrow \\ \text{I.5} \\ \downarrow \\ \angle ABD \cong \angle ADB \\ \downarrow \\ \text{I.16 on } BCD \\ \downarrow \\ \angle BCA < \angle ADB \\ \downarrow \\ \angle BCA < \angle ABD \\ \downarrow \\ \angle ABD < \angle ABC \\ \downarrow \\ \angle BCA < \angle ABC. \end{gathered}$$

18.4 Strict Angle Comparison

The formal statement uses the synthetic relation

`HilbertAngleLess`.

No numerical measure of angles is introduced. Informally,

$$\alpha < \beta$$

means that a congruent copy of α occurs as a proper interior subangle of β.

This relation was already required for Proposition I.16. Proposition I.18 reveals that two further structural properties are needed if strict angle comparison is to behave as a genuine order relation.

The first is that an interior subangle is smaller than the whole angle.

```
hilbert_interior_angle_less
```

Geometrically, if a ray from the vertex of $\angle AOB$ meets the opposite side AB internally, then the angle cut off by that ray is strictly smaller than $\angle AOB$.

The second property is transitivity.

```
hilbert_angleLess_trans
```

It expresses the expected implication

$$\alpha < \beta \quad \text{and} \quad \beta < \gamma \quad \Longrightarrow \quad \alpha < \gamma.$$

Both results are independent of Proposition I.18 and were therefore added to the general Hilbert interface.

18.5 Proof Architecture of the Reconstruction

The proof is carried by one auxiliary point and one chain of angle comparisons:

```
given: AB < AC
        |
choose D on AC with AD ~= AB
        |
triangle ABD is isosceles
        |
I.5: ABD ~= ADB
        |
I.16 in triangle DBC:
ACB < ADB
        |
D lies inside angle ABC:
```

```
ABD < ABC
       |
combine:
ACB < ADB ~= ABD < ABC
       |
therefore ACB < ABC
```

The single classical diagram is sufficient. The different inequalities are not different configurations; they are successive readings of the same configuration.

The formal proof separates naturally into five geometric stages.

18.5.1 Recovering the Interior Point

The hypothesis

$$AB < AC$$

is represented by `HilbertSegmentLess`.

Unpacking it produces a witness D satisfying

$$A - D - C$$

and

$$AB \cong AD.$$

Thus the first step of Euclid's construction is already contained in the definition of strict segment comparison.

18.5.2 The Isosceles Triangle

From

$$AB \cong AD$$

the existing theorem

`hilbert_isosceles_base_angles`

gives

$$\angle ABD \cong \angle ADB.$$

This is precisely Proposition I.5 in the form required by the reconstruction.

18.5.3 The Exterior-Angle Comparison

Since

$$A - D - C,$$

we also have

$$C - D - A.$$

Apply Proposition I.16 to triangle BCD with DA as the extension of DC. This yields

$$\angle BCD < \angle BDA.$$

The ray from C through D is the same ray as the ray from C through A. Hence the first angle can be normalized:

$$\angle BCD = \angle BCA.$$

Thus

$$\angle BCA < \angle BDA.$$

Using the isosceles-angle congruence and the symmetry of the unordered angle representation gives

$$\angle BCA < \angle ABD.$$

18.5.4 The Interior Subangle

The point D lies on the segment AC. Consequently the ray BD meets the interior of that segment.

Therefore BD lies inside $\angle ABC$, and `hilbert_interior_angle_less` gives

$$\angle ABD < \angle ABC.$$

18.5.5 Transitivity

We now have

$$\angle BCA < \angle ABD$$

and

$$\angle ABD < \angle ABC.$$

Applying `hilbert_angleLess_trans` gives

$$\angle BCA < \angle ABC.$$

The final public statement writes the first angle as $\angle ACB$. Since the formal angle representation is insensitive to reversal of its two rays, this is a final orientation normalization rather than an additional geometric argument.

18.6 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_18
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C : Geo.Point)
(hABC : not PrimCollinear Geo A B C)
(hABltAC : HilbertSegmentLess Geo A B A C) :
HilbertAngleLess Geo A C B A B C := by
```

The hypothesis `hABltAC` is immediately unpacked:

```
rcases hABltAC with
<D, hADC, hAB_AD>
```

Thus `hADC` records

$$A - D - C,$$

while `hAB_AD` records

$$AB \cong AD.$$

The isosceles part of the argument is then:

```
have hIso :
Geo.AngleCongruent A B D A D B :=
hilbert_isosceles_base_angles
Geo
A B D
hABD
hAB_AD
```

The exterior-angle step uses Proposition I.16:

```
have hBCD_BDA :
HilbertAngleLess Geo B C D B D A :=
euclid_proposition_16_second
Geo B C D A hBCD hCDA
```

Here

$$C - D - A$$

is obtained from the original betweenness relation

$$A - D - C.$$

The ray identity $CD = CA$ is expressed by `HilbertSameRay`:

```
have hRayCDA :
HilbertSameRay Geo C D A :=
hilbert_sameRay_of_between
Geo C D A hCDA
```

This permits normalization of

$$\angle BCD$$

to

$$\angle BCA.$$

After transporting the right-hand angle across the isosceles congruence, the proof obtains

```
have hBCA_ABD :
HilbertAngleLess Geo B C A A B D := ...
```

which represents

$$\angle BCA < \angle ABD.$$

For the second comparison, the ray BD meets the segment AC at D:

```
have hInsideBD :
HilbertRayMeetsSegment Geo B D A C :=
<D, hADC, hRayBDD>
```

Hence

```
have hABD_ABC :
HilbertAngleLess Geo A B D A B C :=
hilbert_interior_angle_less
Geo
B D A C
hABC
hInsideBD
```

which is exactly

$$\angle ABD < \angle ABC.$$

Transitivity now gives

```
have hFinal :
HilbertAngleLess Geo B C A A B C :=
hilbert_angleLess_trans
Geo
B C A
A B D
A B C
hBCA_ABD
hABD_ABC
```

The final lines merely reverse the first angle from

$$\angle BCA$$

to

$$\angle ACB,$$

matching the public orientation of the proposition.

18.7 Relationship with Earlier Propositions

18.7.1 Proposition I.5

Proposition I.5 supplies the equality of the base angles in the auxiliary isosceles triangle ABD:

$$AB \cong AD \implies \angle ABD \cong \angle ADB.$$

This is the bridge between the comparison of sides and the comparison of angles.

18.7.2 Proposition I.16

Proposition I.16 supplies the strict inequality. Applied to triangle BCD, it compares the remote interior angle at C with the exterior angle at D:

$$\angle BCD < \angle BDA.$$

Thus I.18 depends essentially on the exterior-angle theorem proved two propositions earlier.

18.7.3 Proposition I.17

Proposition I.17 is not required by the proof of I.18. The formal dependency follows Euclid's direct argument through I.5 and I.16.

This is useful structurally: I.18 does not rely on an angle-sum theory developed for I.17. Both propositions instead reuse the synthetic strict-angle relation introduced for I.16.

18.8 Hilbert and Book Zero Components Used

18.8.1 `HilbertSegmentLess`

The hypothesis that one side is shorter than another already contains the point required by Euclid's construction. Unpacking

$$AB < AC$$

produces

$$A - D - C \qquad \text{and} \qquad AB \cong AD.$$

Thus no separate segment-layoff construction is required inside Proposition I.18.

18.8.2 `hilbert_isosceles_base_angles`

This theorem supplies the I.5 step for the auxiliary triangle ABD.

18.8.3 `HilbertSameRay`

Same-ray machinery is used to recognize that, because $A-D-C$, the rays CD and CA determine the same side of the angle at C.

This turns the I.16 result about $\angle BCD$ into the required comparison involving the original angle $\angle BCA$.

18.8.4 `hilbert_interior_angle_less`

This new interface theorem expresses the elementary principle that a proper interior part of an angle is smaller than the whole angle.

In I.18 it gives

$$\angle ABD < \angle ABC$$

from the fact that BD meets AC internally.

18.8.5 `hilbert_angleLess_trans`

This new interface theorem provides transitivity of strict angle comparison.

It combines

$$\angle BCA < \angle ABD$$

and

$$\angle ABD < \angle ABC$$

into the final inequality.

18.9 Formalization Notes

Proposition I.18 is an important transition point in the reconstruction. The classical proof is short, but its formal version depends on several different notions of strict geometric order.

The hypothesis that one side is greater than another is represented by `HilbertSegmentLess`. It is not treated as a numerical inequality between lengths. Instead, the relation supplies the geometric data needed to recover an interior point on the longer side.

The proof then changes domains. Segment comparison produces a point configuration; the isosceles construction produces angle congruence; Proposition I.16 supplies a strict exterior-angle comparison; and `hilbert_interior_angle_less` supplies the second strict angle comparison required for transitivity.

Thus the decisive chain is

$$\text{strict segment comparison} \longrightarrow \text{betweenness} \longrightarrow \text{isosceles angle congruence} \longrightarrow \text{I.16} \longrightarrow \text{interior-angle compari}$$

This explains why I.18 is more than a local theorem about one triangle. It is the first proposition in this part of Book I where the developed segment-order and angle-order interfaces interact in a sustained way.

It also prepares the formal structure of Proposition I.19. Once I.18 provides the direct implication from side order to angle order, I.19 can be proved as a converse by segment trichotomy and elimination of the two impossible alternatives.

18.10 Dependency Summary

The formal dependency structure is

Hilbert incidence + order + congruence
$\downarrow$
`HilbertSegmentLess` `HilbertAngleLess`
$\downarrow$
$A - D - C, \quad AB \cong AD$
$\downarrow$
`hilbert_isosceles_base_angles` (I.5)
$\downarrow$
$\angle ABD \cong \angle ADB$
$\downarrow$
Euclid I.16
$\downarrow$
$\angle BCA < \angle ABD$
$\downarrow$
`hilbert_interior_angle_less`
$\downarrow$
$\angle ABD < \angle ABC$
$\downarrow$
`hilbert_angleLess_trans`
$\downarrow$
`euclid_proposition_18`.

No new geometric axiom is introduced.

The two results added to the Hilbert interface are structural properties of the already existing strict-angle relation rather than assumptions specific to Proposition I.18.

18.11 Conclusion

Proposition I.18 is a particularly clear example of how Euclid's construction survives formal reconstruction.

The classical proof introduces a point D on the longer side, creates an isosceles triangle, applies the exterior-angle theorem, and finally observes that a part of an angle is smaller than the whole. The Lean proof follows exactly this geometry.

What changes is not the mathematical argument but the amount of structure that must be made explicit. The informal steps

$$\angle BCA < \angle ABD < \angle ABC$$

require a formal notion of strict angle comparison, a theorem asserting that an interior subangle is smaller than its containing angle, and a proof of transitivity.

Consequently, Proposition I.18 does more than add another theorem of Book I. It strengthens the reusable language of synthetic angle order. The new results `hilbert_interior_angle_less` and `hilbert_angleLess_trans` are available for subsequent propositions independently of I.18.

The completed Lean development compiles cleanly and contains no `sorry`.

Chapter 19

Proposition I.19

19.1 Introduction

Euclid's Proposition I.19 is the converse of Proposition I.18. In a nondegenerate triangle, the greater angle is opposite the greater side.

For a triangle ABC, if

$$\angle ACB < \angle ABC,$$

then

$$AB < AC.$$

The classical proof is indirect. One compares the two opposite sides AB and AC. Exactly one of the following three possibilities must occur:

$$AB \cong AC, \qquad AB < AC, \qquad AC < AB.$$

The first case contradicts Proposition I.5, because equal sides would imply congruent base angles. The third case contradicts Proposition I.18, because the greater side AB would force the opposite angle at C to be greater than the angle at B. Hence only

$$AB < AC$$

remains.

The formal reconstruction makes this trichotomy explicit through the new general theorem

`hilbert_segment_trichotomy`.

It also isolates a general irreflexivity theorem for strict angle comparison:

`hilbert_angleLess_irrefl`.

These two results are reusable structural components of the Hilbert layer rather than local devices specific to Proposition I.19.

19.2 The Classical Configuration

Let ABC be a nondegenerate triangle and suppose

$$\angle ACB < \angle ABC.$$

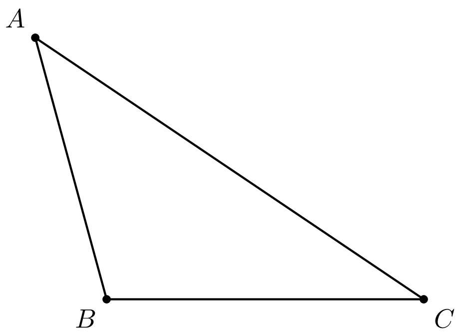


Figure 19.1: Classical configuration for Proposition I.19. The hypothesis is $\angle ACB < \angle ABC$; the conclusion is $AB < AC$.

The theorem asserts that the side opposite the larger angle must be larger:

$$AB < AC.$$

No new auxiliary point is required. The proof is entirely comparative: it uses the trichotomy of the two sides and eliminates the two alternatives incompatible with the assumed angle inequality.

For this reason the single triangle is the right diagram. Adding separate pictures for the three side-comparison branches would falsely suggest three different geometric configurations. The branches are logical alternatives for the same fixed triangle.

19.3 Classical Proof

Assume

$$\angle ACB < \angle ABC.$$

Compare the two sides AB and AC.

If

$$AB = AC,$$

then triangle ABC is isosceles. Proposition I.5 gives

$$\angle ABC = \angle ACB,$$

contradicting the hypothesis that

$$\angle ACB < \angle ABC.$$

If instead

$$AC < AB,$$

then Proposition I.18, applied to triangle ACB, yields

$$\angle ABC < \angle ACB,$$

which again contradicts the original hypothesis.

The only remaining possibility is

$$AB < AC.$$

Thus the greater angle is subtended by the greater side.

The classical dependency pattern is therefore

$$\begin{array}{ccc} & \angle ACB < \angle ABC & \\ & \downarrow & \\ & AB \cong AC \ \vee \ AB < AC \ \vee \ AC < AB & \\ & \downarrow & \\ AB \cong AC & AB < AC & AC < AB \\ \downarrow & & \downarrow \\ \text{I.5 contradiction} & \text{desired case} & \text{I.18 contradiction} \end{array}$$

19.4 Segment Trichotomy

The key structural theorem introduced for the formal proof is

`hilbert_segment_trichotomy`.

Its conclusion has the form

$$AB \cong CD \ \vee \ AB < CD \ \vee \ CD < AB.$$

The proof does not introduce a new axiom. It is derived from segment construction and the order theory of collinear points.

A copy of AB is laid off on the ray CD, producing a point X such that

$$CX \cong AB.$$

If $X = D$, then

$$AB \cong CD.$$

Otherwise, C, D, X are distinct collinear points. The Hilbert betweenness trichotomy gives three possibilities. The case

$$D - C - X$$

is incompatible with the fact that X lies on the ray CD. The remaining two orders give respectively

$$CD < CX$$

and

$$CX < CD.$$

Transporting these inequalities through

$$CX \cong AB$$

yields the required segment trichotomy.

Thus the theorem packages a general completeness property of strict segment comparison.

This helper does have its own useful diagrammatic content, but it belongs to the Hilbert interface rather than to Proposition I.19 itself. A copy of one segment is laid off on the ray of the other, and the order of the copied endpoint relative to the original endpoint determines the three alternatives. For the I.19 documentation, the ASCII dependency diagram above is enough; a second geometric figure would mostly duplicate the meaning of `hilbert_segment_trichotomy`.

19.5 Irreflexivity of Strict Angle Comparison

The second structural theorem used in I.19 is

`hilbert_angleLess_irrefl`.

It states

$$\neg(\alpha < \alpha).$$

In the synthetic definition of `HilbertAngleLess`, the relation

$$\alpha < \beta$$

means that a congruent copy of α occurs as a proper interior subangle of β.

Therefore

$$\alpha < \alpha$$

would imply that an angle is congruent to one of its own proper interior subangles. This is excluded by the previously proved theorem

`hilbert_interior_subangle_not_congruent_whole`.

The irreflexivity result provides exactly the contradiction mechanism needed in both non-target branches of Proposition I.19.

19.6 Proof Architecture of the Reconstruction

The Lean proof divides naturally into three stages.

```
given: ACB < ABC
        |
        v
segment trichotomy for AB and AC
        |
   +----+----+
   |    |    |
   v    v    v
AB~=AC  AB<AC  AC<AB
   |    |      |
   |    |      +--> I.18
```

```
|    |           -> ABC < ACB
|    |           -> contradiction
|    |
|    +----------> goal
|
+--> I.5
     -> ABC ~= ACB
     -> contradiction

therefore AB < AC
```

The geometry itself does not branch. What branches is the order relation between the two opposite sides.

19.6.1 Nondegeneracy of the Compared Side

From the noncollinearity hypothesis

$$\neg\,\mathrm{Collinear}(A, B, C)$$

the proof first derives

$$A \neq C.$$

This is the nondegeneracy condition required to apply segment construction in `hilbert_segment_trichotomy`.

19.6.2 Apply Segment Trichotomy

The theorem

```
hilbert_segment_trichotomy
```

is applied to the segments AB and AC.

This produces three branches:

$$AB \cong AC, \qquad AB < AC, \qquad AC < AB.$$

The middle branch is exactly the desired conclusion.

19.6.3 Eliminate the Two Impossible Branches

In the congruent-side branch, Proposition I.5 gives

$$\angle ABC \cong \angle ACB.$$

The assumed strict inequality

$$\angle ACB < \angle ABC$$

is transported across that congruence, producing

$$\angle ACB < \angle ACB.$$

This contradicts `hilbert_angleLess_irrefl`.

In the branch

$$AC < AB,$$

Proposition I.18 applied to triangle ACB gives

$$\angle ABC < \angle ACB.$$

Combining this with the original inequality by transitivity yields

$$\angle ACB < \angle ACB,$$

again contradicting irreflexivity.

Only

$$AB < AC$$

remains.

19.7 Lean Formalization

The public theorem is:

```
theorem euclid_proposition_19
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C)
    (hAngle :
      HilbertAngleLess Geo A C B A B C) :
    HilbertSegmentLess Geo A B A C := by
```

The proof first derives the nondegeneracy of AC:

```
  have hAC : A != C :=
    hilbert_noncollinear_ne_first
      Geo A C B
      (by
        intro h
        exact hABC
          (PrimCollinearRotate Geo A C B h))
```

Then segment trichotomy is applied:

```
  rcases
      hilbert_segment_trichotomy
        Geo
        A B
        A C
        hAC with
    hEq | hABltAC | hACltAB
```

The middle branch closes immediately:

```
$\cdot$ exact hABltAC
```

For the congruent-side branch, Proposition I.5 gives:

```
have hIso :
    Geo.AngleCongruent A B C A C B :=
  hilbert_isosceles_base_angles
    Geo
    A B C
    hABC
    hEq
```

The strict inequality is transported to the congruent target angle:

```
have hSame :
    HilbertAngleLess Geo A C B A C B :=
  hilbert_angleLess_transport_right
    Geo
    A C B
    A B C
    A C B
    hAngle
    hACB
    hIso
```

This contradicts irreflexivity:

```
exact
  False.elim
    ((hilbert_angleLess_irrefl
        Geo A C B)
      hSame)
```

In the final branch, Proposition I.18 gives the opposite strict inequality:

```
have hOpp :
    HilbertAngleLess Geo A B C A C B :=
  euclid_proposition_18
    Geo
    A C B
    hACB
    hACltAB
```

Transitivity then creates an impossible strict self-comparison:

```
have hCycle :
    HilbertAngleLess Geo A C B A C B :=
  hilbert_angleLess_trans
```

```
      Geo
      A C B
      A B C
      A C B
      hAngle
      hOpp
```

and finally:

```
  exact
    False.elim
      ((hilbert_angleLess_irrefl
          Geo A C B)
        hCycle)
```

19.8 Relationship with Earlier Propositions

19.8.1 Proposition I.5

I.5 is used to eliminate the equality case. If

$$AB \cong AC,$$

then the base angles are congruent:

$$\angle ABC \cong \angle ACB.$$

This is incompatible with the assumed strict angle inequality.

19.8.2 Proposition I.18

I.18 is used to eliminate the reversed side inequality. If

$$AC < AB,$$

then applying I.18 to triangle ACB gives

$$\angle ABC < \angle ACB.$$

Together with the hypothesis

$$\angle ACB < \angle ABC,$$

this produces a strict cycle and therefore a contradiction.

19.8.3 Logical Relation between I.18 and I.19

The two propositions form a converse pair:

$$AB < AC \quad \Longrightarrow \quad \angle ACB < \angle ABC$$

for I.18, and

$$\angle ACB < \angle ABC \implies AB < AC$$

for I.19.

Together they establish the equivalence between ordering of opposite sides and ordering of opposite angles in a nondegenerate triangle.

19.9 Hilbert and Book Zero Components Used

19.9.1 `hilbert_segment_trichotomy`

This theorem provides the complete three-way comparison of two segments and is the main new structural component required by I.19.

19.9.2 `hilbert_angleLess_irrefl`

This theorem excludes strict self-comparison of an angle. It is used in both contradictory branches of the proof.

19.9.3 `hilbert_isosceles_base_angles`

This is the formal version of Proposition I.5 needed to eliminate the equality case.

19.9.4 `hilbert_angleLess_transport_right`

This theorem transports a strict angle comparison across congruence of the containing angle. It converts the equality branch into an impossible self-comparison.

19.9.5 `hilbert_angleLess_trans`

Transitivity combines the original strict inequality with the opposite inequality supplied by I.18.

19.10 Formalization Notes

Proposition I.19 is formally much shorter than Proposition I.18, but its proof is logically revealing.

The argument does not construct a new geometric configuration. Instead, it performs a case analysis on the two sides whose order is to be determined. Segment trichotomy gives three possibilities. Two of them are incompatible with the assumed strict angle inequality.

The equality branch is eliminated by the isosceles-triangle theorem: equal sides force the corresponding angles to be congruent, contradicting strict angle comparison.

The reverse-inequality branch is eliminated by Proposition I.18. If the supposedly smaller side were actually larger, I.18 would force the opposite angle inequality. Together with the hypothesis this would produce a strict angle inequality from an angle to itself, contradicting irreflexivity.

Thus the proof is a characteristic formal converse argument:

$$\text{trichotomy} + \text{direct theorem I.18} + \text{irreflexivity} \implies \text{I.19}.$$

This organization is useful beyond Proposition I.19. It shows that converse comparison theorems need not be proved by reproducing the geometric construction of the direct theorem. Once strict order has trichotomy and irreflexivity, the converse can often be obtained by eliminating the impossible alternatives.

19.11 Dependency Summary

The formal dependency structure is

$$\begin{array}{ccc}
 & \text{Hilbert incidence} + \text{order} + \text{congruence} & \\
 & \downarrow & \\
\texttt{hilbert_segment_trichotomy} & & \texttt{hilbert_angleLess_irrefl} \\
 & \downarrow & \\
 & AB \cong AC \ \lor \ AB < AC \ \lor \ AC < AB & \\
 & \downarrow & \\
\text{I.5} & & \text{I.18} \\
\downarrow & & \downarrow \\
\text{contradiction} & AB < AC & \text{contradiction} \\
 & \downarrow & \\
 & \texttt{euclid_proposition_19}. &
\end{array}$$

No new geometric axiom is introduced.

19.12 Conclusion

Proposition I.19 is formally shorter than I.18, but it exposes an important structural issue: the proof requires a complete comparison principle for segments.

Once the theorem `hilbert_segment_trichotomy` is available, the Euclidean argument becomes exactly the expected three-case proof. Equal sides are ruled out by I.5, the reversed inequality is ruled out by I.18, and the remaining case is the desired conclusion.

The proof also clarifies the role of strict angle order. Instead of introducing an explicit asymmetry theorem, the development uses the more fundamental statement `hilbert_angleLess_irrefl` together with transitivity. This keeps the order theory modular and avoids unnecessary duplication.

Thus I.19 strengthens the reusable Hilbert layer in two directions: completeness of segment comparison and irreflexivity of strict angle comparison. The Euclidean theorem itself then becomes a concise application of those general principles.

The completed Lean development compiles cleanly and contains no `sorry`.

Chapter 20

Proposition I.20

20.1 Introduction

Euclid's Proposition I.20 is the triangle inequality in purely synthetic form. For a nondegenerate triangle ABC, Euclid proves that any two sides taken together are greater than the third. For one of the three cyclic cases the statement is

$$BA + AC > BC.$$

The Hilbert reconstruction does not introduce a numerical length function and does not define an arithmetic sum of real numbers. Instead, the sum $BA + AC$ is represented geometrically. Extend BA beyond A to a point D such that

$$B - A - D \qquad \text{and} \qquad AD \cong AC.$$

Then the whole segment BD represents the concatenation of BA with a copy of AC. The required inequality becomes

$$BC < BD.$$

Thus the formal theorem asserts the existence of such a point D together with the strict segment comparison $BC < BD$.

The completed proof remains entirely inside neutral Hilbert geometry. It uses incidence, order, congruence, Proposition I.5, the synthetic angle-order relation, and Proposition I.19. No parallel postulate, numerical angle measure, or numerical segment length is used.

20.2 The Classical Configuration

Let ABC be a nondegenerate triangle. Extend BA beyond A to a point D and choose D so that

$$AD \cong AC.$$

The construction records

$$B - A - D, \qquad AD \cong AC.$$

The target is

$$BC < BD.$$

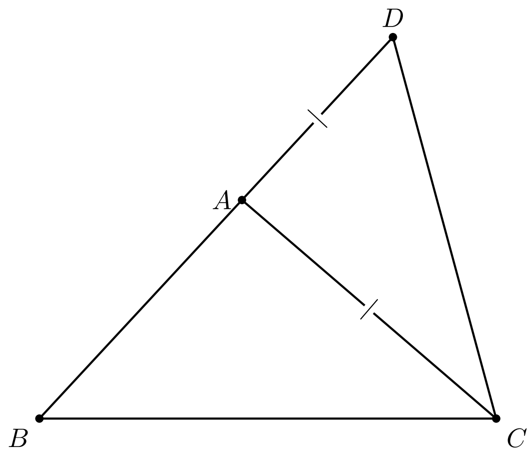


Figure 20.1: Classical configuration for Proposition I.20. The side BA is produced beyond A to D, with $AD \cong AC$.

Geometrically, this is exactly the inequality

$$BC < BA + AC.$$

20.3 Classical Proof

After extending BA to D so that $AD = AC$, join CD.

The auxiliary triangle ADC is isosceles. Proposition I.5 therefore gives

$$\angle ADC = \angle ACD.$$

Since A lies between B and D, the ray CA lies inside the angle BCD. Hence

$$\angle ACD < \angle BCD.$$

Combining this with the isosceles-triangle equality gives

$$\angle BDC < \angle BCD.$$

Now apply Proposition I.19 to triangle BCD. The greater angle is opposite the greater side, so

$$BC < BD.$$

Finally, because $B - A - D$ and $AD = AC$, the segment BD is the geometric concatenation of BA and AC. Therefore

$$BA + AC > BC.$$

The classical dependency pattern is

$$\begin{gathered} B - A - D,\ AD \cong AC \\ \downarrow \\ \angle ADC \cong \angle ACD \quad \text{(I.5)} \\ \downarrow \\ \angle BDC < \angle BCD \\ \downarrow \quad \text{(I.19)} \\ BC < BD. \end{gathered}$$

The single diagram above already contains every geometric ingredient needed for this proof. There is no case split to visualize. In particular, the important graphical information is not merely the shape of triangle BCD, but the collinear order $B - A - D$ together with the marked equality $AD \cong AC$. These two facts explain why BD represents the synthetic sum $BA + AC$.

Wilkins also records an ancient alternative proof, attributed by Proclus to Heron or Porphyry, which bisects $\angle BAC$ and uses I.16 and I.19. That proof has a genuinely different diagram, but it is not the proof formalized here, so it is not added to the main reconstruction.

20.4 Synthetic Form of the Triangle Inequality

The formal reconstruction deliberately avoids introducing an arithmetic addition operation on segments. The public result is expressed by an extension point D:

$$\exists D\ (B - A - D\ \wedge\ AD \cong AC\ \wedge\ BC < BD).$$

This is the natural Hilbert form of Euclid's phrase "two sides taken together are greater than the remaining one."

The representation has two advantages. First, it remains purely geometric. Second, it makes explicit which part of the statement belongs to order and congruence geometry: the "sum" is a concatenated segment, not an externally imposed numerical quantity.

20.5 Proof Architecture of the Reconstruction

The formal proof follows Euclid's geometric strategy closely, but it separates the argument into reusable logical stages. An extension point turns a sum of two sides into a single segment; an interior-angle comparison and an isosceles-triangle transport produce the angle inequality required by Proposition I.19; finally the resulting strict segment comparison is transported back to the original sides.

```
triangle ABC
     |
extend BA beyond A
     |
construct D with B-A-D and AD ~= AC
     |
     +-------------------------------+
     |                               |
     v                               v
BD represents BA+AC         triangle ACD is isosceles
                                     |
                                    I.5
                                     |
                           ADC ~= ACD
                                     |
                    CA lies inside angle DCB
                                     |
                           ACD < DCB
```

```
                             |
                  congruence transport
                             |
                      BDC < BCD
                             |
                           I.19
                             |
                         BC < BD
    |                        |
    +--------------+---------------+
                   |
                   v
              BC < BA+AC
```

Unlike I.21–I.24, no second geometric picture is needed here: the formal architecture is a refinement of the same Euclidean configuration rather than a different case structure or a stronger primitive construction.

20.5.1 Construction of the Extension Point

The point D is constructed in two stages.

First, Hilbert's order axiom extends the line beyond A:

$$B - A - R.$$

In Lean this is obtained from

`HilbertOrder.between_extension`.

Second, a copy of AC is laid off from A on the ray AR using

`HilbertCongruence.segment_construction`.

This produces a point D satisfying

$$AD \cong AC$$

and placing D on the same ray from A as R.

The theorem

`hilbert_between_transport_sameRays`

then transports the original betweenness relation $B - A - R$ to

$$B - A - D.$$

Thus the geometric "sum" segment BD is obtained entirely from existing order and congruence infrastructure.

20.5.2 The Interior-Angle Step

The key order-theoretic observation is that the ray CA meets the open segment DB at the point A itself. Since

$$D - A - B,$$

the witness for

`HilbertRayMeetsSegment Geo C A D B`

is simply A.

The strict angle comparison is then introduced directly with

`hilbert_angleLess_intro`.

The first angle is $\angle ACD$. The corresponding proper subangle of $\angle DCB$ is represented by $\angle DCA$, which is the same unoriented angle as $\angle ACD$. The representation-level symmetry of angles is supplied by

`angle_congruent_reverse_second`

together with reflexivity of angle congruence.

This yields

$$\angle ACD < \angle DCB.$$

No auxiliary half-plane or perpendicular construction is required. The entire strict comparison follows from the ray meeting the opposite segment.

20.5.3 Transport Through the Isosceles Triangle

From

$$AD \cong AC$$

and the noncollinearity of A, D, C, Proposition I.5 is available in the form

`hilbert_isosceles_base_angles`.

It gives

$$\angle ADC \cong \angle ACD.$$

For Proposition I.20 we also introduced the reusable theorem

`hilbert_angleLess_transport_left`.

It expresses invariance of strict angle comparison under replacing the smaller angle by a congruent angle.

Applied here, it transports

$$\angle ACD < \angle DCB$$

across

$$\angle ADC \cong \angle ACD,$$

producing

$$\angle ADC < \angle DCB.$$

Since $B - A - D$, the rays DA and DB coincide. The theorem

`hilbert_angle_eq_of_sameRay_first`

therefore identifies the angle $\angle ADC$ with $\angle BDC$. Hence

$$\angle BDC < \angle DCB.$$

Finally, reversal of the two boundary rays at C identifies

$$\angle DCB \cong \angle BCD.$$

The existing theorem

`hilbert_angleLess_transport_right`

then gives

$$\angle BDC < \angle BCD.$$

20.5.4 Application of Proposition I.19

At this point the proof has produced exactly the hypothesis needed for Proposition I.19 in triangle BCD:

$$\angle BDC < \angle BCD.$$

Applying

`euclid_proposition_19`

yields

$$BC < BD.$$

This is the synthetic triangle inequality for the chosen pair of sides. The other two cyclic cases follow by renaming the vertices.

20.6 Lean Formalization

The public theorem is:

```
theorem euclid_proposition_20
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
```

```
    exists D : Geo.Point,
      Geo.Between B A D /\
      Geo.Congruent A D A C /\
      HilbertSegmentLess Geo B C B D := by
```

The first stage derives $A \neq B$ and extends BA beyond A:

```
  have hAB : A != B :=
    hilbert_noncollinear_ne_first
      Geo A B C hABC

  have hBA : B != A :=
    hAB.symm

  rcases
      HilbertOrder.between_extension
        B A hBA with
    <R, hBAR>
```

A copy of AC is then laid off on the ray AR:

```
  have hAR : A != R :=
    (HilbertOrder.between_incidence
      B A R hBAR).2.1

  rcases
      HilbertCongruence.segment_construction
        (Geo := Geo)
        A C
        A R
        hAR with
    <D, hRayARD, hAD_AC>
```

Betweenness is transported to the actual constructed point D:

```
  have hRayABA :
      HilbertSameRay Geo A B B :=
    hilbert_sameRay_refl
      Geo A B hBA

  have hBAD :
      Geo.Between B A D :=
    hilbert_between_transport_sameRays
      Geo
      B A R
      B D
      hBAR
      hRayABA
      hRayARD
```

The theorem then packages the construction and continues with the strict segment comparison:

```
  refine <D, hBAD, hAD_AC, ?_>
```

The interior ray is represented directly by the point A:

```
  have hDAB :
      Geo.Between D A B :=
    (HilbertOrder.between_incidence
      B A D hBAD).2.2.2.2

  have hRayCAA :
      HilbertSameRay Geo C A A :=
    hilbert_sameRay_refl
      Geo C A hAC

  have hInside :
      HilbertRayMeetsSegment Geo C A D B :=
    <A, hDAB, hRayCAA>
```

Angle reversal is obtained through the representation-level congruence laws:

```
  have hAngleRefl :
      Geo.AngleCongruent A C D A C D :=
    Geometry.Geo.angle_congruent_reflexive
      Geo A C D

  have hAngleSwap :
      Geo.AngleCongruent A C D D C A :=
    (Geo.angle_congruent_reverse_second
      A C D
      A C D).mp
      hAngleRefl
```

This gives the strict subangle relation:

```
  have hLessACD_DCB :
      HilbertAngleLess Geo A C D D C B :=
    hilbert_angleLess_intro
      Geo
      A C D
      D C B
      A
      hACD
      hDCB
      hInside
      hAngleSwap
```

Proposition I.5 and left transport give the corresponding comparison with the base angle at D:

```
  have hIso :
      Geo.AngleCongruent A D C A C D :=
    hilbert_isosceles_base_angles
      Geo
```

```
      A D C
      hADC
      hAD_AC

  have hLessADC_DCB :
      HilbertAngleLess Geo A D C D C B :=
    hilbert_angleLess_transport_left
      Geo
      A C D
      A D C
      D C B
      hLessACD_DCB
      hADC
      hIso
```

The ray DA is then replaced by the same ray DB:

```
  have hRayDAB :
      HilbertSameRay Geo D A B :=
    hilbert_sameRay_of_between
      Geo D A B hDAB

  have hRayDBA :
      HilbertSameRay Geo D B A :=
    hilbert_sameRay_symm
      Geo D A B hRayDAB

  have hBDC_ACD :
      Geo.AngleCongruent B D C A C D := by
    unfold Geometry.Geo.AngleCongruent at hIso |-
    rw [hilbert_angle_eq_of_sameRay_first
      Geo D B A C hRayDBA]
    exact hIso
```

Using left transport again gives

```
  have hLessBDC_DCB :
      HilbertAngleLess Geo B D C D C B :=
    hilbert_angleLess_transport_left
      Geo
      A C D
      B D C
      D C B
      hLessACD_DCB
      hBDC
      hBDC_ACD
```

Finally, the right-hand angle is reversed and Proposition I.19 closes the proof:

```
  have hAngleReflDCB :
      Geo.AngleCongruent D C B D C B :=
    Geometry.Geo.angle_congruent_reflexive
      Geo D C B
```

```
  have hDCB_BCD :
      Geo.AngleCongruent D C B B C D :=
    (Geo.angle_congruent_reverse_second
      D C B
      D C B).mp
      hAngleReflDCB

  have hLessBDC_BCD :
      HilbertAngleLess Geo B D C B C D :=
    hilbert_angleLess_transport_right
      Geo
      B D C
      D C B
      B C D
      hLessBDC_DCB
      hBCD
      hDCB_BCD

  exact
    euclid_proposition_19
      Geo
      B C D
      hBCD
      hLessBDC_BCD
```

20.7 Relationship with Earlier Propositions

20.7.1 Proposition I.5

The construction creates an isosceles triangle. Its base-angle congruence is the mechanism that transports the relevant angle from the auxiliary configuration back toward the original triangle.

20.7.2 Proposition I.16

The exterior-angle theorem supplies the strict angle comparison needed inside the auxiliary triangle. This is the order-theoretic part of the argument: the diagram alone is not used to infer that one angle is larger than another.

20.7.3 Proposition I.19

Proposition I.19 converts the strict angle comparison into a strict comparison of the opposite sides. In this sense I.20 is not an independent segment-order argument: its decisive inequality is obtained through the angle-order theory developed immediately before it.

20.7.4 Transition to Proposition I.21

Proposition I.20 proves the triangle inequality by representing the sum of two sides as a single constructed segment. Proposition I.21 develops this same synthetic segment-arithmetic viewpoint further. Thus I.20 is the first clear bridge from the angle-comparison block I.16–I.19 to the more explicit segment arithmetic used in I.21 and I.22.

20.8 Hilbert and Book Zero Components Used

Proposition I.20 required one new general theorem in the Hilbert interface:

`hilbert_angleLess_transport_left`.

Its role is symmetric to the previously available

`hilbert_angleLess_transport_right`.

Together they express that strict synthetic angle comparison is invariant under congruent replacement of either compared angle. This is not specific to the triangle inequality and is expected to be reusable in later propositions.

The proof also reuses the existing construction pattern

`between_extension` $\longrightarrow$ `segment_construction` $\longrightarrow$ `between_transport_sameRays`,

which already occurs elsewhere in the Hilbert development.

20.9 Dependency Summary

The formal dependency structure of Proposition I.20 can be summarized as

$$\begin{array}{c}
\text{Hilbert incidence + order + congruence} \\
\downarrow \\
\text{between extension + segment construction} \\
\downarrow \\
B - A - D, \quad AD \cong AC \\
\downarrow \\
\text{I.5 + HilbertAngleLess} \\
\downarrow \\
\angle BDC < \angle BCD \\
\downarrow \quad \text{I.19} \\
BC < BD \\
\downarrow \\
BA + AC > BC \quad \text{(synthetic interpretation).}
\end{array}$$

No Euclidean parallel axiom appears in this chain. Proposition I.20 is therefore a theorem of neutral geometry in the present Hilbert reconstruction.

20.10 Conclusion

Proposition I.20 is the first place in this sequence where Euclid's language of adding lengths is naturally reconstructed without introducing numerical length. The formal statement replaces the expression $BA + AC$ by an explicit extension segment BD satisfying

$$B - A - D, \qquad AD \cong AC.$$

The rest of the argument is purely synthetic. Proposition I.5 converts the copied side into an angle congruence, the interior-ray definition produces a strict angle inequality, congruence transport normalizes the two angles, and Proposition I.19 converts the angle inequality back into the desired side inequality.

The resulting Lean proof therefore reproduces the classical geometry while exposing its exact logical structure. No new axiom is introduced, and the only new general infrastructure is the left-transport theorem for strict angle comparison.

Chapter 21

Proposition I.21

21.1 Introduction

Euclid's Proposition I.21 concerns two straight lines drawn from the ends of one side of a triangle to a point inside the triangle.

Let ABC be a nondegenerate triangle. Let D be an interior point, and let the line BD meet the side AC at E, so that

$$A - E - C \qquad \text{and} \qquad B - D - E.$$

Euclid asserts simultaneously that

$$BD + DC < BA + AC$$

and

$$\angle BAC < \angle BDC.$$

The proposition is therefore the first result in this part of Book I whose reconstruction naturally separates into two substantial comparison arguments: one for segments and one for angles.

The formal reconstruction preserves this separation. The segment inequality is obtained by combining Proposition I.20 with a synthetic "addition of unequal segments" argument. The angle inequality is proved independently, using Proposition I.16 and the strict angle comparison machinery developed in the preceding propositions.

A particularly important feature of the formalization is that no numerical length function is introduced. Expressions such as $BA+AC$ and $BD+DC$ are represented by ordinary constructed segments.

21.2 The Classical Configuration

Let ABC be a nondegenerate triangle. Choose a point D inside the triangle and let the line BD meet AC at E.

Thus

$$A - E - C, \qquad B - D - E.$$

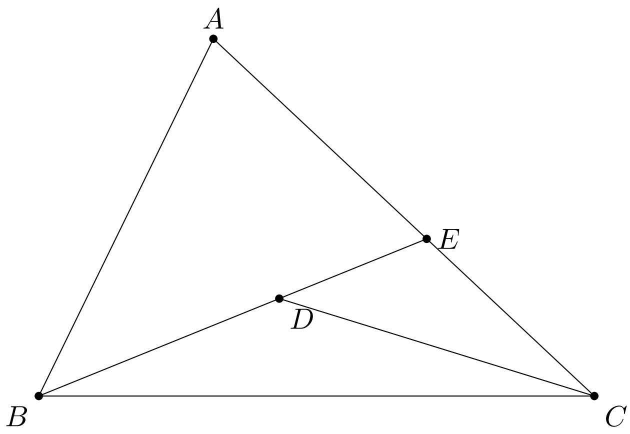


Figure 21.1: Classical configuration for Proposition I.21. The point D lies inside triangle ABC; the line BD meets AC at E.

For the segment inequality, Euclid first compares the broken line $BA + AE$ with BE by applying Proposition I.20 to triangle BAE. The remaining pieces EC and the corresponding copied segment are then added to obtain the comparison with $BA + AC$.

For the angle inequality, the point E allows the exterior-angle theorem to be applied to appropriate auxiliary triangles. The resulting strict angle comparisons are transported along the collinear rays determined by A, E, C and B, D, E.

The formal proof makes these two uses of the same configuration explicit.

Wilkins's presentation is especially useful here because the same diagram supports two logically independent chains. For segments, Proposition I.20 gives

$$DC < DE + EC, \qquad DE < DA + AE,$$

and therefore

$$BD + DC < BD + DE + EC = BE + EC < BA + AE + EC = BA + AC.$$

For angles, the same point E exposes a clean two-step exterior-angle chain:

$$\angle BDC > \angle BEC > \angle BAC.$$

The single classical diagram above therefore serves both parts of the proof. There is no mathematical gain in repeating the same triangle a second time. I.21 is instead a useful example of one configuration supporting two distinct arguments: a segment-sum comparison and an angle-order comparison.

21.3 Classical Proof

Since D lies inside triangle ABC, the line BD meets AC at an interior point E.

Apply Proposition I.20 to triangle ABE. It gives

$$BE < BA + AE.$$

Add EC to the two sides in the geometric sense. Since

$$AE + EC = AC,$$

we obtain

$$BE + EC < BA + AC.$$

Now apply Proposition I.20 to triangle CDE. Since $B - D - E$,

$$BD + DE = BE.$$

Proposition I.20 gives

$$DC + DE > CE,$$

or equivalently

$$CE < DC + DE.$$

Combining the appropriate segment comparisons in the classical configuration yields

$$BD + DC < BA + AC.$$

For the angle part, the exterior-angle theorem I.16 is applied to the triangles determined by the interior point and the two transversals. It yields strict comparisons which, after identifying the collinear rays, give

$$\angle BAC < \angle BDC.$$

Thus both assertions of I.21 follow:

$$BD + DC < BA + AC \qquad \text{and} \qquad \angle BAC < \angle BDC.$$

The classical proof is extremely compact because segment addition, subtraction, and replacement of collinear rays are read directly from the diagram. These are precisely the operations that must be made explicit in the formal reconstruction.

21.4 From Euclid's Diagram to the Formal Argument

The classical proof compresses several different kinds of reasoning into a single diagram. The formal reconstruction separates them.

First, statements such as "the whole is greater than the part" are represented by the strict synthetic relation `HilbertSegmentLess`. Second, sums and differences of segment lengths are not numerical expressions: they are realized by actual collinear configurations and transported by congruence. Third, the angle comparison needed in the proof is obtained from the already formalized exterior-angle machinery rather than inferred from the appearance of the figure.

Thus the reconstruction preserves Euclid's geometric strategy while making explicit the segment arithmetic and order-theoretic steps that the classical presentation leaves implicit.

21.5 Synthetic Segment Arithmetic

The segment part of I.21 exposes an important feature of the Hilbert reconstruction.

The formal development does not introduce a numerical length function, nor does it define an algebraic addition operation on segments. Instead, a sum is represented by a constructed segment.

For example, a point X satisfying

$$B - A - X \qquad \text{and} \qquad AX \cong AC$$

makes BX a representative of

$$BA + AC.$$

Similarly, a point Y satisfying

$$B - D - Y \qquad \text{and} \qquad DY \cong DC$$

makes BY a representative of

$$BD + DC.$$

Consequently the desired inequality is represented formally by

$$BY < BX.$$

The formal representatives can be pictured independently of the classical triangle:

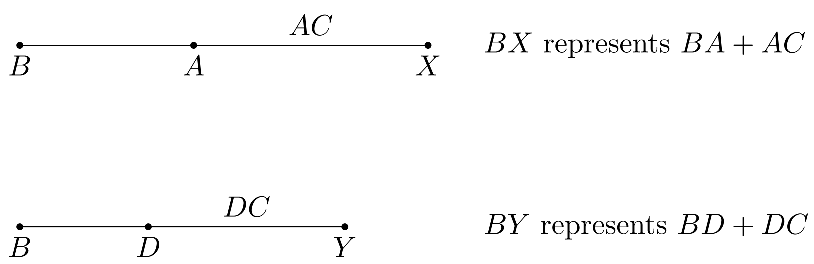


Figure 21.2: Synthetic representatives of the two segment sums in the Lean statement. The target inequality is the ordinary strict segment comparison $BY < BX$.

In Lean this is

```
HilbertSegmentLess Geo B Y B X
```

This observation made an explicit predicate for segment sums unnecessary. The existing language of betweenness, congruence, and strict segment comparison is sufficient.

This is not merely a simplification of I.21. It is evidence that a useful fragment of segment arithmetic can be expressed internally in the synthetic language of Book Zero.

21.6 Proof Architecture of the Reconstruction

The Lean proof divides naturally into three major components.

```
                  common configuration
               A-E-C,  B-D-E
                       |
           +-----------+-----------+
           |                       |
           v                       v
     segment branch           angle branch
           |                       |
        I.20 twice            I.16 twice
           |                       |
 construct representatives    BAC < BEC < BDC
    of BA+AC, BD+DC                |
           |                       |
           v                       v
      BY < BX              angle BAC < BDC
           \                       /
            \                     /
             +---------+---------+
                       |
                       v
               euclid_proposition_21
```

The diagram is deliberately architectural rather than metric. In particular, the points X and Y in the formal theorem are not additional features of Euclid's printed diagram; they are witnesses that reify the two segment sums inside the synthetic language.

21.6.1 The Segment Helper

The first reusable component is

`euclid_proposition_21_segment_helper`

It begins with

$$A - E - C, \qquad B - A - H,$$

together with

$$AH \cong AE$$

and

$$BE < BH.$$

A point F is constructed on the ray AH with

$$AF \cong AC.$$

Since $AE < AC$, congruence transport gives

$$AH < AF.$$

The Book Zero order machinery then yields

$$A - H - F.$$

Together with $B - A - H$, this produces

$$B - A - F \qquad \text{and} \qquad B - H - F.$$

Thus BF represents $BA + AC$.

Next a point G is constructed beyond E with

$$EG \cong EC.$$

Hence

$$B - E - G,$$

and BG represents $BE + EC$.

From

$$A - E - C, \qquad A - H - F,$$

and

$$AE \cong AH, \qquad AC \cong AF,$$

the theorem

```
bookZero_differenceOfParts
```

gives

$$EC \cong HF.$$

The inequality $BE < BH$ can therefore be extended by congruent terminal parts. The theorem

```
bookZero_53_lessThanAdditive
```

yields

$$BG < BF.$$

This is the synthetic form of

$$BE + EC < BA + AC.$$

21.6.2 The Angle Branch

The angle comparison is isolated as

```
euclid_proposition_21_angle
```

with conclusion

```
HilbertAngleLess Geo B A C B D C
```

representing

$$\angle BAC < \angle BDC.$$

The proof first reconstructs the noncollinearity information required for the auxiliary triangles. This is a substantial formal task: Euclid simply reads these facts from the diagram.

Proposition I.16 then supplies the strict exterior-angle comparisons. The results are transported through collinear and same-ray configurations, and strict angle comparison is propagated to the required pair of angles.

The essential point is that no numerical angle measure is used. The proof remains entirely inside the synthetic relation `HilbertAngleLess`.

21.6.3 Final Segment Comparison

The main theorem first applies Proposition I.20 to triangle BAE. It obtains a point H with

$$B - A - H, \qquad AH \cong AE,$$

and

$$BE < BH.$$

The segment helper extends this inequality to a comparison of constructed representatives.

The remaining part of the proof constructs a point Y satisfying

$$B - D - Y, \qquad DY \cong DC.$$

The point X on the other side satisfies

$$B - A - X, \qquad AX \cong AC.$$

After a sequence of congruence transports and applications of the Book Zero comparison lemmas, the proof reaches

```
hBY_BX :
    HilbertSegmentLess Geo B Y B X
```

which is exactly

$$BD + DC < BA + AC.$$

21.7 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_21
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D E : Geo.Point)
    (hABC : Not (PrimCollinear Geo A B C))
    (hAEC : Geo.Between A E C)
    (hBDE : Geo.Between B D E) :
    (exists X Y : Geo.Point,
      Geo.Between B A X /\
      Geo.Congruent A X A C /\
      Geo.Between B D Y /\
      Geo.Congruent D Y D C /\
      HilbertSegmentLess Geo B Y B X) /\
    HilbertAngleLess Geo B A C B D C := by
```

The existential part is the exact synthetic encoding of the segment inequality.

The conditions

```
Geo.Between B A X
Geo.Congruent A X A C
```

say that BX represents $BA + AC$.

Likewise

```
Geo.Between B D Y
Geo.Congruent D Y D C
```

say that BY represents $BD + DC$.

The comparison

```
HilbertSegmentLess Geo B Y B X
```

therefore expresses the first assertion of I.21.

The second assertion appears directly as

```
HilbertAngleLess Geo B A C B D C
```

The two branches are deliberately proved separately and assembled only at the end.

21.8 Relationship with Earlier Propositions

21.8.1 Proposition I.16

Proposition I.16 provides the exterior-angle inequalities needed for the angle branch.

This is the principal earlier proposition behind the proof of

$$\angle BAC < \angle BDC.$$

The formal reconstruction also reuses the same strict-angle language that was introduced and strengthened in the development surrounding I.16–I.18.

21.8.2 Proposition I.20

Proposition I.20 is fundamental to the segment branch.

Applied to triangle BAE, it supplies the initial inequality

$$BE < BA + AE.$$

In the formal proof the sum $BA + AE$ is represented by a constructed point H:

$$B - A - H, \qquad AH \cong AE,$$

so that

$$BE < BH.$$

This inequality is then enlarged synthetically by adding congruent parts.

21.8.3 The Cumulative Character of I.21

Proposition I.21 is markedly more cumulative than the immediately preceding propositions.

Its proof is no longer dominated by one construction. Instead it requires a coordinated use of order, congruence, segment comparison, ray transport, angle comparison, I.16, I.20, and numerous Book Zero lemmas.

For this reason I.21 is a useful test of whether the preceding formal development has produced a genuine working language rather than a collection of isolated theorem statements.

21.9 Hilbert and Book Zero Components Used

21.9.1 HilbertSegmentLess

Strict segment comparison supplies a synthetic replacement for numerical inequalities between lengths.

Its witness-based definition also makes it possible to recover interior points and to turn inequalities into betweenness configurations.

21.9.2 Congruence transport

The proof repeatedly changes representatives of the same segment. This requires strict inequalities to be transported through segment congruence.

Among the relevant Book Zero results are

```
bookZero_30_lessThanCongruence
bookZero_32_lessThanCongruence2
```

These operations play the role that substitution of equal numerical lengths would play in an algebraic proof.

21.9.3 Difference of parts

The theorem

```
bookZero_differenceOfParts
```

expresses the synthetic subtraction principle.

From two collinear decompositions with congruent wholes and congruent initial parts, it identifies the remaining parts.

In I.21 it produces

$$EC \cong HF.$$

21.9.4 Addition of inequalities

The theorem

```
bookZero_53_lessThanAdditive
```

is especially significant here.

It permits an existing strict segment inequality to be extended by congruent parts. In I.21 it turns

$$BE < BH$$

into the corresponding comparison after EC and HF have been added.

This is one of the clearest examples so far of Book Zero functioning as a synthetic arithmetic of segments.

21.9.5 Segment-Inequality Transitivity

The theorem

```
bookZero_52_lessThanTransitive
```

allows the several intermediate comparisons to be composed into the final inequality.

21.9.6 Sum of parts

The existing theorem

```
bookZero_sumOfParts
```

is used when two adjacent congruent pieces must be recognized as giving congruent whole segments.

Together with difference-of-parts and inequality transport, this provides much of the elementary segment calculus required by I.21.

21.10 What I.21 Reveals About Book Zero

The reconstruction of I.21 gives a useful answer to a structural question about the project.

It might appear natural to introduce explicit objects representing formal sums of segments. Indeed, an intermediate version of the development contained such a definition.

The completed proof shows that this is not presently necessary.

The combination of

$$\text{betweenness} + \text{congruence} + \text{strict segment comparison}$$

already permits sums to be constructed, compared, enlarged, reduced, and transported.

Thus the expressions

$$AB + CD, \qquad AB < CD,$$

need not force the development prematurely into an algebra of lengths.

The geometric witnesses themselves carry the arithmetic.

This is important for the intended role of Book Zero. Its purpose is not merely to shorten proofs. It supplies the intermediate operations by which ordinary synthetic reasoning can be expressed without abandoning the primitive Hilbert geometry.

I.21 is the strongest example of this role encountered so far.

21.11 Dependency Summary

The segment branch can be summarized schematically as

$$A - E - C, \quad B - D - E, \quad ABC \text{ noncollinear}$$
$$\downarrow$$
$$\text{I.20 on } BAE$$
$$\downarrow$$
$$BE < BA + AE$$
$$\downarrow$$
$$\text{construct congruent remainders}$$
$$\downarrow$$
$$\texttt{bookZero\_differenceOfParts}$$
$$\downarrow$$
$$\texttt{bookZero\_53\_lessThanAdditive}$$
$$\downarrow$$
$$\text{congruence transport and transitivity}$$
$$\downarrow$$
$$BD + DC < BA + AC.$$

The angle branch is

$$A - E - C, \quad B - D - E, \quad ABC \text{ noncollinear}$$
$$\downarrow$$
$$\text{noncollinearity and ray reconstruction}$$
$$\downarrow$$
$$\text{Euclid I.16}$$
$$\downarrow$$
$$\text{strict angle transport}$$
$$\downarrow$$
$$\angle BAC < \angle BDC.$$

Finally,

$$\begin{array}{ccc} BD + DC < BA + AC & & \angle BAC < \angle BDC \\ & \searrow \; \swarrow & \\ & \texttt{euclid\_proposition\_21}. & \end{array}$$

No new geometric axiom is introduced.

21.12 Formalization Notes

Several points that are nearly invisible in the classical proof become substantial in Lean.

First, every assertion that a point lies on the same line, on the same ray, or on the opposite continuation of a ray must be derived explicitly.

Second, the notation $AB + AC$ cannot silently denote a number. A concrete point must be constructed whose position and congruence properties make an ordinary segment represent that sum.

Third, inequalities cannot simply be rewritten by replacing equal lengths. Congruence transport requires explicit theorems.

Fourth, the two assertions of I.21 genuinely have different proof architectures. Keeping them separate in the formal development makes their dependencies visible and produces reusable intermediate results.

The resulting Lean proof is much longer than Euclid's proof, but the additional length is not caused by a different mathematical idea. It records the incidence, order, congruence, and transport operations which the classical diagram suppresses.

21.13 Conclusion

Proposition I.21 is an important transition point in the formalization of Book I.

The classical proposition combines a comparison of broken lines with a comparison of angles. Its proof assumes a mature ability to manipulate segment sums, strict inequalities, collinear configurations, and exterior angles.

The Lean reconstruction preserves this synthetic structure. It does not replace Euclid's argument by coordinates, numerical lengths, or angle measures.

More importantly, the reconstruction demonstrates that the Book Zero layer is beginning to perform its intended function. Results such as difference of parts, addition of inequalities, congruence transport, ray conversion, and strict-comparison transitivity combine into a usable language in which a substantially more complicated Euclidean argument can be expressed.

The fact that no separate `HilbertSegmentSum` object is needed is part of this conclusion. Synthetic segment arithmetic can, at least at this stage, remain geometry rather than become an external algebra of lengths.

The Hilbert reconstruction of Euclid I.21 is complete. It introduces no new geometric axiom, and the complete development through Proposition I.21 compiles successfully.

Chapter 22

Proposition I.22

22.1 Introduction

Euclid's Proposition I.22 is the classical existence theorem for a triangle with three prescribed side lengths.

Three nondegenerate segments are given. Euclid assumes that any two of them, taken together, are greater than the remaining one. Under these hypotheses he constructs a triangle whose three sides are congruent to the three given segments.

In the present reconstruction the given segments are denoted

$$Aa, \qquad Bb, \qquad Cc.$$

The three hypotheses are the strict triangle inequalities

$$Aa < Bb + Cc, \qquad Bb < Aa + Cc, \qquad Cc < Aa + Bb.$$

The formal conclusion is the existence of a point K such that

$$BK \cong Aa \qquad \text{and} \qquad bK \cong Cc.$$

Since Bb is already the third prescribed side, the triangle BbK has precisely the required side lengths.

Proposition I.22 is especially important for the formal development because its proof exposes two pieces of infrastructure which classical notation tends to hide:

1. the phrase "two segments taken together are greater than a third" must receive a synthetic, witness-based meaning;
2. the existence of the vertex K is an application of circle-circle continuity.

No numerical length function is introduced. Segment sums, strict inequalities, and the circle intersection are all expressed in the synthetic language of betweenness and congruence.

The resulting formal architecture is therefore different from the visual shortcut suggested by the classical drawing:

```
triangle inequalities
    |
    +-- synthetic sum witnesses
    |
    +-- construct a point on the G-circle
    |       that lies inside the F-circle
    |
    +-- construct a point on the G-circle
    |       that lies outside the F-circle
    |
    v
circle-circle continuity
    |
    v
intersection point K
    |
    v
BK ~= Aa,  bK ~= Cc
```

This is the hidden logical content of the apparently simple instruction “draw two intersecting circles.”

22.2 The Classical Configuration

Euclid’s construction is best understood in the classical ray configuration used by Wilkins. Let the three prescribed lengths be represented by A, B, and C. On one ray choose consecutive segments

$$DF = A, \qquad FG = B, \qquad GH = C.$$

Draw the circle centered at F with radius A, and the circle centered at G with radius C. Under the three strict triangle inequalities the circles intersect properly in two points K and L. Either triangle FGK or FGL then has side lengths A, B, C.

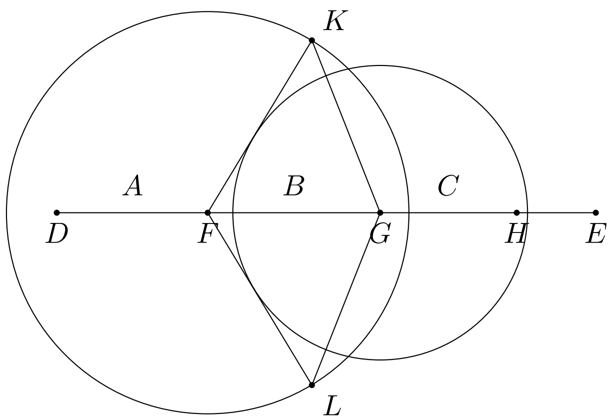


Figure 22.1: Wilkins-style configuration for Euclid I.22. The circles centered at F and G intersect in K and L, producing triangles with side lengths A, B, C.

The important point is that the three inequalities are not decorative side conditions. They control the relative position of the two circles. Wilkins makes this explicit by displaying the six boundary and failure configurations obtained when one of the strict triangle inequalities is replaced by equality or reversed.

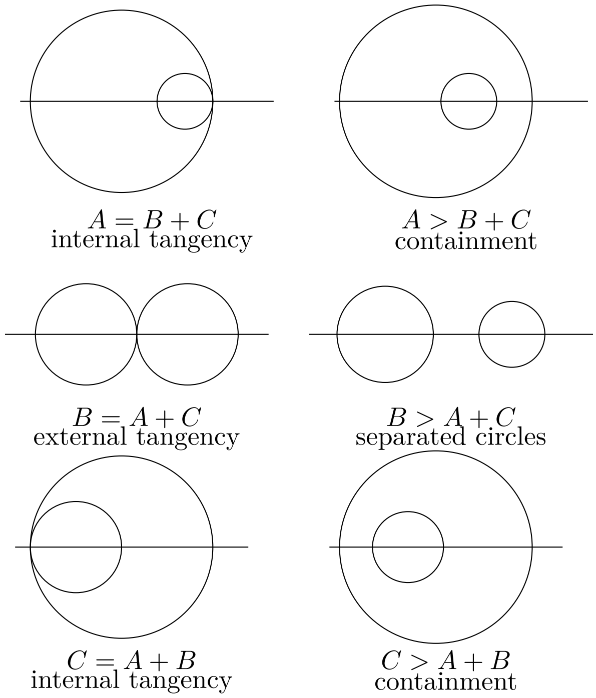


Figure 22.2: The six boundary and failure configurations corresponding to the three triangle inequalities. Equality gives tangency; reversing the inequality destroys proper intersection.

This makes the geometric meaning of the hypotheses visible:

```
A < B + C
B < A + C
C < A + B
      |
      v
two circles intersect properly
      |
      v
triangle with sides A, B, C
```

The formal proof does not use a numerical metric criterion for circle intersection. Instead, it constructs explicit witnesses on one circle that are respectively inside and outside the other circle. Circle-circle continuity then produces the common point.

22.3 Classical Proof

Euclid's proof is a direct ruler-and-compass construction.

Let the three prescribed segments be represented by

$$Aa, \qquad Bb, \qquad Cc,$$

and assume that every two taken together are greater than the remaining one:

$$Aa < Bb + Cc, \qquad Bb < Aa + Cc, \qquad Cc < Aa + Bb.$$

Choose the segment Bb as the base of the required triangle. Draw

- the circle centered at B with radius congruent to Aa;
- the circle centered at b with radius congruent to Cc.

Let K be an intersection point of these two circles. By construction,

$$BK \cong Aa \qquad \text{and} \qquad bK \cong Cc.$$

Together with the base Bb, the triangle BbK therefore has the three prescribed side lengths.

The classical proof may be summarized as

$$\begin{gathered} Aa < Bb + Cc, \quad Bb < Aa + Cc, \quad Cc < Aa + Bb \\ \downarrow \\ \text{draw two circles with centers } B \text{ and } b \\ \downarrow \\ K \text{ is their intersection} \\ \downarrow \\ BK \cong Aa, \qquad bK \cong Cc \\ \downarrow \\ \text{triangle } BbK. \end{gathered}$$

The point at which the axiomatic reconstruction must say more is the existence of K. In the traditional diagram the two circles are simply drawn as intersecting. A formal proof must derive an appropriate intersection theorem from explicit hypotheses.

Thus Proposition I.22 is one of the places where the difference between a construction diagram and an axiomatic existence proof becomes especially visible.

22.4 Statement

The synthetic statement used in the final Lean theorem is:

```
theorem euclid_proposition_22
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A a B b C c : Geo.Point)
    (hAa : A != a)
    (hBb : B != b)
    (hCc : C != c)
    (hAa_lt_BbCc :
      HilbertSegmentSumGreater Geo B b C c A a)
    (hBb_lt_AaCc :
      HilbertSegmentSumGreater Geo A a C c B b)
```

```
    (hCc_lt_AaBb :
      HilbertSegmentSumGreater Geo A a B b C c) :
    exists K : Geo.Point,
      Geo.Congruent B K A a /\
      Geo.Congruent b K C c
```

The three assumptions involving `HilbertSegmentSumGreater` express

$$Bb + Cc > Aa, \qquad Aa + Cc > Bb, \qquad Aa + Bb > Cc.$$

The conclusion gives the third vertex K of the required triangle:

$$BK \cong Aa, \qquad bK \cong Cc,$$

while the base itself is the given segment Bb.

22.5 A Synthetic Predicate for “Two Together Greater”

Proposition I.21 showed that segment sums need not in general be reified as numerical objects. A concrete collinear segment can serve as a representative of a sum.

Proposition I.22 introduces a slightly different requirement. The three triangle inequalities are not merely intermediate steps in the proof: they are part of the public hypotheses of the theorem. Consequently it is useful to package the geometric witness explicitly.

The Book Zero predicate is

```
def HilbertSegmentSumGreater
    [HilbertCongruence Geo]
    (A B C D E F : Geo.Point) : Prop :=
  exists P : Geo.Point,
    Geo.Between A B P /\
    Geo.Congruent B P C D /\
    HilbertSegmentLess Geo E F A P
```

Geometrically, a witness P satisfies

$$A - B - P, \qquad BP \cong CD.$$

Therefore AP represents the sum

$$AB + CD.$$

The final condition

$$EF < AP$$

says precisely

$$AB + CD > EF.$$

For example,

`HilbertSegmentSumGreater Geo B b C c A a`

means that there exists P with

$$B - b - P, \qquad bP \cong Cc, \qquad Aa < BP.$$

Thus BP is a synthetic representative of $Bb + Cc$.

This definition does not introduce an algebra of lengths. It merely packages an ordinary geometric construction together with the strict comparison already provided by `HilbertSegmentLess`.

22.6 Proof Architecture of the Reconstruction

The reconstruction is intentionally layered. The final theorem is short because the difficult circle geometry is isolated in reusable helpers.

The overall dependency is

$$\begin{array}{c}
\text{three triangle inequalities} \\
\downarrow \\
\texttt{HilbertSegmentSumGreater} \\
\downarrow \\
P, Q, R \text{ representing the three sums} \\
\downarrow \\
\texttt{euclid_proposition_22_circle_witnesses} \\
\downarrow \\
N \text{ inside}, \quad H \text{ outside} \\
\downarrow \\
\texttt{euclid_proposition_22_circle_intersection} \\
\downarrow \\
K \\
\downarrow \\
BK \cong Aa, \quad bK \cong Cc.
\end{array}$$

The principal intermediate results are discussed below.

22.6.1 Transporting the outside inequality

The helper

`euclid_proposition_22_outside_helper`

handles the synthetic arithmetic behind the outside point.

Suppose BP represents

$$Bb + Cc,$$

and another collinear segment FH is assembled from pieces

$$FG \cong Bb, \qquad GH \cong Cc.$$

Using `bookZero_sumOfParts`, the proof obtains

$$BP \cong FH.$$

Hence the triangle inequality

$$Aa < BP$$

can be transported to

$$Aa < FH$$

by `bookZero_30_lessThanCongruence`.

This is the formal replacement for the classical step "equal sums may be substituted in an inequality."

22.6.2 Same-ray trichotomy

A point constructed on a ray can lie before, at, or beyond the reference point. The helper

```
euclid_proposition_22_sameRay_trichotomy
```

makes this order distinction explicit.

For points G, F, N on the same ray from G, it proves

$$G - N - F \quad \vee \quad N = F \quad \vee \quad G - F - N.$$

The proof uses strict segment trichotomy together with `bookZero_51_lessThanBetween` and layoff uniqueness.

This case distinction is essential when the point N on the second circle is compared with the first circle.

22.6.3 Cancellation for strict segment inequalities

The helper

```
euclid_proposition_22_lessThan_cancel_right
```

is a synthetic cancellation principle.

If

$$A - B - E, \qquad C - D - F,$$

the terminal parts satisfy

$$BE \cong DF,$$

and

$$AE < CF,$$

then

$$AB < CD.$$

Classically this is an immediate cancellation of equal addends. Formally it requires a proof from the developed strict-order and congruence theory.

This helper is one of the clearest examples in I.22 of Book Zero providing arithmetic-like behavior without numerical lengths.

22.6.4 The inside point

The proof constructs a point N on the ray from b through B such that

$$bN \cong Cc.$$

The two possible order configurations are handled by

```
euclid_proposition_22_inside_left
euclid_proposition_22_inside_right
euclid_proposition_22_inside_helper
```

and the same-ray trichotomy theorem.

The goal is not merely to locate N on the second circle. One must also prove that it is inside, or on the center of, the first circle:

$$N = B \quad \vee \quad BN < Aa.$$

The relevant triangle inequalities are used to derive this comparison in each possible collinear configuration.

22.6.5 Constructing explicit sum representatives

The helper

```
euclid_proposition_22_extend_with_segment
```

constructs the ordinary geometric representative of a sum.

Starting from a nondegenerate segment AB and another segment CD, it extends the ray through B to a point P satisfying

$$A - B - P, \qquad BP \cong CD.$$

Hence

$$AP$$

represents

$$AB + CD.$$

This is the basic geometric operation behind the witnesses P, Q, R used in the core construction.

22.6.6 Circle witnesses

The theorem

```
euclid_proposition_22_circle_witnesses
```

is the main preparatory result.

From the three represented triangle inequalities it constructs points H and N on the circle centered at b with radius congruent to Cc, with the following properties:

$$bH \cong Cc, \qquad Aa < BH,$$

and

$$bN \cong Cc, \qquad N = B \ \vee \ BN < Aa.$$

Thus H is outside the circle centered at B with radius Aa, while N is inside it or coincides with its center.

This is precisely the configuration required by circle-circle continuity.

22.7 Circle-Circle Continuity

The central existence step is isolated in

`euclid_proposition_22_circle_intersection`

The Hilbert circle-continuity theorem does not use the radius Aa directly as a numerical quantity. The proof first lays off a congruent segment BT:

$$BT \cong Aa.$$

The first circle is therefore represented by center B and radius segment BT.

The second circle has center b and radius bN, with

$$bN \cong Cc.$$

The already constructed points satisfy

$$bH \cong bN,$$

so H lies on the second circle, and

$$Aa < BH.$$

After transport through $BT \cong Aa$, this becomes

$$BT < BH,$$

showing that H is outside the first circle.

Similarly,

$$N = B \quad \vee \quad BN < Aa$$

is transported to

$$N = B \quad \vee \quad BN < BT,$$

showing that N is inside the first circle.

The theorem

```
hilbert_circle_circle_intersection
```

then yields a point K satisfying

$$BK \cong BT \qquad \text{and} \qquad bK \cong bN.$$

Finally congruence transitivity returns to the original given segments:

$$BK \cong Aa, \qquad bK \cong Cc.$$

This is the required vertex.

22.8 The Core and the Public Theorem

Once the circle witnesses and the circle-intersection theorem are available, the remaining layers become very short.

The theorem

```
euclid_proposition_22_core
```

accepts explicit representatives P, Q, R of the three sums:

$$BP = Bb + Cc, \qquad AQ = Aa + Cc, \qquad AR = Aa + Bb,$$

together with

$$Aa < BP, \qquad Bb < AQ, \qquad Cc < AR.$$

It invokes the two successive helpers

```
euclid_proposition_22_circle_witnesses
euclid_proposition_22_circle_intersection
```

The intermediate wrapper

```
euclid_proposition_22_from_sum_witnesses
```

packages the same construction when all three sum witnesses are supplied explicitly.

The public theorem finally removes P, Q, R from its interface. It opens the three witnesses stored in `HilbertSegmentSumGreater` and passes them to the internal construction.

Thus the public API states exactly the geometric theorem rather than the implementation details of its proof.

22.9 Lean Formalization

The public theorem deliberately exposes only the three given segments, their nondegeneracy, the three triangle inequalities, and the required vertex:

```
theorem euclid_proposition_22
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A a B b C c : Geo.Point)
    (hAa : A != a)
    (hBb : B != b)
    (hCc : C != c)
    (hAa_lt_BbCc :
      HilbertSegmentSumGreater Geo B b C c A a)
    (hBb_lt_AaCc :
      HilbertSegmentSumGreater Geo A a C c B b)
    (hCc_lt_AaBb :
      HilbertSegmentSumGreater Geo A a B b C c) :
    exists K : Geo.Point,
      Geo.Congruent B K A a /\
      Geo.Congruent b K C c
```

The public assumptions are therefore the exact synthetic counterparts of

$$Aa < Bb + Cc, \qquad Bb < Aa + Cc, \qquad Cc < Aa + Bb.$$

Internally, each occurrence of `HilbertSegmentSumGreater` is unpacked into a concrete witness. Schematically:

```
rcases hAa_lt_BbCc with
  <P, hBbP, hbP_Cc, hAa_BP>

rcases hBb_lt_AaCc with
  <Q, hAaQ, haQ_Cc, hBb_AQ>

rcases hCc_lt_AaBb with
  <R, hAaR, haR_Bb, hCc_AR>
```

The points P, Q, R are not abstract numbers representing sums. They are ordinary points producing collinear segments whose lengths realize the three sums geometrically.

The next layer constructs the circle witnesses. In schematic form:

```
have hWitnesses :=
  euclid_proposition_22_circle_witnesses
    ...

rcases hWitnesses with
  <H, N, hH_on_circle, hH_outside,
         hN_on_circle, hN_inside>
```

The important output is the inside/outside configuration

$$Aa < BH, \qquad N = B \ \vee \ BN < Aa,$$

while both H and N lie on the circle centered at b with radius congruent to Cc.

The actual existence step is then concentrated in

```
euclid_proposition_22_circle_intersection
```

which invokes

```
hilbert_circle_circle_intersection
```

after transporting the radius and inequality data into the orientation expected by the continuity theorem.

Finally, congruence transitivity converts the constructed radius representatives back to the original segments and yields

$$BK \cong Aa, \qquad bK \cong Cc.$$

The Lean proof is therefore longer than Euclid's paper proof not because it changes the construction, but because it makes three suppressed layers explicit:

$$\text{segment arithmetic} \ + \ \text{ray/order geometry} \ + \ \text{circle continuity}.$$

22.10 Relationship with Earlier Propositions

22.10.1 Proposition I.20

Proposition I.20 established the triangle inequality inside an already given triangle.

Proposition I.22 uses the converse direction as an existence hypothesis: three prescribed segment lengths satisfying the three triangle inequalities should determine a triangle.

The logical roles are therefore different. I.20 proves inequalities from a triangle; I.22 assumes the inequalities in order to construct a triangle.

22.10.2 Proposition I.21

Proposition I.21 demonstrated that segment sums can be handled synthetically by constructing ordinary segments representing sums of parts.

I.22 develops this idea further. Because the triangle inequalities are part of the public statement rather than temporary intermediate facts, the witness-bearing predicate `HilbertSegmentSumGreater` is introduced as an interface notion.

Thus I.22 turns the segment-arithmetic techniques of I.21 into reusable public infrastructure.

22.10.3 Proposition I.23

Proposition I.23 follows I.22 in Euclid's sequence and uses it essentially in Euclid's own construction of an angle congruent to a given angle.

In the Hilbert reconstruction, however, I.23 does not formally depend on I.22 because Hilbert III.4 already supplies angle construction.

This makes I.22 a useful reference point for distinguishing the historical dependency graph of the *Elements* from the dependency graph induced by the Hilbert foundation.

22.11 Hilbert and Book Zero Components Used

22.11.1 HilbertSegmentLess

The relation

```
HilbertSegmentLess
```

is the synthetic strict order on segments.

It supplies the comparisons needed to distinguish points lying inside and outside a circle and supports the transport and transitivity operations used throughout the proof.

22.11.2 bookZero_49_layoff

The theorem

```
bookZero_49_layoff
```

copies a given nondegenerate segment onto a chosen ray.

In I.22 it is used to represent the radius Aa by a segment based at the center B:

$$BT \cong Aa.$$

22.11.3 bookZero_sumOfParts

The theorem

```
bookZero_sumOfParts
```

recognizes congruent wholes assembled from congruent adjacent parts.

It is used when a segment representing $Bb + Cc$ must be identified with another segment built from a copy of Bb followed by a copy of Cc.

22.11.4 Congruence transport for inequalities

The theorems

```
bookZero_30_lessThanCongruence
bookZero_32_lessThanCongruence2
```

transport strict segment comparisons through congruent representatives.

They replace the informal algebraic operation of substituting equal lengths into inequalities.

22.11.5 bookZero_31_trichotomy1

This theorem supplies the equality case when neither of two nondegenerate segments is strictly shorter than the other.

Together with layoff uniqueness it provides the middle case

$$N = F$$

in the same-ray trichotomy used by I.22.

22.11.6 bookZero_51_lessThanBetween

The theorem

```
bookZero_51_lessThanBetween
```

converts strict segment comparison on a common ray into strict betweenness.

It is the bridge between the order relation on segment lengths and the incidence-order geometry needed to locate the constructed circle points.

22.11.7 Circle-circle intersection

The theorem

```
hilbert_circle_circle_intersection
```

is the decisive existence principle.

Given a point of one circle lying inside a second circle and another point of the same first circle lying outside the second, it produces a common point of the two circles.

Proposition I.22 is therefore not merely an application of elementary segment arithmetic. It is the point where that arithmetic is used to establish the hypotheses of a genuine continuity theorem.

22.12 Comparison with Euclid's Original Organization

Euclid presents I.22 as an explicit ruler-and-compass construction. He places copies of the three given segments on a straight line and draws two circles whose intersection gives the required triangle.

The Hilbert reconstruction preserves the same geometric idea but changes its logical organization.

First, the base Bb is already available as one of the given segments, so there is no need to reproduce Euclid's preliminary line solely to manufacture a base of the correct length.

Second, the three assumptions "two together greater than the remaining one" are exposed through `HilbertSegmentSumGreater`. Their witnesses give concrete segments representing the required sums.

Third, the existence of the intersection is not treated as visually obvious. Explicit points N and H are constructed on one circle, and they are proved to lie respectively inside and outside the other. Only then is circle-circle continuity invoked.

Thus the formal proof separates three issues which Euclid's diagram compresses:

$$\text{segment arithmetic} \quad + \quad \text{order on rays} \quad + \quad \text{continuity.}$$

The resulting proof is longer, but the additional length identifies the exact logical content required for the construction.

22.13 What Proposition I.22 Adds to Book Zero

I.21 suggested that explicit segment-sum objects were unnecessary: inside a proof, sums could be represented directly by constructed segments.

I.22 refines that conclusion rather than contradicting it.

The new predicate

`HilbertSegmentSumGreater`

does not turn segment sums into algebraic objects. It is a witness-bearing proposition:

$$AB + CD > EF$$

means that a point exists which geometrically realizes $AB + CD$, and that the resulting whole segment is strictly greater than EF.

This distinction is important.

The project still has no numerical length map and no binary addition operation on lengths. What has been added is a convenient *interface proposition* for a classical phrase that occurs as a hypothesis of Euclid's theorem.

Proposition I.22 also shows that the Book Zero layer is now strong enough to prepare the exact inside/outside data required by a continuity theorem. The layer is therefore doing more than abbreviating elementary congruence arguments: it is connecting synthetic segment arithmetic with topological existence principles.

22.14 Dependency Summary

The public assumptions are

$$Aa \neq 0, \quad Bb \neq 0, \quad Cc \neq 0,$$

$$\begin{gathered} Aa < Bb + Cc, \\ Bb < Aa + Cc, \\ Cc < Aa + Bb. \end{gathered}$$

In Lean the three inequalities are encoded by

```
HilbertSegmentSumGreater Geo B b C c A a,
HilbertSegmentSumGreater Geo A a C c B b,
HilbertSegmentSumGreater Geo A a B b C c.
```

The complete proof architecture is

$$\begin{gathered} \text{triangle inequalities} \\ \downarrow \\ P, Q, R \text{ sum witnesses} \\ \downarrow \\ \text{segment transport + same-ray order} \\ \downarrow \\ N = B \ \vee \ BN < Aa \qquad Aa < BH \\ \downarrow \\ bN \cong Cc, \qquad bH \cong Cc \\ \downarrow \\ \texttt{hilbert\_circle\_circle\_intersection} \\ \downarrow \\ BK \cong Aa, \qquad bK \cong Cc \\ \downarrow \\ \texttt{euclid\_proposition\_22}. \end{gathered}$$

No new geometric axiom is introduced in Proposition I.22. The proof uses the already available circle-circle continuity theorem together with the developed Hilbert and Book Zero segment theory.

22.15 Formalization Notes

Several features of I.22 are easy to miss in an informal proof.

First, a circle radius is represented by an ordinary segment, not by a real number. Even when the intended radius is Aa, the circle-continuity theorem is applied using a constructed representative BT with

$$BT \cong Aa.$$

Second, "inside a circle" is expressed synthetically by a strict segment comparison such as

$$BN < BT,$$

with the center case $N = B$ handled separately.

Third, "outside a circle" is likewise a strict segment comparison:

$$BT < BH.$$

Fourth, the fact that N lies on the ray from b through B is not enough to determine its order relative to B. The proof requires an explicit three-way analysis of the possible positions.

Fifth, cancellation of equal terminal pieces from strict inequalities is not automatic. It is isolated as a theorem and proved from the existing segment-order machinery.

Finally, the public theorem is deliberately cleaner than the internal construction. The implementation works with many auxiliary points, but the theorem exposes only the three original segments, their three triangle inequalities, and the required vertex.

22.16 Conclusion

Proposition I.22 is a major structural milestone in the reconstruction of Book I.

Its classical content is elementary: three admissible lengths determine a constructible triangle. Formalizing that statement reveals, however, a precise interaction between three layers of synthetic geometry.

The first layer is segment arithmetic. Betweenness, congruence, strict segment comparison, addition of adjacent parts, cancellation, and transport are used to express the three triangle inequalities and to manipulate their witnesses.

The second layer is ray and order geometry. Constructed points must be located explicitly on rays, and their possible orders must be distinguished.

The third layer is continuity. Once one point of the second circle is proved inside the first and another outside it, `hilbert_circle_circle_intersection` supplies the required common point.

The final theorem therefore has a particularly clear conceptual form: the triangle inequalities are exactly the hypotheses needed to build the inside/outside witnesses for two circles.

No coordinates, numerical lengths, or angle measures enter the proof. The reconstruction remains entirely synthetic, and the complete development through Euclid I.22 compiles successfully.

Chapter 23

Proposition I.23

23.1 Introduction

Euclid's Proposition I.23 asks for the construction, at a prescribed point on a prescribed straight line, of an angle congruent to a given rectilinear angle.

In Euclid's development this is a genuine construction theorem. The construction proceeds indirectly: Euclid chooses points on the arms of the given angle, obtains three segment lengths, uses Proposition I.22 to construct a triangle with those three sides, and then invokes Proposition I.8 (SSS) to identify the required angle.

In the Hilbert reconstruction used by CGJteamLab, the logical status of the proposition is different. Hilbert's congruence axiom III.4 already provides angle transport on a prescribed side of a prescribed ray, together with uniqueness of the resulting ray. Consequently the formal Proposition I.23 is a short wrapper over a stronger foundational principle.

This contrast is mathematically important. Proposition I.23 is an example in which the order of Euclid's propositions and the dependency order induced by the chosen axiomatic foundation no longer coincide.

23.2 The Classical Configuration

Euclid's construction is best understood in the original notation of the classical argument. A straight segment AB and a given angle at C are prescribed. The objective is to construct at A, on the given straight line, an angle equal to the angle at C.

Wilkins's presentation makes the constructional character of the proposition especially clear. The initial data are simply the line segment and the given angle:

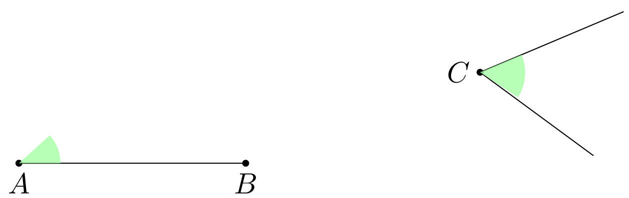


Figure 23.1: Initial data for Euclid I.23: a prescribed segment AB and a given rectilinear angle at C.

Euclid does not transport the angle directly. He first chooses points D and E on the two arms of the given angle and forms the triangle CDE. A second triangle AFG is then constructed with G lying on AB.

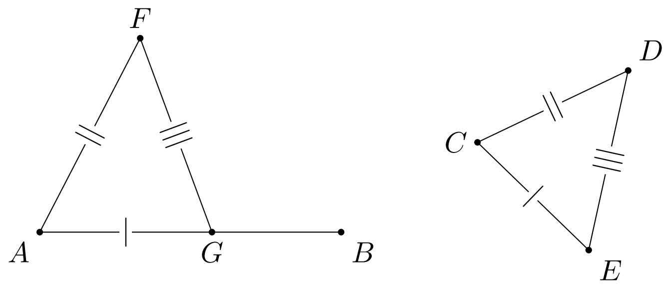


Figure 23.2: Euclid's completed auxiliary construction: $AG \cong CE$, $AF \cong CD$, and $FG \cong DE$.

The three side congruences give

$$\triangle AFG \cong \triangle CDE$$

by Proposition I.8, and hence

$$\angle FAG \cong \angle DCE.$$

The drawings therefore encode the actual Euclidean construction rather than merely displaying two congruent angles.

23.3 Classical Proof

Euclid derives the construction from earlier propositions rather than treating angle transport as primitive.

Choose points A and C on the two arms of the given angle $\angle ABC$. The three segments

$$AB, \qquad BC, \qquad AC$$

determine the triangle ABC.

Now use Proposition I.22 to construct a triangle DEF whose corresponding sides satisfy

$$DE \cong AB, \qquad DF \cong BC, \qquad EF \cong AC.$$

By Proposition I.8, the two triangles are congruent by SSS. Hence their corresponding included angles are congruent, and in particular

$$\angle ABC \cong \angle EDF$$

after the appropriate correspondence of vertices.

Thus the classical dependency pattern is

$$\begin{array}{c}\text{given angle}\\ \downarrow\\ \text{choose a triangle containing the angle}\\ \downarrow\\ \text{I.22}\\ \downarrow\\ \text{construct a triangle with the same three side lengths}\\ \downarrow\\ \text{I.8 (SSS)}\\ \downarrow\\ \text{corresponding angle congruence}\\ \downarrow\\ \text{I.23.}\end{array}$$

In Euclid's route, angle transport is therefore reduced to segment construction and triangle congruence.

23.4 Hilbert III.4

Hilbert reverses the foundational emphasis.

Instead of deriving angle transport from triangle construction, Hilbert includes angle construction among the congruence axioms. In the present formal layer the relevant operation is

```
HilbertCongruence.angle_construction
```

Conceptually, it states that from

- a nondegenerate source angle;
- a prescribed initial ray;
- a choice of one side of the supporting line,

one obtains a ray on that side forming an angle congruent to the source angle.

The Hilbert axiom is stronger than Euclid I.23 because it also contains a uniqueness assertion: on the chosen side of the line, the constructed ray is unique.

This additional information is not needed in the public statement of Proposition I.23, but it remains available in the Hilbert interface.

23.5 Proof Architecture of the Reconstruction

The formal reconstruction separates naturally into four stages.

23.5.1 The Source Angle

The hypothesis

$$\neg\,\text{Collinear}(A, B, C)$$

ensures that $\angle ABC$ is nondegenerate.

This is the source angle that will be transported.

23.5.2 The Prescribed Ray

The points D and E determine the initial ray of the new angle.

The hypotheses assert that

$$D \neq E$$

and that both points lie on a common supporting line. Thus the formal data specify not merely a vertex, but an oriented ray represented by two distinct points.

23.5.3 Selecting the Side

A point S not lying on the supporting line selects one of its two sides.

The constructed point F is required to satisfy

$$\text{SameSide}(F, S, \text{base}).$$

This datum has no explicit counterpart in Euclid's informal formulation, where the intended side is supplied by the diagram and by the instruction to construct the angle "on" a given straight line.

In the axiomatic reconstruction it must be stated.

23.5.4 Applying Angle Construction

The Hilbert angle-construction theorem supplies a point F satisfying the required same-side condition and

$$\angle ABC \cong \angle DEF.$$

Its uniqueness component is discarded because Proposition I.23 asks only for existence.

Thus the entire formal proof is an interface extraction from Hilbert III.4.

The Hilbert configuration is consequently much simpler than the Euclidean construction. The source angle, the prescribed ray DE, and the side selector S determine the required ray directly:

This diagram is intentionally not the Euclidean CDE/AFG construction. It represents the stronger primitive available in the Hilbert congruence layer.

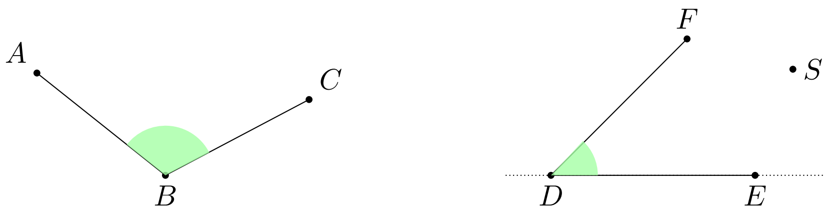


Figure 23.3: Hilbert III.4 configuration. The point S selects the side of the supporting line on which the second ray is constructed.

23.6 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_23
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D E S : Geo.Point)
    (hABC : Not (PrimCollinear Geo A B C))
    (hDE : D != E)
    (base : Geo.Line)
    (hDbase : HilbertIncidence.OnLine D base)
    (hEbase : HilbertIncidence.OnLine E base)
    (hSbase : Not (HilbertIncidence.OnLine S base)) :
    Exists fun F : Geo.Point =>
      HilbertSameSide Geo F S base /\
      Geo.AngleCongruent A B C D E F := by
```

The mathematical meaning of the arguments is:

$$\angle ABC$$

is the source angle,

$$DE$$

is the prescribed initial ray, and S selects the side of the supporting line on which the second ray must be constructed.

The proof then invokes the Hilbert construction principle and extracts the components required by the Euclidean statement. Schematically:

```
rcases HilbertCongruence.angle_construction ... with
  <F, hSameSide, hAngle, hUnique>

exact <F, hSameSide, hAngle>
```

The uniqueness witness `hUnique` is deliberately not returned.

This is the essential feature of the Lean proof: its brevity reflects the strength of the Hilbert interface rather than an omitted geometric argument.

23.7 The Euclidean Route and the Hilbert Route

23.7.1 The Euclidean Route

Euclid reduces angle transport to triangle construction:

$$\text{I.22} + \text{I.8} \longrightarrow \text{I.23}.$$

Proposition I.22 constructs a triangle with prescribed side lengths, while Proposition I.8 identifies the corresponding angle by SSS.

The contrast is especially transparent in dependency form:

```
Euclid
------
given angle at C
    |
choose D,E on its arms
    |
triangle CDE
    |
I.20  -> triangle inequalities
    |
I.22  -> construct AFG
    |
I.8 / SSS
    |
angle FAG ~= angle DCE


Hilbert / Lean
--------------
source angle + prescribed ray + side selector
    |
Hilbert III.4
    |
constructed ray + angle congruence
    |
euclid_proposition_23
```

This route passes through segment arithmetic, circle intersection, continuity, and triangle congruence.

23.7.2 The Hilbert Route

In the Hilbert foundation the corresponding dependency is simply

$$\text{III.4} \longrightarrow \text{I.23}.$$

Angle transport has already been placed in the congruence layer.

Thus Proposition I.23 does not formally depend on Proposition I.22 or Proposition I.8.

23.7.3 Foundational Interpretation

The difference is not merely one of proof length.

Euclid treats angle transport as something to be constructed from more elementary operations. Hilbert treats the ability to copy an angle to a prescribed side of a ray as part of the primitive congruence structure.

The same geometric statement therefore occupies two different logical levels in the two systems.

This is precisely the kind of distinction that becomes visible when Book I is reconstructed over an explicit axiomatic interface.

23.8 Relationship with Earlier Propositions

23.8.1 Proposition I.8

In Euclid's proof, Proposition I.8 is the final congruence theorem. Once the auxiliary triangle has been constructed with the required three side lengths, SSS identifies the transported angle.

In the formal Hilbert proof I.8 is not used.

23.8.2 Proposition I.22

Euclid uses Proposition I.22 to construct the auxiliary triangle with prescribed sides.

The formal Hilbert proof again does not use it. Reintroducing I.22 would reconstruct Euclid's historical dependency chain on top of a foundation that already contains the stronger angle-construction principle.

This is especially noteworthy because I.22 required explicit continuity machinery in the reconstruction, whereas I.23 requires none.

23.8.3 Book I Order versus Formal Dependency

The file for Proposition I.23 naturally follows Proposition I.22 in the Book I sequence.

The logical dependency graph, however, does not contain an edge

$$\text{I.22} \longrightarrow \text{I.23}.$$

The textual order of the *Elements* and the dependency order induced by Hilbert geometry therefore diverge at this point.

23.9 Hilbert and Book Zero Components Used

23.9.1 HilbertCongruence.angle_construction

This is the central ingredient of the proof.

It already contains the existence of the required point, the side condition, the angle congruence, and a uniqueness property stronger than the conclusion of I.23.

23.9.2 HilbertSameSide

The same-side relation makes explicit which of the two mirror-image constructions is intended.

The point S serves as a side selector, and the constructed point F is required to lie on the same side of the supporting line.

23.9.3 HilbertSameRay

The uniqueness clause of Hilbert III.4 is naturally expressed in terms of equality of rays. In the project, rays are represented through points together with the relation `HilbertSameRay` rather than as a separate primitive type.

23.9.4 No New Book Zero Lemma

Proposition I.23 exposes no missing low-level operation.

Unlike several preceding propositions, it does not require a new Book Zero theorem. The needed construction already exists at the Hilbert congruence level.

This negative result is structurally useful: a Book I proposition should not generate helper infrastructure merely because it appears as a separate proposition in Euclid.

23.10 Uniqueness and the Strength of the Hilbert Statement

Euclid asks for a construction.

Hilbert III.4 gives existence together with uniqueness on the selected side of the supporting line.

Suppose another point F' lies on the selected side and determines an angle congruent to the same source angle. The Hilbert uniqueness clause says that the ray from the new vertex through F' coincides with the ray through F.

Thus the formal public theorem intentionally forgets information.

This is a useful interface-design principle: the theorem representing Euclid I.23 should state exactly the Euclidean conclusion, while the stronger foundational theorem remains available separately.

23.11 Dependency Summary

The formal dependency structure is

$$\begin{gathered}\text{Hilbert incidence + congruence}\\ \downarrow\\ \text{nondegenerate source angle}\\ \text{prescribed ray and supporting line}\\ \text{side selector}\\ \downarrow\\ \texttt{HilbertCongruence.angle\_construction}\\ \downarrow\\ F \text{ on the prescribed side}\\ \angle ABC \cong \angle DEF\\ \downarrow\\ \texttt{euclid\_proposition\_23}.\end{gathered}$$

No use is made of

$$\text{I.22}, \qquad \text{I.8},$$

or of a new continuity argument.

No new geometric axiom and no new Book Zero lemma are introduced.

23.12 Formalization Notes

Several features of the formal statement deserve emphasis.

First, the point S is additional explicit data. It selects one side of the line containing D and E. Without this information there are two mirror-image rays satisfying the angle-congruence requirement.

Second, the conditions that D and E lie on the base line and that $D \neq E$ ensure that the prescribed initial ray is nondegenerate.

Third, the hypothesis that A, B, C are noncollinear ensures that the source angle is nondegenerate.

Fourth, the theorem returns a point F, not an abstract ray. This matches the point-based representation of rays used throughout the Hilbert layer.

Finally, the short proof should not be interpreted as evidence that Proposition I.23 is mathematically trivial. Rather, its mathematical content has already been absorbed into the axiomatic foundation.

23.13 Conclusion

Proposition I.23 provides a particularly clean example of the difference between historical construction order and formal axiomatic dependency.

Euclid constructs the transported angle indirectly. Proposition I.22 produces a triangle with prescribed side lengths, and Proposition I.8 identifies the desired angle by SSS.

Hilbert places the corresponding construction directly in the congruence axioms. The formal reconstruction can therefore obtain Euclid's conclusion immediately from III.4.

The important result is not merely that the Lean proof is short. The reconstruction identifies exactly *why* it is short: a theorem in Euclid has become part of the foundational interface in Hilbert geometry.

This also explains why Proposition I.23 introduces no new Book Zero machinery. The proposition does not reveal a missing elementary operation; instead, it reveals a change in foundational organization.

The completed formal theorem therefore preserves Euclid's visible Book I milestone while recording a substantially different dependency graph underneath it.

Chapter 24

Proposition I.24

24.1 Introduction

Euclid's Proposition I.24 is the classical hinge theorem. Two triangles have two corresponding pairs of congruent sides. If the included angle of the first triangle is greater than the included angle of the second, then the side opposite the greater angle is greater than the corresponding side of the second triangle.

In the notation of the formal development, the hypotheses are

$$AB \cong DE, \qquad AC \cong DF,$$

and

$$\angle EDF < \angle BAC.$$

The conclusion is

$$EF < BC.$$

The reconstruction is entirely synthetic. No numerical length function, coordinate system, or numerical angle measure is introduced. Proposition I.24 is nevertheless substantially more delicate than its statement suggests. The classical diagram suppresses a genuine case distinction in the position of the auxiliary point. The Lean reconstruction makes this order-theoretic issue explicit and proves all possible configurations.

This is also a historically significant feature of I.24. The proof printed in the traditional text of Euclid treats only one of three possible configurations. Proclus supplied arguments for the omitted configurations, and later editors and commentators discussed alternative repairs. Thus the case split forced by the formal reconstruction is not an artifact of Lean: it reflects a classical issue in the textual and mathematical history of the proposition.

24.2 The Classical Configuration

Let ABC and DEF be nondegenerate triangles satisfying

$$AB \cong DE, \qquad AC \cong DF, \qquad \angle EDF < \angle BAC.$$

Euclid constructs at D an angle congruent to the larger angle at A and lays off a segment DG congruent to AC (and hence to DF). Thus

$$\angle EDG \cong \angle BAC, \qquad DG \cong AC \cong DF.$$

By SAS,

$$\triangle ABC \cong \triangle DEG,$$

so that

$$BC \cong EG.$$

It remains to prove

$$EF < EG.$$

The historical difficulty is best seen by displaying all three configurations distinguished in modern commentary. Following Wilkins, the distinction is made by the positions of D and E relative to the line through G and F.

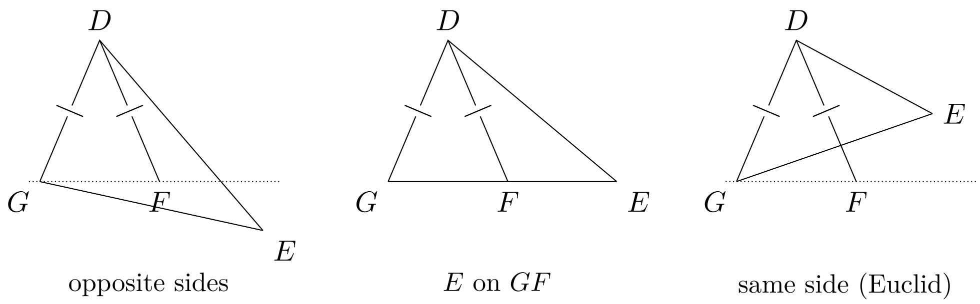


Figure 24.1: The three configurations distinguished in the classical discussion of Proposition I.24. Euclid's printed proof treats only the third.

The equal marks on DG and DF record the isosceles triangle used in the classical argument. The three diagrams are not three Lean branches; they belong to the historical analysis of Euclid's construction.

24.3 Classical Proof

Euclid's argument proceeds as follows. Since

$$AB \cong DE, \qquad AC \cong DG, \qquad \angle BAC \cong \angle EDG,$$

Proposition I.4 gives

$$BC \cong EG.$$

Also

$$DF \cong DG,$$

so triangle DFG is isosceles. Proposition I.5 therefore supplies the corresponding equality of its base angles. In the configuration displayed by Euclid, this angle equality, together with the position of E, yields a strict comparison between the two angles of triangle EFG. Proposition I.19 then gives

$$EF < EG.$$

Since $EG \cong BC$, one obtains

$$EF < BC.$$

The classical dependency pattern is therefore

$$\begin{array}{c}
\angle EDF < \angle BAC \\
\downarrow \\
\text{construct } G: \quad DG \cong AC, \quad \angle EDG \cong \angle BAC \\
\downarrow\ I.4 \\
EG \cong BC \\
\downarrow \\
DF \cong DG \\
\downarrow\ I.5 \\
\text{angle comparison in } \triangle EFG \\
\downarrow\ I.19 \\
EF < EG \\
\downarrow \\
EF < BC.
\end{array}$$

The problem is not the metric or congruence part of this argument. The problem is the unproved assumption about the relative position of the auxiliary points.

24.4 The Classical Configuration Gap

Modern commentary makes the difficulty precise. After the construction used in I.24, three configurations are possible. In Wilkins's formulation they are distinguished by the position of the relevant points with respect to the line through the two auxiliary points. The proof given by Euclid covers only one of these configurations; Proclus supplied proofs for the other two.

This point was also discussed by later editors. Simson modified the setup by adding a restriction on which of the two equal sides is chosen in the construction, with the aim of avoiding the three-case analysis. Heath's commentary records both the added condition and the underlying issue.

The historical point should therefore be stated carefully. Hilbert did not first discover the missing cases of I.24. They were already known in the ancient commentary tradition. What the Hilbert axiomatic framework provides is a language in which the relevant incidence, separation, and betweenness information must be stated and proved rather than inferred from a picture.

The Lean reconstruction follows this axiomatic discipline. It does not assume that the auxiliary point occupies the position suggested by one particular diagram.

For the third configuration, which is the one explicitly treated by Euclid, Wilkins makes the angle comparison visually explicit. Since

$$DG \cong DF,$$

the base angles of the isosceles triangle DGF are congruent, and the displayed configuration gives the chain

$$\angle EFG > \angle DFG = \angle DGF > \angle EGF.$$

Proposition I.19 then yields $EG > EF$.

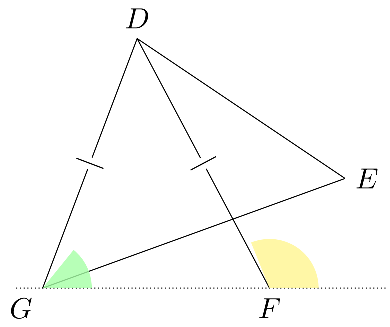


Figure 24.2: Euclid's third configuration, with the two angle-comparison regions highlighted schematically. The colours are explanatory only; the formal proof uses synthetic angle order, not numerical angle measure.

24.5 Strict Angle and Segment Comparison

Two synthetic order relations are central to the reconstruction.

The relation `HilbertAngleLess` expresses strict comparison of angles without introducing numerical angle measure. Informally,

$$\alpha < \beta$$

means that a congruent copy of α occurs as a proper interior subangle of β.

Similarly, `HilbertSegmentLess` expresses strict comparison of segments through betweenness and congruence, without introducing a real-valued length function.

Proposition I.24 combines these two order theories. Its hypothesis is a strict angle inequality, while its conclusion is a strict segment inequality. The hinge construction is the bridge between them.

24.6 Proof Architecture of the Reconstruction

The formal proof separates naturally into seven stages.

24.6.1 The SAS Construction

The helper `euclid_proposition_24_sas_aux` starts from

$$AB \cong DE, \qquad AC \cong DF, \qquad \angle EDF < \angle BAC.$$

The strict angle comparison supplies an interior ray of $\angle BAC$. A point H is laid off on this ray so that

$$AH \cong AC.$$

Since $AC \cong DF$, congruence transport gives

$$AH \cong DF.$$

The auxiliary angle at A is congruent to $\angle EDF$, and SAS applied to ABH and DEF yields

$$BH \cong EF.$$

Thus the original goal $EF < BC$ is reduced to the hinge comparison

$$BH < BC.$$

24.6.2 The Hinge Configuration

The interior ray through H meets the open segment BC at a point Y. The same-ray information does not determine the order of H and Y on the ray from A. The formal proof therefore uses same-ray trichotomy and obtains exactly three possibilities:

$$A - H - Y, \qquad H = Y, \qquad A - Y - H.$$

This is the central order-theoretic decomposition of the reconstruction.

It is important not to identify these three Lean cases with the three historical configurations above. The historical split concerns the position of D and E relative to the line GF. The Lean split concerns the order of H and Y on a single ray from A after the auxiliary hinge point has already been constructed.

The formal geometry can be displayed independently of the historical diagram:

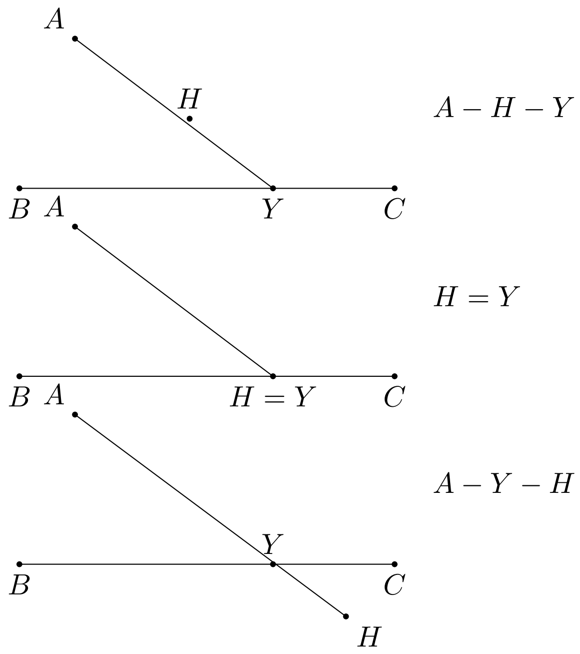


Figure 24.3: The three order configurations used by the Lean hinge helper. In all three, Y lies in the open segment BC; the split concerns only the order of H and Y on their common ray from A.

24.6.3 Case 1: $A - H - Y$

This is the longer nontrivial branch. After reconstructing the necessary noncollinearity facts, Proposition I.16 is used to obtain strict exterior-angle comparisons. The relation

$$AH \cong AC$$

makes triangle AHC isosceles, allowing the relevant angles at H and C to be transported across congruence.

The point Y lies between B and C, so the appropriate ray from H meets the opposite side internally. The theorem `hilbert_interior_angle_less` turns this incidence fact into a strict angle comparison. After the necessary same-ray normalizations and transitivity steps, the proof obtains

$$\angle BCH < \angle BHC.$$

Proposition I.19 applied to triangle BHC now gives

$$BH < BC.$$

24.6.4 Case 2: $H = Y$

If the auxiliary point is itself the intersection point with BC, then

$$B - H - C.$$

The definition of strict segment comparison immediately gives

$$BH < BC.$$

No angle comparison is required.

24.6.5 Case 3: $A - Y - H$

This branch has a different geometry. At C, the ray through Y meets the open segment HA, so

$$\angle HCY < \angle HCA.$$

Because $C - Y - B$, this comparison can be transported from the ray CY to the ray CB.

Again $AH \cong AC$ makes triangle AHC isosceles. The base-angle congruence transfers the comparison to the corresponding angle at H. At H, the ray through Y meets the open segment CB, giving a second proper interior-subangle comparison. Since $H - Y - A$, the ray HY agrees with the ray HA. Transitivity and angle-congruence transport therefore yield

$$\angle BCH < \angle BHC.$$

A final application of I.19 gives

$$BH < BC.$$

The two nontrivial cases therefore reach the same final comparison through different angle-order paths.

24.6.6 The Hinge Helper

The theorem `euclid_proposition_24_hinge_aux` packages the complete case analysis:

$$\begin{array}{ccc}
 & \text{interior ray through } H \text{ meets } BC \text{ at } Y & \\
 & \downarrow & \\
 & \text{same-ray trichotomy for } A, Y, H & \\
 & \downarrow & \\
A - H - Y & H = Y & A - Y - H \\
\downarrow & \downarrow & \downarrow \\
\text{Case 1} & \text{betweenness} & \text{Case 3} \\
 & \downarrow & \\
 & BH < BC. &
\end{array}$$

No location of H is inferred from the diagram.

24.6.7 Completion of I.24

The SAS helper supplies

$$BH \cong EF,$$

and the hinge helper supplies

$$BH < BC.$$

Strict segment comparison is transported through congruence, giving

$$EF < BC.$$

This is exactly the conclusion of Proposition I.24.

24.7 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_24
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C D E F : Geo.Point)
    (hABC : Not (PrimCollinear Geo A B C))
    (hDEF : Not (PrimCollinear Geo D E F))
    (hAB_DE : Geo.Congruent A B D E)
    (hAC_DF : Geo.Congruent A C D F)
    (hAngle :
      HilbertAngleLess Geo E D F B A C) :
    HilbertSegmentLess Geo E F B C := by
  ...
```

The first major helper packages the SAS construction. Its output contains the auxiliary ray and point together with the two metric facts needed later:

$$AH \cong AC, \qquad BH \cong EF.$$

Schematically, the public proof begins by unpacking this construction:

```
rcases
    euclid_proposition_24_sas_aux
      (Geo := Geo)
      A B C D E F
      hABC hDEF hAB_DE hAC_DF hAngle with
  <X, H, hMeet, hRayAXH, hAH_AC, hBH_EF>
```

The hinge comparison is then delegated to the second major helper:

```
have hBH_BC :
    HilbertSegmentLess Geo B H B C :=
  euclid_proposition_24_hinge_aux
    (Geo := Geo)
    A B C H X
    hABC
```

```
    hMeet
    hRayAXH
    hAH_AC
```

Inside that helper the intersection witness is extracted and the point order is split exhaustively:

```
rcases hMeet with <Y, hBYC, hRayAXY>

rcases
    euclid_proposition_22_sameRay_trichotomy
      (Geo := Geo)
      A Y H hAY hAH hRayAYH with
  hAHY | hYH | hAYH
```

The three branches are then dispatched to Case 1, the immediate betweenness case, and Case 3. The public theorem contains none of this order analysis; it sees only the reusable result $BH < BC$.

Finally the segment inequality is transported from BH to the congruent segment EF. Thus the public proof is short because the difficult geometry has been isolated in named synthetic helpers.

24.8 Relationship with Earlier Propositions

24.8.1 Proposition I.5

The isosceles base-angle theorem is used repeatedly after

$$AH \cong AC.$$

It is the main congruence bridge between the angle at C and the angle at H in the hinge triangle.

24.8.2 Proposition I.16

Case 1 uses the exterior-angle theorem to obtain strict angle comparisons which are not available directly from the interior-ray configuration. This is the principal reason that Case 1 is longer than Case 3.

24.8.3 Proposition I.19

Both nontrivial branches end with the same angle inequality in triangle BHC:

$$\angle BCH < \angle BHC.$$

Proposition I.19 converts this angle comparison into the required side comparison

$$BH < BC.$$

Thus I.19 is the final metric step of the hinge argument.

24.8.4 Proposition I.23

The classical Euclidean proof uses I.23 to construct at D an angle equal to the given larger angle. In the Hilbert reconstruction the corresponding construction is expressed through the already available Hilbert angle construction machinery. Proposition I.23 is therefore part of the classical dependency picture, while the formal proof uses the more general interface theorem behind that construction.

24.9 Hilbert and Book Zero Components Used

24.9.1 HilbertAngleLess

Strict angle comparison is the input order relation of I.24. It also provides the interior-ray witness needed for the auxiliary construction.

24.9.2 HilbertSegmentLess

Strict segment comparison is the conclusion of I.24. Its witness-based form permits the middle case $H = Y$ to close directly from betweenness.

24.9.3 HilbertSameRay

Same-ray machinery is used both to represent the auxiliary interior ray and to normalize angles when two syntactically different rays determine the same geometric direction.

24.9.4 Same-ray trichotomy

The trichotomy is the formal mechanism that exposes the hidden case split. It produces the exhaustive alternatives

$$A - H - Y, \qquad H = Y, \qquad A - Y - H.$$

This is the order-theoretic core of the reconstruction.

24.9.5 hilbert_interior_angle_less

This theorem turns a ray meeting the opposite segment internally into a strict comparison between the resulting subangle and the whole angle. It is used repeatedly in the two nontrivial hinge cases.

24.9.6 Angle-congruence transport

The formal proof must explicitly transport strict inequalities across congruent angles and across coincident rays. These steps replace informal phrases such as “this is the same angle” which are normally read directly from a diagram.

24.10 Historical Note: Euclid, Proclus, Simson, and Hilbert

Proposition I.24 is a useful example of the distinction between a classical geometric argument and a complete axiomatic treatment of its diagram.

The traditional Euclidean proof presents one configuration. As emphasized in modern study notes by David R. Wilkins, three configurations can arise, and Euclid's printed argument covers only one of them. Proclus supplied proofs for the two omitted configurations, and later commentators reproduced or reorganized these supplements.

Robert Simson pursued a different repair. He added a condition governing which of the two equal sides is to be used in the construction, thereby seeking to force the auxiliary point into the configuration required by the standard proof. Heath records this modification in his commentary on I.24.

Hilbert's significance is different. His axiomatic treatment of order and incidence supplies a framework in which such positional assumptions cannot remain implicit. Betweenness, collinearity, ray identity, and separation must be justified formally. The Lean reconstruction inherits precisely this discipline.

Accordingly, the three-way split in the present proof should not be read as a peculiarity of proof-assistant programming. It is a machine-checked form of a genuine classical issue in Proposition I.24.

24.11 Dependency Summary

The formal dependency structure is

Hilbert incidence + order + congruence

↓

`HilbertAngleLess`, `HilbertSegmentLess`, `HilbertSameRay`

↓

angle construction, interior-ray theory, same-ray trichotomy

↓

I.5, I.16, I.19 and Book Zero transport lemmas

↓

`euclid_proposition_24_sas_aux`
`euclid_proposition_24_hinge_case1`
`euclid_proposition_24_hinge_case3`
`euclid_proposition_24_hinge_aux`

↓

`euclid_proposition_24`.

No numerical metric, numerical angle measure, circle continuity, or additional hinge axiom is introduced.

24.12 Conclusion

Proposition I.24 is one of the clearest examples so far of what formal reconstruction can reveal about a classical synthetic proof.

The metric core is simple. The auxiliary SAS construction gives

$$BH \cong EF,$$

and the final application of Proposition I.19 converts an angle comparison into

$$BH < BC.$$

The real difficulty is positional: the auxiliary point does not have a single order relation forced by the hypotheses.

The Lean proof therefore separates the possibilities

$$A - H - Y, \qquad H = Y, \qquad A - Y - H$$

and proves each one. This exhaustive split corresponds to a difficulty already recognized in the classical commentary tradition. The formal proof does not replace Euclid's synthetic geometry; it supplies the order and incidence information that the traditional diagram leaves implicit.

The completed development compiles cleanly and contains no `sorry`. Proposition I.24 introduces no new geometric axiom. It uses the Hilbert framework, the Book Zero proof language, and the synthetic results established earlier in Book I.

Chapter 25

Proposition I.25

25.1 Introduction

Euclid's Proposition I.25 is the converse comparison theorem to Proposition I.24.

Suppose two triangles have two corresponding sides congruent:

$$AB \cong DE, \qquad AC \cong DF.$$

If the third side of the first triangle is greater than the third side of the second,

$$EF < BC,$$

then the included angle of the first triangle is greater:

$$\angle EDF < \angle BAC.$$

The formal reconstruction is strikingly short. Unlike Proposition I.24, no auxiliary geometric construction is needed inside the proposition itself. The proof is an elimination argument based on the previously established trichotomy for strict angle comparison.

The difficult geometric work needed for I.25 has already been absorbed into the Hilbert interface, in particular into `angle_trichotomy`. Once that theorem is available, Proposition I.25 becomes almost purely logical.

25.2 The Classical Configuration

Let $\triangle ABC$ and $\triangle DEF$ satisfy

$$AB \cong DE, \qquad AC \cong DF,$$

and suppose

$$EF < BC.$$

The conclusion is

$$\angle EDF < \angle BAC.$$

No new point is constructed in Proposition I.25. The same two triangles are compared throughout the proof.

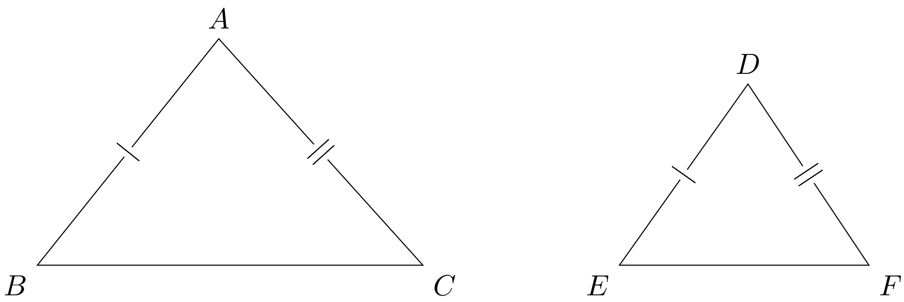


Figure 25.1: The configuration of Proposition I.25.

25.3 Classical Proof

Euclid excludes the two alternatives incompatible with the required conclusion.

First suppose

$$\angle BAC \cong \angle EDF.$$

Then Proposition I.4 (SAS), applied to the two pairs of congruent sides, gives

$$BC \cong EF,$$

contradicting $EF < BC$.

Next suppose

$$\angle BAC < \angle EDF.$$

Apply Proposition I.24 with the two triangles interchanged. Since the included angle of $\triangle DEF$ is greater, I.24 gives

$$BC < EF,$$

again contradicting $EF < BC$.

Therefore the only remaining possibility is

$$\angle EDF < \angle BAC.$$

25.4 Angle Trichotomy

The decisive interface result is `angle_trichotomy`. For two nondegenerate angles it provides exactly the three alternatives

$$\angle BAC \cong \angle EDF, \qquad \angle BAC < \angle EDF, \qquad \angle EDF < \angle BAC.$$

This theorem was established in the Hilbert layer immediately before the formalization of I.25. Its proof uses Hilbert angle construction, same-side information, the order of rays in a half-plane, and the principle that two collinear points lying on the same side of a line through the vertex determine the same ray.

Thus angle trichotomy is not an additional axiom introduced for I.25. It is a derived structural theorem about the already defined relation `HilbertAngleLess`.

25.5 Proof Architecture of the Reconstruction

The formal proof is a three-branch elimination argument. Its complete architecture is short enough to display directly in ASCII:

```
angle_trichotomy
    |
    +-- angles congruent
    |      -> SAS
    |      -> BC ~= EF
    |      -> contradiction with EF < BC
    |
    +-- BAC < EDF
    |      -> Proposition I.24
    |      -> BC < EF
    |      -> contradiction with EF < BC
    |
    +-- EDF < BAC
           -> goal
```

This is not merely a summary of the Lean tactics. It is the actual mathematical dependency structure of the reconstructed proof.

The first branch is eliminated by triangle congruence. The second is eliminated by the direct hinge theorem I.24. The third branch is already the desired statement.

Consequently, after angle trichotomy has been established, I.25 contains no hidden construction and no additional geometric case analysis.

25.6 Hilbert Reconstruction

25.6.1 The Congruent-Angle Branch

Assume

$$\angle BAC \cong \angle EDF.$$

Together with

$$AB \cong DE, \qquad AC \cong DF,$$

Hilbert SAS yields

$$BC \cong EF.$$

The interface theorem `hilbert_segmentLess_not_congruent` states that a strictly shorter segment cannot be congruent to the longer one. Hence this branch is impossible.

25.6.2 The Wrong Strict-Inequality Branch

Assume

$$\angle BAC < \angle EDF.$$

Interchange the roles of the triangles and reverse the side congruences:

$$DE \cong AB, \qquad DF \cong AC.$$

Then Proposition I.24 gives

$$BC < EF.$$

But the original hypothesis is

$$EF < BC.$$

The asymmetry theorem `hilbert_segmentLess_asymm` excludes these simultaneous strict inequalities.

25.6.3 The Desired Branch

The remaining case is exactly

$$\angle EDF < \angle BAC,$$

so no further argument is needed.

25.7 Lean Formalization

The public theorem has the form

```
theorem euclid_proposition_25
    [HilbertCongruence Geo]
    (A B C D E F : Geo.Point)
    (hABC : not (PrimCollinear Geo A B C))
    (hDEF : not (PrimCollinear Geo D E F))
    (hAB_DE : Geo.Congruent A B D E)
    (hAC_DF : Geo.Congruent A C D F)
    (hEF_BC : HilbertSegmentLess Geo E F B C) :
    HilbertAngleLess Geo E D F B A C
```

The proof begins by invoking `angle_trichotomy` on the two included angles.

In the equality branch, `SAS` produces a `TriangleCongruenceResult`; its field `sideBC` gives $BC \cong EF$. After reversing the congruence, the contradiction is discharged by `hilbert_segmentLess_not_congruent`.

In the second branch, Proposition I.24 is called with the two triangles interchanged. This gives $BC < EF$, contradicting the hypothesis through `hilbert_segmentLess_asymm`.

The third branch is returned directly.

25.8 Relationship with Earlier Propositions

25.8.1 Proposition I.4

Proposition I.4 supplies the SAS criterion used to eliminate the equality branch:

$$\angle BAC \cong \angle EDF, \quad AB \cong DE, \quad AC \cong DF \quad \Longrightarrow \quad BC \cong EF.$$

25.8.2 Proposition I.24

Proposition I.24 is the central direct dependency of I.25. I.24 states that, under two pairs of congruent sides, a greater included angle implies a greater third side. I.25 reverses the comparison: a greater third side implies a greater included angle.

The formal proof of I.25 uses I.24 exactly once, to eliminate the wrong direction of angle inequality.

Together they give, under

$$AB \cong DE, \qquad AC \cong DF,$$

the order correspondence

$$\angle EDF < \angle BAC \quad \iff \quad EF < BC.$$

25.9 Hilbert and Book Zero Components Used

25.9.1 `angle_trichotomy`

This is the principal structural result exposed by I.25. It converts the comparison of two nondegenerate angles into an exhaustive three-way split: congruent, smaller in one direction, or smaller in the other direction.

25.9.2 SAS

The existing SAS theorem converts equality of the included angles and the two side congruences into congruence of the third sides.

25.9.3 `hilbert_segmentLess_not_congruent`

This theorem expresses incompatibility between strict segment order and segment congruence. It eliminates the first trichotomy branch.

25.9.4 `hilbert_segmentLess_asymm`

This theorem expresses asymmetry of strict segment comparison:

$$AB < CD \quad \implies \quad \neg(CD < AB).$$

It eliminates the second branch after I.24 produces the opposite inequality.

25.10 Formalization Notes

Proposition I.25 provides a useful contrast with I.24.

I.24 required substantial geometric work. Its formal reconstruction had to expose point-order cases hidden by Euclid's diagram and develop the corresponding hinge argument in each configuration.

I.25 does not repeat any of that geometry. Instead, the reconstruction first strengthens the reusable Hilbert interface by proving angle trichotomy. Once this has been done, I.25 is a direct logical consequence of

angle trichotomy, SAS, I.24.

This separation is important for the architecture of the project. A long foundational or order-theoretic argument should not be buried inside every later Euclidean proposition that needs it. Once identified, it becomes part of the intermediate synthetic language and later proofs become correspondingly transparent.

I.25 is therefore a particularly clean example of the role of the Hilbert interface: difficult geometric infrastructure is proved once, after which a classical theorem can recover its natural short proof.

25.11 Dependency Summary

The underlying dependency graph is

```
Hilbert incidence + order + congruence
        |
        v
same-side and ray-order theory
        |
        v
angle_trichotomy
        |
        +-------------------+
        |                   |
        v                   v
       SAS            Proposition I.24
        |                   |
        +---------+---------+
                  |
                  v
       euclid_proposition_25
```

At the level of the final theorem:

```
angle_trichotomy
    |
    +-- BAC ~= EDF
    |     -> SAS
    |     -> BC ~= EF
    |     -> contradiction with EF < BC
    |
    +-- BAC < EDF
    |     -> I.24
    |     -> BC < EF
    |     -> contradiction with EF < BC
    |
```

```
+-- EDF < BAC
      -> exact
```

No new geometric axiom, continuity principle, coordinate system, or numerical angle measure is introduced.

25.12 Conclusion

Proposition I.25 is the converse comparison theorem to Proposition I.24.

Its formal reconstruction is exceptionally concise, but the concision is not accidental. The theorem becomes short only after the underlying angle-order theory has been completed.

The proof begins with angle trichotomy. Congruence of the included angles is excluded by SAS, because it would force congruence of the third sides. The wrong strict inequality is excluded by Proposition I.24, because it would force the opposite strict comparison of the third sides. The remaining branch is precisely the desired conclusion.

Thus I.25 gives a particularly clear instance of the architecture

$$\text{foundational Hilbert theory} \longrightarrow \text{reusable synthetic interface} \longrightarrow \text{short Euclidean theorem.}$$

Chapter 26

Proposition I.26

26.1 Introduction

Euclid's Proposition I.26 contains two triangle-congruence criteria.

For two nondegenerate triangles ABC and DEF, assume the two corresponding angles

$$\angle ABC \cong \angle DEF, \qquad \angle BCA \cong \angle EFD.$$

Euclid then considers two possibilities for the given corresponding side.

1. In the ASA case,
$$BC \cong EF.$$
2. In the SAA case,
$$AB \cong DE.$$

In either case the conclusion is full triangle congruence:

$$AB \cong DE, \qquad BC \cong EF, \qquad AC \cong DF,$$

together with

$$\angle BAC \cong \angle EDF.$$

The two parts have very different formal behavior.

The ASA part is almost entirely absorbed by the existing Hilbert congruence theory. The SAA part is not. Its proof still requires the essential contradiction argument of Euclid: a hypothetical strict inequality of the corresponding sides is converted into an auxiliary SAS configuration and then contradicted by Proposition I.16.

Thus Proposition I.26 provides a useful boundary case. One half is already available as interface-level geometry; the other still exposes genuine Book I proof structure.

26.2 Classical Configurations

Wilkins emphasizes that Proposition I.26 establishes both the ASA and the SAA congruence rules. The same basic pair of triangles is used for both arguments, but the auxiliary point lies on a different side in the two cases.

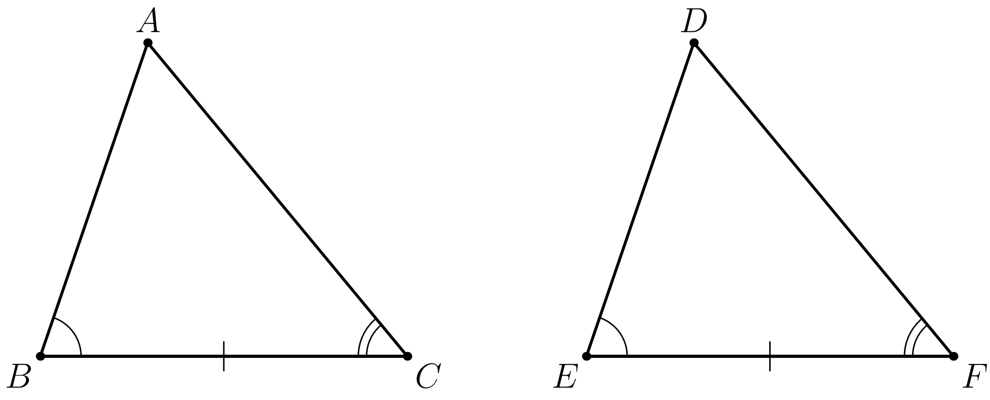


Figure 26.1: The common input configuration for Proposition I.26. The two pairs of corresponding angles are marked by one and two arcs respectively, and in the ASA case the corresponding side $BC \cong EF$ is marked by a single tick.

26.2.1 ASA auxiliary configuration

Assume first

$$BC \cong EF.$$

Euclid supposes, for contradiction, that the corresponding sides AB and DE are unequal. If AB is the greater side, choose a point G between A and B such that

$$GB \cong DE,$$

and join G to C.

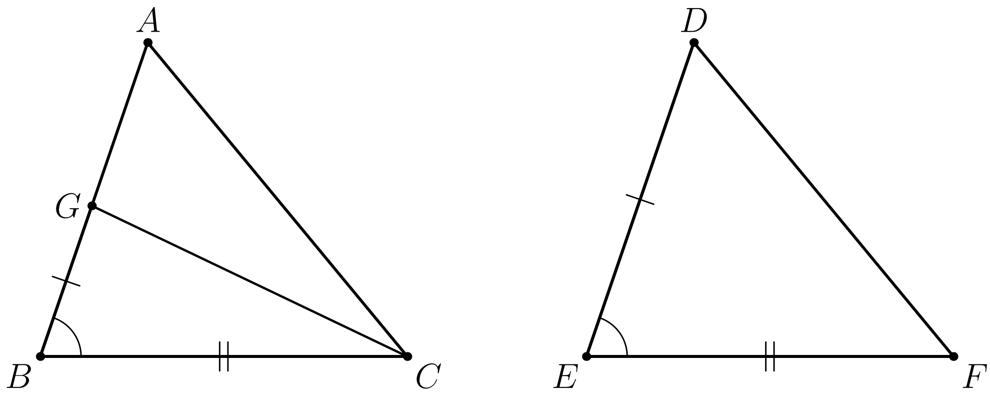


Figure 26.2: Euclid's ASA reductio configuration. The point G lies between A and B. The diagram marks exactly the local SAS data $GB \cong DE$, $BC \cong EF$, and $\angle GBC \cong \angle DEF$.

SAS gives

$$\angle BCG \cong \angle EFD.$$

Together with the original hypothesis

$$\angle EFD \cong \angle BCA,$$

this would imply

$$\angle BCG \cong \angle BCA.$$

But G lies inside the side AB, so $\angle BCG$ is a proper part of $\angle BCA$. This contradiction forces

$$AB \cong DE.$$

A final SAS application to the original triangles gives the remaining side and angle congruences.

26.2.2 SAA auxiliary configuration

Now assume instead

$$AB \cong DE.$$

Suppose, for contradiction, that $EF < BC$. Then there is a point H between B and C such that

$$BH \cong EF.$$

Join A to H.

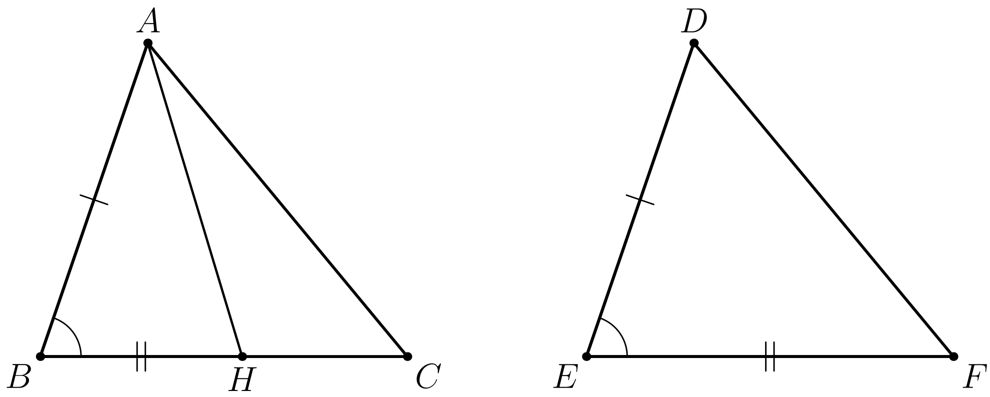


Figure 26.3: Euclid's SAA reductio configuration. The point H lies between B and C. The diagram marks exactly the local SAS data $AB \cong DE$, $BH \cong EF$, and $\angle ABH \cong \angle DEF$. Proposition I.16 is then applied to $\triangle AHC$.

Since B, H, C are collinear on the same ray from B,

$$\angle ABH \cong \angle ABC.$$

Hence SAS on $\triangle BAH$ and $\triangle EDF$ gives

$$\angle BHA \cong \angle EFD.$$

Using the original angle hypothesis,

$$\angle EFD \cong \angle BCA,$$

we obtain

$$\angle BHA \cong \angle BCA.$$

On the other hand, $C - H - B$, so $\angle BHA$ is an exterior angle of triangle AHC. Proposition I.16 gives the strict inequality

$$\angle BCA < \angle BHA,$$

contradicting their congruence.

Thus

$$\neg(EF < BC).$$

Interchanging the two triangles and repeating the same argument gives

$$\neg(BC < EF).$$

Segment trichotomy therefore yields

$$BC \cong EF.$$

The SAA case has now been reduced to the ASA case.

26.3 Diagrammatic Audit

Three figures are retained because they have three genuinely different jobs.

Figure 26.1 records the common input configuration. Figure 26.2 isolates the auxiliary SAS configuration used in the ASA reductio. Figure 26.3 isolates the corresponding SAS configuration used in the SAA reductio.

The last two diagrams deliberately do not mark every hypothesis of Proposition I.26. They mark only the congruences used in the local SAS step. This follows the logic of Wilkins's presentation and avoids overloading a single figure with relations that belong to different stages of the proof.

The auxiliary points are constructed geometrically: G lies genuinely on the segment AB, and H lies genuinely on the segment BC. Thus the incidence information used by the proof is represented directly rather than being approximated by hand-chosen coordinates.

These are not successive snapshots of one construction. They represent the two different reductio mechanisms contained in Proposition I.26.

26.4 Hilbert Reconstruction

The two halves of Proposition I.26 behave differently over the present Hilbert interface.

26.4.1 ASA

The theorem `hilbert_asa_sides` already contains the central mathematical content of the ASA case.

The given side in Euclid's formulation is

$$BC \cong EF.$$

The theorem is applied after cyclically viewing the triangles as

$$\triangle BCA \qquad \text{and} \qquad \triangle EFD.$$

After the required angle and endpoint reorientations it gives

$$BA \cong ED, \qquad CA \cong FD.$$

These are transported to

$$AB \cong DE, \qquad AC \cong DF.$$

At that point all three pairs of sides are congruent. The existing `HilbertSSS` theorem supplies the remaining corresponding angle:

$$\angle BAC \cong \angle EDF.$$

Thus the formal ASA proof does not reconstruct Euclid's reductio.

26.4.2 SAA

No permutation of `hilbert_asa_sides` directly solves the SAA problem. With the side $AB \cong DE$ fixed, ASA would require the angles at A and B, while the hypotheses supply the angles at B and C.

The formal reconstruction therefore isolates Euclid's essential contradiction argument in the helper

```
euclid_proposition_26_saa_not_less
```

The helper proves

$$\neg(EF < BC).$$

Its architecture is

```
EF < BC
   |
witness H with B-H-C and BH ~= EF
   |
same-ray transport:
angle ABH ~= angle ABC
   |
SAS on BAH and EDF
   |
angle BHA ~= angle EFD ~= angle BCA
   |
I.16 on triangle ACH
   |
angle BCA < angle BHA
   |
contradiction
```

The same helper is then called again with the two triangles interchanged, producing

$$\neg(BC < EF).$$

Book Zero segment trichotomy yields

$$BC \cong EF,$$

and the public SAA theorem finishes by calling the already proved ASA case.

26.5 Proof Architecture of the Reconstruction

The two formal paths can be summarized separately.

```
I.26 ASA
--------

two angle congruences
+ BC ~= EF
      |
      v
hilbert_asa_sides
on BCA and EFD
      |
      v
AB ~= DE
AC ~= DF
      |
      v
HilbertSSS
      |
      v
angle BAC ~= angle EDF

I.26 SAA
--------

two angle congruences
+ AB ~= DE
      |
      +----------------------+
      |                      |
      v                      v
not (EF < BC)          not (BC < EF)
SAS + I.16             same helper,
                       triangles swapped
      |                      |
      +----------+-----------+
                 |
                 v
      bookZero_31_trichotomy1
                 |
                 v
             BC ~= EF
                 |
                 v
          I.26 ASA case
                 |
                 v
AC ~= DF
angle BAC ~= angle EDF
```

The contrast with Euclid is precise.

For ASA, the Hilbert interface has already absorbed the reductio into a general theorem.

For SAA, Proposition I.16 remains a genuine formal dependency.

26.6 Lean Formalization

The complete implementation is available in the repository. Only the decisive interface steps are reproduced here.

For ASA, the first substantial call is

```
have hASA :=
  hilbert_asa_sides
    Geo
    B C A
    E F D
    hBCA
    hEFD
    hBC
    hAngleB'
    hAngleC
```

After endpoint reversal this gives the two missing sides. The final angle is then obtained from

```
have hSSS :=
  HilbertSSS
    Geo
    A B C
    D E F
    hABC
    hAB
    hBC
    hAC
```

For SAA, the public proof first excludes both strict segment inequalities:

```
have hNotEF_BC :=
  euclid_proposition_26_saa_not_less
    Geo A B C D E F
    hABC hDEF hAngleB hAngleC hAB
```

The same helper is applied with the two triangles interchanged. Segment trichotomy then produces the missing side congruence, and the theorem reduces to the already established ASA case.

This organization keeps the long exterior-angle argument inside one named helper rather than duplicating it in the public theorem.

26.7 Relationship with Earlier Propositions

26.7.1 Proposition I.4

Euclid uses SAS in both halves of I.26.

In the ASA proof, SAS is used after cutting off a segment congruent to the corresponding side of the second triangle.

In the SAA proof, SAS is applied to the auxiliary triangle BAH and $\triangle EDF$.

The formal SAA helper preserves this use of SAS directly.

26.7.2 Proposition I.16

Proposition I.16 is the decisive difference between the ASA and SAA arguments.

The SAA helper requires the exterior-angle inequality

$$\angle BCA < \angle BHA.$$

This strict inequality contradicts the angle congruence obtained from SAS. Without I.16 the helper would not close.

26.7.3 Segment trichotomy

The formal SAA proof separates the two possible strict segment inequalities:

$$EF < BC \qquad \text{and} \qquad BC < EF.$$

After both are excluded, `bookZero_31_trichotomy1` returns the remaining possibility

$$BC \cong EF.$$

Thus Book Zero supplies the final order-theoretic bridge from the two contradiction arguments to the ASA case.

26.8 Hilbert and Book Zero Components Used

26.8.1 `hilbert_asa_sides`

This is the main interface theorem for the ASA half. It converts a common corresponding side together with the two adjacent angle congruences into the remaining two side congruences.

Its availability explains why the formal ASA proof is much shorter than Euclid's proof.

26.8.2 `HilbertSSS`

Once all three corresponding side congruences have been recovered, SSS provides the remaining angle congruence required by the public statement.

26.8.3 euclid_proposition_26_saa_not_less

This local helper is the formal core of the SAA proof. It packages the classical argument

$$\text{strict side inequality} \longrightarrow \text{SAS} \longrightarrow \text{angle congruence} \longrightarrow \text{I.16 contradiction.}$$

The helper is reusable under interchange of the two triangles, so the symmetric strict-inequality branch does not duplicate the proof.

26.8.4 bookZero_31_trichotomy1

This theorem converts the exclusion of both strict comparisons into segment congruence.

It is the precise formal replacement for the classical conclusion that, if neither segment is greater than the other, the two segments are equal.

26.9 Formalization Notes

Proposition I.26 is a particularly useful diagnostic theorem.

The ASA part shows what happens when a classical Euclidean proof has already been absorbed into the lower Hilbert interface. The Book I wrapper is then short and need not reproduce the historical reductio.

The SAA part shows the opposite situation. The available ASA theorem cannot be reached merely by permuting the vertices, because the given side is not the side between the two supplied angles. A genuine argument is still required.

The formal proof therefore mirrors an important part of Euclid's historical structure: it constructs an interior point, applies SAS, and invokes the exterior-angle theorem I.16.

At the same time, the proof is reorganized to expose the reusable logical unit. Instead of proving two symmetric strict-inequality branches independently, one helper is applied twice.

This gives the architecture

$$\text{classical geometric contradiction} \longrightarrow \text{named helper} \longrightarrow \text{segment trichotomy} \longrightarrow \text{existing ASA interface.}$$

No new geometric axiom, numerical length, coordinate system, or numerical angle measure is introduced.

26.10 Dependency Summary

At the classical level:

```
ASA:
I.3-style segment cut
   |
```

```
I.4 (SAS)
   |
proper-angle contradiction
   |
equal corresponding side
   |
I.4
   |
triangle congruence


SAA:
segment cut
   |
I.4 (SAS)
   |
I.16
   |
contradiction
   |
equal corresponding side
   |
I.4
   |
triangle congruence
```

At the formal level:

```
ASA:
hilbert_asa_sides
      |
      v
two missing sides
      |
      v
HilbertSSS
      |
      v
euclid_proposition_26_asa


SAA:
SAS + Proposition I.16
      |
      v
euclid_proposition_26_saa_not_less
      |
      +-- exclude EF < BC
      |
      +-- exclude BC < EF
      |
```

```
      v
bookZero_31_trichotomy1
      |
      v
BC ~= EF
      |
      v
euclid_proposition_26_asa
      |
      v
euclid_proposition_26_saa
```

Thus Proposition I.26 exhibits both modes of the current reconstruction in a single theorem: one half collapses to established interface geometry, while the other still requires a substantial Euclidean argument.

26.11 Conclusion

Proposition I.26 completes the elementary triangle-congruence theory of the first part of Book I by establishing the ASA and SAA criteria.

The formal reconstruction makes the distinction between the two cases sharper than the classical presentation.

ASA is already essentially available in the Hilbert interface. The Book I proof only has to orient the triangles correctly, recover the missing sides, and use SSS for the remaining angle.

SAA cannot be reduced by permutation alone. Its proof requires the exterior-angle theorem I.16 to eliminate a hypothetical strict inequality of the corresponding sides. Once both strict inequalities are excluded, Book Zero segment trichotomy reduces SAA to ASA.

The proposition therefore gives a compact example of the layered architecture of the project:

$$\text{Hilbert interface} \longrightarrow \text{Book Zero order theory} \longrightarrow \text{Euclidean congruence theorem.}$$

Chapter 27

Proposition I.27

27.1 Introduction

Proposition I.27 is the first proposition in Book I whose conclusion is parallelism. Its mathematical content is the converse direction of the familiar alternate-angle criterion:

$$\text{equal alternate interior angles} \quad \Longrightarrow \quad \text{parallel straight lines.}$$

No form of Euclid's Fifth Postulate is required. The proposition belongs to neutral geometry: Euclid proves it by showing that any hypothetical intersection of the two lines would contradict Proposition I.16.

The present formalization is especially interesting because the Hilbert development already contains the alternate-angle implication as a general theorem. At the same time, translating the classical transversal diagram into the hypotheses of that theorem exposes additional incidence, order, ray-orientation, and side-of-line information that is suppressed in the traditional picture.

For this reason the Lean file deliberately retains two formulations of Proposition I.27. Their precise proof-theoretic relation is left visible rather than being eliminated by premature refactoring.

27.2 Classical Statement

Let a straight line meet the straight lines AB and CD at E and F, respectively. Suppose that the alternate interior angles

$$\angle AEF \qquad \text{and} \qquad \angle EFD$$

are equal. Then

$$AB \parallel CD.$$

The basic configuration is shown in Figure 27.1.

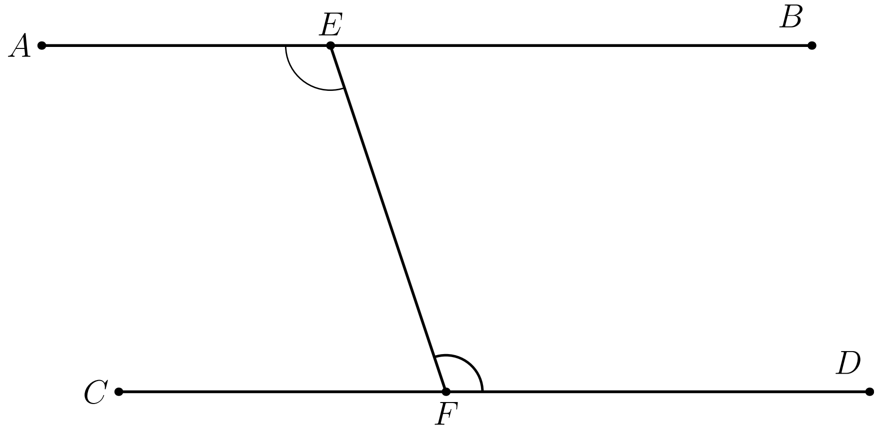


Figure 27.1: The transversal configuration of Proposition I.27. The alternate interior angles $\angle AEF$ and $\angle EFD$ are congruent.

27.3 Euclid's Proof

The proof is by contradiction.

Suppose first that the straight lines AB and CD, when produced beyond B and D, meet at a point G. Then $\angle AEF$ is an exterior angle of the triangle EFG. By Proposition I.16,

$$\angle EFD < \angle AEF.$$

This contradicts the hypothesis

$$\angle AEF \cong \angle EFD.$$

The corresponding reductio configuration is shown in Figure 27.2.

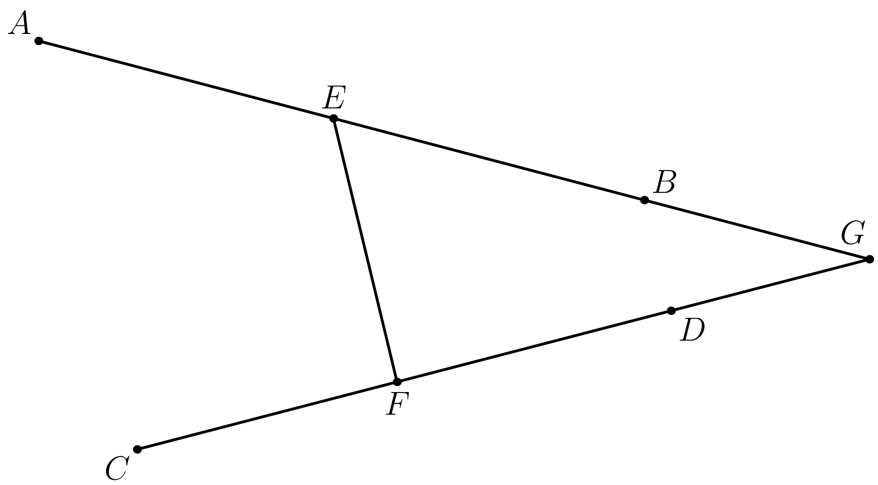


Figure 27.2: First reductio branch. If the two lines meet beyond B and D at G, Proposition I.16 makes $\angle AEF$ strictly greater than $\angle EFD$.

There is a second possible direction of intersection. Suppose that the lines, when produced beyond A and C, meet at a point H. Then $\angle EFD$ is an exterior angle of the triangle EFH. Again Proposition I.16 gives

$$\angle AEF < \angle EFD,$$

which contradicts the same angle-congruence hypothesis.

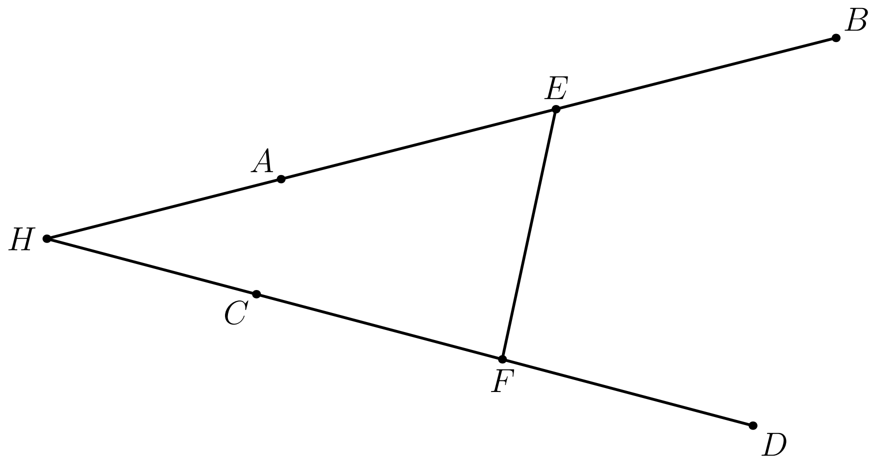


Figure 27.3: Second reductio branch. If the two lines meet beyond A and C at H, Proposition I.16 makes $\angle EFD$ strictly greater than $\angle AEF$.

Both possible directions of intersection therefore lead to contradictions. Hence the two straight lines do not meet:

$$AB \parallel CD.$$

This is exactly the classical content of Proposition I.27.

27.4 Diagrammatic Audit

Three figures are retained because they represent three logically distinct configurations.

Figure 27.1 records only the hypothesis of the proposition: a transversal and a pair of congruent alternate interior angles.

Figure 27.2 represents the first hypothetical intersection, beyond B and D. Figure 27.3 represents the second hypothetical intersection, beyond A and C.

The last two figures are therefore not construction stages and are not simultaneously realizable with the original angle-congruence hypothesis. They are diagrams of the two reductio branches. Their purpose is to make the two applications of Proposition I.16 geometrically transparent.

As in Proposition I.26, the figures should encode the local geometric step being used rather than accumulate every relation occurring elsewhere in the proof.

27.5 Hilbert Reconstruction

The Hilbert development reaches the mathematical core of I.27 before the Euclidean sequence reaches Proposition I.27.

At the synthetic-interface level the implication

$$\text{equal alternate angles} \quad \Longrightarrow \quad \text{parallel lines}$$

is already available through

```
parallel_from_equal_angles
```

The first Lean formulation therefore exposes I.27 directly at this level. Its proof is only an application of the existing theorem.

This is not an appeal to the Euclidean parallel postulate. The alternate-angle implication is a theorem of the neutral Hilbert development. The logical direction is important: I.27 proves a sufficient condition for parallelism; the converse direction used in Euclidean geometry appears later, in Proposition I.29, where the Fifth Postulate enters.

27.6 Two Formal Formulations

The current Lean file intentionally retains two formulations.

27.6.1 The interface-level formulation

The theorem

```
euclid_proposition_27
```

uses the ordered configuration already expected by `parallel_from_equal_angles`. Its proof is:

```
exact
  parallel_from_equal_angles
    Geo
    A C D
    B E F
    hADC
    hCEB
    hDEF
    hCED
    hAngle
```

At this level the central mathematical content of Proposition I.27 has already been packaged by the Hilbert theory. The Book I theorem merely exposes that result in the Euclidean sequence.

27.6.2 The explicit transversal formulation

The second theorem is

```
euclid_proposition_27_transversal
```

and begins with a configuration closer to the classical diagram:

$$A - E - B, \qquad C - F - D,$$

with E and F on an explicitly supplied transversal *trans*. It also records that the selected interior arms EA and FD lie on opposite sides of the transversal and assumes

$$\angle AEF \cong \angle EFD.$$

This version reveals information that the traditional phrase "alternate interior angles" leaves implicit.

27.7 Normalization of the Transversal Configuration

The explicit proof first chooses a point M between E and F:

$$E - M - F.$$

This provides explicit representatives for the two directions of the transversal. Betweenness gives the corresponding same-ray relations. After reversing the first angle and transporting its transversal arm along these rays, the original congruence

$$\angle AEF \cong \angle EFD$$

is normalized to

$$\angle MEA \cong \angle MFD.$$

The theorem

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

then yields the local parallelism

$$EA \parallel FD.$$

The remainder of the proof is transport along the two original straight lines. Since

$$A - E - B$$

the first local line is extended from EA to AB. Since

$$C - F - D,$$

the second is transported from FD to CD. Thus

$$AB \parallel CD.$$

27.8 Proof Architecture

The classical and formal routes may be displayed schematically as follows.

```
Euclid
------
angle AEF ~= angle EFD
        |
assume AB and CD meet
        |
        +-----------------------------+
        |                             |
 meet beyond B,D                meet beyond A,C
        |                             |
 exterior angle AEF             exterior angle EFD
        |                             |
       I.16                          I.16
        |                             |
 EFD < AEF                      AEF < EFD
        |                             |
 contradiction                   contradiction
        +-------------+---------------+
                      |
                  AB || CD


Hilbert interface
-----------------
ordered alternate-angle configuration
        |
parallel_from_equal_angles
        |
     parallel


Explicit transversal formalization
----------------------------------
A-E-B, C-F-D
+ transversal EF
+ opposite-side information
+ angle AEF ~= angle EFD
        |
choose M with E-M-F
        |
same-ray and angle normalization
```

```
        |
angle MEA ~= angle MFD
        |
hilbert_parallel_of_alternate_angles_oppositeSide_lines
        |
EA || FD
        |
transport along A-E-B and C-F-D
        |
AB || CD
```

27.9 Proof-Theoretic Status

The coexistence of the two Lean formulations is deliberate.

The short formulation makes the mathematical core of I.27 unusually visible: the proposition coincides closely with an already established general theorem of the Hilbert layer.

The explicit transversal formulation answers a different question: what formal information is required to pass from the classical diagram to that general theorem? Its additional work concerns incidence, betweenness, same rays, side-of-line data, orientation of angles, and transport of parallelism.

At present we do not force a final interpretation of this distinction. It may eventually be preferable to regard the short theorem as the canonical Book I interface and the transversal theorem as a normalization result. It may instead prove useful to retain the explicit configuration as part of the public Euclidean layer.

Keeping both versions preserves this question for later comparison with the proof corpus and with the dependency graph of the Hilbert reconstruction.

27.10 Relationship with Proposition I.16

Euclid's classical proof depends directly on Proposition I.16. Each possible intersection converts one of the supposedly equal alternate angles into an exterior angle of a triangle, and I.16 makes that angle strictly greater than the remote interior angle.

The present Lean theorem does not invoke `euclid_proposition_16` directly at the Book I level because the corresponding alternate-angle theorem has already been established inside the Hilbert theory.

Thus the difference is one of proof organization:

$$\begin{aligned} \text{Euclid} &: \text{I.16} \longrightarrow \text{I.27}, \\ \text{Hilbert reconstruction} &: \text{order and angle theory} \longrightarrow \text{alternate-angle parallelism} \longrightarrow \text{I.27}. \end{aligned}$$

The Book I order and the formal dependency order therefore need not coincide.

27.11 Neutral Geometry and the Fifth Postulate

Proposition I.27 is a useful logical boundary marker.

It establishes

$$\text{equal alternate interior angles} \Longrightarrow \text{parallel lines}$$

without using the Fifth Postulate. This direction is valid in neutral geometry.

The converse statement,

$$\text{parallel lines} \Longrightarrow \text{equal alternate interior angles},$$

is part of Proposition I.29 and requires Euclidean parallelism.

Consequently I.27 should not be classified as a theorem depending on Hilbert's Euclidean parallel axiom merely because its conclusion contains the word "parallel". Its proof uses the definition of parallelism as nonintersection, together with the neutral theory developed before it.

27.12 Lean Formalization

The complete implementation is contained in `Proposition27.lean`. The public file currently records both the interface-level theorem and the explicit transversal theorem.

The latter proof has the following principal stages:

1. choose M with $E - M - F$;
2. obtain the two same-ray relations along the transversal;
3. normalize the given angle congruence;
4. apply the Hilbert alternate-angle parallelism theorem;
5. transport the resulting local parallelism along the collinear triples $A - E - B$ and $C - F - D$.

No Euclidean parallel axiom is used.

27.13 Conclusion

Proposition I.27 is formally small at the level of the developed Hilbert interface but conceptually revealing.

Euclid proves parallelism by excluding the two possible intersection directions through Proposition I.16. The Hilbert development has already abstracted this reasoning into a reusable alternate-angle theorem, so the core Book I formulation becomes a thin wrapper.

The explicit transversal theorem shows what that compression hides: the classical diagram silently carries order, orientation, side-of-line, and ray information that must be made explicit before the abstract theorem can be applied.

For the moment both formalizations are retained. Their coexistence is not duplication to be removed immediately, but a useful record of the difference between the mathematical content of I.27 and the formal normalization of its classical diagram.

Chapter 28

Proposition I.28

28.1 Introduction

Proposition I.28 gives two sufficient angle conditions for parallelism. A transversal meets two straight lines. Parallelism follows either from equality of a pair of corresponding angles or from the condition that the two interior angles on the same side are together equal to two right angles.

Both branches reduce to Proposition I.27. In each case the essential target is the same equality of alternate interior angles.

No form of the Fifth Postulate is used. Proposition I.28 belongs to neutral geometry.

28.2 Classical Configuration

Let the straight line EF meet AB at G and CD at H. The configuration is

$$A - G - B, \qquad C - H - D, \qquad E - G - H - F.$$

The proposition has two alternative hypotheses.

28.2.1 Corresponding-angle criterion

Assume

$$\angle EGB \cong \angle GHD.$$

Since $A - G - B$ and $E - G - H$, the angles

$$\angle EGB \qquad \text{and} \qquad \angle AGH$$

are vertical angles. By Proposition I.15,

$$\angle AGH \cong \angle EGB.$$

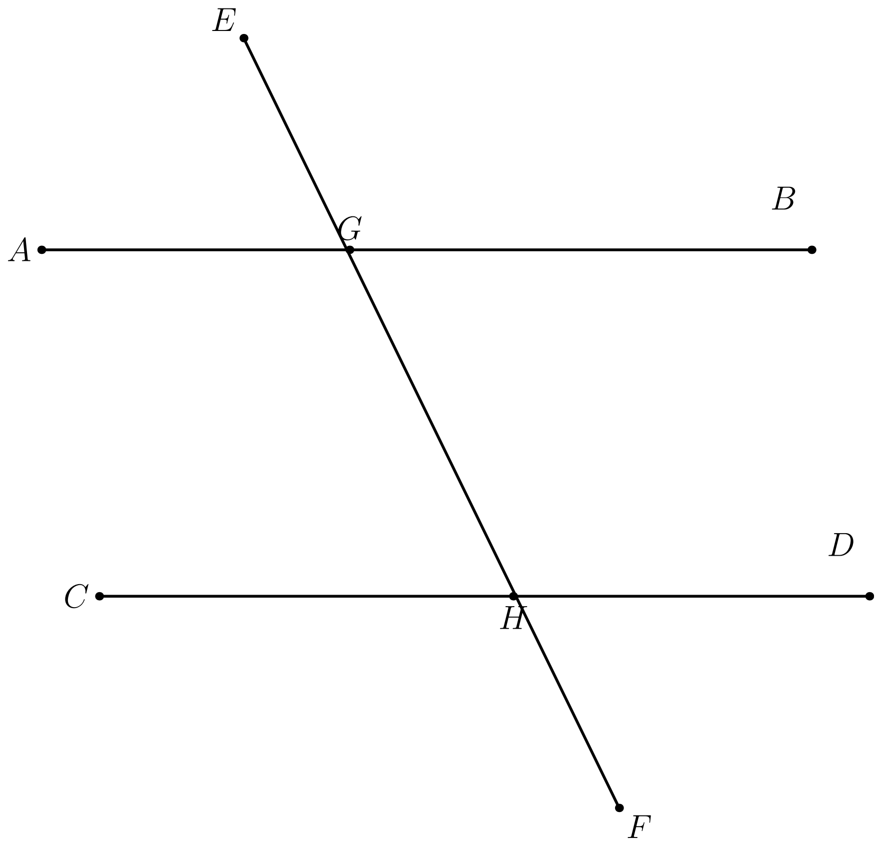


Figure 28.1: The common transversal configuration for Proposition I.28.

Hence

$$\angle AGH \cong \angle GHD.$$

These are alternate interior angles, so Proposition I.27 gives

$$AB \parallel CD.$$

28.2.2 Same-side interior-angle criterion

Assume that

$$\angle BGH \quad \text{and} \quad \angle GHD$$

are together equal to two right angles.

Because $A - G - B$, the angles $\angle AGH$ and $\angle BGH$ form a supplementary pair. Proposition I.13 gives the second equality to two right angles. Subtracting the common angle $\angle BGH$ leaves

$$\angle AGH \cong \angle GHD.$$

Again these are alternate interior angles, and Proposition I.27 gives

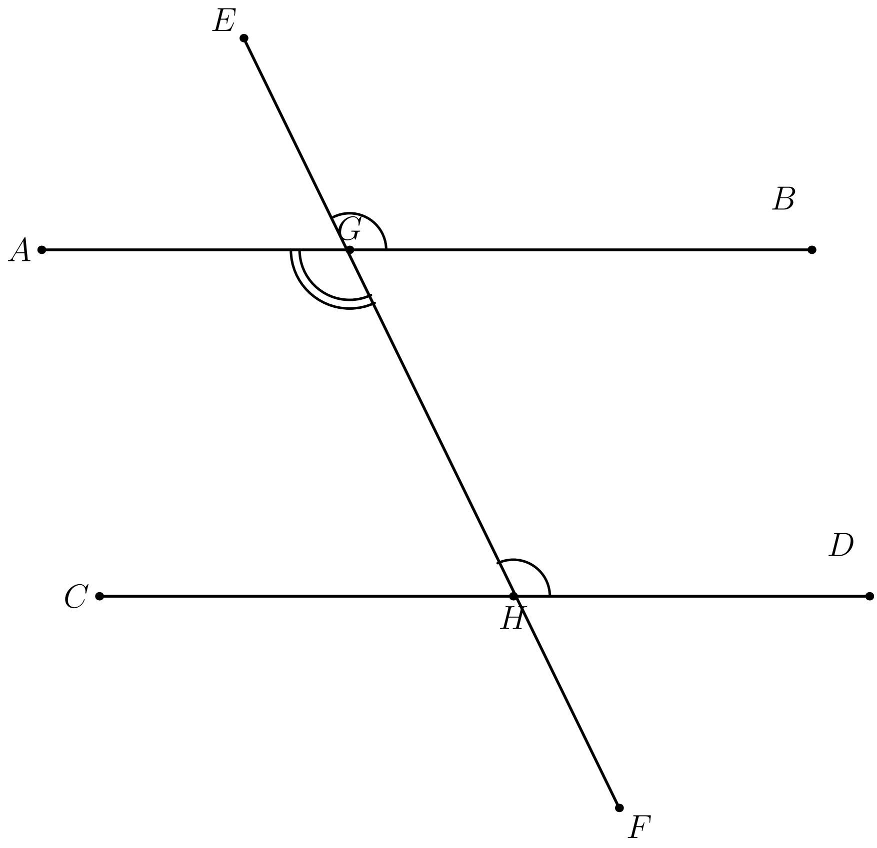


Figure 28.2: The first criterion. The corresponding angles $\angle EGB$ and $\angle GHD$ are congruent. Proposition I.15 identifies $\angle EGB$ with the vertical angle $\angle AGH$, reducing the argument to Proposition I.27.

$$AB \parallel CD.$$

28.3 Hilbert Reconstruction

The formal reconstruction preserves the common geometric core of the two branches:

$$\angle AGH \cong \angle GHD \implies AB \parallel CD.$$

This final implication is exactly the explicit transversal form of Proposition I.27.

28.3.1 First branch

The Lean theorem for the corresponding-angle criterion first applies `VerticalAngles`. From the two betweenness relations at G it obtains the vertical-angle congruence. A reversal of the second angle and transitivity with the given corresponding-angle congruence yield

$$\angle AGH \cong \angle GHD.$$

The proof then invokes

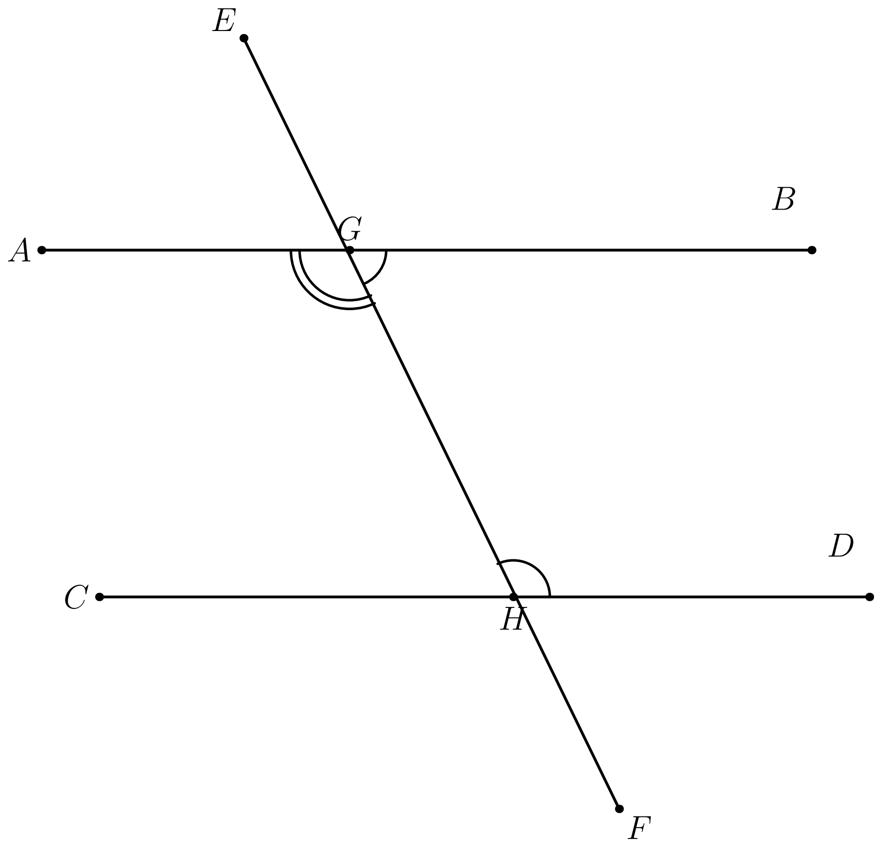


Figure 28.3: The second criterion. The interior angles $\angle BGH$ and $\angle GHD$ on the same side are together equal to two right angles. The supplementary angle to $\angle BGH$ is $\angle AGH$, so the condition reduces to the alternate-angle hypothesis of Proposition I.27.

`euclid_proposition_27_transversal`

with G and H as the two intersection points of the transversal.

Schematically,

$$\begin{gathered} \angle EGB \cong \angle GHD \\ \angle AGH \cong \angle EGB \quad \text{(vertical angles)} \\ \downarrow \\ \angle AGH \cong \angle GHD \\ \downarrow \\ \texttt{euclid\_proposition\_27\_transversal} \\ \downarrow \\ AB \parallel CD. \end{gathered}$$

28.3.2 Second branch

The project does not introduce numerical angle measure or an operation of angle addition. The statement that two angles are together equal to two right angles is represented synthetically by an explicit supplementary ray.

The local predicate used in Proposition I.28 is

```
def HilbertAnglesEqualTwoRightAnglesWithSupplement
    (A O B C P D E : Geo.Point) : Prop :=
  Geo.Between D P E /\
  Geo.AngleCongruent A O B C P E
```

For the I.28 configuration it is instantiated as

```
HilbertAnglesEqualTwoRightAnglesWithSupplement
  Geo G H D H G B A
```

and therefore supplies

$$B - G - A$$

together with

$$\angle GHD \cong \angle HGA.$$

Betweenness symmetry recovers $A - G - B$. Symmetry and reversal of angle congruence then give

$$\angle AGH \cong \angle GHD.$$

The proof again ends with

```
euclid_proposition_27_transversal
```

and hence with $AB \parallel CD$.

28.4 Neutral Geometry

The logical direction in both parts of Proposition I.28 is

$$\text{angle condition} \quad \Longrightarrow \quad \text{parallel lines.}$$

This direction does not require Hilbert's Group IV. The proof ultimately reduces to the neutral alternate-angle criterion already formalized in Proposition I.27.

Thus Propositions I.27 and I.28 belong to the neutral part of the theory. The reverse direction, from parallelism to the corresponding angle relations, appears in Proposition I.29 and is the point at which Euclidean parallelism enters.

28.5 Dependency Summary

The two formal branches are

$$
\begin{array}{ccc}
\text{corresponding-angle congruence} & & \text{same-side angles equal to } 2R \\
\downarrow & & \downarrow \\
\text{vertical angles} & & \text{synthetic supplementary encoding} \\
\downarrow & & \downarrow \\
\text{alternate interior angles congruent} & & \text{alternate interior angles congruent} \\
\downarrow & & \downarrow \\
\text{I.27} & & \text{I.27} \\
\downarrow & & \downarrow \\
AB \parallel CD & & AB \parallel CD.
\end{array}
$$

No new parallel axiom is introduced.

28.6 Conclusion

Proposition I.28 adds two practical recognition criteria for parallel lines, but it does not introduce a new mechanism of parallelism. Both criteria are converted into the alternate-angle condition of Proposition I.27.

The first conversion uses vertical angles. The second uses the synthetic supplementary-angle language already developed in the reconstruction. Consequently the formal proof remains entirely within Hilbert incidence, order, congruence, and the neutral theory of parallelism.

Chapter 29

Proposition I.29

29.1 Introduction

Proposition I.29 is the first proposition in Book I whose proof genuinely uses the Euclidean parallel axiom.

The preceding Propositions I.27 and I.28 establish implications of the form

$$\text{angle condition} \longrightarrow \text{parallel lines.}$$

Those implications belong to neutral geometry. Proposition I.29 reverses the direction:

$$\text{parallel lines} \longrightarrow \text{angle relations.}$$

This is the point at which Hilbert's Group IV enters the reconstruction.

The proposition has three conclusions. If a transversal cuts two parallel straight lines, then:

$$\angle AGH \cong \angle GHD,$$

the alternate interior angles are congruent;

$$\angle EGB \cong \angle GHD,$$

the exterior angle is congruent to the interior and opposite angle on the same side; and the same-side interior angles

$$\angle BGH \qquad \text{and} \qquad \angle GHD$$

are together equal to two right angles.

Only the first conclusion contains the genuinely Euclidean step. The other two are consequences of the first by vertical-angle and supplementary-angle geometry.

29.2 Classical Configuration

Let the straight line EF meet the parallel straight lines AB and CD at G and H, respectively:

$$A - G - B, \qquad C - H - D, \qquad E - G - H - F.$$

Assume

$$AB \parallel CD.$$

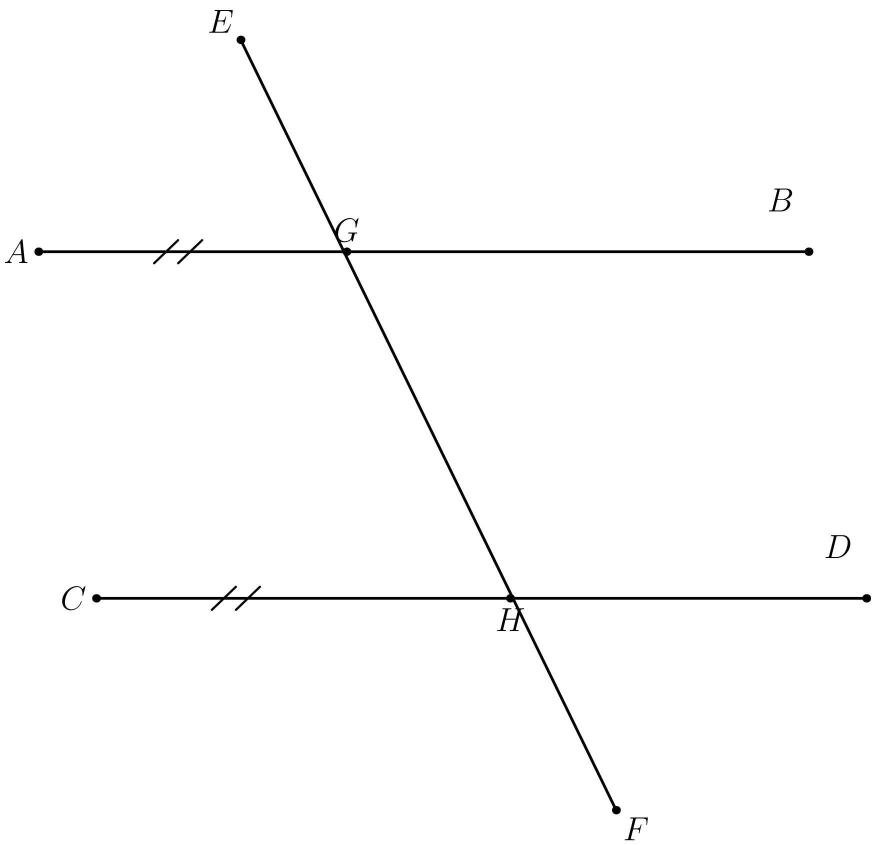


Figure 29.1: The transversal configuration of Proposition I.29. The straight lines AB and CD are parallel.

29.3 First Conclusion: Alternate Interior Angles

The principal assertion is

$$\angle AGH \cong \angle GHD.$$

This is the converse direction of Proposition I.27.

In the Hilbert reconstruction the corresponding general theorem is the Euclidean direction of Hilbert's Theorem 30. The proof constructs through the second intersection point a line making the required alternate angle. The neutral alternate-angle criterion shows that this constructed line is parallel to the first given line. Hilbert's parallel axiom then identifies the constructed line with the already given parallel line.

Structurally, the first part is exactly the construction mechanism isolated later in Proposition I.31: copy an arbitrary angle at the external point and use I.27 to obtain a parallel. No Euclidean parallel axiom is involved in this existence step.

Schematically,

$$\begin{gathered}
AB \parallel CD \\
\downarrow \\
\text{construct through } H \text{ a line with the required alternate angle} \\
\downarrow \\
\text{neutral alternate-angle criterion} \\
\downarrow \\
\text{constructed line is parallel to } AB \\
\downarrow \text{ Hilbert Group IV} \\
\text{constructed line} = CD \\
\downarrow \\
\angle AGH \cong \angle GHD.
\end{gathered}$$

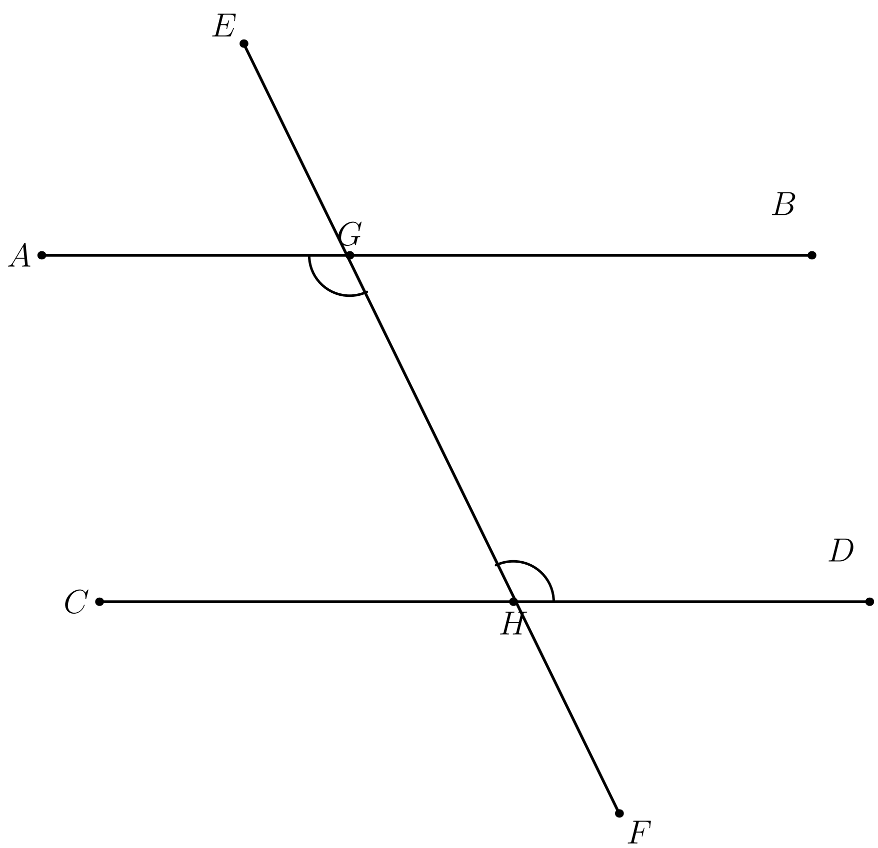


Figure 29.2: The Euclidean core of Proposition I.29: parallelism implies congruence of the alternate interior angles.

29.4 Second Conclusion: Corresponding Angles

Once

$$\angle AGH \cong \angle GHD$$

has been established, the second conclusion is neutral.

Because $A - G - B$ and $E - G - H$, the angles

$$\angle AGH \qquad \text{and} \qquad \angle EGB$$

are vertical angles. Proposition I.15 therefore gives

$$\angle EGB \cong \angle AGH.$$

Combining this with the alternate-angle conclusion yields

$$\angle EGB \cong \angle GHD.$$

Thus the second branch has the dependency pattern

$$\text{I.29 alternate} \longrightarrow \text{I.15 vertical angles} \longrightarrow \text{corresponding-angle congruence.}$$

29.5 Third Conclusion: Same-Side Interior Angles

The third conclusion states that

$$\angle BGH \qquad \text{and} \qquad \angle GHD$$

are together equal to two right angles.

The reconstruction does not introduce numerical angle measure or a binary operation of angle addition. Instead it uses the synthetic predicate

```
HilbertAnglesEqualTwoRightAnglesWithSupplement
```

already placed in the Hilbert interface.

Because

$$A - G - B,$$

the rays GA and GB are opposite. Hence $\angle AGH$ is the supplementary angle associated with $\angle BGH$.

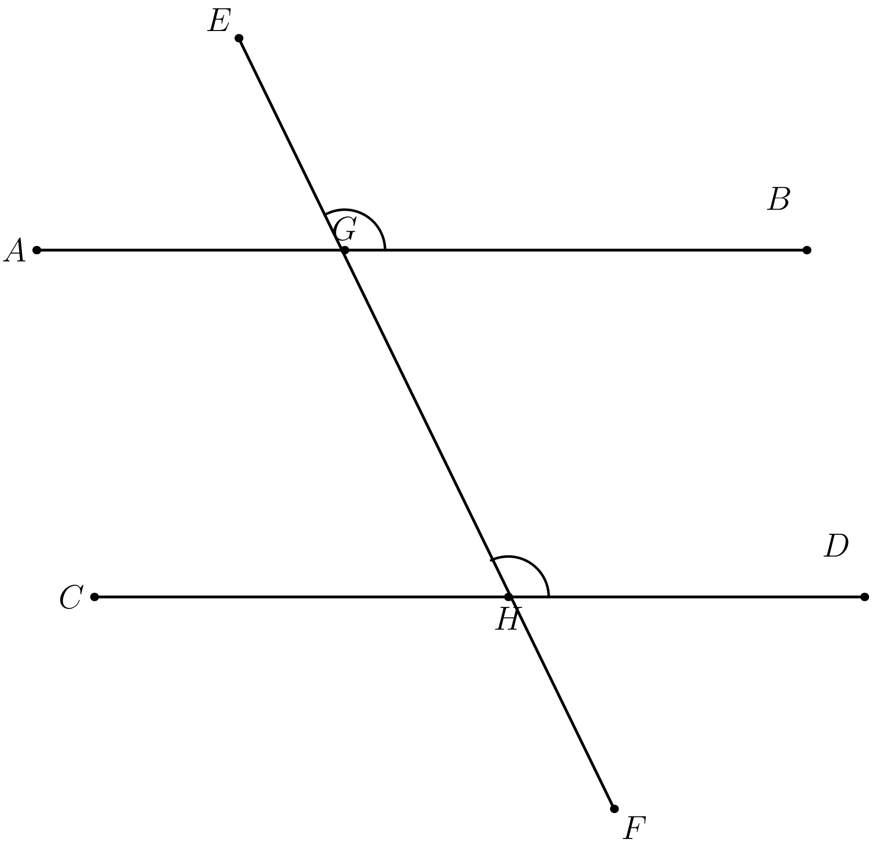


Figure 29.3: The corresponding-angle conclusion. The exterior angle $\angle EGB$ is vertical to $\angle AGH$, which is already congruent to $\angle GHD$.

From the first conclusion,

$$\angle AGH \cong \angle GHD.$$

Therefore the same-side pair satisfies the synthetic condition expressing that it is together equal to two right angles.

The dependency is

$$\begin{gathered}\angle AGH \cong \angle GHD \\ A - G - B \\ \downarrow \\ \text{supplementary configuration} \\ \downarrow \\ \angle BGH + \angle GHD = 2R \quad \text{(synthetically)}.\end{gathered}$$

29.6 Lean Formalization

The final Lean development contains five theorems.

The interface-level form is

```
euclid_proposition_29
```

and is a direct wrapper around

`equal_angles_from_parallel`

which is the Euclidean direction of Hilbert's Theorem 30.

The explicit transversal form is

`euclid_proposition_29_transversal`

with the conclusion

$$\angle AGH \cong \angle GHD.$$

Its proof first transports the global parallel relation

$$AB \parallel CD$$

to the local lines determined by the actual angle arms,

$$GA \parallel HD.$$

A point M is then chosen with

$$G - M - H.$$

The line-level Hilbert theorem gives

$$\angle MGA \cong \angle MHD.$$

Angle reversal and same-ray transport along the transversal convert this into

$$\angle AGH \cong \angle GHD.$$

The second theorem,

`euclid_proposition_29_corresponding`

combines the alternate-angle theorem with `VerticalAngles` to prove

$$\angle EGB \cong \angle GHD.$$

The third theorem,

`euclid_proposition_29_same_side`

combines the alternate-angle theorem with the supplementary relation $A - G - B$ and returns

`HilbertAnglesEqualTwoRightAnglesWithSupplement`

for the same-side interior angles.

Finally,

`euclid_proposition_29_all`

packages the three conclusions in one theorem.

29.7 Proof Architecture

The formal dependency structure is

$$\begin{array}{c} AB \parallel CD \\ \downarrow \\ \texttt{hilbert\_alternate\_angles\_of\_parallel\_oppositeSide\_lines} \\ \downarrow \text{ Hilbert Group IV} \\ \angle AGH \cong \angle GHD \end{array}$$

and then

$$\begin{array}{ccc} & \angle AGH \cong \angle GHD & \\ \swarrow & & \searrow \\ \text{vertical angles} & & \text{supplementary configuration} \\ \downarrow & & \downarrow \\ \angle EGB \cong \angle GHD & & \angle BGH + \angle GHD = 2R. \end{array}$$

Thus the Euclidean axiom is needed only once. After the alternate-angle conclusion has been obtained, the remaining two conclusions are neutral.

29.8 The Logical Boundary at I.29

Propositions I.27–I.29 form a particularly clear logical block.

$$\begin{aligned} \text{I.27} &: \text{ equal alternate angles} \longrightarrow \text{parallel lines,} \\ \text{I.28} &: \text{ other angle conditions} \longrightarrow \text{parallel lines,} \\ \text{I.29} &: \text{ parallel lines} \longrightarrow \text{angle relations.} \end{aligned}$$

The first two implications are neutral. The third is Euclidean.

This distinction is visible directly in the Lean typeclasses: I.27 and I.28 do not require the Euclidean plane structure, while I.29 uses

```
[HilbertEuclideanPlane Geo]
```

at the theorem that converts parallelism into alternate-angle congruence.

29.9 Existence and Uniqueness Behind I.29

The first conclusion of I.29 admits a useful structural decomposition.

The neutral part of the argument constructs through H a line having the required direction. Angle construction supplies the copied angle, and I.27 proves that the constructed line is parallel to AB. This is the same existence mechanism that appears explicitly in Proposition I.31.

At this stage neutral geometry gives only

$$\text{there exists a line through } H \text{ parallel to } AB.$$

But the hypotheses of I.29 already contain another line through H, namely CD, with

$$AB \parallel CD.$$

To conclude that the newly constructed line is actually CD, one needs uniqueness of the parallel through H. This is precisely the point at which Hilbert's Group IV, equivalently the Playfair-type uniqueness principle used in the reconstruction, enters.

Thus the logical core of the first conclusion can be displayed as

$$\begin{array}{c}
\text{angle construction} \\
\downarrow \\
\text{equal alternate angles} \\
\downarrow \\
\text{I.27} \\
\downarrow \\
\text{a constructed parallel through } H \\
\downarrow \quad \text{Group IV / parallel uniqueness} \\
\text{the constructed line is } CD \\
\downarrow \\
\text{the required angle congruence of I.29.}
\end{array}$$

This makes precise the observation that the Euclidean axiom is needed only once. It is not needed to construct a parallel direction; it is needed to identify the constructed parallel with the parallel line already present in the hypotheses.

Wilkins' discussion of the parallel propositions highlights this existence–uniqueness separation. Read retrospectively after I.31, the mechanism can be summarized schematically as

$$\text{I.31-type construction} + \text{I.27} + \text{Playfair uniqueness} \Longrightarrow \text{the angle conclusion of I.29.}$$

This is a structural analysis of I.29, not a claim that Euclid's original proof of I.29 depends on the later Proposition I.31.

29.10 Conclusion

Proposition I.29 is the first genuine Euclidean-parallelism theorem in the Book I reconstruction.

Its essential mathematical content is the single implication

$$AB \parallel CD \longrightarrow \angle AGH \cong \angle GHD.$$

That implication is Hilbert's Euclidean direction of the alternate-angle theorem and depends on Group IV.

The remaining statements of Euclid's proposition are consequences of this result. Corresponding angles are obtained through vertical-angle congruence, and the same-side two-right-angles condition is represented synthetically by a supplementary configuration.

The resulting formalization therefore isolates the exact point at which the parallel axiom enters and avoids attributing additional Euclidean strength to the two corollary branches.

Chapter 30

Proposition I.30

30.1 Introduction

Proposition I.30 establishes a structural property of Euclidean parallelism:

$$AB \parallel EF, \qquad CD \parallel EF$$

imply

$$AB \parallel CD.$$

In Euclid's proof the conclusion is obtained by drawing a transversal through the three lines, applying Proposition I.29 twice, using transitivity of angle congruence, and then applying Proposition I.27.

In the Hilbert reconstruction the same theorem is more naturally organized through the uniqueness of a parallel line through a point. This is exactly the content of Hilbert's Group IV.

A formal subtlety must be made explicit. In the project, `Geo.Parallel` is a strict relation: parallel lines are distinct nondegenerate lines with disjoint point-line carriers. Therefore the statement must exclude the case in which the first and third lines are actually the same line.

The formal hypothesis is

$$\text{PointLine}(A, B) \neq \text{PointLine}(C, D).$$

This distinctness is implicit in Euclid's classical diagram, where the three straight lines are drawn as three separate lines.

30.2 Classical Configuration

Let AB, CD, and EF be three distinct straight lines such that

$$AB \parallel EF$$

and

$$CD \parallel EF.$$

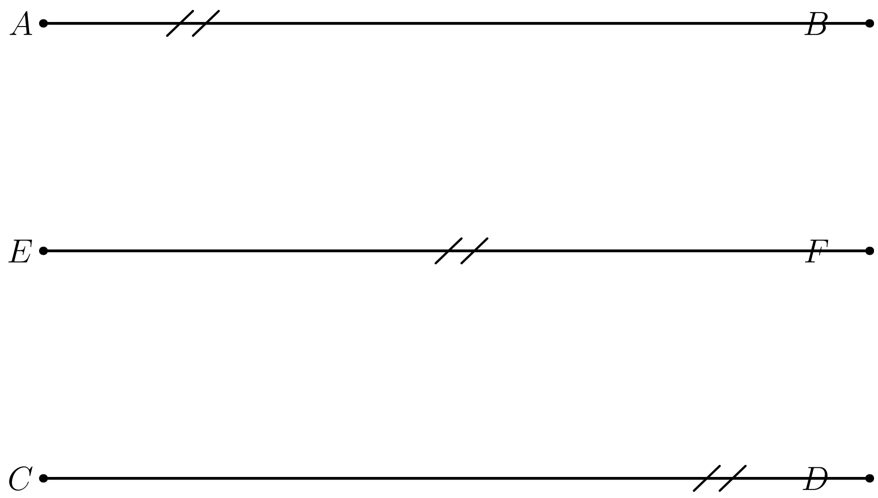


Figure 30.1: The configuration of Proposition I.30. The two outer lines are each parallel to the same middle line.

The conclusion is

$$AB \parallel CD.$$

30.3 Euclid's Proof Architecture

Euclid introduces a transversal meeting the three parallel lines.

The first application of Proposition I.29 gives an equality of alternate angles for the first pair of parallel lines. A second application of Proposition I.29 gives the corresponding equality for the second pair.

Thus one obtains a chain of angle congruences

$$\angle AGK \cong \angle GHF \cong \angle GKD.$$

By transitivity,

$$\angle AGK \cong \angle GKD.$$

These are alternate interior angles for the first and third lines. Proposition I.27 therefore gives

$$AB \parallel CD.$$

The classical dependency graph is therefore

$$
\begin{array}{cccc}
AB \parallel EF & & CD \parallel EF & \\
\downarrow \text{I.29} & & \downarrow \text{I.29} & \\
\angle AGK \cong \angle GHF & & \angle GHF \cong \angle GKD & \\
& \searrow & \downarrow & \swarrow \\
& & \angle AGK \cong \angle GKD & \\
& & \downarrow \text{I.27} & \\
& & AB \parallel CD. &
\end{array}
$$

30.4 Hilbert Reconstruction

The Hilbert reconstruction isolates the more general structural theorem

`hilbert_parallel_transitive_distinct`

with the mathematical content

$$AB \parallel EF, \qquad CD \parallel EF, \qquad \text{PointLine}(A, B) \neq \text{PointLine}(C, D)$$

imply

$$AB \parallel CD.$$

The proof does not pass through angle congruences.

Assume, for contradiction, that the first and third lines meet at a point P. Then both lines pass through P, and both are disjoint from EF.

Since P cannot lie on EF, there are two lines through the same point P, each parallel to EF.

Hilbert's Group IV says that through a point outside a line there is at most one line disjoint from the given line. Therefore the two lines through P must coincide.

Hence

$$\text{PointLine}(A, B) = \text{PointLine}(C, D),$$

contradicting the distinctness hypothesis.

Thus the first and third lines have no common point, and therefore

$$AB \parallel CD.$$

30.5 Lean Formalization

The reusable Hilbert-level theorem is

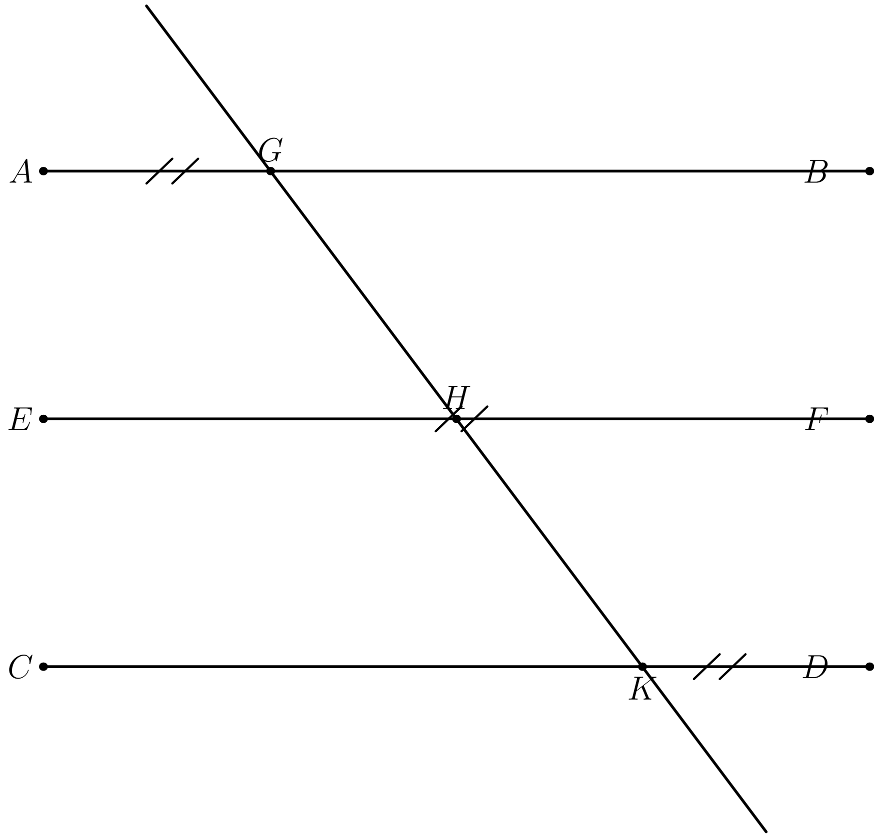


Figure 30.2: The Hilbert proof by contradiction. If the two outer lines met at P, they would be two distinct parallels through the same point to the middle line EF, contradicting Group IV.

```
theorem hilbert_parallel_transitive_distinct
    [HilbertEuclideanPlane Geo]
    (A B C D E F : Geo.Point)
    (hAB_EF : Geo.Parallel A B E F)
    (hCD_EF : Geo.Parallel C D E F)
    (hDistinct :
      Geo.PointLine A B != Geo.PointLine C D) :
    Geo.Parallel A B C D
```

The proof first obtains incidence lines through the three endpoint pairs:

$$AB, \qquad CD, \qquad EF.$$

The two hypotheses of parallelism are converted into disjointness of the corresponding incidence lines.

To prove $AB \parallel CD$, the proof assumes a common point P of the first and third point-line carriers. Membership in the carriers is transported to incidence on the chosen Hilbert lines.

The two parallel hypotheses imply that P is outside the line EF. The call

```
HilbertEuclideanPlane.parallel_unique
```

then identifies the two lines through P.

Finally,

```
hilbert_pointLine_eq_of_points_on_line
```

converts equality of the incidence carriers into equality of the project point-line carriers, contradicting `hDistinct`.

The Book I theorem

```
euclid_proposition_30
```

is consequently only a wrapper around this reusable Hilbert theorem.

30.6 Logical Position of Proposition I.30

The sequence I.27–I.30 now has a clear structure:

$$\begin{aligned} \text{I.27} &: \text{ equal alternate angles} \longrightarrow \text{parallel}, \\ \text{I.28} &: \text{ other angle conditions} \longrightarrow \text{parallel}, \\ \text{I.29} &: \text{ parallel} \longrightarrow \text{angle relations}, \\ \text{I.30} &: \text{ parallel to the same line} \longrightarrow \text{parallel to one another}. \end{aligned}$$

The first two propositions are neutral.

Proposition I.29 introduces the Euclidean direction by using Hilbert's parallel axiom.

Proposition I.30 is then another direct manifestation of the same Euclidean principle: uniqueness of a parallel through a point.

Thus I.30 does not introduce a new independent parallel mechanism. It packages a structural consequence of Group IV.

30.7 Conclusion

Proposition I.30 becomes particularly transparent in the Hilbert reconstruction.

Euclid proves the theorem indirectly through angle relations:

$$\text{I.29} \longrightarrow \text{angle transitivity} \longrightarrow \text{I.27}.$$

The formal Hilbert development instead exposes the structural content:

$$\text{two distinct lines parallel to the same line} \longrightarrow \text{they cannot meet},$$

because any intersection point would produce two different parallels through one point to the same given line.

The resulting theorem

`hilbert_parallel_transitive_distinct`

is reusable beyond Proposition I.30, while

`euclid_proposition_30`

records the corresponding result in the sequence of Book I.

Chapter 31

Proposition I.31

31.1 Introduction

Proposition I.31 is a construction theorem:

$$\text{given a straight line } AB \quad \text{and a point } P \notin AB,$$

construct a straight line through P parallel to AB.

At first sight the proposition belongs naturally to the Euclidean theory of parallel lines developed in I.29–I.30. Its logical status is different, however. Proposition I.31 does not require Hilbert's Group IV. It is an existence theorem in neutral geometry.

The construction uses an arbitrary transversal through the given point, copies an angle by Hilbert's angle-construction axiom, and then applies the neutral alternate-angle criterion corresponding to Euclid I.27.

Thus the essential mechanism is

$$\text{angle construction} \longrightarrow \text{equal alternate angles} \longrightarrow \text{parallel lines.}$$

No numerical angle measure and no construction of a right angle are needed.

31.2 Classical Configuration

Let AB be the given straight line and let P be a point outside it.

The goal is to construct a point $Q \neq P$ such that

$$AB \parallel PQ.$$

Choose the line AP as a transversal. Then choose a point E with

$$A - E - P.$$

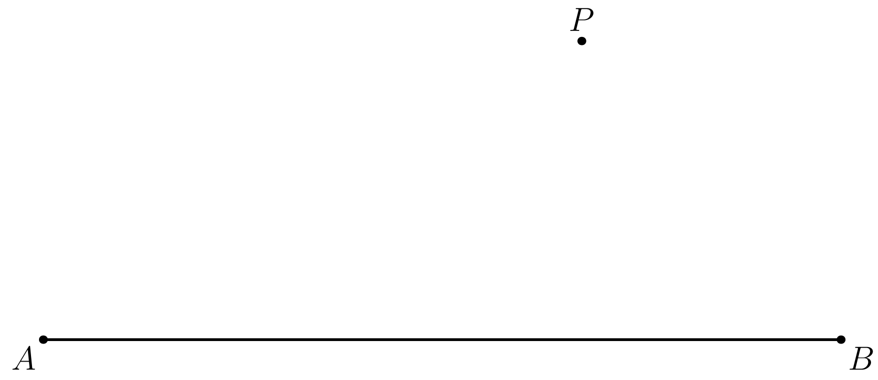


Figure 31.1: The initial data for Proposition I.31: the given straight line AB and the external point P.

The source angle is

$$\angle EAB.$$

At P, with PE as the first ray, construct a ray PQ on the appropriate side of the transversal such that

$$\angle EAB \cong \angle EPQ.$$

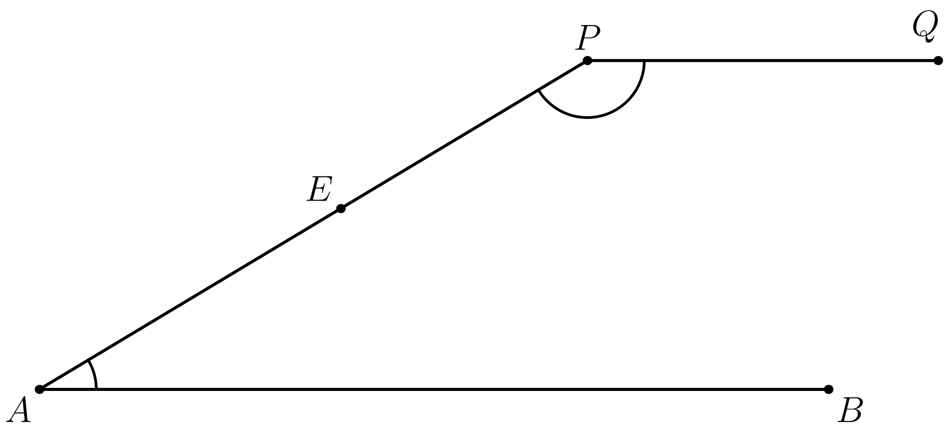


Figure 31.2: The transversal AP, the point E with $A - E - P$, and the copied angle at P.

The two congruent angles are alternate angles for the lines AB and PQ. Proposition I.27 therefore gives

$$AB \parallel PQ.$$

31.3 Why No Right Angle Is Needed

The construction does not proceed through perpendicular lines.

The angle

$$\angle EAB$$

is completely arbitrary. Its numerical size is irrelevant. Hilbert's angle-construction principle allows this angle to be copied at P, with the new ray placed on a prescribed side of the transversal.

Thus the direction of the new line is transferred synthetically by angle congruence:

$$\angle EAB \cong \angle EPQ.$$

The parallelism then follows from the converse alternate-angle theorem, that is, Proposition I.27.

This is conceptually important. The construction of a parallel does not require a prior theory of perpendiculars or a notion of 90°. It requires only the ability to reproduce a given angle and control its position relative to the transversal.

31.4 The Hidden Configuration

A classical diagram makes the side relation almost invisible. In the formal proof it must be explicit.

The equality

$$\angle EAB \cong \angle EPQ$$

alone is not sufficient. The point Q must be chosen so that B and Q lie on opposite sides of the transversal AP.

The formal construction therefore contains three distinct ingredients:

$$A - E - P,$$

which fixes the directions of the transversal rays;

$$B \notin AP,$$

which guarantees that the source angle is nondegenerate; and

$$B,\ Q \quad \text{on opposite sides of } AP,$$

which identifies the copied angles as genuine alternate angles.

This is exactly the type of configuration information that is compressed into an ordinary Euclidean picture but becomes explicit in a Hilbert-style formalization.

31.5 Hilbert Reconstruction

The substantial geometric content is isolated in the reusable theorem

```
hilbert_parallel_through_point_exists
```

with the formal statement

```
theorem hilbert_parallel_through_point_exists
    [HilbertCongruence Geo]
    (A B P : Geo.Point)
    (hAB : A != B)
    (hABP : not Collinear Geo A B P) :
    exists Q : Geo.Point,
      P != Q /\
      Geo.Parallel A B P Q
```

The proof first constructs the incidence line through A and B, and then the transversal through A and P.

Next it chooses E with

$$A - E - P.$$

The non-collinearity assumption implies that B lies off the transversal, so the angle $\angle EAB$ is genuine.

A point S is then chosen on the side of the transversal opposite to B. Hilbert's angle-construction principle constructs Q on the side of S such that

$$\angle EAB \cong \angle EPQ.$$

The side relation is transported from S to Q, yielding the required opposite-side configuration for B and Q.

Finally,

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

applies the neutral alternate-angle criterion and returns

$$AB \parallel PQ.$$

31.6 Lean Formalization

The Book I theorem is

```
theorem euclid_proposition_31
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B P : Geo.Point)
```

```
    (hAB : A != B)
    (hABP : not Collinear Geo A B P) :
    exists Q : Geo.Point,
      P != Q /\
      Geo.Parallel A B P Q := by

  exact
    hilbert_parallel_through_point_exists
      Geo
      A B P
      hAB
      hABP
```

Thus the Book I file is only a wrapper. The reusable geometric theorem belongs to the Hilbert interface, while the Proposition file records the corresponding result in Euclid's sequence.

The dependency graph of the actual Lean proof is

$$\begin{array}{c} \text{line through } A, P \\ \downarrow \\ A - E - P \\ \downarrow \\ \text{opposite-side choice} \\ \downarrow \\ \texttt{angle\_construction} \\ \downarrow \\ \angle EAB \cong \angle EPQ \\ \downarrow \\ \texttt{hilbert\_parallel\_of\_alternate\_angles\_oppositeSide\_lines} \\ \downarrow \\ AB \parallel PQ. \end{array}$$

31.7 Neutral Existence Versus Euclidean Uniqueness

Proposition I.31 establishes the existence of a parallel through an external point.

It does not establish uniqueness.

This distinction separates I.31 from the immediately preceding Euclidean parallel propositions. In the Hilbert reconstruction, uniqueness is the additional content of Group IV:

$$\text{through a point outside a line there is at most one parallel.}$$

Hence

$$\boxed{\text{I.31: existence}}$$

belongs to neutral geometry, while

$$\boxed{\text{Group IV: uniqueness}}$$

is the Euclidean addition.

This also explains a useful structural relation with Proposition I.29. Once a parallel has been constructed by the mechanism of I.31, a uniqueness principle such as Playfair's axiom can identify that constructed line with an already given parallel through the same point. This observation belongs to the logical analysis surrounding I.29; it is not an additional step in the proof of I.31 itself.

31.8 Final Configuration

The construction produces a point $Q \neq P$ satisfying

$$AB \parallel PQ.$$

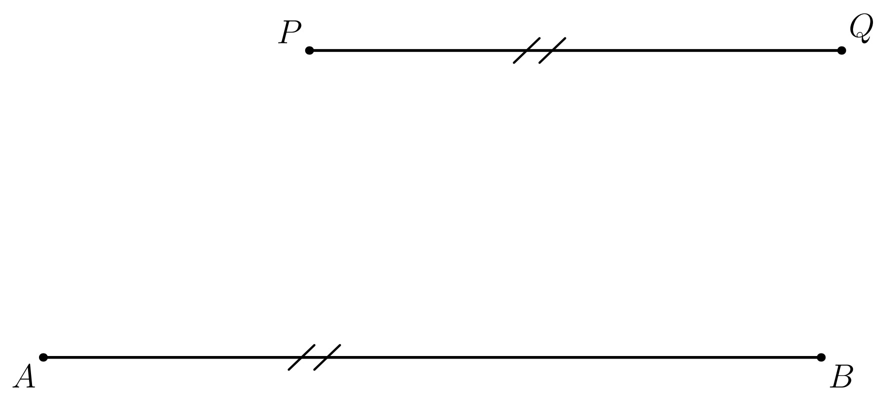


Figure 31.3: The final configuration of Proposition I.31: the constructed line through P is parallel to the given straight line AB.

31.9 Conclusion

Proposition I.31 is a particularly clear example of the difference between existence and uniqueness in the theory of parallels.

Its proof requires no Euclidean parallel axiom. A transversal is chosen, an arbitrary angle is copied at the external point, the side configuration is controlled explicitly, and Proposition I.27 proves that the resulting line is parallel to the given line.

The Hilbert reconstruction therefore isolates the reusable theorem

```
hilbert_parallel_through_point_exists
```

as a neutral construction result.

The Book I theorem

```
euclid_proposition_31
```

is then a thin wrapper around this more general piece of the developed Hilbert theory.

Chapter 32

Proposition I.32

32.1 Introduction

Proposition I.32 is the first proposition in Book I that applies the Euclidean parallel theory developed in I.29–I.31 directly to a triangle.

Let ABC be a nondegenerate triangle and extend BC through C to D. Euclid asserts first that the exterior angle at C is equal to the two remote interior angles:

$$\angle ACD = \angle BAC + \angle ABC.$$

He then concludes that the three interior angles of the triangle are together equal to two right angles.

In the present reconstruction no numerical angle measure is introduced. Thus the familiar modern formula

$$\angle BAC + \angle ABC + \angle ACB = 180^\circ$$

is not the literal Lean statement. Instead the proposition is represented synthetically: a ray inside the exterior angle decomposes it into two angles congruent to the two remote interior angles, while $B - C - D$ records the straight-angle configuration at C.

This distinction is mathematically important. The formal proof establishes the synthetic content of the 180° theorem without introducing degrees or an arithmetic operation of angle addition.

32.2 Classical Statement

Let ABC be a triangle and let BC be produced to D. Proposition I.32 has two conclusions.

First,

$$\angle ACD = \angle BAC + \angle ABC.$$

Second,

$$\angle BAC + \angle ABC + \angle ACB = 2R,$$

where $2R$ denotes two right angles.

The classical Euclidean proof constructs through C a line parallel to AB. Proposition I.29 then identifies the two parts of the exterior angle with the two remote interior angles.

32.3 The Formal Configuration

The Lean proof refines the classical diagram because the formal argument must record the orientation of rays and the incidence data needed by the parallel-angle theorem.

The principal points satisfy

$$A - M - C, \qquad B - M - E, \qquad A - R - D, \qquad B - C - D.$$

The auxiliary construction gives

$$AB \parallel CE.$$

Moreover E and R determine the same ray from C. Thus the ray CR lies inside the exterior angle ACD and decomposes it into the two angles used in the formal statement.

For the explicit application of Proposition I.29 the proof also extends the two parallel lines:

$$A - B - X, \qquad E - C - F.$$

Hence

$$AX \parallel EF.$$

32.4 Synthetic Representation of the Exterior-Angle Equality

The first conclusion is represented by the predicate

```
HilbertExteriorAngleEqualsRemoteAngles
```

defined by

```
def HilbertExteriorAngleEqualsRemoteAngles
    [HilbertIncidence Geo]
    (A B C D : Geo.Point) : Prop :=
  exists R : Geo.Point,
    Geo.Between A R D /\
    Geo.AngleCongruent B A C A C R /\
    Geo.AngleCongruent A B C R C D
```

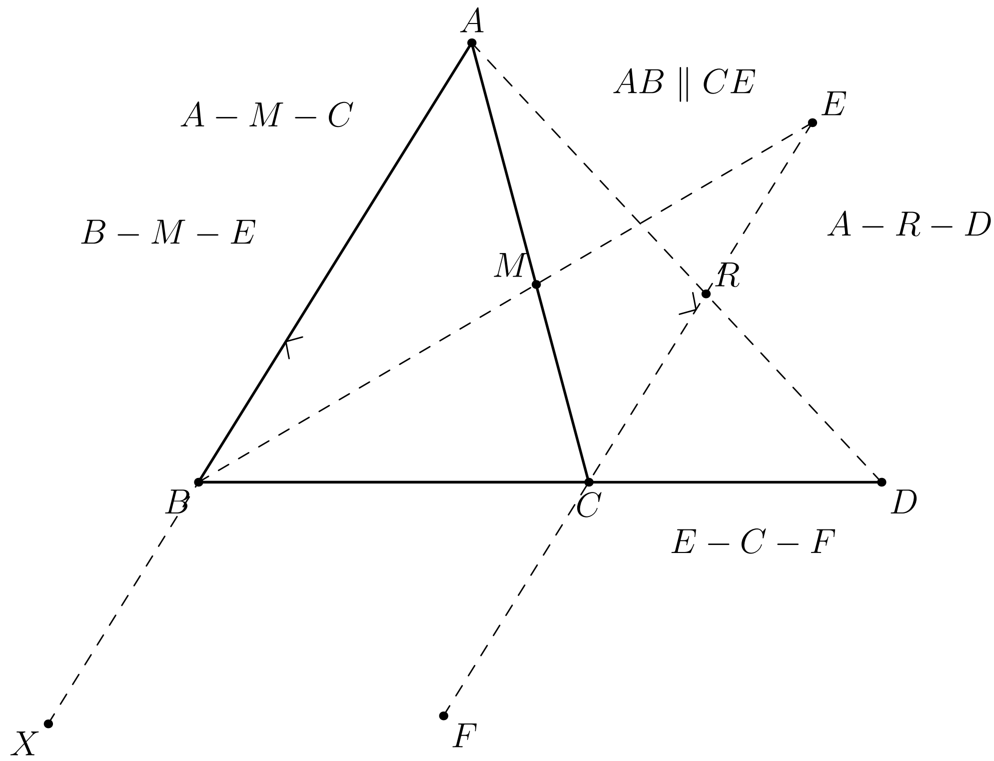


Figure 32.1: The formal configuration of Proposition I.32. The essential parallel relation is $AB \parallel CE$. The points X and F are technical extensions used to put the configuration into the explicit transversal form of Proposition I.29.

Thus the equation

$$\angle ACD = \angle BAC + \angle ABC$$

is encoded without angle addition. Instead there exists a point R with

$$A - R - D$$

such that

$$\angle BAC \cong \angle ACR$$

and

$$\angle ABC \cong \angle RCD.$$

The ray CR is therefore the synthetic "division point" of the exterior angle.

32.5 Construction of the Correctly Oriented Parallel Ray

A naive use of Proposition I.31 would produce a line through C parallel to AB, but a line alone does not determine which of its two rays lies inside the exterior angle ACD.

This orientation issue is genuine in the formal proof.

The reconstruction therefore uses the already established auxiliary configuration

```
hilbert_exterior_angle_aux
```

which supplies points M, E satisfying

$$A - M - C, \qquad B - M - E,$$

together with

$$\angle BAC \cong \angle MCE.$$

The theorem

```
hilbert_exterior_angle_aux_parallel
```

then yields

$$AB \parallel CE.$$

The important point is that E is not an arbitrary point on the parallel through C. Its orientation is inherited from the auxiliary exterior-angle construction.

The helper

```
hilbert_exterior_angle_inside
```

uses this oriented configuration, together with $B - C - D$, to show that the ray CE meets the open segment AD. Hence there is a point R such that

$$A - R - D$$

and E, R lie on the same ray from C.

This is the formal crossbar step of the proof.

32.6 First Remote Interior Angle

From

$$A - M - C$$

the points M and A lie on the same ray from C. Therefore

$$\angle MCE = \angle ACE.$$

Since E and R lie on the same ray from C,

$$\angle ACE = \angle ACR.$$

Combining these ray transports with the congruence supplied by the auxiliary construction gives

$$\boxed{\angle BAC \cong \angle ACR}.$$

In Lean this is the intermediate result

```
hFirst :
    Geo.AngleCongruent B A C A C R
```

and it establishes the first half of the exterior-angle decomposition.

32.7 Preparing Proposition I.29

The second congruence is obtained from the parallel relation, but the explicit theorem

```
euclid_proposition_29_transversal
```

expects the two intersections to lie between endpoints of the parallel lines.

The proof therefore extends AB beyond B:

$$A - B - X,$$

and extends EC beyond C:

$$E - C - F.$$

Parallelism is transported along the corresponding collinear triples to obtain

$$AX \parallel EF.$$

The transversal is the line BD, with intersection points B and C.

There remains an orientation condition. Since $A - R - D$, the points A and R lie on the same side of BD. From the same-ray relation between E and R, together with $E - C - F$, the proof obtains

$$R - C - F.$$

Thus R and F lie on opposite sides of BD. Transporting this relation through the same-side relation between A and R gives

$$A \text{ and } F \quad \text{on opposite sides of } BD.$$

At this point all hypotheses of the explicit transversal form of I.29 are available.

32.8 Second Remote Interior Angle

Proposition I.29 applied to

$$AX \parallel EF$$

with transversal BD gives

$$\angle ABC \cong \angle BCF.$$

At C the two straight lines are represented by

$$B - C - D \qquad \text{and} \qquad E - C - F.$$

Proposition I.15, through the theorem `VerticalAngles`, therefore gives

$$\angle BCF \cong \angle ECD.$$

By transitivity,

$$\angle ABC \cong \angle ECD.$$

Finally E and R lie on the same ray from C, so the first arm of the second angle may be transported from E to R:

$$\boxed{\angle ABC \cong \angle RCD}.$$

This is the Lean result

```
hSecond :
    Geo.AngleCongruent A B C R C D
```

and completes the exterior-angle theorem.

32.9 Proof Architecture of the Exterior-Angle Part

The actual formal dependency structure is

$$\begin{gathered} ABC \text{ noncollinear}, \quad B - C - D \\ \downarrow \\ \texttt{hilbert\_exterior\_angle\_aux} \\ \downarrow \\ A - M - C, \quad B - M - E, \quad \angle BAC \cong \angle MCE \\ \downarrow \\ AB \parallel CE \\ \downarrow \\ \texttt{hilbert\_exterior\_angle\_inside} \\ \downarrow \\ A - R - D, \quad \text{SameRay}(C, E, R) \end{gathered}$$

and then the proof splits:

$$\begin{array}{ccc} & A - R - D, \quad AB \parallel CE & \\ \swarrow & & \searrow \\ \text{same-ray transport} & & \text{extend the two parallels} \\ \downarrow & & \downarrow \\ \angle BAC \cong \angle ACR & & AX \parallel EF \\ & & \downarrow \\ & & \text{I.29 + vertical angles} \\ & & \downarrow \\ & & \angle ABC \cong \angle RCD. \end{array}$$

The two branches are then packaged as

$$\exists R, \quad A - R - D, \quad \angle BAC \cong \angle ACR, \quad \angle ABC \cong \angle RCD.$$

32.10 Where the Euclidean Parallel Axiom Enters

The construction and order-theoretic parts of the proof use the Hilbert incidence, order, congruence, side-of-line and crossbar machinery.

The specifically Euclidean step is the use of Proposition I.29:

$$AX \parallel EF \longrightarrow \angle ABC \cong \angle BCF.$$

As established in the reconstruction of I.29, this direction depends on Hilbert's Group IV, or equivalently on uniqueness of the parallel through a point.

Thus I.32 inherits its Euclidean character from I.29.

This is structurally different from I.27 and I.28, where angle conditions imply parallelism in neutral geometry. In I.32 the proof needs the converse direction:

$$\text{parallelism} \longrightarrow \text{angle congruence.}$$

32.11 The Triangle-Angle-Sum Statement

The second formal predicate is

```
HilbertTriangleAnglesEqualTwoRightAngles
```

defined by

```
def HilbertTriangleAnglesEqualTwoRightAngles
    [HilbertIncidence Geo]
    (A B C : Geo.Point) : Prop :=
  exists D : Geo.Point,
    Geo.Between B C D /\
    HilbertExteriorAngleEqualsRemoteAngles
      A B C D
```

The theorem

```
euclid_proposition_32
```

is consequently short. From noncollinearity one obtains $B \neq C$, extends BC through C to a point D, and applies

```
euclid_proposition_32_exterior
```

to that extension.

The proof architecture is simply

$$\begin{gathered} ABC \text{ noncollinear} \\ \downarrow \\ B \neq C \\ \downarrow \\ \text{extend } BC \text{ through } C \\ \downarrow \\ B - C - D \\ \downarrow \\ \texttt{euclid\_proposition\_32\_exterior} \\ \downarrow \\ \texttt{HilbertTriangleAnglesEqualTwoRightAngles}. \end{gathered}$$

32.12 What "Equal to Two Right Angles" Means Here

There is an important distinction between the classical mathematical consequence and the literal formal statement.

In ordinary angle-measure notation, I.32 yields

$$\angle BAC + \angle ABC + \angle ACB = 180^\circ.$$

The present Hilbert layer, however, does not assign numerical measures to angles. Nor does it define a general arithmetic operation

$$\alpha + \beta + \gamma.$$

Instead the formalization records the synthetic configuration from which the two-right-angles conclusion follows:

$$B - C - D$$

makes the rays CB and CD opposite, so $\angle ACB$ and $\angle ACD$ form a straight-angle configuration. The exterior angle is then decomposed by CR into angles congruent to $\angle BAC$ and $\angle ABC$.

Consequently the formal theorem proves the geometric content

$$\angle BAC + \angle ABC + \angle ACB = 2R$$

without translating $2R$ into the numerical value 180°.

Strictly speaking, the current predicate `HilbertTriangleAnglesEqualTwoRightAngles` stores the extension $B - C - D$ and the exterior-angle decomposition; the supplementary interpretation of the straight line at C is implicit in this data rather than represented by a separate angle-addition object.

This is not merely a coding detail. It identifies exactly what must be formalized in a synthetic geometry before the familiar numerical 180° formulation can even be stated.

32.13 Lean Formalization

The principal theorem is

```
euclid_proposition_32_exterior
```

with hypotheses

```
(A B C D : Geo.Point)
(hABC : Not (PrimCollinear Geo A B C))
(hBCD : Geo.Between B C D)
```

and conclusion

```
HilbertExteriorAngleEqualsRemoteAngles A B C D
```

The proof uses the following main components:

```
hilbert_exterior_angle_aux
hilbert_exterior_angle_aux_parallel
hilbert_exterior_angle_inside
hilbert_angle_eq_of_sameRay_first
hilbert_angle_eq_of_sameRay_second
hilbert_between_points_sameSide_transversal
euclid_proposition_29_transversal
VerticalAngles
```

The second theorem,

```
euclid_proposition_32
```

is a corollary obtained by extending BC and invoking the exterior-angle theorem.

The typeclass

```
[HilbertEuclideanPlane Geo]
```

appears because the proof ultimately uses Proposition I.29 and hence the Euclidean parallel axiom.

32.14 Relation to the Classical Proof

At the level of a paper proof, I.32 is short:

$$\text{construct a parallel through } C \longrightarrow \text{use I.29 twice} \longrightarrow \text{add the adjacent angles.}$$

The Lean reconstruction exposes several pieces hidden by this description.

First, a parallel *line* through C is insufficient: the correct orientation of the ray must be established.

Second, the ray used to split the exterior angle must be shown to meet the segment AD. This is the crossbar component.

Third, the explicit I.29 theorem carries side-of-line hypotheses that must be proved from the betweenness configuration.

Fourth, the proof does not "add angles" numerically. It records a synthetic decomposition of the exterior angle.

Thus the expanded formal proof is not evidence that Euclid's argument is mathematically complicated. It is an audit of the incidence, order and orientation information suppressed by the classical diagram.

32.15 Conclusion

Proposition I.32 combines the parallel theory of I.29–I.31 with the synthetic theory of triangles.

Its essential Euclidean step is

$$AB \parallel CE \longrightarrow \text{the required angle congruence,}$$

which is supplied by I.29.

The formal proof then shows that the correctly oriented parallel ray through C cuts the exterior angle into two parts congruent to the remote interior angles:

$$\angle BAC \cong \angle ACR, \qquad \angle ABC \cong \angle RCD.$$

Together with the straight-line relation $B - C - D$, this is the synthetic content of the classical theorem

$$\boxed{\angle BAC + \angle ABC + \angle ACB = 180^\circ}.$$

The reconstruction therefore reaches the familiar 180° theorem without introducing degrees: the numerical formulation appears only as the modern interpretation of a purely synthetic Hilbert-style configuration.

Chapter 33

Proposition I.33

33.1 Introduction

Euclid I.33 states that if two equal and parallel straight segments are joined towards the same parts, then the joining segments are themselves equal and parallel. In the notation used here, the hypotheses are

$$AB \parallel CD, \qquad AB \cong CD,$$

and the conclusion is

$$AC \parallel BD, \qquad AC \cong BD.$$

The proposition is Euclidean rather than neutral. The genuinely Euclidean step is the passage from parallel lines to congruent alternate angles, corresponding to Euclid I.29 and Hilbert's parallel axiom.

The formal reconstruction is particularly instructive because Euclid's phrase "towards the same parts" contains geometric orientation information which must be made explicit in Lean.

33.2 The Classical Configuration

For the explicit reconstruction we use the configuration occurring in Beeson's formal proof. The cross-joins AD and BC meet internally at a point M:

$$A - M - D, \qquad B - M - C.$$

Thus M lies between A and D, and also between B and C. The two given segments satisfy

$$AB \parallel CD, \qquad AB \cong CD.$$

The required conclusion is

$$AC \parallel BD, \qquad AC \cong BD.$$

The betweenness assumptions are not merely a convenient way of drawing the figure. They encode the orientation information hidden in Euclid's phrase "towards the same parts."

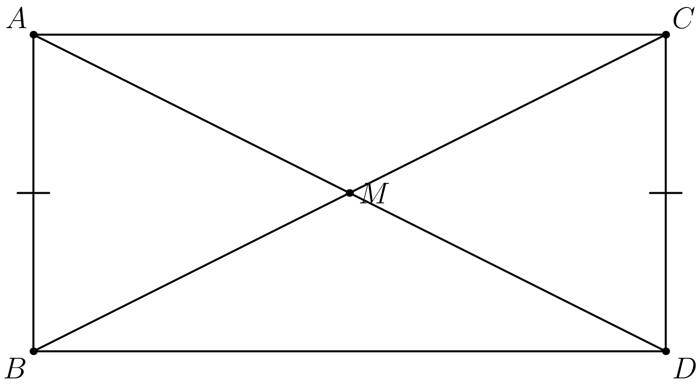


Figure 33.1: The configuration for Euclid I.33. The segments AB and CD are equal and parallel, while AD and BC meet internally at M.

33.3 Classical Proof

The classical dependency chain is short:

$$\text{I.29} \longrightarrow \text{SAS (I.4)} \longrightarrow \text{I.27}.$$

The line BC is a transversal of the parallel lines AB and CD. By I.29 the appropriate alternate angles are congruent.

Together with

$$AB \cong CD$$

and the common side BC, this gives the hypotheses of SAS for the two triangles determined by the configuration.

SAS gives, in particular,

$$AC \cong BD,$$

and also a second pair of congruent alternate angles.

The converse theorem I.27 can then be applied to this second pair of angles, yielding

$$AC \parallel BD.$$

Hence the joining segments are both equal and parallel.

33.4 The Hidden Orientation Data

The classical diagram makes several facts visually immediate. The formal proof cannot use them implicitly.

From

$$A - M - D, \qquad B - M - C,$$

the Lean proof first obtains a line through B, M, C. It then proves that neither A nor D lies on this transversal.

Since M lies on the transversal and is between A and D, the open segment AD meets the transversal. Consequently A and D lie on opposite sides of the line BC.

This produces the formal separation datum

```
hOppositeAD :
HilbertOppositeSide Geo A D trans
```

which is required by the line-level alternate-angle theorem.

The proof also chooses an auxiliary point N between B and C. Thus

$$B - N - C$$

and, by symmetry of betweenness,

$$C - N - B.$$

These relations produce the corresponding same-ray data:

```
hRayBNC :
HilbertSameRay Geo B N C

hRayCNB :
HilbertSameRay Geo C N B
```

They are used to normalize the unoriented angles appearing in the Hilbert interface.

Thus the apparently simple phrase "towards the same parts" expands formally into incidence, betweenness, opposite-side, and same-ray information.

33.5 Proof Architecture of the Reconstruction

The Lean file contains two levels of I.33.

The first theorem is

```
euclid_proposition_33
```

It is a thin theorem at the level of the developed Hilbert API. Instead of starting from the explicit crossing configuration, it accepts an orientation witness suitable for the existing one-pair-parallel-and-congruent criterion.

The relevant reusable theorem is

```
OnePairParallelCongruentCriterion
```

which recognizes the corresponding quadrilateral as a parallelogram. Equality of the remaining opposite sides is then obtained from

```
ParallelogramOppositeSidesCongruent
```

The second theorem is

```
euclid_proposition_33_crossing
```

This is the configuration-faithful reconstruction. Its hypotheses contain the explicit point M and the two betweenness relations used by Beeson. The proof then reconstructs the classical I.29–SAS–I.27 argument.

The distinction between the two theorems is deliberate. The first shows that the mathematical content of I.33 is already available at a higher level in the Hilbert interface. The second shows how the classical Euclidean configuration is translated into the lower-level data required by that theory.

33.6 Lean Formalization

The explicit theorem has the following statement:

```
theorem euclid_proposition_33_crossing
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
(A B C D M : Geo.Point)
(hParallel : Geo.Parallel A B C D)
(hCongruent : Geo.Congruent A B C D)
(hAMD : Geo.Between A M D)
(hBMC : Geo.Between B M C) :
Geo.Parallel A C B D /\
Geo.Congruent A C B D := by
```

The first stage extracts the incidence information associated with $B - M - C$, including a transversal through B, M, C.

Parallelism of AB and CD is then used to prove that A and D cannot lie on this transversal. Together with $A - M - D$, this gives

```
hOppositeAD :
HilbertOppositeSide Geo A D trans
```

An interior point N of BC is introduced to obtain the same-ray normalizations needed by the angle interface.

The Euclidean direction of the alternate-angle theorem is invoked through the following interface theorem:

```
hilbert_alternate_angles_of_parallel_
oppositeSide_lines
```

After normalization of the rays, this gives the included-angle congruence required for SAS.

The two triangles are then compared by

```
SAS
```

and the resulting congruence object is stored as

```
hTriangles :
TriangleCongruenceResult
Geo B A C C D B
```

One component gives the required segment congruence:

```
hAC_BD :
Geo.Congruent A C B D
```

Another component gives the angle congruence from which the second parallelism is obtained.

After same-ray normalization the relevant angle statement is

```
hAlternate :
Geo.AngleCongruent N C A N B D
```

The converse alternate-angle theorem is then applied:

```
hilbert_parallel_of_alternate_angles_
oppositeSide_lines
```

and yields the required parallelism of AC and BD.

33.7 Relationship with Earlier Propositions

Proposition I.33 combines two directions of the parallel-angle theory developed immediately before it.

The first direction is the Euclidean one:

$$\text{parallel lines} \Longrightarrow \text{congruent alternate angles.}$$

This is the content represented by I.29. It is the point at which the Euclidean parallel axiom enters the proof.

The second direction is

$$\text{congruent alternate angles} \Longrightarrow \text{parallel lines.}$$

This is the converse represented by I.27 and belongs to the neutral part of the theory.

I.33 therefore provides a compact test of the architecture developed for I.27–I.29. The proof begins with the Euclidean direction, passes through SAS, and ends with the neutral converse.

33.8 Hilbert and Book Zero Components Used

No new axiom is introduced for Proposition I.33.

The explicit proof uses existing infrastructure for:

- incidence and the line through two distinct points;
- betweenness and its incidence consequences;
- separation by a line;
- opposite-side relations;
- same-ray relations;
- segment-congruence symmetries;
- non-collinearity;
- SAS triangle congruence;
- the Euclidean alternate-angle theorem;
- the converse alternate-angle theorem.

The API-level theorem additionally uses the existing parallelogram machinery. In particular:

```
OnePairParallelCongruentCriterion
```

and

```
ParallelogramOppositeSidesCongruent
```

No helper specific to Proposition I.33 is added to `HilbertInterface` or to Book Zero.

33.9 Formalization Notes

The main formal issue in I.33 is not SAS itself. It is the orientation of the configuration.

In an ordinary diagram it is immediately apparent which sides of the transversal contain A and D, and which rays at B and C determine the relevant angles. None of this information is implicit in the formal language.

The crossing formulation makes the required information explicit through

$$A - M - D, \qquad B - M - C.$$

From these two relations the proof reconstructs the separation and ray data required by the Hilbert interface.

This also explains why the Lean proof is longer than the classical three-step argument. The additional code does not represent new Euclidean mathematics. It reconstructs configuration information which the classical proof obtains from the diagram and from the phrase “towards the same parts.”

The formal proof therefore separates two layers clearly:

$$\begin{gathered}\text{configuration normalization}\\ \downarrow\\ \text{I.29}\\ \downarrow\\ \text{SAS}\\ \downarrow\\ \text{I.27.}\end{gathered}$$

33.10 Dependency Summary

The dependency structure of the explicit reconstruction is:

$$\begin{gathered}\text{Hilbert incidence, order, and congruence axioms}\\ \downarrow\\ \text{betweenness / incidence / separation / SameRay}\\ \downarrow\\ \text{Euclidean parallel axiom (Group IV)}\\ \downarrow\\ \text{parallel} \Rightarrow \text{congruent alternate angles}\\ \downarrow\\ \text{SAS}\\ \downarrow\\ \text{congruent alternate angles} \Rightarrow \text{parallel}\\ \downarrow\\ \text{Euclid I.33.}\end{gathered}$$

New axiom: no.

New helper: no.

Build clean: yes.

Sorry: no.

33.11 Conclusion

Euclid I.33 is mathematically short but formally revealing. Once the orientation hidden in the classical configuration is exposed, the original proof architecture survives:

$$\text{I.29} \longrightarrow \text{SAS} \longrightarrow \text{I.27.}$$

The explicit crossing theorem shows how the classical statement is recovered from betweenness and separation data. The shorter API-level theorem shows, independently, that the underlying parallelogram principle is already available in the developed Hilbert theory.

No new axiom and no Proposition-specific helper are required.

Chapter 34

Proposition I.34

34.1 Introduction

Euclid's Proposition I.34 collects the fundamental metric and angular properties of a parallelogram. For a parallelogram $ABCD$, it asserts that opposite sides are equal, opposite angles are equal, and a diagonal divides the parallelogram into two equal parts.

The present reconstruction separates these three assertions according to the language actually available in the CGJteamLab Hilbert development. The first two are represented directly by congruence of segments and angles. The final claim is represented by congruence of the two triangles cut off by the diagonal. No numerical area function is introduced.

This Proposition is therefore a useful example of a recurring distinction in the project: the classical theorem and the formal theorem have the same synthetic content, but the Lean proof uses a different proof architecture from Euclid's original argument.

34.2 The Classical Configuration

Let $ABCD$ be a parallelogram, with

$$AB \parallel CD, \qquad BC \parallel DA.$$

Draw the diagonal AC. The diagonal separates the parallelogram into the triangles ABC and CDA.

The required conclusions are

$$AB \cong CD, \qquad BC \cong DA,$$
$$\angle DAB \cong \angle BCD, \qquad \angle ABC \cong \angle CDA,$$

and, in the language of the present formalization,

$$\triangle ABC \cong \triangle CDA.$$

34.3 Classical Proof

Euclid's proof uses the diagonal and the theory of parallel lines established immediately before Proposition I.34. Since $AB \parallel CD$, Proposition I.29 gives one pair of alternate angles. Since

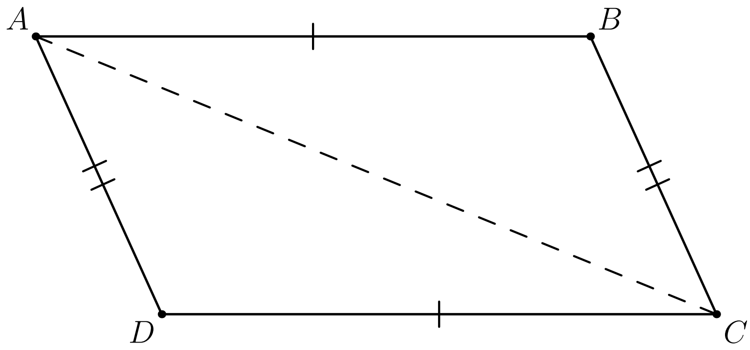


Figure 34.1: The configuration for Proposition I.34. The quadrilateral $ABCD$ is a parallelogram and AC is the diagonal used in the reconstruction.

$BC \parallel DA$, a second application of Proposition I.29 gives another pair of alternate angles. The common diagonal then allows Proposition I.26 to identify the two triangles. The corresponding opposite sides and angles are therefore equal. The final statement that the diagonal bisects the parallelogram follows because the two triangles are congruent.

A compact classical dependency graph is

$$\mathrm{I.29} + \mathrm{I.29} \longrightarrow \mathrm{I.26} \longrightarrow \begin{array}{c}\text{opposite sides equal,}\\ \text{opposite angles equal,}\\ \text{diagonal gives equal triangles.}\end{array}$$

The Beeson–Narboux–Wiedijk proof file makes explicit several configuration facts hidden by the classical diagram: non-collinearity of the relevant triangles, opposite-side data for the transversal configuration, and endpoint order needed to apply the formal versions of I.29 and I.26. These details are part of the classical reconstruction problem, even though they are not visible in Euclid's short paper proof.

34.4 The Main Formalization Issue: Reusing Parallelogram Theory

The current CGJteamLab proof does not replay the classical I.29–I.26 route. By the time Proposition I.34 is reached, the Hilbert interface already contains a reusable Euclidean theory of parallelograms. In particular, it contains a theorem stating that the opposite sides of a Euclidean parallelogram are congruent.

This changes the proof DAG substantially. Once

$$AB \cong CD \qquad \text{and} \qquad BC \cong DA$$

are available, the diagonal AC supplies the third side congruence

$$AC \cong CA.$$

The triangles ABC and CDA are therefore congruent by the already proved Hilbert SSS theorem. The required opposite-angle congruence is extracted from that triangle-congruence result.

Thus the formal proof uses the same diagonal as Euclid, but it enters the triangle argument through SSS rather than through two alternate-angle relations and ASA.

34.5 Proof Architecture of the Reconstruction

The reusable proof architecture is

$$
\begin{gathered}
AB \parallel CD, \quad BC \parallel DA \\
\Downarrow \\
\text{IsParallelogram } ABCD \\
\Downarrow \\
AB \cong CD, \quad BC \cong DA \\
AC \cong CA \\
\Downarrow \quad \text{SSS} \\
\triangle ABC \cong \triangle CDA \\
\Downarrow \\
\angle ABC \cong \angle CDA.
\end{gathered}
$$

The second pair of opposite angles is obtained by applying the same reusable angle theorem to the cyclically relabelled parallelogram $BCDA$ and then using symmetry of angle congruence.

This produces two interface-level helpers:

```
ParallelogramOppositeAngleCongruent
ParallelogramOppositeAnglesCongruent
```

The first proves one pair of opposite angles. The second packages both pairs. They belong to `HilbertInterface` because they are general properties of every Euclidean parallelogram, not facts specific to Proposition I.34.

34.6 Lean Formalization

The main Proposition theorem is deliberately thin:

```
theorem euclid_proposition_34
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D : Geo.Point)
    (hParallelogram : IsParallelogram Geo A B C D) :
    And
      (OppositeSidesCongruent Geo A B C D)
      (OppositeAnglesCongruent Geo A B C D)
```

Its two branches are discharged by the established interface theorems

```
ParallelogramOppositeSidesCongruent
ParallelogramOppositeAnglesCongruent
```

The third part of Euclid's statement is represented separately:

```
theorem euclid_proposition_34_diagonal
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D : Geo.Point)
    (hParallelogram : IsParallelogram Geo A B C D) :
    TriangleCongruenceResult Geo A B C C D A := by
```

The proof first derives non-collinearity of A, B, C from the disjointness of the parallel lines AB and CD. It then obtains the two pairs of opposite side congruences from the parallelogram theorem, uses reflexivity of congruence for the diagonal, and applies SSS:

```
  have hSides :
      OppositeSidesCongruent Geo A B C D :=
    ParallelogramOppositeSidesCongruent
      Geo A B C D hParallelogram

  have hAC_CA :
      Geo.Congruent A C C A :=
    CongruentSwapSecond
      Geo A C A C
      (hilbert_congruent_reflexive Geo A C)

  have hSSS :=
    HilbertSSS
      Geo
      A B C
      C D A
      hABC
      hSides.1
      hSides.2
      hAC_CA

  exact hSSS.2
```

No area object occurs in this theorem. The formal counterpart of Euclid's "diameter bisects the areas" is the congruence of the two triangles cut off by the diagonal.

34.7 Relationship with Earlier Propositions

Classically, Proposition I.34 depends directly on I.29 and I.26. In the present Lean development these earlier Euclidean consequences have already been absorbed into the reusable parallelogram theory of `HilbertInterface`. Consequently the final Proposition file does not call `euclid_proposition_29` or `euclid_proposition_26` directly.

This is an intentional architectural compression. The classical dependency has not disappeared mathematically; it has been moved into general Hilbert- level theorems that are proved once and then reused.

The relation with I.33 is especially close. Proposition I.33 recognized a parallelogram from one pair of opposite sides that were both parallel and congruent, together with orientation data.

Proposition I.34 starts after the quadrilateral has already been recognized as a parallelogram and extracts its standard Euclidean consequences.

34.8 Hilbert and Book Zero Components Used

The formal Proposition uses the following established components:

- `IsParallelogram`, defined by two pairs of opposite parallel sides;
- `ParallelogramOppositeSidesCongruent`, the Euclidean theorem giving the two opposite-side congruences;
- `HilbertSSS`, the derived SSS congruence theorem;
- incidence uniqueness and the extensional representation of parallel lines, used to prove that A, B, C are non-collinear;
- reflexivity and endpoint symmetry of segment congruence;
- symmetry of angle congruence, used when packaging the second pair of opposite angles.

No new Book Zero theorem is required for Proposition I.34.

34.9 Formalization Notes

34.9.1 Axiomatic status

Proposition I.34 is Euclidean in the present formulation. The hypothesis is

```
[HilbertEuclideanPlane Geo]
```

The Euclidean strength is consumed in the general theorem asserting congruence of opposite sides of a parallelogram. This is not a neutral fact for a parallelogram defined only by parallel opposite sides: in hyperbolic geometry such a quadrilateral need not have congruent opposite sides.

After those side congruences are available, the local SSS step itself belongs to the congruence theory and does not require an additional use of the parallel axiom.

34.9.2 Hidden configuration data

The concise classical diagram suppresses a non-degeneracy issue. Lean must show explicitly that A, B, C are not collinear before applying SSS. The proof uses the fact that AB and CD are parallel and hence their extensional point-lines are disjoint. If C lay on AB, then C would belong to both parallel lines, a contradiction.

No new `SameSide`, `OppositeSide`, or `Between` witness is needed in the final Proposition I.34 proof. Those notions occur in the lower-level derivation of the reusable Euclidean parallelogram theory, but the API-level proof has already hidden them behind the established theorem.

34.9.3 Reusable helpers introduced

Two reusable parallelogram helpers were added while proving I.34:

```
ParallelogramOppositeAngleCongruent
ParallelogramOppositeAnglesCongruent
```

The first derives one pair of opposite angles from SSS on the two triangles cut by a diagonal. The second obtains both pairs by cyclic relabelling. These are not local Proposition helpers: they belong to `HilbertInterface` and are available to later propositions.

The packaging predicate

```
OppositeAnglesCongruent
```

records the two required angle congruences.

34.10 Dependency Summary

The classical dependency is

$$\mathrm{I.29} + \mathrm{I.29} \longrightarrow \mathrm{I.26} \longrightarrow \mathrm{I.34}.$$

The actual Lean dependency is

$$\begin{gathered}\text{Hilbert axioms}\\ \Downarrow\\ \text{HilbertInterface Euclidean parallel theory}\\ \Downarrow\\ \text{ParallelogramOppositeSidesCongruent}\\ \Downarrow\\ \text{HilbertSSS}\\ \Downarrow\\ \text{ParallelogramOppositeAngleCongruent}\\ \Downarrow\\ \text{ParallelogramOppositeAnglesCongruent}\\ \Downarrow\\ \text{Euclid I.34.}\end{gathered}$$

For the diagonal theorem the last part is even shorter:

$$\text{opposite sides congruent} + AC \cong CA \longrightarrow \text{SSS} \longrightarrow \triangle ABC \cong \triangle CDA.$$

New axiom:	no
New reusable helper:	yes
Book Zero helper added:	no
Build clean:	yes
`sorry`:	no

34.11 Conclusion

Proposition I.34 is short at the Proposition-file level because the general Euclidean theory of parallelograms has already been established in the Hilbert interface. This is a deliberate gain from the layered architecture of the project.

Euclid obtains the theorem from parallel-angle relations and ASA. The present Lean reconstruction instead reuses the already derived congruence of opposite sides and applies SSS across the diagonal. The resulting proof is not a mechanical transcription of Euclid, but it preserves the same synthetic geometry while exposing where the Euclidean content actually resides.

The third classical assertion is represented without introducing area: the diagonal AC is proved to cut the parallelogram into the congruent triangles ABC and CDA.

Chapter 35

Proposition I.35

35.1 Introduction

Euclid's Proposition I.35 is the first proposition in Book I in which the word "equal" no longer refers merely to congruent segments or congruent angles. The proposition states that two parallelograms on the same base and in the same parallels are equal to one another. The two parallelograms need not be congruent. What Euclid uses is a cut-and-paste comparison of rectilinear figures.

This makes I.35 a genuine transition point in the reconstruction. Up to I.34, the main formal objects are points, lines, betweenness, segment congruence, angle congruence, and parallelism. At I.35 one must decide what "equal figures" means before the classical argument can even be stated faithfully.

The present development does not introduce numerical area. Instead it introduces a small formal scissors calculus. Triangles are the basic pieces, finite formal sums of triangles represent decompositions, and a generated relation records the operations of splitting a triangle, replacing it by a congruent triangle, adding decompositions, and composing such steps.

The final Lean theorem therefore concludes not a numerical area equality but formal equicomplementability:

```
theorem euclid_proposition_35
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D E F : Geo.Point)
    (hABCD : IsParallelogram Geo A B C D)
    (hEBCF : IsParallelogram Geo E B C F)
    (hADF : Geo.Between A D F)
    (hAEF : Collinear Geo A E F) :
    HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo A B C D)
      (hilbertParallelogramTerm Geo E B C F)
```

The extra order and collinearity hypotheses are not cosmetic. They make explicit part of the configuration which is left implicit by the classical diagram and are essential to a complete formal case analysis.

35.2 The Classical Configuration

Let $ABCD$ and $EBCF$ be parallelograms on the common base BC. Their opposite sides AD and EF lie on one straight line which is parallel to BC. Euclid's statement is therefore

$$ABCD \quad \text{and} \quad EBCF \quad \text{on the same base } BC \quad \Longrightarrow \quad ABCD = EBCF$$

in Euclid's language of equality of rectilinear figures.

A complete synthetic proof has to distinguish three possible positions of the two upper sides. Wilkins isolates exactly these configurations:

1. the upper sides overlap, with order A, E, D, F;
2. the upper sides meet at a common endpoint, so $E = D$;
3. the upper sides are disjoint, with order A, D, E, F.

Only the third configuration is treated explicitly in Euclid's proof. The first two are not exceptional from the formal point of view: they are genuine cases which a theorem prover cannot recover from the picture.

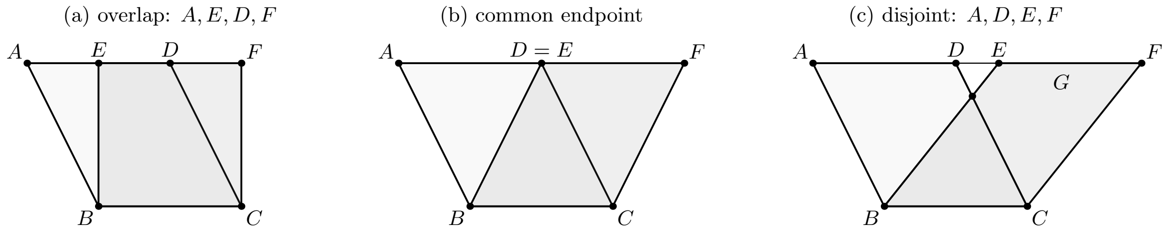


Figure 35.1: The three upper-side configurations relevant to Proposition I.35. The point G is required only in the disjoint case.

The current Lean theorem uses $A - D - F$ as an outer container:

$$A - D - F, \qquad A, E, F \text{ collinear.}$$

From the parallelogram hypotheses and this data, the reconstruction first proves $A - E - F$. The relative order of the two interior points D and E then yields precisely the three cases shown in Figure 35.1.

35.3 Classical Proof

Euclid's explicit proof is the disjoint case. Since $ABCD$ and $EBCF$ are parallelograms, Proposition I.34 gives

$$AD = BC = EF, \qquad AB = DC.$$

Because DE is common to the two longer upper segments, Common Notions 1 and 2 yield

$$AE = DF.$$

The lines AB and DC are parallel, and the upper line AF is a transversal. Proposition I.29 therefore gives

$$\angle EAB = \angle FDC.$$

Euclid then uses Proposition I.4 to conclude that the outer triangles are equal:

$$\triangle EAB = \triangle FDC.$$

In the disjoint configuration the lines EB and DC meet at G. Euclid subtracts the common triangle DGE from the two equal outer triangles, obtaining equal trapezia, and then adds the common triangle GBC. This produces the two parallelograms.

The classical dependency graph is therefore

$$\begin{gathered}
\text{I.34} + \text{I.29} \\
\Downarrow \\
AE = DF, \quad AB = DC, \quad \angle EAB = \angle FDC \\
\Downarrow \quad \text{I.4} \\
\triangle EAB = \triangle FDC \\
\Downarrow \quad \text{C.N.3} \\
\text{equal remainders} \\
\Downarrow \quad \text{C.N.2} \\
ABCD = EBCF.
\end{gathered}$$

The overlap and common-endpoint cases use the same outer-triangle comparison but a different addition pattern. Wilkins' three-case analysis is therefore a useful completion of Euclid's single drawn configuration.

35.4 Equidecomposition and Equicomplementability

The decisive modern control source for I.35 is Hilbert's theory of plane area. In Chapter IV, Section 18 of the *Foundations of Geometry*, Hilbert defines two polygons to be equidecomposable when they can be triangulated into finitely many pairwise congruent triangles. He then defines equicomplementability by allowing pairs of equidecomposable polygons to be adjoined before comparing the resulting figures by equidecomposition.

In Section 19 Hilbert states Theorem 44: two parallelograms with the same bases and the same altitudes are equicomplementable. This is the Hilbertian counterpart of the Euclidean area argument which begins at I.35.

The distinction between the two notions is essential. Equidecomposability is a direct finite scissors congruence. Equicomplementability is weaker: the figures may become equidecomposable only after suitable complements have been adjoined. Hilbert explicitly notes that in non-Archimedean geometry figures with equal bases and altitudes may be equicomplementable without being equidecomposable.

Hartshorne makes the same distinction under the name *equal content*. His definition requires figures P, Q and P', Q' to be nonoverlapping, requires Q and Q' to be equidecomposable, and requires the two enlarged figures to be equidecomposable. His Example 22.1.2 identifies Euclid I.35 as an example of equal content.

This matters for the present Lean development because the formal calculus is not yet a complete model of polygonal regions. It records formal sums of triangles, but it does not yet carry an interior or a nonoverlap predicate. Consequently

`HilbertScissorsEquicomplementable`

should be read exactly as its source comment says: a *term-level analogue* of Hilbert's equicomplementability relation. The theorem proved in I.35 is therefore a faithful algebraic scissors certificate, but a later semantic layer would still be needed to identify every formal sum with an actual nonoverlapping polygonal union.

35.4.1 Why numerical area is deliberately avoided

A modern proof could assign an area to each parallelogram and write

$$\text{area} = \text{base} \cdot \text{height}.$$

That would be the wrong logical direction here. Numerical area is developed later than the elementary scissors theory in both Hilbert and Hartshorne. Using it at I.35 would replace Euclid's synthetic argument by a stronger and logically downstream theorem.

The present reconstruction therefore stays at the level of finite cutting, congruence, and recombination.

35.5 The Hilbert Scissors Calculus

The file `HilbertScissors.lean` introduces two formal types:

```
abbrev HilbertTriangleTerm :=
  Multiset Geo.Point

abbrev HilbertScissorsTerm :=
  Multiset (HilbertTriangleTerm Geo)
```

A triangle is represented by an unordered multiset of its three vertices. A scissors term is an unordered multiset of triangle terms. This choice has two useful formal consequences:

- the order of the three names used to write a triangle is not geometric data;
- the order in which the triangular pieces are listed is not geometric data either.

Thus much of the bookkeeping which would otherwise be expressed by explicit list permutations disappears at the representation level.

The central generated relation is

```
inductive HilbertScissorsEq
    (Geo : Geometry.Geo) :
    HilbertScissorsTerm Geo ->
    HilbertScissorsTerm Geo ->
    Prop

  | refl
      (P : HilbertScissorsTerm Geo) :
      HilbertScissorsEq Geo P P

  | symm
      {P Q : HilbertScissorsTerm Geo}
      (h : HilbertScissorsEq Geo P Q) :
      HilbertScissorsEq Geo Q P

  | trans
      {P Q R : HilbertScissorsTerm Geo}
      (hPQ : HilbertScissorsEq Geo P Q)
      (hQR : HilbertScissorsEq Geo Q R) :
      HilbertScissorsEq Geo P R

  | add
      {P Q R S : HilbertScissorsTerm Geo}
      (hPQ : HilbertScissorsEq Geo P Q)
      (hRS : HilbertScissorsEq Geo R S) :
      HilbertScissorsEq Geo
        (P + R)
        (Q + S)

  | split
      (A B C X : Geo.Point)
      (hBXC : Geo.Between B X C) :
      HilbertScissorsEq Geo
        (hilbertScissorsTriangle Geo A B C)
        (hilbertScissorsTriangle Geo A B X +
         hilbertScissorsTriangle Geo A X C)

  | congruent
      (A B C D E F : Geo.Point)
      (hCong :
        TriangleCongruenceResult
          Geo A B C D E F) :
      HilbertScissorsEq Geo
        (hilbertScissorsTriangle Geo A B C)
        (hilbertScissorsTriangle Geo D E F)
```

The two geometric generators are therefore exactly the expected ones:

- `split`: cut a triangle at a point lying between two vertices;
- `congruent`: replace one triangle by a congruent triangle.

The remaining constructors make this relation an equivalence relation compatible with formal addition.

For formal equicomplementability, the implementation asks for formal complements R and S such that

$$R \sim S \qquad \text{and} \qquad P + R \sim Q + S,$$

where $\sim$ denotes `HilbertScissorsEq`. This is the algebraic pattern of Hilbert's definition, while the geometric nonoverlap conditions remain outside the present term language.

35.5.1 Triangulation lemmas

Two reusable lemmas are particularly important:

```
parallelogram_two_triangulations
crossing_quadrilateral_two_triangulations
```

The first proves that the two diagonal triangulations of a parallelogram are scissors-equivalent. It uses the existing theorem that the diagonals of a Euclidean parallelogram meet internally.

The second is more general. If the diagonals AC and BD of a quadrilateral meet internally at M, then

$$ABC + ACD \sim ABD + BCD.$$

The scissors move is easier to read as a three-stage refinement.

$$ABC + ACD$$

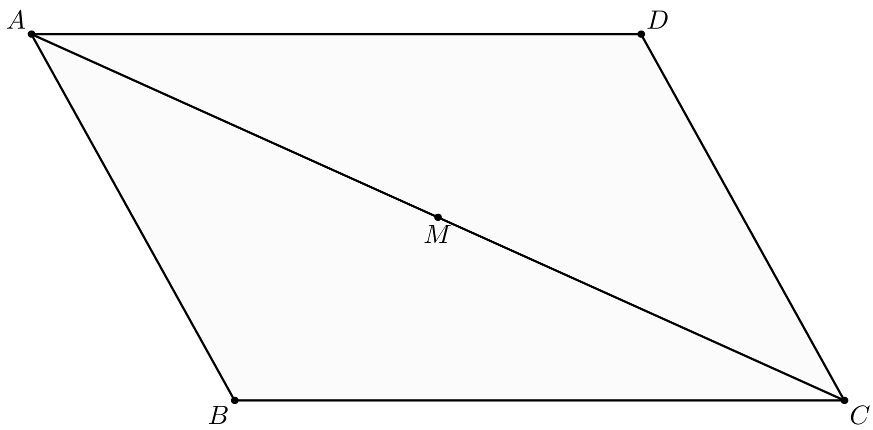


Figure 35.2: First triangulation of the quadrilateral, along the diagonal AC.

First, split the two triangles along the intersection point M of the diagonals. This refines the decomposition into the same four elementary triangles.

The four pieces can then be regrouped along the other diagonal BD.

Its proof is pure scissors algebra after the two betweenness witnesses are provided. This lemma is what makes the overlap case short: the common trapezoid $EBCD$ can be flipped from one diagonal triangulation to the other without introducing a special area theorem.

$$ABM + BMC + CMD + DMA$$

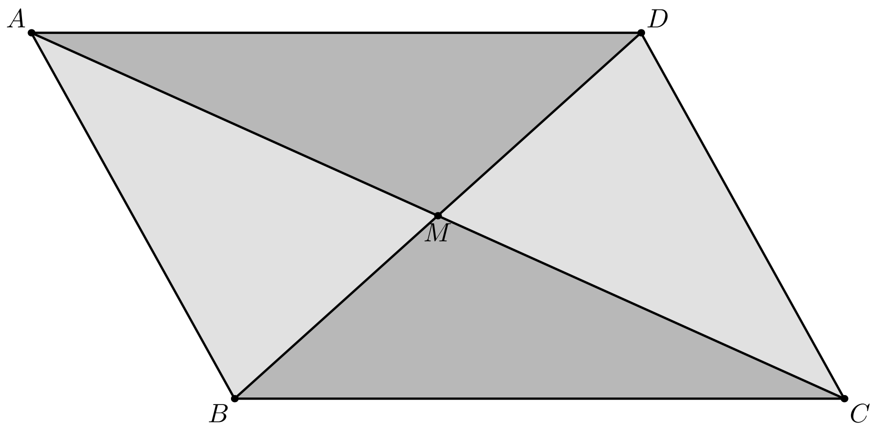


Figure 35.3: Common refinement at the diagonal intersection M.

$$ABD + BCD$$

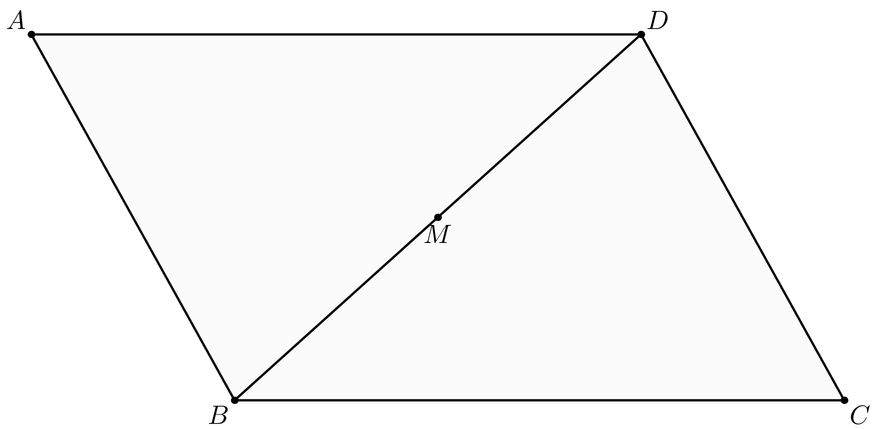


Figure 35.4: Second triangulation of the same quadrilateral, along the diagonal BD.

35.6 Proof Architecture of the Reconstruction

The actual Lean proof is not a mechanical transcription of Euclid's proof. It has two logically distinct stages.

35.6.1 Stage 1: recover the upper-line order

The final theorem is given

$$A - D - F \qquad \text{and} \qquad A, E, F \text{ collinear.}$$

The theorem `i35_upper_order` proves

$$A - E - F.$$

The proof uses opposite-side congruence for the two parallelograms to obtain $AD \cong EF$. Hilbert's betweenness trichotomy on A, E, F then gives three possible orders. The two bad orders are excluded separately.

The first exclusion,

`i35_not_EAF`,

is a segment-order argument. From $A - D - F$ one gets $AD < AF$; congruence transports this to $EF < AF$. The bad order $E - A - F$ would imply the reverse strict inequality, contradicting asymmetry.

The second exclusion,

`i35_not_AFE`,

is genuinely planar. The helper `i35_cross_intersection` first constructs a point where BF crosses AC. The diagonal theorem for the second parallelogram gives another point where BF crosses EC. Under the bad order $A - F - E$, the same line would also meet the third side AE of the triangle AEC. The line-separation theorem `hilbert_line_avoids_third_triangle_side` forbids one line from meeting all three sides in the required interior configuration.

Once $A - E - F$ is known, the pure order theorem `i35_right_cases_of_common_container` compares the two interior points D and E and produces exactly

$$A - E - D - F, \qquad E = D, \qquad \text{or} \qquad A - D - E - F.$$

35.6.2 Stage 2: solve the three scissors cases

The three cases are dispatched by `i35_from_three_cases`.

Overlap case

The outer triangles EAB and FDC are shown congruent. The common trapezoid $EBCD$ is then re-triangulated by `crossing_quadrilateral_two_triangulations`. No nonzero complement is needed. The two parallelogram terms are directly scissors-equivalent, so they are equicomplementable with zero complements.

Common-endpoint case

The two parallelograms share the upper endpoint. Again no auxiliary point G is required. The proof changes the triangulations and replaces the two outer triangles by congruent ones. The resulting parallelogram terms are directly scissors-equivalent, hence equicomplementable with zero complements.

Disjoint case

This is Euclid's explicit configuration. Here the point G is essential. The helper `i35_disjoint_intersection`

constructs G with

$$E - G - B, \qquad D - G - C.$$

The existence proof is not read off from the diagram. It uses Hilbert order and Pasch-type intersection machinery, followed by parallel/order transport to place the intersection on both required segments.

Instead of literally subtracting DGE as Euclid does, the formal proof uses the equivalent completion pattern. It is useful to display the four stages separately.

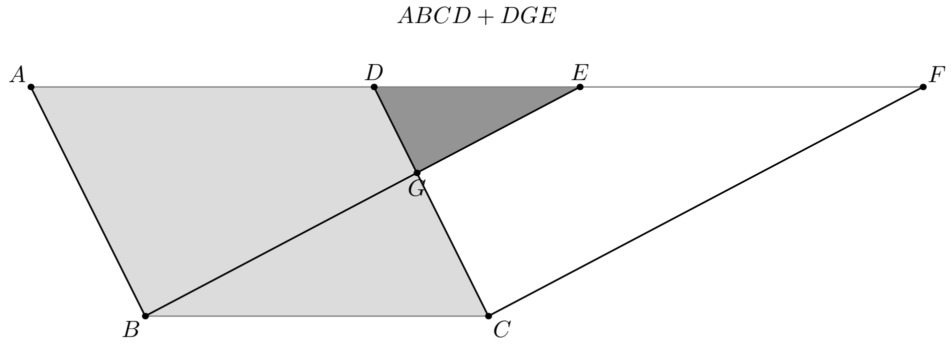


Figure 35.5: Step 1. Adjoin the triangle DGE to the parallelogram $ABCD$.

Using the betweenness relations $E - G - B$ and $D - G - C$, split and regroup the augmented figure. The result is the outer triangle EAB together with the common triangle GBC.

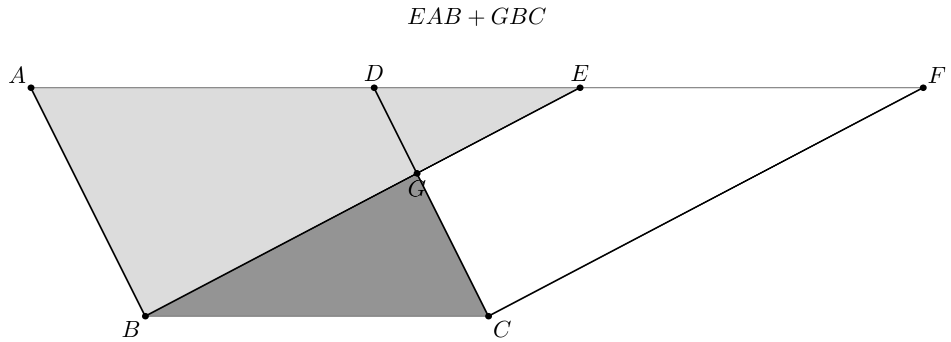


Figure 35.6: Step 2. The augmented left-hand figure is rearranged as $EAB + GBC$.

The outer triangles EAB and FDC are congruent. The scissors calculus therefore replaces the first outer triangle by the second while leaving the common triangle GBC unchanged.

Finally, reverse the splitting and regrouping on the right-hand side.

Thus the complete formal chain is

$$\begin{aligned} ABCD + DGE &\sim EAB + GBC \\ &\sim FDC + GBC \\ &\sim EBCF + DGE. \end{aligned}$$

This is exactly the formal equicomplementability witness needed for I.35.

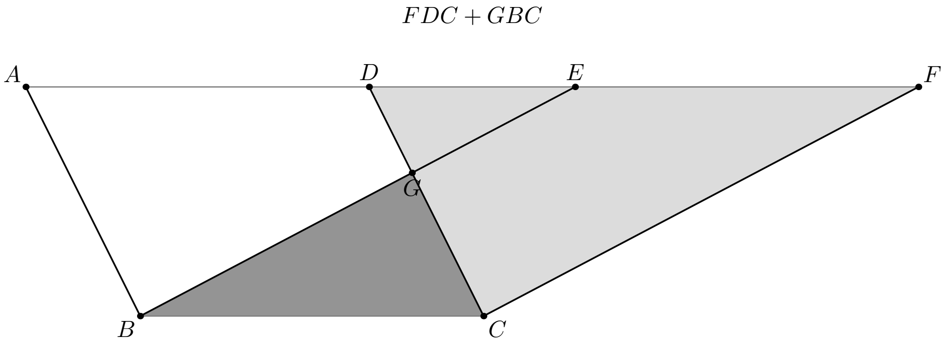


Figure 35.7: Step 3. Replace the congruent outer triangle: $EAB + GBC \sim FDC + GBC$.

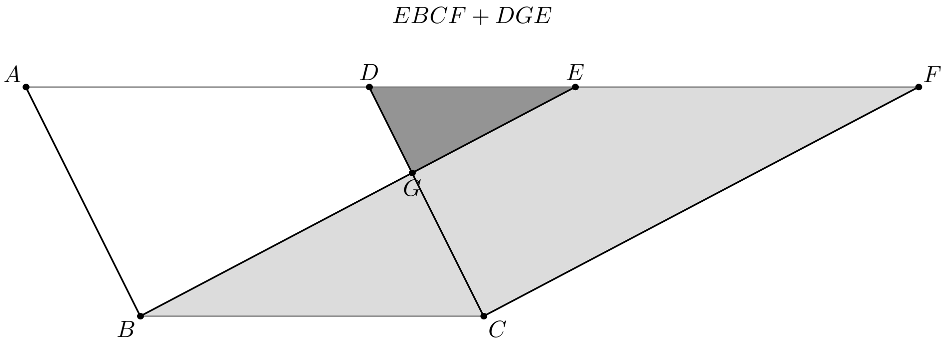


Figure 35.8: Step 4. Regroup the pieces as the second augmented parallelogram $EBCF + DGE$.

35.6.3 The complete Lean DAG

The reconstruction can be summarized as

$$\text{two parallelograms on } BC + A - D - F + \text{Collinear}(A, E, F)$$

$$\Downarrow$$

$$AD \cong EF + \text{segment order} + \text{line separation}$$

$$\Downarrow$$

$$A - E - F$$

$$\Downarrow$$

$$\boxed{A - E - D - F} \quad \boxed{E = D} \quad \boxed{A - D - E - F}$$

$$\Downarrow$$

overlap scissors proof   common-endpoint proof   disjoint proof with G

$$\Downarrow$$

`HilbertScissorsEquicomplementable`

$$\Downarrow$$

`euclid_proposition_35`.

35.7 Lean Formalization

The final Proposition file is intentionally short. It imports the separate scissors module and exposes only the Book I theorem:

```
import CGJteamLab.HilbertScissors

namespace Geometry

universe u

variable (Geo : Geometry.Geo)

theorem euclid_proposition_35
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D E F : Geo.Point)
    (hABCD : IsParallelogram Geo A B C D)
    (hEBCF : IsParallelogram Geo E B C F)
    (hADF : Geo.Between A D F)
    (hAEF : Collinear Geo A E F) :
    HilbertScissorsEquicomplementable Geo
      (hilbertParallelogramTerm Geo A B C D)
      (hilbertParallelogramTerm Geo E B C F) := by
  exact
    i35_complete
      Geo
      A B C D E F
      hABCD
      hEBCF
      hADF
      hAEF

end Geometry
```

All substantial work is therefore isolated in `HilbertScissors.lean`. This is deliberate. Proposition I.35 is the first client of the scissors language, but the formal content calculus is not part of the statement of Euclid I.35 itself.

The endpoint theorem `i35_complete` is also thin. Its role is to package the established ordered proof:

$$\begin{gathered}\texttt{i35\_upper\_order}\\ \Downarrow\\ \texttt{i35\_right\_cases\_of\_common\_container}\\ \Downarrow\\ \texttt{i35\_from\_three\_cases}\\ \Downarrow\\ \texttt{i35\_complete}.\end{gathered}$$

35.8 Relationship with Earlier Propositions

35.8.1 Proposition I.29

Euclid uses I.29 explicitly to identify the corresponding angles $\angle EAB$ and $\angle FDC$. The current Lean proof does not replay this exact SAS route at the top level. Parallelism enters through the developed Hilbert theory and through the parallelogram infrastructure.

35.8.2 Proposition I.34

I.34 is the immediate metric input. Opposite sides of the two parallelograms are congruent, so

$$AD \cong BC \cong EF$$

and the second pair of opposite sides provides the remaining outer-triangle side congruence. The Lean helper `ParallelogramOppositeSidesCongruent` packages this information.

The current reconstruction then proves the outer triangles by the already available Hilbert SSS theorem rather than by Euclid's SAS argument. This is another example of the project deliberately reusing a stronger developed Hilbert API rather than mechanically replaying the historical DAG.

35.8.3 Transition to Proposition I.36

I.36 is a direct generalization from the same base to equal bases. Euclid's classical proof uses I.33 to construct an intermediate parallelogram and then applies I.35 twice. For the formal development this means that I.35 is not an isolated experiment: the scissors/equicomplementability language is now a real dependency of the next area propositions.

35.9 Hilbert and Book Zero Components Used

The I.35 reconstruction uses a much broader part of the developed Hilbert interface than the final 38-line Proposition file suggests.

- `IsParallelogram` and the Euclidean parallelogram theory;
- `ParallelogramOppositeSidesCongruent`;
- `ParallelogramDiagonalIntersectionExists`;
- the derived `HilbertSSS` triangle-congruence theorem;
- Hilbert betweenness trichotomy and inner/outer transitivity;
- synthetic segment comparison and transport under congruence;
- Pasch/intersection machinery for the disjoint case;
- line separation through `hilbert_line_avoids_third_triangle_side`;
- transport of betweenness from collinearity and parallelism;
- the newly introduced formal scissors layer in `HilbertScissors.lean`.

No new Book Zero theorem is introduced by the final Proposition file. The new reusable material is instead concentrated in the scissors module.

The source hierarchy also changes emphasis at this point. Borsuk remains a useful control source for order, congruence, and parallelogram geometry, but the new notion of equality of figures is controlled more directly by Hilbert's Chapter IV and Hartshorne's theory of content. The Beeson proof corpus remains useful primarily as an audit of hidden incidence and order conditions rather than as the architecture of the Lean proof.

35.10 Formalization Notes

35.10.1 Axiomatic status

The final theorem assumes

```
[HilbertEuclideanPlane Geo]
```

and is therefore Euclidean in the current project API. This is consistent with Hilbert's Chapter IV treatment, which assumes the parallel axiom in addition to incidence, order, and congruence. The present file does not claim a sharper minimal axiom boundary for every internal helper.

The Euclidean strength is already visible in the reusable parallelogram results. Opposite-side congruence and the developed intersection/parallel machinery are called through the Euclidean-plane instance used by the project. Once the relevant congruences and betweenness data are available, individual scissors rewrites themselves are purely formal.

35.10.2 Hidden configuration data

I.35 exposes more hidden diagrammatic information than the immediately preceding propositions. The formal proof must explicitly manage:

- the outer order $A - D - F$;
- collinearity of A, E, F;
- the derived order $A - E - F$;
- the trichotomy between overlap, common endpoint, and disjoint upper sides;
- existence and exact betweenness position of the point G in the disjoint case;
- internal diagonal intersections used for re-triangulation;
- non-collinearity needed for Pasch and line-separation arguments.

The point G is especially instructive. It is not a universal witness for all forms of I.35. It appears only in the disjoint configuration. Trying to force one G-based proof onto every configuration would obscure the geometry and create unnecessary order obligations.

35.10.3 The scissors language is formal, not yet semantic

The nested `Multiset` representation is a formal algebra of triangle terms. It intentionally forgets vertex order and piece order, but it also forgets geometric interiors. Therefore formal addition is total even when two actual geometric pieces would overlap.

This is the main semantic limitation of the present module. A future polygonal semantics could add a nonoverlap predicate, partial composition, or a realization theorem connecting valid formal scissors derivations to nonoverlapping rectilinear figures. None of that is required to complete the current I.35 certificate, but the distinction is documented here so that the formal relation is not silently identified with a stronger geometric notion.

35.10.4 Reusable helpers introduced

The most obviously reusable scissors helpers are

```
triangle_swap12
triangle_swap23
triangle_cycle
scissors_triangle_swap12
scissors_triangle_swap23
scissors_triangle_cycle
parallelogram_two_triangulations
crossing_quadrilateral_two_triangulations
equicomplementable_refl
equicomplementable_symm
```

The remaining helpers whose names begin with `i35_` are deliberately left proposition-specific for now. Some may later reveal a more general Hilbert statement, but no architectural promotion is required for I.35.

35.11 Dependency Summary

The classical dependency is

$$\text{I.34} + \text{I.29} + \text{I.4} + \text{Common Notions 1–3} \longrightarrow \text{I.35}.$$

The actual Lean dependency is better represented as

$$\begin{array}{c}
\text{Hilbert incidence + order + congruence + Euclidean parallel theory} \\
\Downarrow \\
\texttt{HilbertInterface} \\
\Downarrow \\
\text{parallelogram congruence and diagonal theory} \\
+\ \text{betweenness / Pasch / line separation} \\
\Downarrow \\
\texttt{HilbertScissorsEq} + \texttt{HilbertScissorsEquicomplementable} \\
\Downarrow \\
\texttt{i35_upper_order} \\
\texttt{i35_right_cases_of_common_container} \\
\texttt{i35_overlap}\quad \texttt{i35_common_endpoint}\quad \texttt{i35_equicomplementable_core} \\
\Downarrow \\
\texttt{i35_complete} \\
\Downarrow \\
\texttt{euclid_proposition_35}.
\end{array}$$

New axiom:	no
New formal framework:	yes — `HilbertScissors.lean`
New Book Zero helper:	no
Build clean:	yes
`sorry`:	no

35.12 Conclusion

Proposition I.35 is a structural turning point in the formal reconstruction of Book I. The geometry is still elementary, but the proposition introduces an equality of figures which cannot be represented faithfully by segment or angle congruence alone.

Hilbert's distinction between equidecomposability and equicomplementability provides the correct conceptual guide. The present Lean development follows that guide by introducing a formal scissors calculus rather than a numerical area function. The calculus is intentionally modest: it is a term-level algebra of triangular decompositions, not yet a full semantic theory of polygonal regions.

The proof itself also exposes a classical omission which is invisible on the usual diagram. There are three upper-side configurations, and the auxiliary intersection point G belongs only to the disjoint one. Once this case split is made at the correct level, the proof decomposes naturally into order geometry, triangle congruence, and formal scissors recombination.

The resulting Proposition file is short because the complexity has been placed in a reusable and explicitly named layer. This is exactly the kind of separation needed for the next area propositions, beginning with I.36.

Appendix A

Euclid I.1–I.32 in Greenberg's Hilbert-Style Development

A.1 Purpose and scope

This appendix records a source audit of Euclid Book I, Propositions I.1–I.32, against Marvin Jay Greenberg's *Euclidean and Non-Euclidean Geometries: Development and History*, fourth edition (2008) [4].

The purpose of the audit is deliberately stricter than identifying mathematically related results. For each proposition we ask:

- whether Greenberg explicitly identifies the Euclidean proposition;
- whether he supplies a complete proof, a proof skeleton, a hint, an exercise, an axiom, or only related infrastructure;
- whether the argument is genuinely synthetic or is presented using numerical measurement;
- what additional foundational issue Greenberg explicitly records, such as continuity, betweenness, separation, or the status of SAS;
- how the result relates to the present Lean reconstruction.

No observation from the source audit is intentionally discarded merely because it is not needed for the present proof. Such remarks may become important later when Book I is extended beyond the present checkpoint or when the Hilbert and Book Zero interfaces are reorganized.

A.2 Why the fourth edition matters

The fourth edition is not a bibliographical reprint of the third. Greenberg explicitly reports substantial changes to Chapters 3 and 4. In particular, the proof of the crossbar theorem was moved from the exercises into the text; the terminology of Hilbert planes was aligned with Hartshorne; Dedekind's axiom was no longer assumed as part of the basic Euclidean setup; the existence proof for midpoints was moved from the exercises into the text; and the discussion of measurement was rewritten to stress that real-number measurement is convenient rather than mathematically necessary.

These changes directly affect the status of several propositions in I.1–I.32, especially I.9, I.10, I.16, I.17, and I.20.

A.3 Comparison table

Table A.1: Euclid I.1–I.32 in Greenberg's fourth edition

Euclid	**Greenberg counterpart**	**Status in Greenberg**	**Source-audit remarks and relation to the Lean reconstruction**
I.1	Construction of an equilateral triangle on a given segment.	Greenberg explicitly treats Euclid's first proposition as containing a continuity gap. The gap is filled by the circle-circle continuity principle.	This is a foundational warning, not merely a proof detail. Greenberg emphasizes that a weak quadratic continuity principle is needed here. Thus I.1 has a different status from the later neutral results derived from incidence, betweenness, and congruence alone. This should be retained when the formal dependency graph of Book I is documented.
I.2	Euclid's second postulate; extension of a segment and laying off a congruent copy.	Greenberg asks the reader explicitly to prove Euclid's second postulate and later states that it follows easily from Axioms C-1 and B-2.	This is the primitive mechanism behind geometric segment addition. Greenberg later uses exactly this observation in his discussion of I.20. The point is important for Book Zero: "addition" of segments need not be numerical; it is implemented synthetically by extension plus segment construction.
I.3	Cutting off from a greater segment a segment congruent to a given smaller segment.	No separate Greenberg proposition labelled as Euclid I.3 was located in the fourth-edition text. Its content is absorbed into segment construction, segment ordering, and betweenness.	The absence of a named counterpart is itself informative. Modern Hilbert-style organization decomposes I.3 into lower-level operations rather than treating it as an independent theorem. The Lean proof should therefore be compared with C-1, Proposition 3.12, and Proposition 3.13 rather than with a single Greenberg theorem.
I.4	SAS triangle congruence.	SAS is Congruence Axiom C-6. Greenberg explicitly rejects Euclid's superposition argument as a proof in Euclid's stated system.	This proposition becomes foundational in the Hilbert-style system. Greenberg stresses that SAS cannot be proved from the preceding incidence, betweenness, and congruence axioms without adding an appropriate congruence axiom. This remains an important distinction between Euclid's proposition order and the dependency order of the formal Hilbert layer.
I.5	Proposition 3.10: base angles of an isosceles triangle.	A complete synthetic proof is printed, using SAS.	This is a direct Hilbert-style reconstruction of the mathematical content of I.5. It is one of the cleanest reference proofs for the formal development.

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Table A.1 – continued from previous page

Euclid	Greenberg counterpart	Status in Greenberg	Source-audit remarks and relation to the Lean reconstruction
I.6	Proposition 3.18: converse of Proposition 3.10.	The proposition is stated, but the proof is assigned as Exercise 27.	Greenberg confirms the theorem and its dependency location but does not supply the completed derivation in the main text. The formal Lean proof therefore fills a genuine proof-level omission in the textbook presentation, without claiming a new mathematical theorem.
I.7	Uniqueness configuration for triangles on the same base and on the same side.	No explicit fourth-edition counterpart labelled as Euclid I.7 was located in the source audit. Related uniqueness is distributed through triangle laying-off, congruence axioms, side-of-line machinery, and subsequent congruence criteria.	This row should not be simplified to "absent." The relevant mathematics is present as infrastructure, but Greenberg does not appear to preserve I.7 as a named theorem. For future work it is worth recording exactly which formal uniqueness lemma replaces I.7 in the Hilbert interface.
I.8	Proposition 3.22: SSS criterion for congruence.	SSS is stated as a proposition. Its proof is left as Exercise 32, with a substantial hint reducing the argument by the corollary to SAS and a same/opposite-side configuration.	Greenberg supplies the mathematical route but not a completed printed proof. This is especially relevant to the formal project because SSS was previously a foundational debt in the Hilbert interface. The exercise shows that Greenberg regards SSS as derivable from the Hilbert congruence machinery rather than as an additional axiom.
I.9	Proposition 4.4(a): every angle has a unique bisector.	The proposition is stated in the text; Exercise 6(b) asks for the proof of existence and uniqueness. Greenberg says the angle-bisector construction follows easily from the midpoint construction.	Greenberg explicitly comments on Euclid's own I.9: Euclid uses I.1 and also relies tacitly on a betweenness fact visible in the diagram, namely that a constructed point lies on the required opposite side. This note should be retained because it identifies the hidden order/separation content that a Lean proof must expose.
I.10	Proposition 4.3: every segment has a unique midpoint.	A major change from the third edition: the existence proof is now in the text as an explicit 18-step proof skeleton. The reader is asked to justify the steps; uniqueness remains Exercise 6(a).	Greenberg explicitly says that Euclid's proof of I.10 tacitly requires the crossbar theorem. His replacement construction avoids using an equilateral triangle and therefore avoids importing the continuity issue from I.1. This distinction is valuable for future optimization of the formal dependency graph.
I.11	Perpendicular erected at a point on a line; part of Proposition 3.16.	Proposition 3.16 gives a complete proof of existence of a perpendicular through an arbitrary point, with a separate case when the point lies on the line.	The modern theorem merges the content of I.11 and I.12. When the point lies on the line, Greenberg uses the existence of a right angle and angle laying-off. The result is obtained without invoking Euclid's circle construction.

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Table A.1 – continued from previous page

Euclid	Greenberg counterpart	Status in Greenberg	Source-audit remarks and relation to the Lean reconstruction
I.12	Perpendicular dropped from a point to a line; part of Proposition 3.16.	A complete proof is printed for the case in which the point is not on the line.	Greenberg explicitly warns that Euclid's original construction of I.12 tacitly assumes line-circle continuity. His proof of Proposition 3.16 avoids that route; indeed, he notes that using line-circle continuity there would be circular because his later derivation of that continuity principle uses Proposition 3.16. This is a particularly important dependency note for any future foundational cleanup.
I.13	Supplementary-angle configuration when a line stands on another line.	No direct proposition labelled I.13 was located. The relevant theory is distributed among the definition of supplementary angles, betweenness, Proposition 3.14 on supplements of congruent angles, and angle addition (Proposition 3.19).	The source audit therefore treats I.13 as an example of reorganization: Greenberg proves and uses more general structural facts rather than retaining the Euclidean proposition as a named unit. In Lean, the corresponding proof obligations naturally live in the angle and ray interface.
I.14	Converse straight-line criterion from adjacent supplementary angles.	No direct proposition labelled I.14 was located in the fourth-edition text. Its ingredients occur in angle laying-off uniqueness, supplementary-angle theory, ray order, and angle addition/subtraction.	As with I.13, the mathematical content is absorbed into a stronger modern infrastructure. This should be preserved as a source-audit observation rather than recorded simply as "not found."
I.15	Proposition 3.15(a): vertical angles are congruent.	The theorem is stated in the text. Exercise 25 asks the reader to deduce it from Proposition 3.14.	Thus the result is present but the proof is not fully written out in the main text. The dependency on supplements of congruent angles matches the synthetic route used in formal reconstruction.
I.16	Exterior Angle Theorem 4.2.	A complete proof is printed. Greenberg then separately analyzes Euclid's original proof and fills its diagrammatic gap with betweenness and plane separation.	This is one of the richest entries in the audit. Greenberg stresses that the missing assumption is not that "lines are infinite in extent." What is needed is the betweenness theory, especially plane separation B-4 and its consequences. He also notes that the theorem fails in elliptic geometry because the relevant separation/parallel structure fails. These remarks are potentially important well beyond I.16.

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Table A.1 – continued from previous page

Euclid	Greenberg counterpart	Status in Greenberg	Source-audit remarks and relation to the Lean reconstruction
I.17	Corollary 2 to the Exterior Angle Theorem: the sum of the degree measures of any two angles of a triangle is less than 180 degrees.	A complete short proof is printed, but it is stated in the language of numerical angle measure.	This is not the purely synthetic presentation used in the Lean reconstruction. Greenberg is explicit, however, that the introduction of numbers is a convenience: Hartshorne's treatment by congruence classes is mathematically preferable and valid in all Hilbert planes. Thus the numerical wording should not be mistaken for a foundational need for real numbers.
I.18–I.19	Proposition 4.5: in a triangle, the greater side lies opposite the greater angle and conversely.	Greenberg combines I.18 and I.19 into one proposition. The proof is Exercise 8, with a specific synthetic hint: choose a point on a side, use an isosceles triangle, apply the exterior angle theorem, and use trichotomy for the converse.	This is strong confirmation of the dependency structure reconstructed in Lean. The textbook does not provide the full formal derivation, so the Lean development must supply all order, noncollinearity, ray, and angle-transport details suppressed by the hint.
I.20	Triangle inequality.	A complete seven-step proof is printed. It is written using segment length notation and addition, but Greenberg immediately explains that the numerical language is only a convenience.	This row changes significantly relative to a superficial reading. Greenberg says that Euclid's proof is the same proof, with "two sides taken together" interpreted as geometric addition of segment congruence classes. He explains that Euclid II is obtained from C-1 and B-2 and that C-3 makes segment-class addition well-defined. Hence Archimedes' axiom and the full measurement theorem are not required. This is very close in mathematical structure to the present synthetic Lean proof. Greenberg also records Heron's alternative proof as an exercise: bisect the angle, use the crossbar theorem, the exterior angle theorem, and Proposition 4.5.
I.21	Interior-point comparison in a triangle; related to the exterior-angle theorem, Proposition 4.5, segment order, and triangle inequality infrastructure.	No separate fourth-edition proposition explicitly identified as Euclid I.21 was located in the present audit. Its ingredients belong to the neutral comparison theory developed around the exterior-angle theorem, side–angle comparison, and the triangle inequality.	This is another case in which Greenberg's organization is more structural than Euclid's proposition sequence. The Lean reconstruction is especially informative here because it exposes the exact betweenness and angle-order transports needed to turn the interior-point configuration into the two strict segment comparisons.

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Table A.1 – continued from previous page

Euclid	Greenberg counterpart	Status in Greenberg	Source-audit remarks and relation to the Lean reconstruction
I.22	Converse to the triangle inequality: construction of a triangle from three segments satisfying the pairwise triangle inequalities.	Greenberg explicitly ties the converse to the triangle inequality to circle-circle continuity. In his foundational organization this is a continuity result rather than a consequence of neutral incidence, betweenness, and congruence alone.	This is the principal new foundational observation beyond I.20. The Lean reconstruction likewise isolates the continuity-dependent existence step from the synthetic segment arithmetic expressing that each pair of sides taken together is greater than the third. The comparison confirms that the continuity assumption is genuine and should remain visible in the dependency graph.
I.23	Construction of an angle congruent to a given angle on a prescribed side of a given ray; Greenberg's angle-laying-off infrastructure.	The mathematical content is already part of the Hilbert-style angle construction machinery used by Greenberg. It is therefore not naturally presented as a derived Euclidean construction depending on I.22.	This sharply illustrates the difference between Euclid's historical dependency order and the Hilbert foundation. Euclid derives I.23 using earlier constructions, whereas the present Lean theorem is an interface wrapper around the stronger angle-construction principle already available in the congruence layer.
I.24	Comparison of third sides from two equal-side configurations with an inequality between the included angles; related to angle laying-off, SAS, Proposition 4.5, and neutral order theory.	No separate fourth-edition proposition explicitly identified as Euclid I.24 was located in the present audit. The required mathematical ingredients occur in Greenberg's general triangle-congruence and side–angle comparison infrastructure.	The formal reconstruction adds an important point not captured by a mere theorem-name correspondence: Euclid's displayed construction has a configuration gap. The Lean proof must analyze the possible relative positions of the auxiliary points rather than reading one configuration from the diagram. Thus I.24 is a particularly clear example of the project's role in exposing hidden order information.
I.25	Converse comparison for two triangles with two corresponding sides congruent; Greenberg Proposition 4.6.	Proposition 4.6 states, for two triangles with two corresponding sides congruent, that inequality of the included angles is equivalent to inequality of the third sides. Its proof is assigned as Exercise 22, with a detailed hint using the crossbar theorem, Proposition 4.5, isosceles-triangle arguments, and the Exterior Angle Theorem.	Greenberg packages the mathematical content of Euclid I.24 and I.25 into a single biconditional comparison theorem. Thus I.25 is not a separate stage in his organization: it is the converse direction of the same neutral comparison result. This strongly confirms the formal project's decision to treat the two propositions as neighboring directions of one comparison mechanism rather than as unrelated constructions.

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Table A.1 – continued from previous page

Euclid	Greenberg counterpart	Status in Greenberg	Source-audit remarks and relation to the Lean reconstruction
I.26	ASA and SAA triangle congruence; Proposition 3.17 and Proposition 4.1.	Proposition 3.17 is the ASA criterion. Its proof is supplied as Exercise 26 in Chapter 3. Proposition 4.1 gives the SAA criterion after the Exterior Angle Theorem; Greenberg leaves its proof as an exercise, and later Exercise 10 gives a correct proof skeleton.	Greenberg separates the two cases that Euclid combines in I.26. The ASA case belongs directly to the Hilbert congruence development, whereas the SAA case is placed after the neutral Exterior Angle Theorem. This is useful for the Lean dependency graph: the two formal branches need not have identical internal proofs even though Euclid states them together.
I.27	Theorem 4.1, Alternate Interior Angle Theorem.	A complete proof is printed: congruent alternate interior angles imply that the two lines are parallel. Greenberg proves this in neutral geometry, before assuming Hilbert's parallel postulate.	This is the precise modern Hilbert-style counterpart of the essential content of I.27. Greenberg explicitly warns immediately afterward not to use its converse. Hence the Lean reconstruction's placement of I.27 on the neutral side of the parallel-postulate boundary is confirmed directly by the source.
I.28	Theorem 4.1 together with the corresponding-angle reduction recorded in Chapter 4, Exercise 32.	The alternate-interior criterion is Theorem 4.1. Exercise 32 asks the reader to prove that corresponding angles are congruent if and only if alternate interior angles are congruent. The supplementary-angle machinery supplies the same-side-interior formulation.	Greenberg does not preserve Euclid I.28 as one named theorem, but its two classical criteria reduce to the neutral Alternate Interior Angle Theorem and elementary angle relations. This agrees with the Lean reconstruction: I.28 remains neutral and should not be confused with the converse parallel-to-angle direction used in I.29.
I.29	Proposition 4.8: the converse to Theorem 4.1.	Greenberg states that the converse Alternate Interior Angle Theorem is logically equivalent to Hilbert's parallel postulate. Proposition 4.8 is assigned as Exercise 5; the hint proves both directions of the equivalence, using uniqueness of parallels and angle laying-off.	This is the exact foundational boundary reached by Euclid at I.29. Greenberg also remarks earlier that Euclid postponed use of the fifth postulate until his twenty-ninth proposition. The Lean theorem therefore correctly requires the Euclidean-plane structure precisely when parallelism is used to recover angle congruence.
I.30	Transitivity of parallelism; consequence/equivalent form of the parallel postulate in Greenberg's Chapter 4.	Immediately after Theorem 4.5 Greenberg lists equivalent forms of the parallel postulate. Proposition 4.7 concerns a transversal meeting both of two parallel lines, and its exercise explicitly asks the reader to deduce that transitivity of parallelism is equivalent to Hilbert's parallel postulate.	I.30 is therefore genuinely Euclidean in Greenberg's logical analysis, not a neutral theorem about the definition of parallel lines. This matches the formal reconstruction, where the theorem is downstream of the Euclidean uniqueness principle rather than merely of neutral existence of parallels.

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Table A.1 – continued from previous page

Euclid	Greenberg counterpart	Status in Greenberg	Source-audit remarks and relation to the Lean reconstruction
I.31	Corollary 2 to Theorem 4.1: existence of a parallel through a point not on a given line.	Greenberg proves existence in neutral geometry by dropping a perpendicular from the point to the given line and then erecting a second perpendicular. He emphasizes that existence is available without the parallel postulate; what cannot be proved neutrally is uniqueness.	This distinction is especially important for I.31. Euclid's construction asks only for a parallel through the point, so its existence content is neutral in Greenberg's Hilbert-style theory. Hilbert's Group IV becomes relevant only when one asks that this parallel be the unique parallel. The source therefore confirms the existence-versus-uniqueness distinction that must remain explicit in the Lean documentation.
I.32	Proposition 4.11: Hilbert's parallel postulate implies that every triangle has angle sum 180°; together with Proposition 4.8, the converse Alternate Interior Angle Theorem.	Greenberg states Proposition 4.11 as a consequence of Hilbert's parallel postulate and assigns its proof as Exercise 8. He then defines triangle defect and stresses that exact angle sum 180° cannot be obtained in neutral geometry merely from the preceding theory. Elsewhere he identifies the converse Alternate Interior Angle Theorem as equivalent to the parallel postulate.	This is the closest Greenberg counterpart to the second half of Euclid I.32 and confirms its genuinely Euclidean status. Greenberg formulates the result numerically using degree measure, whereas the Lean reconstruction deliberately keeps the synthetic content: extend a side, split the exterior angle by a correctly oriented parallel ray, and represent "two right angles" without introducing degree arithmetic. Thus the source and the formalization agree on the dependency while using different languages for angle addition.

A.4 Structural observations from the audit

The fourth edition makes several general points that are directly relevant to the formal reconstruction.

First, Greenberg states that the main purpose of Chapter 3 is to fill gaps in Euclid's presentation, while also acknowledging that not every gap is filled there. He emphasizes that almost all elementary synthetic Euclidean geometry can nevertheless be developed from his axiomatic system.

Second, Greenberg separates three kinds of foundational phenomena that are easy to blur in an informal reading of Euclid:

1. **incidence, betweenness, and congruence**, which define a Hilbert plane and support a large neutral fragment;
2. **quadratic continuity**, needed for constructions such as Euclid I.1 and, later, the converse to the triangle inequality;
3. **measurement by real numbers**, which is convenient but not required for the elementary synthetic content under study.

Third, the fourth edition repeatedly points out that statements which look obvious in diagrams encode nontrivial order information. The crossbar theorem is the most visible example. Its proof was promoted from an exercise to the main text in this edition, which makes the book especially useful as a reference for the formal project.

Fourth, Greenberg's organization confirms that proposition numbering is not a dependency structure. Several Euclidean propositions disappear as named theorems because their content is absorbed into more general machinery; conversely, some single Greenberg propositions combine two Euclidean propositions.

A.5 Notes that should be retained for future work

The following source observations should be retained because they may matter in later Book I work.

- Greenberg defines a Hilbert plane as a model of the thirteen incidence, betweenness, and congruence axioms.
- His Euclidean setup adds Hilbert's Euclidean parallel axiom and circle-circle continuity; Dedekind's axiom is deliberately not assumed in the fourth edition.
- He explicitly distinguishes circle-circle continuity from Archimedes' axiom and from Dedekind completeness.
- The converse to the triangle inequality, Euclid I.22, is tied directly to circle-circle continuity. This is now part of the completed I.1–I.32 block and provides the clearest new continuity issue after I.1.
- Greenberg repeatedly refers to Hartshorne for the technically complete treatment of elementary geometry without real numbers, especially for congruence classes of angles and their addition.
- The fourth edition states that numerical measurement is introduced in Chapter 4 to shorten the exposition, not because the geometry requires real numbers.
- Greenberg explicitly rejects the widespread claim that Euclid I.16 needs an additional assumption that lines are "infinite in extent." His diagnosis is that the missing machinery is betweenness and plane separation.
- The existence proof for a midpoint in Proposition 4.3 is not Euclid's own construction. Greenberg deliberately gives a route which avoids dependence on equilateral triangles and therefore avoids importing the continuity requirement of I.1.
- Greenberg gives Heron's alternative proof of the triangle inequality as an exercise. This provides a second synthetic route to I.20 that may later be useful for comparing proof graphs or designing tactics.

A.6 Result of the fourth-edition query

The fourth edition does not provide a continuous sequence of thirty-two fully written synthetic proofs corresponding one-for-one to Euclid I.1–I.32. It provides something more useful for foundational analysis: a reorganized Hilbert-style theory in which the Euclidean results have different statuses.

The extension from I.25 to I.32 makes the logical boundary at I.29 particularly sharp. I.25 is absorbed together with I.24 into Greenberg's biconditional triangle-comparison Proposition 4.6. I.26 is split between the ASA criterion of Proposition 3.17 and the SAA criterion of Proposition 4.1. I.27 and I.28 remain on the neutral side: the Alternate Interior Angle Theorem proves angle congruence implies parallelism, and the corresponding-angle formulation is reduced to it.

At I.29 the direction reverses. Greenberg explicitly identifies the converse Alternate Interior Angle Theorem as logically equivalent to Hilbert's parallel postulate. Transitivity of parallelism, corresponding to I.30, is likewise listed as an equivalent form. By contrast, the existence statement in I.31 is already available in neutral geometry; only uniqueness of the parallel requires Hilbert's parallel postulate. Finally Proposition 4.11 states that Hilbert's parallel postulate implies that every triangle has angle sum 180°, giving Greenberg's numerical counterpart to the synthetic I.32 reconstruction.

For the present project Greenberg should therefore be used as a foundational and dependency reference, not as a source from which the Lean proofs can simply be transcribed. The formal development exposes the intermediate operations that textbook prose suppresses: permutations of collinearity, orientation of betweenness, same-ray transport, side-of-line arguments, normalization of unoriented angles, transport of segment and angle order under congruence, and repeated noncollinearity obligations.

A.7 Current checkpoint

The present source audit now concerns Euclid Book I, Propositions I.1–I.32.

The I.25–I.32 block adds the clearest structural transition encountered so far. Propositions I.25–I.28 remain within neutral comparison and angle-to-parallel theory. Proposition I.29 is the exact point at which the converse direction, parallel-to-angle, introduces the Euclidean parallel principle. Proposition I.30 inherits the same Euclidean status through transitivity of parallelism. Proposition I.31 then separates existence from uniqueness: Greenberg proves that a parallel through an external point exists already in neutral geometry, while uniqueness is Hilbert's parallel postulate. Proposition I.32 returns to the Euclidean direction and yields the angle-sum theorem.

Greenberg's treatment of I.32 is also a useful warning about language. His Proposition 4.11 is stated numerically as angle sum 180°, after a measurement theory has been introduced. The Lean reconstruction does not need that numerical layer. It records instead the synthetic exterior-angle decomposition and the straight-line configuration expressing two right angles. The agreement is therefore logical rather than notational: both developments locate the theorem beyond the neutral boundary, but they package its conclusion differently.

This appendix should be treated as a durable research record for the I.1–I.32 checkpoint. When later propositions are formalized, new Greenberg observations should be appended rather than replacing or silently deleting the present notes.

Appendix B

Euclid I.1–I.32 in Hartshorne's Hilbert-Plane Development

B.1 Purpose and scope

This appendix records a source audit of Euclid Book I, Propositions I.1–I.32, against Robin Hartshorne's *Geometry: Euclid and Beyond* [5].

The audit has a specific purpose. Hartshorne begins with Euclid's own proofs and constructions, analyzes where their logical structure is incomplete, and then reconstructs the relevant part of Book I inside a Hilbert-plane framework. For each proposition I.1–I.32, we record:

- whether Hartshorne regards Euclid's proof as valid as it stands;
- whether the proposition is replaced by an axiom or by a more general Hilbert-plane theorem;
- whether Hartshorne supplies a complete proof, a proof sketch, or an exercise;
- which hidden order, betweenness, separation, congruence, or existence issue he explicitly identifies;
- which points are especially relevant to a fully formal synthetic reconstruction.

The purpose is not to claim novelty for the classical mathematics. Rather, the audit identifies the exact level at which a conventional synthetic proof still leaves obligations that must be made explicit in a formal proof assistant.

B.2 Hartshorne's organization of the problem

Hartshorne separates the discussion into two stages. Chapter 1 studies Euclid's own geometry and repeatedly asks whether a proof depends on a diagram, an unstated positional assumption, or a construction whose existence has not yet been justified. Chapter 2 then introduces Hilbert's axioms of incidence, betweenness, and congruence and reconstructs the early propositions inside that framework.

The key structural statement appears in the section on Hilbert planes: Theorem 10.4 states that all of Euclid I.1–I.28, except I.1 and I.22, can be proved in an arbitrary Hilbert plane. The first exception is the circle-intersection problem in I.1; the second is the prescribed-side triangle construction I.22. Beginning with I.29 Hartshorne moves to a Hilbert plane with the parallel axiom (P), and Theorem 12.1 recovers Euclid's theory of parallels I.29–I.34.

B.3 Comparison table

Table B.1: Euclid I.1–I.32 in Hartshorne's Hilbert-plane development

Euclid	**Hartshorne counterpart**	**Status in Hartshorne**	**Source-audit remarks and relevance to formalization**
I.1	Construction of an equilateral triangle on a given segment.	Hartshorne treats I.1 as the first genuine exception to the Hilbert-plane recovery of Book I. The intersection of the two circles used by Euclid cannot be derived from the basic Hilbert-plane axioms alone.	Chapter 1 already isolates the hidden existence problem: the diagram shows two circles meeting, but Euclid's stated postulates do not guarantee that they do. Hartshorne therefore requires an additional intersection or continuity principle. This is a foundational dependency, not a minor gap in presentation.
I.2	Transporting a given segment to a specified point.	Hartshorne states that I.2 is effectively replaced by the segment congruence construction axiom C1.	The Euclidean proposition becomes primitive infrastructure in the Hilbert system. This is important for later segment arithmetic and for the synthetic interpretation of "adding" segments by extending a ray and laying off a congruent copy.
I.3	Cutting off from a greater segment a segment equal to a given smaller segment.	Hartshorne groups I.3 with I.2 as effectively replaced by C1 together with the order theory of segments.	The proposition is no longer structurally fundamental as a named theorem. Its mathematical content is decomposed into segment construction, uniqueness, betweenness, and segment ordering.
I.4	SAS triangle congruence.	SAS is taken as axiom C6. Hartshorne analyzes Euclid's superposition argument and concludes that it relies on an unstated theory of rigid motions.	This is one of the sharpest differences between Euclid's proposition order and a Hilbert-style dependency order. The proof by superposition is not accepted as a derivation from Euclid's explicit postulates and common notions.
I.5	Base angles of an isosceles triangle.	Hartshorne explicitly reinterprets Euclid's proof in the Hilbert system and gives the proof in detail using segment construction and SAS.	This is a direct example of the program of recovering Euclid inside a Hilbert plane. Hartshorne also points out the hidden betweenness facts needed to justify some of Euclid's positional claims.
I.6	Converse of the isosceles base-angle theorem.	Hartshorne says that Euclid's argument is essentially valid, but one point must be rewritten using the formally defined order relation on angles.	The phrase "less than" cannot be left intuitive. The proof depends on having an explicit angle-order relation and on using its asymmetry and trichotomy correctly. This is precisely the kind of obligation that becomes visible in Lean.

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Euclid	Hartshorne counterpart	Status in Hartshorne	Source-audit remarks and relevance to formalization
I.7	Uniqueness configuration for triangles on the same base and on the same side.	Hartshorne identifies a hidden dependence on the relative position of the lines and points. The missing betweenness justification is assigned for completion in the Hilbert-plane exercises.	This proposition is a canonical example of a proof whose diagram suppresses case distinctions. Hartshorne stresses that one must know where the lines and intersection points lie before the angle inequalities used in the proof are legitimate.
I.8	SSS triangle congruence.	Hartshorne replaces Euclid's superposition proof with a new theorem, Proposition 10.1 (SSS), proved from the Hilbert congruence axioms.	A complete Hilbert-style proof is supplied. The construction creates a copy of one triangle on the opposite side of a side of the other and uses angle addition and SAS. The positional cases are explicitly noted.
I.9	Construction of an angle bisector.	Hartshorne reconstructs I.9 using Hilbert tools. Since I.1 is not available in an arbitrary Hilbert plane, he replaces Euclid's equilateral-triangle step by the construction of an isosceles triangle.	He explicitly notes that it is not enough to perform the construction: one must prove that the constructed ray really lies in the interior of the original angle. That requires the order/separation machinery.
I.10	Construction of the midpoint of a segment.	Hartshorne reconstructs the proposition by the same strategy: replace Euclid's use of I.1 by an isosceles triangle. The corresponding Hilbert-tools construction is left as Exercise 10.2.	The proposition is therefore recoverable in a Hilbert plane without using the circle-intersection principle required by I.1. This is an important alternative dependency route.
I.11	Erecting a perpendicular at a point of a line.	Hartshorne explains that the construction can be recovered using Hilbert tools and assigns the explicit construction as Exercise 10.3.	The reconstruction also establishes the existence of right angles, which is not taken for granted beforehand. The proof is therefore part of the basic congruence-and-angle infrastructure rather than a compass construction in Euclid's original sense.
I.12	Dropping a perpendicular from a point to a line.	Hartshorne explicitly says that Euclid's original compass construction does not work in an arbitrary Hilbert plane. A new existence proof is required; the explicit construction is Exercise 10.4.	This is a significant foundational note. The failure is not that the theorem is false, but that Euclid's chosen construction imports an intersection principle unavailable in the bare Hilbert system.
I.13	Angles formed when a straight line stands on another straight line.	Hartshorne replaces the Euclidean proposition by Proposition 9.2 on supplements of congruent angles.	He remarks explicitly that Proposition 9.2 may be viewed as the proper Hilbert-style replacement for I.13. The modern statement is more general and fits directly into the angle-congruence infrastructure.

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Euclid	**Hartshorne counterpart**	**Status in Hartshorne**	**Source-audit remarks and relevance to formalization**
I.14	Converse straight-line criterion from adjacent supplementary angles.	Hartshorne treats I.14 as an easy consequence of the preceding supplement theory and asks for a careful Hilbert-plane formulation in Exercise 10.7.	The theorem is mathematically routine only after the modern notions of rays, supplementary angles, and angle congruence have been made precise. The formal version therefore belongs naturally to the low-level angle interface.
I.15	Vertical angles are equal.	Corollary 9.3: vertical angles are congruent. A complete short proof is given from the theory of supplementary angles.	This is a clean example in which Hartshorne replaces Euclid's common notion argument by an explicitly stated congruence theorem.
I.16	Exterior Angle Theorem.	Proposition 10.3 gives a full Hilbert-plane proof.	Hartshorne returns to the exact point he had criticized in Chapter 1: the constructed ray must really lie inside the relevant angle. He closes this gap using betweenness and plane separation. Thus the proof makes explicit the hidden positional content of Euclid's diagram.
I.17	Any two angles of a triangle are together less than two right angles.	Hartshorne says that I.17–I.21 are valid in a Hilbert plane after reinterpretation. For I.17 he avoids treating "the sum of two angles" as a primitive object and reformulates the statement using supplements and angle order.	This is especially relevant to formalization: the familiar informal statement must be replaced by a proposition expressible in the available angle language.
I.18	The greater side of a triangle subtends the greater opposite angle.	Hartshorne regards the Euclidean argument as recoverable but assigns a careful Hilbert-plane proof as Exercise 10.6.	The exercise explicitly calls attention to betweenness and inequalities. These are exactly the two classes of hidden obligations that dominate a formal reconstruction of I.18.
I.19	The greater angle of a triangle subtends the greater opposite side.	Included in Hartshorne's statement that I.17–I.21 are valid in a Hilbert plane after the preceding reconstruction of order and congruence.	Hartshorne does not single out a new main-text proof comparable to Proposition 10.3. Its validity rests on the now-formalized order relations for sides and angles and on the preceding comparison theorem.
I.20	Triangle inequality.	Hartshorne says that I.17–I.21 are valid in a Hilbert plane and assigns a careful proof of I.20 as Exercise 10.8.	The theorem therefore belongs to neutral Hilbert geometry. Hartshorne's exercise status is important: the mathematics is declared recoverable, but the complete chain of synthetic justifications is left to the reader. A formal proof must expose all those omitted steps explicitly.

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Table B.1 – continued from previous page

Euclid	**Hartshorne counterpart**	**Status in Hartshorne**	**Source-audit remarks and relevance to formalization**
I.21	Comparison theorem for two triangles on a common base, with one vertex inside the other triangle.	Hartshorne includes I.21 in his statement that I.17–I.21 are valid in an arbitrary Hilbert plane.	The proposition remains neutral. Its two conclusions depend on the already developed order theory for segments and angles; no new continuity or parallel principle is introduced.
I.22	Construction of a triangle from three prescribed side lengths satisfying the strict triangle inequalities.	This is Hartshorne's second explicit exception. In a bare Hilbert plane the required two-circle intersection need not exist, even when the side inequalities hold. Section 11 restores the construction by adding the intersection axiom (E); Theorem 11.7 then validates I.22.	This is not a defect in the inequality theory but a genuine existence gap. It sharply separates neutral order/congruence results from circle-intersection continuity.
I.23	Transport of a given angle to a prescribed point and side.	Hartshorne does not reconstruct Euclid's I.22-based proof in a bare Hilbert plane; instead I.23 is replaced directly by congruence axiom (C4), the angle transporter.	A long Euclidean construction becomes primitive Hilbert infrastructure. This is a major dependency compression and explains why later neutral results can proceed even though I.22 itself fails without (E).
I.24	Two triangles with two corresponding sides congruent: the greater included angle has the greater third side.	Hartshorne states that the remaining results before the parallel postulate, including I.24, are valid as they stand in a Hilbert plane.	The proposition is a neutral comparison theorem. Its formal burden is therefore configuration and strict order, not Euclidean parallelism.
I.25	Converse comparison: with two corresponding sides congruent, the greater third side has the greater included angle.	Included among the remaining pre-parallel results valid in a Hilbert plane.	Together I.24 and I.25 form a neutral comparison pair. The source supports treating them as neighboring directions of one order-theoretic mechanism.
I.26	ASA and AAS triangle congruence.	Hartshorne explicitly identifies I.26 with ASA and AAS and includes it among the results valid in an arbitrary Hilbert plane.	The theorem is neutral. For formalization the two cases may still be separated internally because their proof obligations need not be identical.
I.27	Equal alternate interior angles imply that the two lines are parallel.	Hartshorne treats I.27 as a neutral theorem and uses it repeatedly before introducing the parallel axiom. In Chapter 1 he derives it from I.16 by excluding an intersection on either side.	This direction, angle relation to parallelism, is available without Playfair. It is the neutral half of the later converse pair with I.29.
I.28	Corresponding-angle or supplementary-interior-angle criteria imply parallelism.	Hartshorne says that I.28 follows from I.27 using vertical angles I.15 or supplementary angles I.13, and Theorem 10.4 includes it in the Hilbert-plane fragment.	I.28 completes the neutral parallel criteria. Its conclusion is parallelism, but no uniqueness-of-parallels principle is needed.

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Table B.1 – continued from previous page

Euclid	**Hartshorne counterpart**	**Status in Hartshorne**	**Source-audit remarks and relevance to formalization**
I.29	A transversal of two parallel lines gives equal alternate interior angles and the associated angle relations.	This is the first proposition where Hartshorne requires the parallel axiom. With Playfair axiom (P), construct through the relevant point a line making the required angle by (C4); I.27 proves that the constructed line is parallel to the second given line; uniqueness from (P) identifies it with the original parallel.	Hartshorne isolates the Euclidean step exactly: construction and I.27 are neutral, while identification of two parallels through the same point is the non-neutral uniqueness step.
I.30	Lines parallel to the same line are parallel.	Hartshorne states that I.30 is essentially equivalent to the parallel axiom (P).	Transitivity of parallelism is therefore not merely a formal property of the definition. In Hartshorne's organization it is an equivalent interface form of Euclidean parallel uniqueness.
I.31	Construction of a parallel through a given external point.	Hartshorne explicitly notes that existence already follows in a Hilbert plane from the angle transporter (C4) and I.27, without (P). After (P) is assumed, I.31 can be strengthened to existence and uniqueness.	This is the clearest existence-versus-uniqueness split in the source. The proposition's construction is neutral; only the stronger unique parallel statement is Euclidean.
I.32	Exterior angle equals the two remote interior angles; equivalently, the angles of a triangle are together equal to two right angles.	Hartshorne places I.32 among the remaining propositions following from (P). He also warns that literal angle addition is not primitive in his Hilbert system and proposes the scrupulous reformulation: the sum of any two angles is supplementary to the third.	The theorem is genuinely Euclidean through I.29. Hartshorne's reformulation is especially relevant to Lean, where the synthetic supplementary-angle statement can be preferable to numerical or additive angle language.

B.4 Foundational observations retained from Hartshorne

Several remarks in Hartshorne are important beyond the immediate I.1–I.32 audit and should be preserved for later work.

- Hartshorne repeatedly uses a proof from a diagram as a warning sign. When a step depends on the relative position of points, lines, or rays, that position must be derived from the axioms rather than read from the figure.
- Betweenness and plane separation are not secondary bookkeeping devices. They are the mechanism that repairs several of the early Euclidean proofs, especially I.7, I.9, and I.16.
- Hartshorne's Hilbert plane is built from incidence, betweenness, and congruence. The parallel postulate is kept separate so that one can identify which propositions belong to neutral geometry.

- The first major exception is I.1. The basic Hilbert-plane axioms do not guarantee the intersection of the two circles used by Euclid. Hence an additional intersection/continuity principle is required.
- Euclid's constructions and Hilbert's existence axioms should not be conflated. A Euclidean compass construction may fail as a proof in an arbitrary Hilbert plane even when the corresponding geometric object can still be proved to exist by another route.
- Hartshorne replaces Euclid's superposition method by axiomatic SAS. He interprets superposition as requiring a theory of rigid motions, which is not supplied by Euclid's stated postulates.
- Segment addition is formalized synthetically. Hartshorne defines the sum of two segments by laying off a congruent copy on a ray and proves that this construction is well-defined up to congruence.
- Segment order is then defined using betweenness and congruence, and Hartshorne proves its compatibility with congruence and its trichotomy properties.
- The angle theory is developed in parallel: supplementary angles, vertical angles, angle addition, and angle order are all made explicit before the later Euclidean propositions are recovered.
- I.17 illustrates an important language issue. An informal sum of angles need not correspond directly to a primitive formal object, so the theorem is reformulated using inequalities and supplements.
- Hartshorne explicitly identifies I.22 as the second exception in Theorem 10.4. The continuity/intersection issue encountered at I.1 therefore returns inside the present checkpoint.
- I.23 is replaced by axiom (C4), the angle transporter. Thus the Hilbert reconstruction can recover later neutral propositions without reconstructing Euclid's I.22-dependent angle-copy construction.
- Theorem 10.4 ends at I.28. The endpoint is structural: I.27–I.28 derive parallelism from angle relations in neutral geometry, whereas I.29 reverses the implication and requires uniqueness of parallels.
- Hartshorne states that I.30 is essentially equivalent to (P), making transitivity of parallelism an equivalent Euclidean interface form.
- I.31 separates existence from uniqueness. Existence of a parallel is neutral, from (C4) and I.27; uniqueness is supplied only after (P).
- For I.32 Hartshorne explicitly recommends replacing informal angle addition by a supplementary-angle formulation. This is directly relevant to a synthetic Lean statement without numerical angle measure.

B.5 Result of the Hartshorne audit

Hartshorne reorganizes Euclid I.1–I.32 according to the logical layers of a Hilbert plane.

The first layer is neutral geometry. Theorem 10.4 states precisely that all propositions I.1–I.28 except I.1 and I.22 can be recovered in an arbitrary Hilbert plane. The two exceptions are both existence problems caused by missing circle-intersection continuity. Hartshorne later adds axiom (E), and Theorem 11.7 restores exactly the Euclidean constructions I.1 and I.22.

Within the neutral layer Hartshorne also changes the dependency graph. I.2 and I.3 are absorbed into segment transporter axiom (C1), I.4 becomes SAS axiom (C6), and I.23 is replaced by the angle transporter axiom (C4). Several Euclidean constructions are therefore compressed into primitive Hilbert infrastructure. Other propositions, especially I.5, I.8, I.15, and I.16, receive explicit Hilbert-style reconstructions in which betweenness, plane separation, angle interiors, and strict order are made visible.

The second layer begins at I.29. Hartshorne explicitly identifies this as the first use of the parallel axiom. His Hilbert-style proof of I.29 is particularly informative: construct a candidate line with the desired angle by (C4), use neutral I.27 to show that it is parallel, and then use Playfair uniqueness (P) to identify it with the given parallel. Thus the non-neutral step is isolated exactly at uniqueness.

This immediately clarifies I.30 and I.31. Hartshorne says that I.30 is essentially equivalent to (P). By contrast, the existence content of I.31 is already neutral, following from (C4) and I.27; only the stronger existence-and-uniqueness statement needs (P). Theorem 12.1 then packages I.29–I.34 as the Euclidean theory of parallels.

Finally, I.32 illustrates a language issue already visible at I.17. Hartshorne notes that literal sums of angles are not primitive objects in his system. The careful Hilbert-style statement is that the sum of any two angles of a triangle is supplementary to the third. This gives a direct source precedent for a synthetic formal statement that avoids degree arithmetic.

For formal work, the central contribution of Hartshorne is therefore not merely classification into neutral and Euclidean theorems. He repeatedly identifies the exact hidden obligation suppressed by a classical diagram: intersection existence, betweenness, side-of-line data, interior-of-angle claims, order, and uniqueness. These are precisely the obligations that become explicit proof terms in Lean.

B.6 Current checkpoint

This appendix now concerns Euclid Book I, Propositions I.1–I.32.

The structural picture at this checkpoint is sharp:

$$\begin{array}{ll} \text{Hilbert plane:} & I.1\text{–}I.28,\ \text{except I.1 and I.22;} \\ \text{Hilbert plane + (E):} & \text{restores the constructions I.1 and I.22;} \\ \text{Hilbert plane + (P):} & I.29\text{–}I.34\ \text{(the theory of parallels).} \end{array}$$

Within that classification, I.31 deserves separate emphasis: its existence statement belongs to the neutral layer even though its position in Euclid comes after I.29. Only uniqueness belongs to the Euclidean layer.

This appendix should be treated as the durable Hartshorne source audit for the I.1–I.32 checkpoint. Later propositions should extend the record rather than silently revise the distinctions established here.

Appendix C

Euclid I.27–I.32 in Wilkins's Study Notes

C.1 Purpose and scope

This appendix records a pilot source audit of David R. Wilkins's *Study Notes on Euclid's Elements*, Book I, Propositions I.27–I.32. Unlike the Greenberg and Hartshorne appendices, its primary object is the internal Euclidean dependency DAG: earlier propositions and Common Notions used, converse statements, alternative proofs, equivalences, and configuration variants.

C.2 Proposition-by-proposition audit

Prop.	**Euclidean dependency DAG**	**Wilkins additions**	**Significance for formalization**
I.27	I.16 $\rightarrow$ I.27. If the two lines met on either side of the transversal, one of the equal alternate angles would be an exterior angle of a triangle and hence greater than the remote interior angle.	Wilkins explicitly treats both possible sides of intersection and identifies I.27 as the converse of the Alternate Angles Theorem obtained later from I.29.	The proof DAG is exceptionally small. Parallelism is obtained from the neutral exterior-angle inequality; the formal burden is chiefly the two possible intersection orientations.
I.28	First branch: I.15 $\rightarrow$ I.27. Corresponding-angle equality is converted by vertical angles into alternate-angle equality. Second branch: I.13 + Common Notion 3 $\rightarrow$ I.27.	Wilkins treats the two criteria separately. The second argument displays the Common Notion cancellation step rather than hiding it in angle arithmetic.	I.28 naturally splits into two wrappers over I.27 after different angle normalizations. The geometric engine remains I.27.

Prop.	**Euclidean dependency DAG**	**Wilkins additions**	**Significance for formalization**
I.29	Fifth Postulate + I.13 → same-side interior angles equal two right angles → alternate and corresponding angle equalities.	Wilkins explicitly states that I.29 is the first proposition in Book I requiring the Fifth Postulate. He relates its conclusions to the converses already proved in I.27 and I.28.	This is the exact reversal of the preceding neutral direction: I.27–I.28 infer parallelism from angle relations; I.29 infers angle relations from parallelism.
I.30	Euclid's route: two uses of I.29 → equality of suitable angles → I.27 → transitivity of parallelism. A detailed variant inserts I.15.	Wilkins gives another route using corresponding angles and I.28. More importantly, he proves that I.30 is logically equivalent to Playfair's Axiom and shows that Playfair + I.27 implies I.29.	I.30 becomes a bridge between Euclid's DAG and an axiomatic interface for Euclidean parallelism, rather than merely the next theorem in the linear sequence.
I.31	I.23 → construction of an equal alternate angle; I.27 → the constructed line is parallel.	Wilkins separately notes that I.29 gives uniqueness of the infinite line through the point parallel to the given line.	The DAG isolates existence from uniqueness: I.23 + I.27 suffice for existence; I.29 enters only for uniqueness. This distinction should remain explicit in a Hilbert-style interface.
I.32	I.31 → auxiliary parallel; I.29 twice → the two required angle equalities; I.13 → the two-right-angles conclusion.	Wilkins also presents the proof attributed by Eudemus to the Pythagoreans and explains how its configuration is related to Euclid's by extension, reorientation, and relabelling.	The invariant mechanism is clearer than the particular diagram: construct a parallel, transport two angles using I.29, and close the straight angle. Equivalent configurations may have very different formal costs in Lean.

C.3 Dependency structure exposed by the pilot

The six Study Notes reveal the local structure

$$I.16 \longrightarrow I.27,$$

with I.28 obtained from I.27 after angle normalization using I.15 or I.13 together with Common Notion 3. The Euclidean transition is

$$\text{Fifth Postulate} \longrightarrow I.29.$$

Proposition I.29 then becomes the principal parallel-to-angle transport theorem. It feeds I.30, supplies the uniqueness observation attached to I.31, and is used twice in I.32.

A schematic local DAG is

$$\begin{array}{ccccc}
I.16 & \longrightarrow & I.27 & \longrightarrow & I.28 \\
 & & \downarrow & & \\
\text{Fifth Postulate} & \longrightarrow & I.29 & \longrightarrow & I.30 \\
 & & \downarrow & & \\
I.23 & \longrightarrow & I.31 & \longrightarrow & I.32.
\end{array}$$

This is intentionally schematic: I.31 does not depend on I.29 for existence, while I.32 depends on both I.31 and I.29.

C.4 Converses and equivalences

- I.27 is the converse of the Alternate Angles Theorem contained in I.29.
- The corresponding-angle part of I.28 is the converse of the Corresponding Angles Theorem contained in I.29.
- I.30 is logically equivalent to Playfair's Axiom.
- Playfair's Axiom together with I.27 implies I.29.
- I.31 proves existence using I.23 and I.27, whereas I.29 supplies uniqueness.
- The Euclidean and Pythagorean proofs associated with I.32 use different diagrams but share the same parallel-angle transport mechanism.

C.5 Pilot conclusion

The I.27–I.32 sample is sufficient to justify a separate Wilkins appendix. The strongest evidence is I.30: its Study Note gives alternative proof paths, proves equivalence with Playfair's Axiom, and reconstructs I.29 from Playfair plus I.27. Thus the linear sequence of propositions becomes a network of implications and equivalences.

For the formal project, the role of the Wilkins audit is to record the Euclidean dependency DAG, alternative proof paths, converses, equivalences, and configuration variants, and ultimately to compare these with the dependency graph emerging from the Lean/Hilbert reconstruction.

The next extension should proceed backward through I.1–I.26 without changing the structure established by this pilot. Later propositions can then be appended as the Book I reconstruction advances.

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